RZWQM2

Agricultural Systems Model

User Guide

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Chapter 1:
INTRODUCTION

1.1 **Welcome to RZWQM2**

The Root Zone Water Quality Model 2 (RZWQM2) is a windows application that includes the RZWQM2 Windows Interface (RZWQM2.exe) and the science simulation model (RZWQMrelease.exe). The RZWQM2 science code is a research level simulation model for Agricultural Cropping Systems. The RZWQM2 Science integrates many physical, chemical, and biological modules to accomplish this simulation. Details on the original science modules can be found in the Root Zone Water Quality Model (RZWQM): Modeling Management Effects on Water Quality and Crop Production (Ahuja et al., 2000). RZWQM2 enhancements include: the DSSAT 4.0 CROPGRO and CERES modules for crop growth, profile simulation to 30 meters, Quick-plant and Quick-turf module upgrades, addition of a Quick-tree module and some new simulation options. As with the original RZWQM, the RZWQM2 Science, can also be run in a DOS mode independent of the interface given the required input data files *(See Appendix 1).* However, the RZWQM2 Interface facilitates data management and user interaction with the simulation model through dialogs, spreadsheets, wizards, help and output visualization. This user manual is developed and expanded from an early version of RZWQM98 (Rojas et al., 2000).

1.2 **Minimum System Requirements**

- An x86 architecture compatible, CPU.
- Microsoft Windows 7 or superior.
- Display Resolution of 1024 X 768 pixels. When it is operated in this resolution, all entry screens, plot windows, and output viewer will appear as expected. Although the system can operate in lower resolutions, it is not recommended and some displays may not show correctly.
- Some interface functions may not work as expected in Windows Vista
- See Appendix 2 for minimum data requirements.
- 300 MB Free Disk space for the installation.
- Additional projects have variable space requirements based on number of scenarios and selected outputs.

1.3 **Program Availability**

RZWQM2 is available for download from:
USDA ARS Research Tab Scientific Software/Models site:
https://www.ars.usda.gov/research/software/download/?softwareid=412#downloadForm

Save the installation file “RZWQM SETUP.EXE” to your computer.

1.4 **Installation**

1.4.1 New Users

Run the RZWQM SETUP.EXE and follow the installation prompts to install RZWQM2. A "What's New" file will display at the start detailing latest features. Users may also specify the directory path for the install program.

1.4.2 Users Upgrading from Previous Versions

Previous users have two steps to installing the most current version: upgrading the RZWQM2 Software itself and upgrading previous scenarios if needed.

> **Important**: If you work in your install directory, backup the "RZWQM2\Projects" directory as well as any master databases you may have customized from the "RZWQM2\Data" or "RZWQM2\Databases" directories to an alternate location before uninstalling

Previous RZWQM Software must be uninstalled before installing the latest version. Use the uninstaller provided with your windows operating system.

On Windows Systems follow these steps:

- Go to START
- Control Panel
- "Add or Remove Programs", or "Uninstall a Program", or "Programs and Features"
- Find RZWQM2 in the list
- Right-click on RZWQM2 and select uninstall/change, or double click on it.

All files that were part of the initial installation will be removed from the directory where RZWQM2 was installed. After uninstallation is complete, navigate to the RZWQM2 install path and remove any installation folders that still exist from the "RZWQM2" install path.

You are now ready to run the install file of the latest RZWQM2 version "RZWQM SETUP.EXE." Follow the installation prompts changing install path if desired. Application installation is complete when FINISH is chosen.
After installing the RZWQM2 application, older scenarios may need to be updated as well. Scenario upgrades are only necessary when new science features require a change in input and its defaults. The install "WhatsNew.rtf" displayed at install time and located in the <install directory>/Bin will indicate if an import "for versions older than 1.x" is required. The user interface will also prompt the user for importing when an older scenario is opened. Import is located on the Scenario Menu under INPUT| IMPORT RZWQM *.DAT files. If you have used RZWQM2 version 1.0 with specifying DSSAT 3.5 crops to simulate, there is additional steps to update the DSSAT4.0 databases supplied with software for your previous crops. Instructions are located in the "RZWQMReadme.doc" in the DOCUMENTATION directory for the application. (See Chapter 5: How to Import Legacy Scenarios)

Tip: If you have modified RZWQM.DAT or RZINIT.DAT outside the user interface with a text editor, the INPUT|IMPORT function must be used to load the file changes into the interface screens for use in the RZWQM2 windows application.

Important Note: Running the model without importing the scenario when prompted will result in data corruption. Import reads many of the ".DAT" files (ASCII files) and updates older version files to the new and current standard. Before running the model, the RZWQM2 interface reads these updated files and writes new science model input files.

1.4.3 Users Upgrading from Previous Versions Who Prefer to Run in DOS mode without the RZWQM2 Interface

Follow the applications uninstall and install directions above to have a copy on your system. Copy the RZWQMRELEASE.EXE from the RZWQM2<INSTALL>\BIN directory to the scenario directory you have all the required input files (*.DAT) in *(See Appendix 1)* and you are ready to run in DOS mode. Or, you may write a DOS batch file to call the RZWQMrelease.exe from the \BIN\ folder.

For versions that require a scenario upgrade (INPUT IMPORT), There are two options to upgrade your scenario *.DAT files. Manually edit your files *.DAT files to add changes highlighted in the RZWQM2README.PDF "What's New Section" for the version; or, use the RZWQM2 interface to open/create a project with those scenarios in it and use ***Input | Import*** to bring the old RZWQM.DAT and RZINIT.DAT into the interface and ***Run*** and ***File | Save Scenario*** to produce new ".DAT" files for the science.

Note: If you copy a project to another name outside the user interface or you run RZWQM2 in the DOS mode, please be sure that the IPNAMES.DAT file directory paths are accessing the *.DAT input you want to run in that scenario directory. Since the RZWQM2 interface writes paths to the IPNAMES.DAT and *DSSAT.RZX file on RUN just before executing RZWQMRELEASE.EXE science, you may want to use the interface to update after you copy your project.

Quick Start

RZWQM2 provides an example project called "Projects" and a scenario called "New Scenario". You may open and save this project/scenario under a new name to run a scenario quickly; however, you are encouraged to build your own project with the ***FILE | NEW PROJECT*** function, locate it anywhere on your computer, and use the "default" example scenario to create it (See Chapter 4). Creating a "blank" new project or scenario is discouraged because the model needs basic parameters which can loaded from the default scenario/project. Remember to specify the climate files (*.met, *.brk, *.sno) from the scenario view drop down boxes before executing the run. You can always replace them with your own weather data once you have developed your study site climate files (See Chapter 4 on how to build weather files).

1.5 **Getting Help**

Apart from the information available in this manual, you can learn more about a particular command or feature in RZWQM2 by clicking on **Help | Help Topics** or clicking  on the Standard toolbar. The help system (developed with RoboHelp software) provides program navigation, data descriptions, and parameter estimations for all sections of the program. To get help while in a particular section, press the F1 key on your keyboard.

Help on the newest features not available with F1 can be found in the RZWQMREADME.DOC or in this guide located in <Install Dir>\DOCUMENTATION directory.

The Root Zone Water Quality Model Book is available at:
http://www.wrpllc.com/books/rzwqm.html

RZWQM Publications List can be found in <Install Dir>\DOCUMENTATION and example application publications can be found in <Install Dir>\DOCUMENTATION\Publications

Email Support can be reached at RZWQMsupport@ars.usda.gov
Training is provided on a need basis and travel to our location is at the user's expense.

1.6 **RZWQM2 New Changes**

What's New? : Version 4.2

- Hermes Sucros Crop Model and databases, N fixing nodules in Hermes and Generic Crops
- DSSAT CSM beet sugar (BS),sunflower (SU),alfalfa (AL),Bermuda Grass (BM)
- Generic Model Update
- Alfalfa with multiple harvest events (event 2 thru n has 0.0 plant population)
- Alternative to S-W PET Option to use ASCE Standardized reference crop (short or tall)
- SHAW AET as a plant water stress option
- GUI Bug fixes in Pesticides Ancestry Application Parent, Daughter, Granddaughter (WARNING No Copy Paste Allowed in Pesticides), experimental data below ground biomass input , and P-EST with CSM Cropgro
- Updates project summary, ana output, et deficit based on active rooting zone

What's New? : Version 3.41

- Project level parameter estimation/optimization (P-EST) across scenarios
- Update Science for plastic mulch

What's New? : Version 3.40

- Generic Alfalfa crop
- SHAW ET and plastic mulch

What's New? : Version 3.29

- Project summary updates in statistics computation and annual water/N mass balance output added to seasonal button.
- Harvest efficiency implementation for total aboveground biomass change
- Soil water redistribution occurs 24 hours a day

What's New? : Version 3.28

- GUI error fix project summary statistics
- Science update residue output

What's New? : Version 3.27

- GUI error fix on Meteorology|Edit *.met file and save
- Science update for over 100 years climate

What's New? : Version 3.25

- Science update for resetting soil water content in simulation and proso millet in rotation
- RZWQM2 P-EST update to optimize Potato and Millets
- GUI Site Description Horizon update for initializing P-EST horizons

What's New? : Version 3.20

- Science update for potato module
- GUI Project level Meteorology Create/Edit redesign

What's New? : Version 3.10

- Science update in comp2exp.out and add file horizon.out
- GUI vector variable output changed from Sat Hydr Conductivity to Hyd Conductivity**;** customize horizon.out depths in experiment data input.

What's New? : Version 3.00

- Public Version includes: Input Parameter Optimization, Non-Linear uncertainty analysis and running scenarios with uncertainty for confidence band statistics for output. Irrigation Rules with daily, seasonal, monthly or over year limitations. Sub irrigation available. Mass balance of Carbon, Nitrogen, Water and Pesticides. GHG emissions and CO2 effects. Options added SHAW ET model running with observed canopy or model, ability to reset to observed soil water during the run and new plant water stress algorithms. Sediment Erosion. Storm data smoothing tools. Multiple Crop Growth models. Observed vs. Predicted Performance statistics and graphics.

What's New? : Version < 3.00

- See RZWQMReadme.pdf in the RZWQM2/Documentation Directory

Chapter 2:
RZWQM2 Workspace

Overview of the Workspace

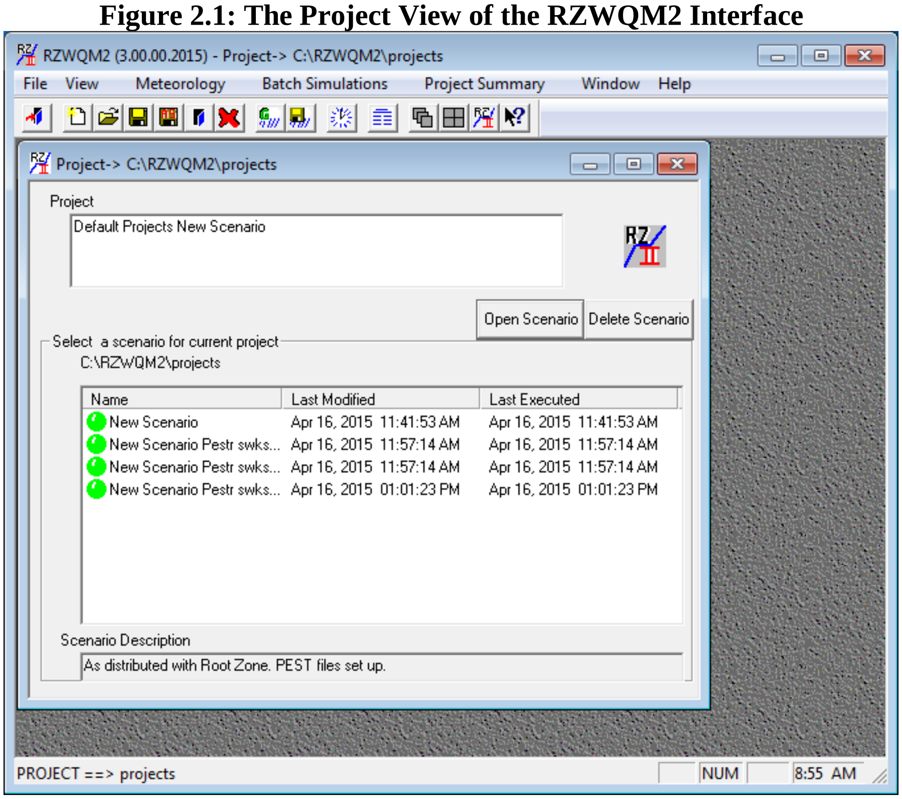
Before you begin any work in RZWQM2, it is essential to be familiar with the program workspace and the project/scenario paradigm. This will help you navigate the interface and understand the directory structure. Starting RZWQM2, the project window opens the default project ("projects") or the last opened project in the project view (Figure 2.1).

2.1 **Project & Scenario Paradigm**

A project can be named after a study you want to model (e.g. farm, experiment station, an experiment). A scenario is a particular treatment (site-specific resources and management) under your project. It may also be a phase of a treatment (representing each crop in rotation for a given year) under your project.

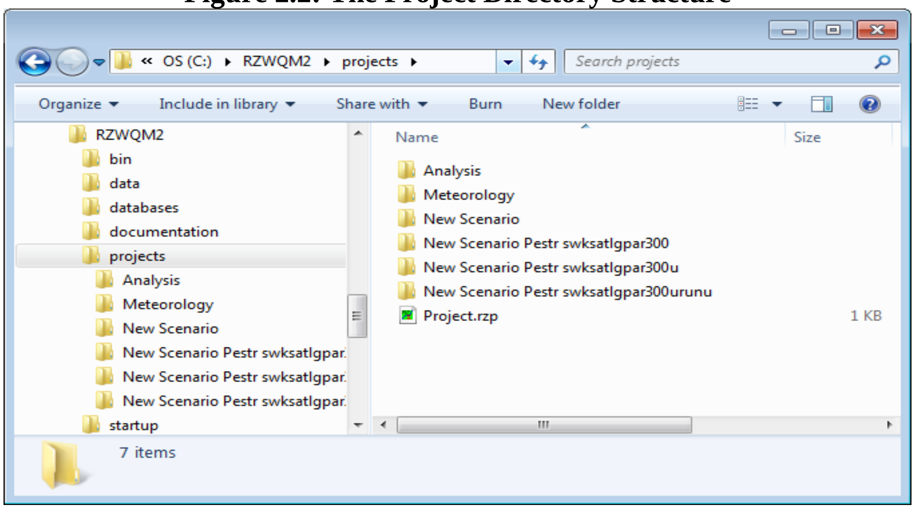
Important Note: Project and Scenario file names can include interior space, upper and lowercase letters, and digits 0 to 9 as well as [ ] { } ′ −. Do not use any of the following system reserved characters / \ : ? * " < > | or any other special characters in a filename.

Figure 2.1: The Project View of the RZWQM2 Interface



The project is a structure that contains all meteorology files (Meteorology folder), analysis output files for each scenario (Analysis Folder), and individual scenarios (subfolder for each named scenario). Since meteorology files are shared by all the scenarios and analysis output files (“<Scenario-Name>.ana”) are saved in for statistical analysis across scenarios, they are under the heading of project-level information. A project may have one or many scenarios (folders). The “PROJECT.RZP” file present in the *project directory* will be recognized by RZWQM2 to open the directory as a project (Figure 2.2).

Figure 2.2: The Project Directory Structure

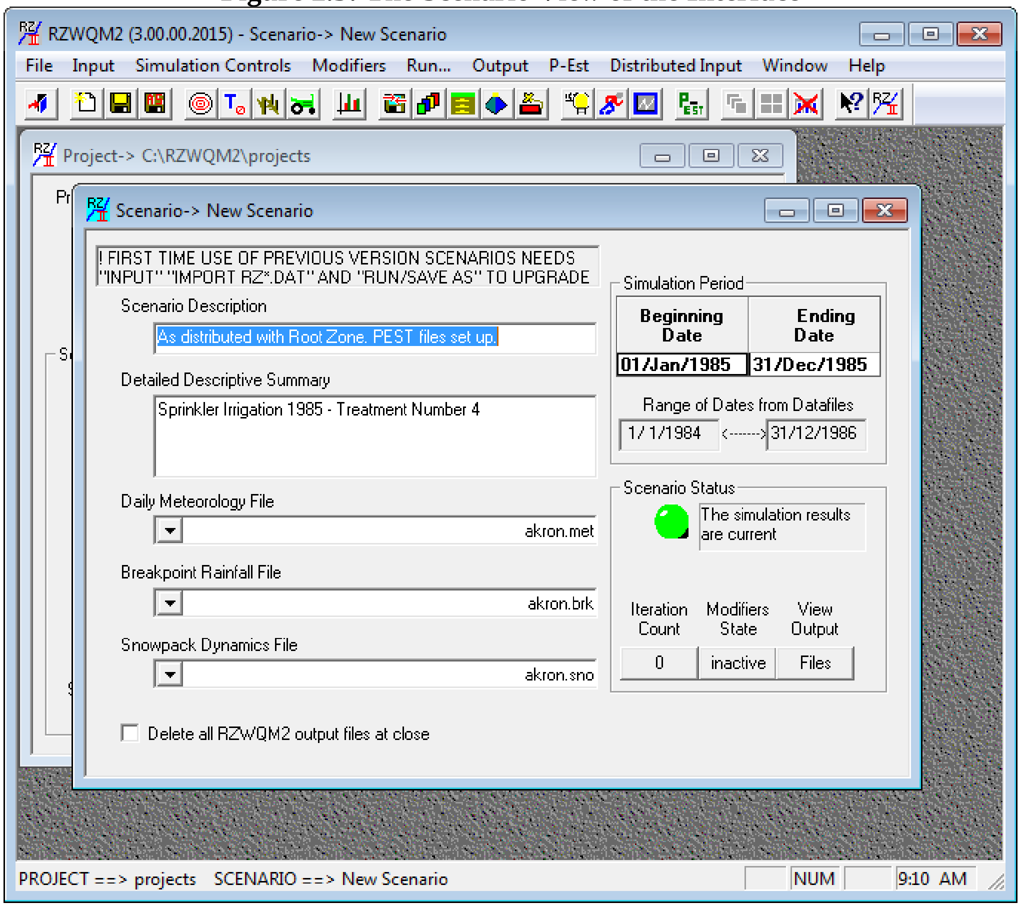


Note: Projects are independent directories of the application and should be located outside the install directory (it may be on any drive).

Warning: Double clicking *project.rzp file in the windows explorer can have unexpected results among XP and Win7*. It is recommended to open projects using the RZWQM2 user interface with FILE|Open Project and select the *project.rzp* file.

Selecting a scenario name and clicking ***OPEN SCENARIO*** (or double clicking on the scenario) results in a scenario view (Figure 2.3). The scenario represents a treatment of the study (with respect to site description, initial system state, residue state of the system, and management practices). Model input and output files are stored within the scenario directory. Graphical output can be accessed by the scenario menu ***OUTPUT|AUTOPLOT*** function and tabular from **OUTPUT|VIEW FILES** menu item or ***VIEW OUTPUT FILES*** button on the scenario window (Figure 2.3).

Figure 2.3: The Scenario View of the Interface

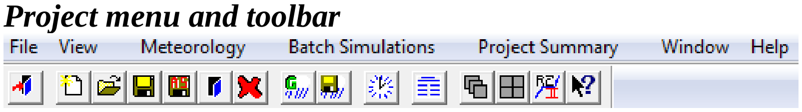


Important Note: Only 1 project and 1 scenario should be open at a time in the RZWQM2 workspace. Use ***File | Exit*** on the scenario view to return to the project workspace view. It is discouraged to pop between project and scenario windows. On the project view screen, use ***File | Exit*** to close your project and Exit the program. Closing with "X" on the upper right corner will prompt the user to save their information.

2.2 **Menus And Toolbars**

The menus provide access to all the commands available when in the project or scenario windows. These are used the same way as menus in other Windows programs. The most commonly used menu items are also reflected by toolbar icons. Tool tips display the function of each toolbar icon. There are two menus in RZWQM2: the project menu and the scenario menu.

- ***Project menu and toolbar***



Contains project file commands such as: ***New, Open, Save, Save As, Close, Delete and Exit*** the RZWQM2 application. View toggles the status bar. Meteorology accesses the

Cligen90 weather generator and editing/creating data files (*.brk, *.hmt and *.met) from scratch. Batch Simulations allows scenario execution batch mode. ***Project Summary*** gives access to analysis tools for scenario comparisons. There is also window arrangement and help.

- ***Scenario menu and toolbar***

Contains commands specific to Scenario: ***New, Save, Save As, Exit/Save, and Exit/No Save***. Input required from the user include: site description, initial state, crop residue state, management operations, PRMS Snow parameters (default file). Input also includes optional entering of Experiment data and importing (upgrading legacy files). *Simulation Controls* access simulation output variables, breakthrough curve nodes, debugging output selection, and simulation iteration control. ***Modifiers*** are an interface "what if" tool that allows quick relative changes in driving variables and management practices. Final menu items access to running simulations and viewing output graphs and files.

Note: The IPNAMES.DAT file in the scenario directory has path locations for all files to be used as input by the science. For DSSATCSM 4.0 users, the **.RZX* file specifies input files for the crop module. Both these path files can only be updated by opening the scenario and saving it in the RZWQM2 interface. Run is also recommended.

Descriptions of Projects and Scenarios

Both projects and scenarios allow you to enter brief and/or detailed descriptions to help differentiate one from another. The scenario brief description is shown on the scenario selection when the name is highlighted with the cursor.

Scenario Selection List on the Project View

All available scenarios are listed and available for selection. The scenario name followed by the last modification date and last execution date are also listed to show how up-to-date the output results are compared to the input parameters (Figure 2.1). Scenario control functions are available when you click on the right button while the mouse pointer is in this list.

Simulation Dates on the Scenario View

The simulation start and ending dates are selected in this section (Figure 2.3). A handy calendar is displayed when you double click on the date selection. The date range in the meteorology files is displayed in the fields below the date selections. The beginning and ending date of the run will default to the range of dates in the meteorology file selected. The user may select any shorter time period.

Meteorology File Selection on the Scenario View

This selects the meteorology files to be used such as breakpoint rainfall (*.BRK), daily temperature, wind, and radiation (*.MET), and parameters related to snow melt (*.SNO) (Figure 2.3). Hourly temperature, wind and radiation files (*.HMT) may be used instead of daily files in RZWQM2 Version 2 and later. Meteorology files may be used by several scenarios in the same project. A scenario may use any set of meteorology files located in the projects METEOROLOGY folder (Figure 2.2).

Simulation Status and Power Buttons

The simulation status is displayed as a green or red ball to show delinquent simulation results. A green ball indicates that the output is current to the input provided, while the red ball indicates out of date output (Figure 2.1 and 2.3). The simulation status is not applicable if you edit *.DAT files outside the interface and run scenarios in DOS mode.

The power buttons in the lower right of the scenario view allow quick access to iteration control, modifiers section, and viewing of ASCII output files (Figure 2.3). The text label on the power buttons changes to reflect the current state of the iteration control and modifier usage.

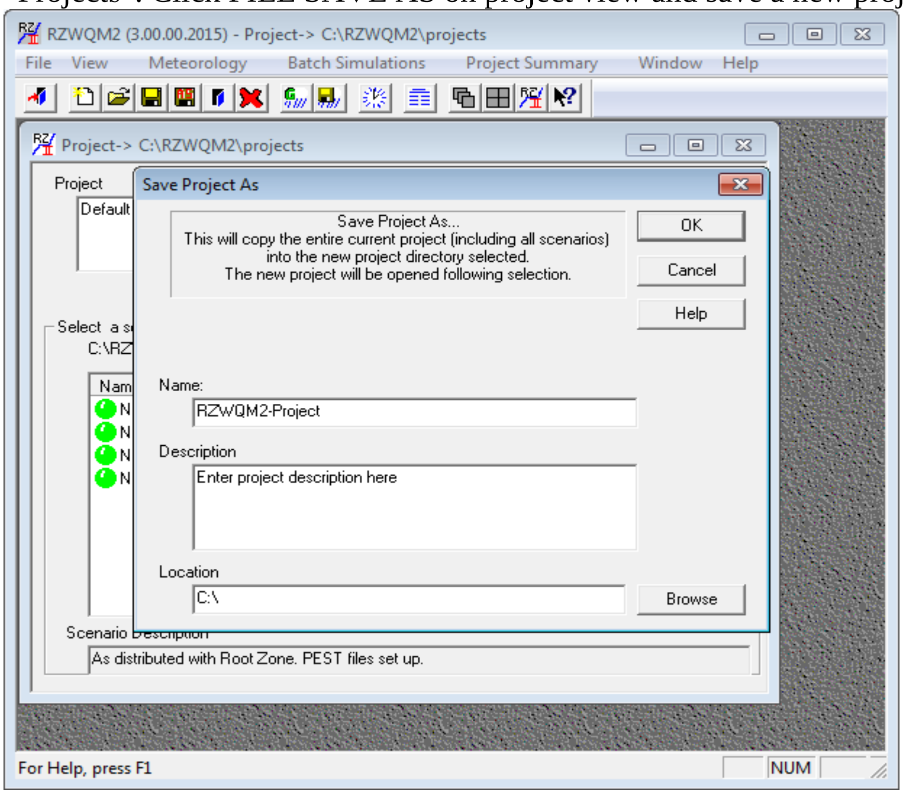
Status Bar

This bar displays at the bottom of the RZWQM2 application window and lists messages in the lower left corner and the time in the lower right corner. This can be shut off through the project menu under ***VIEW*** (Figure 2.1).

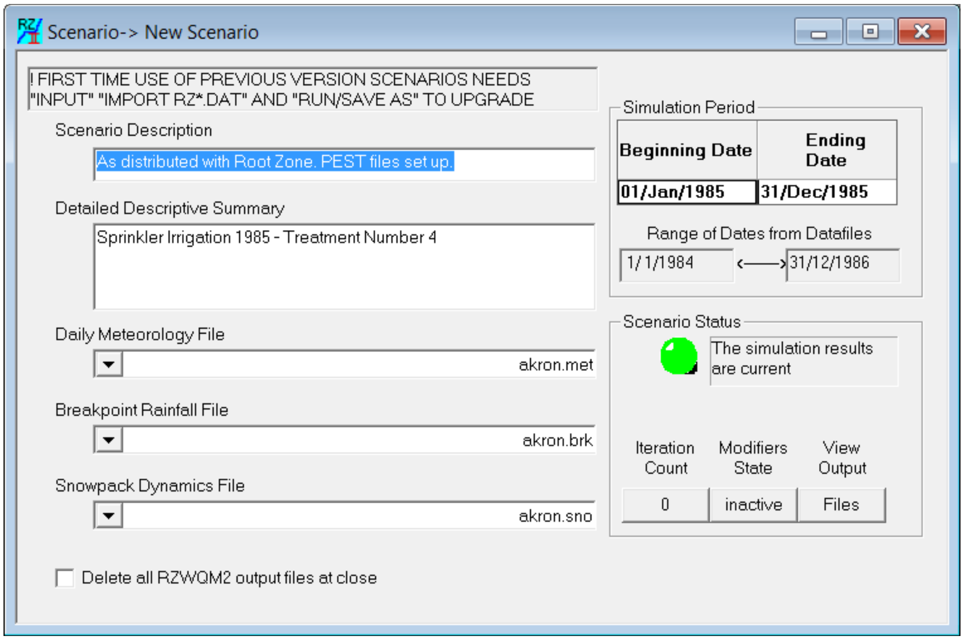
Chapter 3:
Getting Started

Now that you have become familiar with the program workspace and the project/scenario paradigm, you are ready to work with RZWQM2. When you install RZWQM2, it includes a default project called "projects" and scenario called "New Scenario" for you to use in creating your own. In a few easy steps you can start RZWQM2, save a new project from the example, save a new scenario from the default, view scenario level input and output, run the simulation model, save scenario result and compare among scenarios at the project level. Other example projects are located in the "DATA" directory.

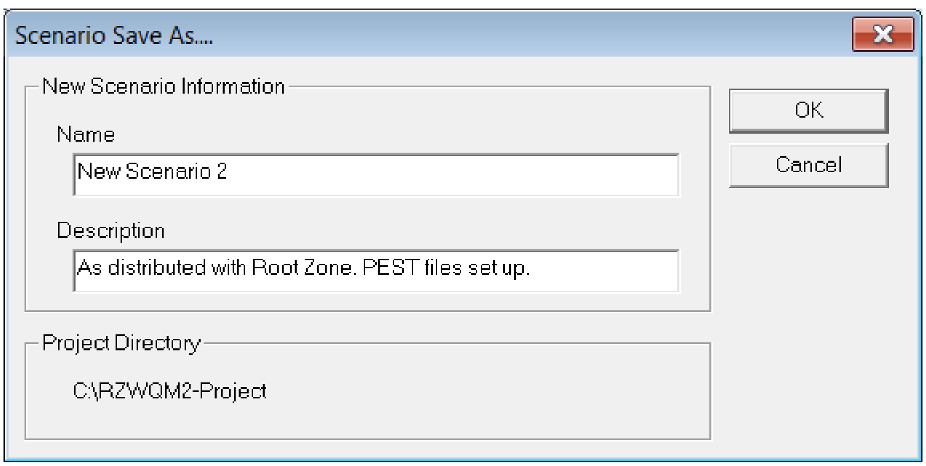
1. From the desktop, execute RZWQM2. The program will open the default "Projects". Click FILE SAVE AS on project view and save a new project.



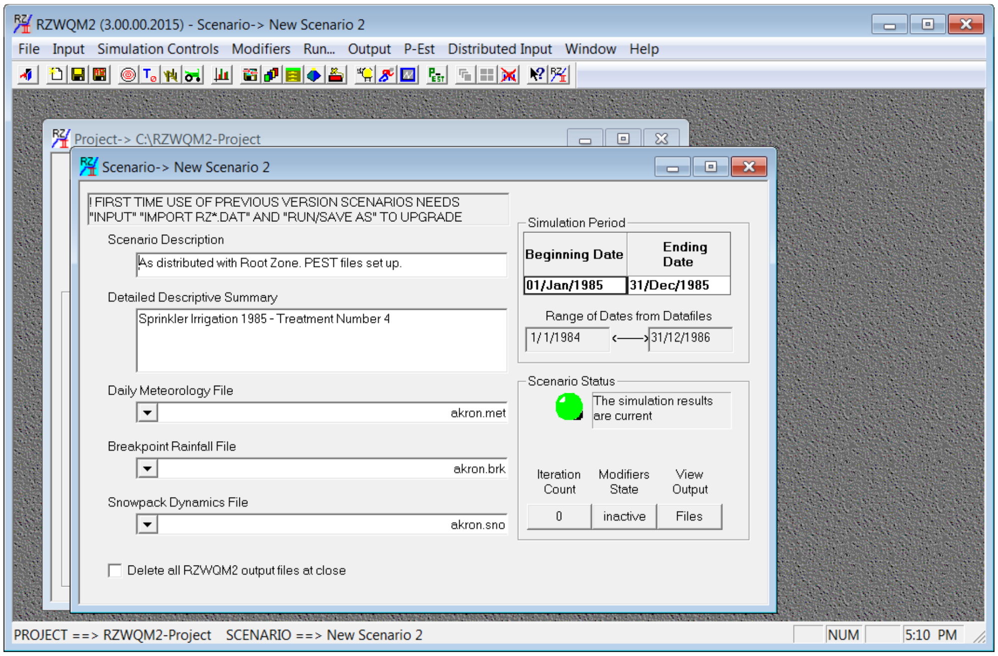
2. Click on ***New Scenario*** in the column labeled Name, then click on the ***Open Scenario*** button to open the scenario under your new RZWQM-Project**.** **Tip:** Double Click in the same region will select and open the scenario.



3. On the menu click ***FILE | SAVE SCENARIO AS*** to New Scenario 2. ***OK*** saves input files for New Scenario to New Scenario 2.

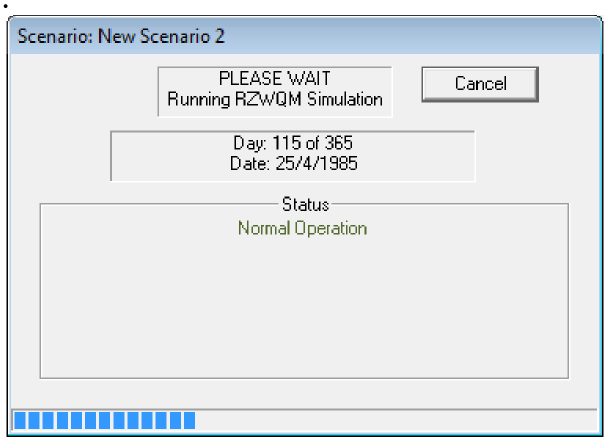


4. The Scenario view screen will then appear with the default simulation period, descriptions, and meteorology, breakpoint rainfall, or snowpack files selected. Notice the Application Window Title and the Scenario View Window title is New Scenario 2.



5. Make sure that input data files are selected in each of the boxes labeled Daily Meteorology File, Breakpoint Rainfall File, and Snowpack Dynamics File. If initially there is none selected, click on the arrow at the right of the field for options. **Note:** Snowpack Dynamics (*.SNO) contains parameters for the PRMS snow model (Leavesley et. al., 2006) not daily snow data.

6. Click on ***RUN SCENARIO*** or icon  on the scenario toolbar. You should see a running progress window at the center of your monitor notifying the status of the execution at the selected time intervals. **Note:** *.DAT files will be rewritten at this time and the science model executed. If you have changed *.DAT files outside the user interface and wish to run them within the interface you must use INPUT| IMPORT to bring them into the interface before run. Progress of run is reported in the RZWQM2.LOG file and any mass balance problem is reported in MASSBAL.OUT.

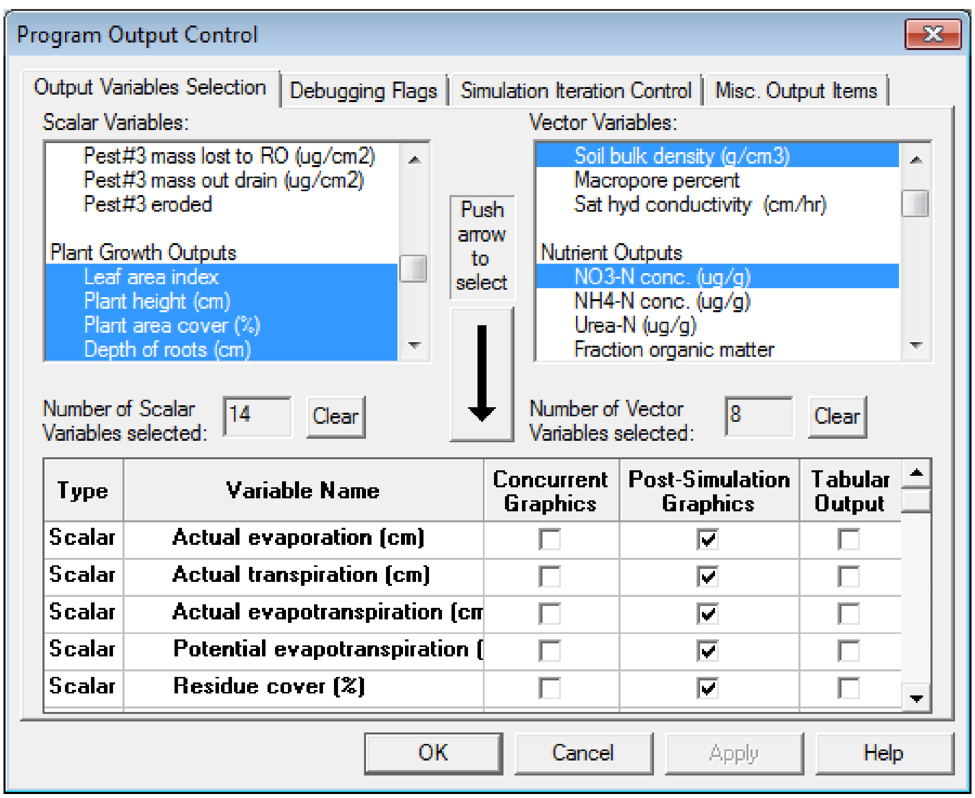


7. Click on Scenario view menu ***OUTPUT | VIEW OUTPUT FILES*** or use the ***VIEW OUTPUT FILES*** button on the Scenario view screen and select file type *.LOG.

8. Select **RZQWM2.LOG** and View. The log tells you how the run executed day by day. NORMAL TERMINATION appears at the bottom of a successful run that has an expdata.dat file to compare observed versus simulated results. "Observed data are not available ---model run successful" appears at the bottom of a successful run if expdata.dat was not included for the scenario. Any error messages will display in the RZWQM2.LOG file as well.

9. View other file types of interest to make sure the run went as expected. MANAGE.OUT lists the implementation of all management practices in the run. MASSBAL.OUT records mass balance problems; whereas MBLNIT.OUT, MBLWAT.OUT, and MBLCARBON.OUT detail nitrogen, water and carbon balances, respectively.

10. View *.OUT files for results or use "Output Auto plot" for graphics.
- RZWQM.OUT and if DSSAT Crop Chosen OVERVIEW.OUT. If experiment data were compared, view COMP2EXP.OUT. (See Appendix 8)
- Output Auto Plot allows you to select files (*.PLT) to plot to the window. DAILY.PLT has scalar values and LAYER.PLT has vector values (i.e. soil layers).



Note: Plot files were generated based on the users input selection of output variables under “Simulation Controls” and the check for Post-Simulation Graphics.

11. ***SAVE*** the run with ***FILE | EXIT and “Yes” save***.
Important Note: Changes and the new run will not be saved until you ***FILE | EXIT and “Yes” save***.

12. You are now returned to the project view of the program. Project Summary will now have access to the general RZWQM2 output files for the scenario called <scenario name>.ANA to compare results across scenarios. Use control click to select multiple scenarios, select the statistics and summary time period, then ***GENERATE SUMMARY***. For example, minimum soil water will be the same among the two.

Note: It is recommended to exit the scenario before doing activity at the project level.

Chapter 4:
Creating a Project for the study site

In RZWQM2, the project is the file drawer where you store all your simulation information. You can edit/create meteorology data, setup multiple scenarios, analyze simulation results across scenarios, and have access to all the scenarios.

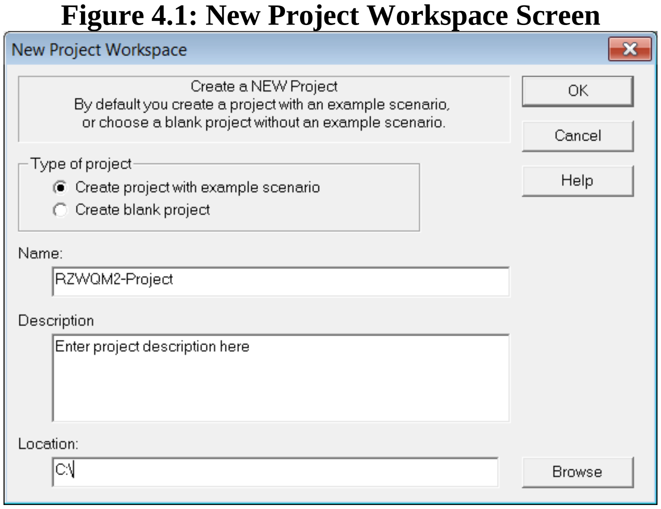
4.1 **How to Create a New Project**

A default or the last project used will display as open when RZWQM2 starts.

1. Click ***FILE*** on project menu or on the toolbar, ***and then CLOSE PROJECT***.
2. Click ***FILE | NEW PROJECT***.
3. Create the project with example scenario (default scenario) (Recommended Figure 4.1). The default scenario is copied from its location in your RZWQM2 <INSTALL DIR>/STARTUP directory to your new project. A project can be created blank but this is not recommended as it will be completely empty of required data as it creates just the project subdirectory structure.
4. Fill in the project name (i.e., Study Area)
5. Enter the project general description (i.e. Location) and it will always show in your project view.
6. Enter or browse to select the location you will store the project.
   Important Note: Do not create a project under an existing RZWQM2 project directory
7. Click ***OK*** to complete.
8. Click ***FILE | SAVE PROJECT*** and proceed to opening scenario created.

Tip: A new project may also be created by ***File | Open Project*** of your choice then ***File | Save Project As*** with a new name. You can then customize *.MET and *.BRK files as well as scenario content.

Figure 4.1: New Project Workspace Screen

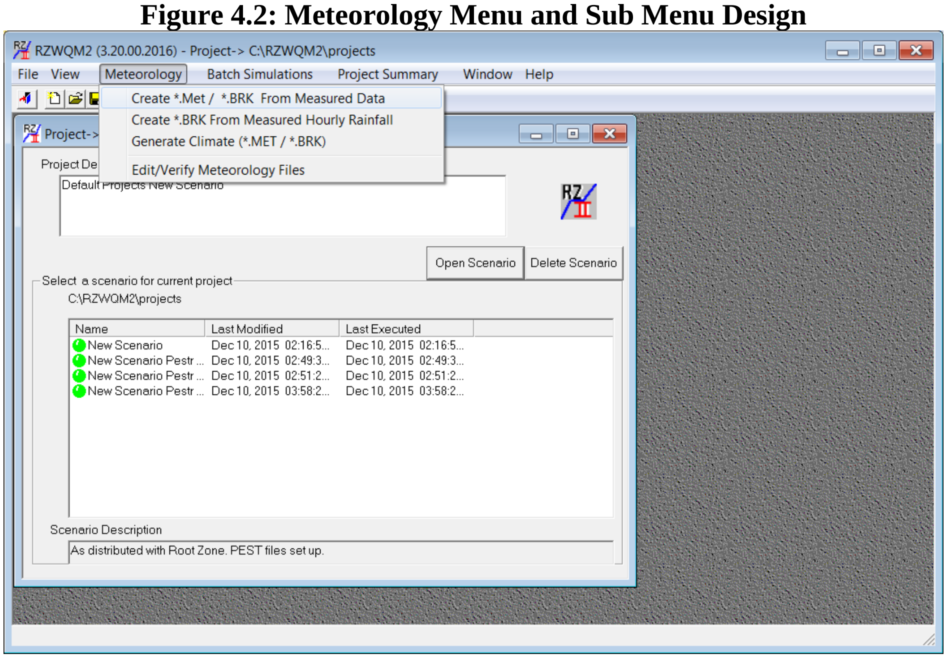


4.2 **How to Generate/Create/Edit Meteorological Files**

Meteorology files can only be created or edited at the project level. Created files may contain generated or measured data. Figure 4.2 ***Create | Generate Climate*** will use the CLIGEN90 weather generator at the user specified U.S. weather station to generate data based on historical statistics. Generated weather files use the same root name with different extensions: (.MET) for daily meteorology, hourly meteorology (.HMT), and (.BRK) for breakpoint rainfall. In Addition, a snow file (.SNO) for snowpack dynamics is required. The snow file **(***.SNO) is copied directly from the default scenario and given the same root name matching the generated *.MET and *.BRK files.

Important Note: The snow file is not daily snow data but an input parameter file for the PRMS snow module which can be accessed via WordPad from the scenario menu INPUT| PRMS SNOW PARAMETERS. Daily snow data and snow pack is derived from your daily precipitation input, temperature below freezing and the snow pack module.
It is not recommended to use generated climate, measured weather data is preferred.

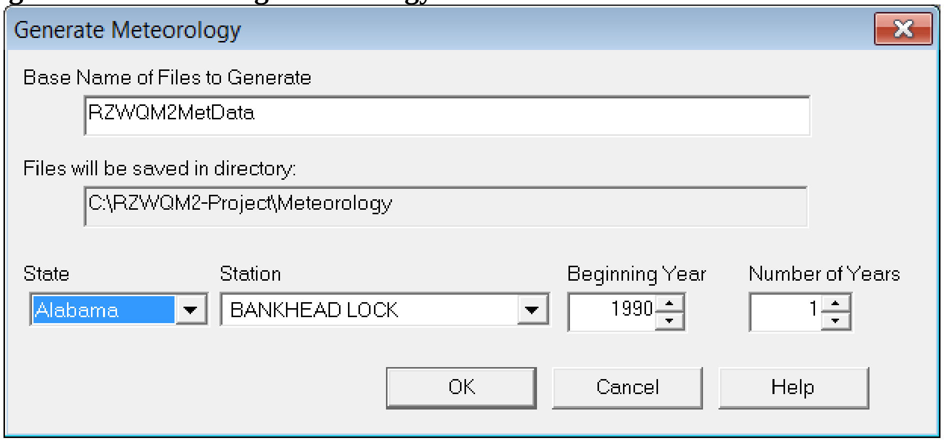
Figure 4.2: Meteorology Menu and Sub Menu Design



How to Generate:

1. Open the project you want to generate files for.
2. Click ***METEOROLOGY | GENERATE*** on project menu or  on the toolbar.
3. Input the Base Name for the weather files (Figure 4.3)
4. Select state and weather station for GLIGEN90 weather generator.
5. Enter the starting year and number to generated. Created files are automatically saved in the Meteorology folder.
   Tip: Select the closest weather station to your site and include the starting year of your experiment or study. Generated data can be used until better values are available.

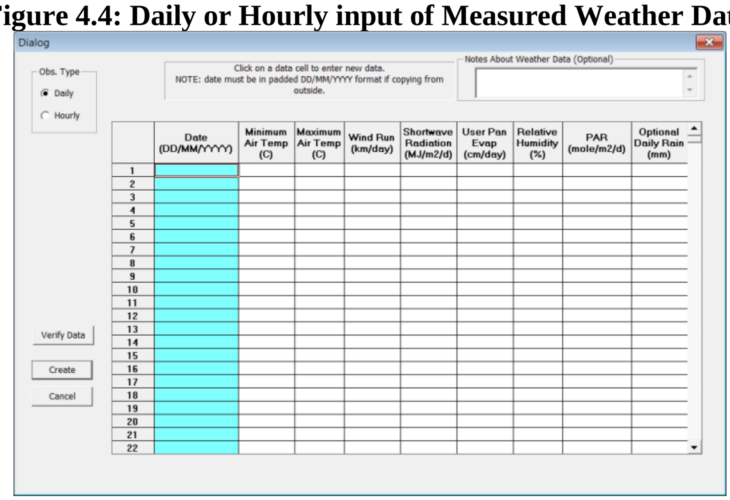
Figure 4.3: Generating Meteorology File with Cli90 Climate Generator Screen



4.3 **How to Create Meteorology files from Measured Weather Data:**

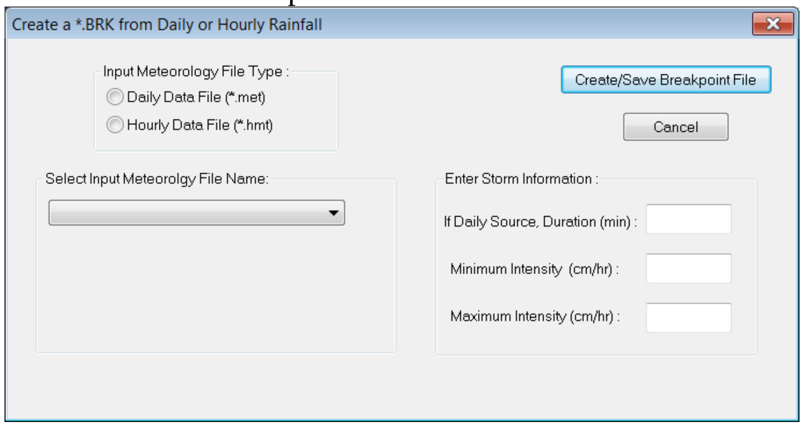
1. Given you have opened the project you want.
2. Click ***METEOROLOGY | CREATE *.MET / *.BRK FROM MEASTURED DATA*** on the project menu.
3. Select whether you want to enter daily or hourly weather

Figure 4.4: Daily or Hourly input of Measured Weather Data



4. Select from an external Excel sheet the rows and columns you wish to copy and paste them into the spreadsheet (Figure 4.4). Entering Rainfall as >=0.0 amounts will still allow you to auto-create your storm level file (*.brk) in step 8.
5. You can optionally enter notes on the weather data and source.
6. Select ***Verify Data*** to check the data for appropriate ranges and consecutive dates.
7. Select ***Create***, enter a name and the *.met (daily) or *.hmt (hourly) file will be created for you.
8. The next step prompts you for information to create the breakpoint storm data file. In RZWQM2 Version 3.0 and earlier (Figure 4.5) the item "Create Breakpoint from Daily or hourly rainfall" was a standalone menu item. It has been integrated to the meteorology file creation for your convenience.

Figure 4.5: Create a storm breakpoint file from measured data.



1. The radial button will default to the type of weather data entered previously daily meteorology file (*.MET) or hourly meteorology (*.HMT) and the file name.
2. Storm Intensity Information is defaulted and can be over ridden by the user. Minimum and Maximum storm intensity is used whether you are generating from daily or hourly rainfall to determine the range of intensity acceptable in the breakpoint file. Storm duration is used only when daily data is provided and 1 storm is set for the day from 0.0 minutes after midnight to the specified duration. Storm durations are calculated from hourly data and more than one storm can occur within the day.
3. Click **CREATE/ SAVE** to save the breakpoint file created. The name will default to the file name of origin with BRK extension; but can be changed by the user as long as “BRK” remains the extension for a breakpoint file.
4. Click **CANCEL** to exit the screen without creating the file.

Important Note: Meteorology input files may not contain blank lines, missing dates or non-consecutive dates. If a rainfall column is null or 0.0 in the *.MET or *.HMT file selected a break point file will be created but empty of storm information. If rainfall is >0.0 the breakpoint will be created from measured data.

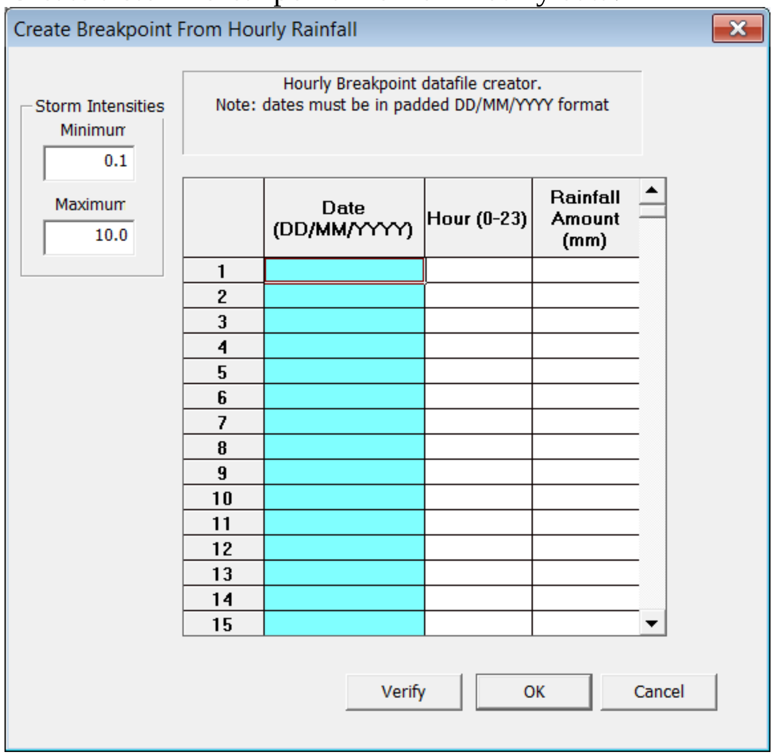
4.4 **How to Create Breakpoint files from Measured Hourly Rainfall:**

This menu item allows the user to take hourly rainfall from another source and use the create breakpoint file for storms. Therefore, a user can create a daily meteorology file and not create a storm a day; but rather, enter more detailed rainfall (mm) by hour to create the breakpoint file.

1. Click **METEOROLOGY | CREATE BRK.* FROM MEASURED HOURLY RAINFALL**.
2. Figure 4.6 spread allows you to copy from an external EXCEL and paste hourly rainfall.
3. Enter the minimum and maximum allowable cm/hr storm intensity.
4. Verify Data, which checks dates consecutive 24 readings per day and amount >=0
5. Days with no rain must be fully present as 0.0 for 24 readings.

6. OK will create the breakpoint (*.brk) file.

Figure 4.6: Create a storm breakpoint file from hourly data.



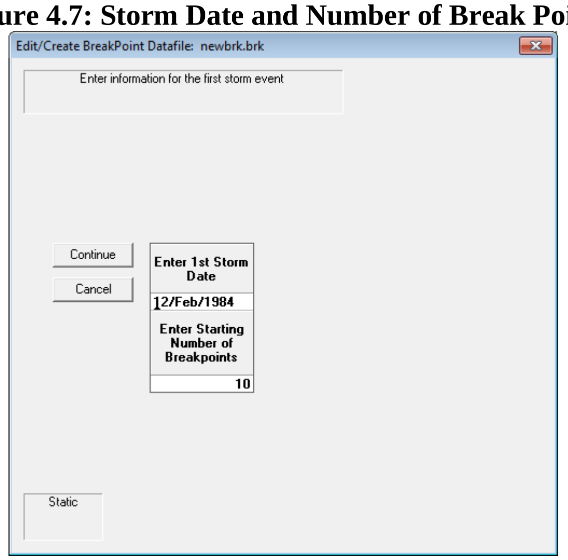
Important Note: If your application of RZWQM2 is addressing erosion issues with storms, the most detailed storm level information is critical. Automated breakpoint file creation from daily or hourly data may not be sufficient. We have a VBA utility program external to RZWQM2 that can read time stamp tipping bucket or weighing plate data and create *.BRK file with assumptions.
We also still allow manual creation of the *.BRK under **METEOROLOGY | EDIT** in the RZWQM2 GUI.

4.5 **How to Create Breakpoint Rainfall Data Manually Entered Storms:**

In addition to automated breakpoint file creation from daily or hourly rainfall we have left a manual creation of the breakpoint file in the RZWQM2 functionality although it is rarely used for its tedious nature it fully explains how to enter storms to become *.brk files.

1. Given you have opened the project you want the created files for.
2. Click ***METEOROLOGY | EDIT*** on project menu or on the toolbar. Display names for breakpoint rainfall (*.BRK) file names which you can select to edit OR
3. To create a new file, simply enter a new name and a blank file will be created for you.
4. ***Open*** on the *.BRK file type created and enter data as follows:
   A. Enter the first storm date and the number of point you measured the storm then ***CONTINUE***.

Figure 4.7: Storm Date and Number of Break Points



   B. Breakpoint rainfall consists of points (time followed by the amount on the entry screen). The starting time of the storm is measured in minutes after midnight. There are a total of 1440 minutes in a day. The amount should be an accumulated amount throughout the storm. Multiple storms may occur on the same day.

   Figure 4.8 shows an event starting 300 minutes after midnight with 0.5 inches at 350 minutes after midnight and 0.1 inch coming between 350 and 400 minutes after midnight. The storm event ended at 600 minutes after midnight for a total of 2.6 inches of rain received from the storm.

   Figure 4.9 summarizes the total for the storm over 300 minutes. A second storm could be inserted below for the same day February 12th, but the first break point time would need to be 601 minutes after midnight as the first storm in Figure 4.8 ended at 600 minutes after midnight.

Figure 4.8: Curve for a Rain Event with Break Points

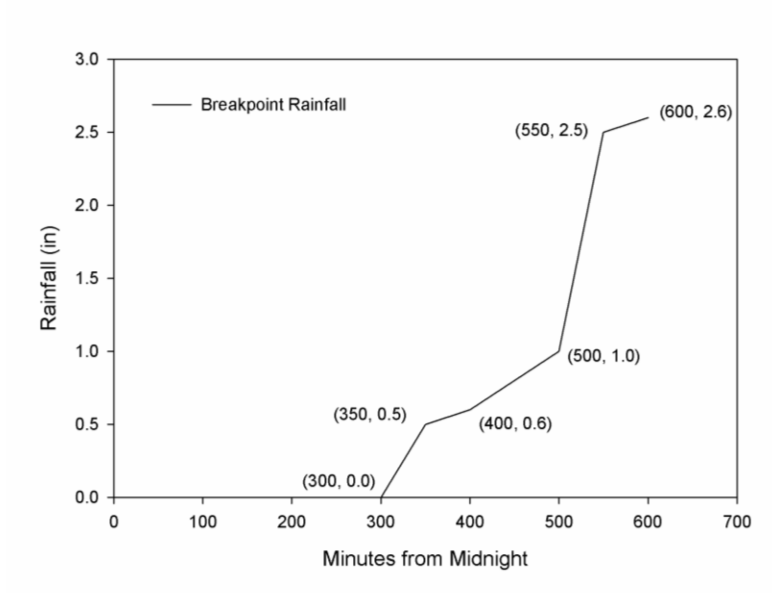
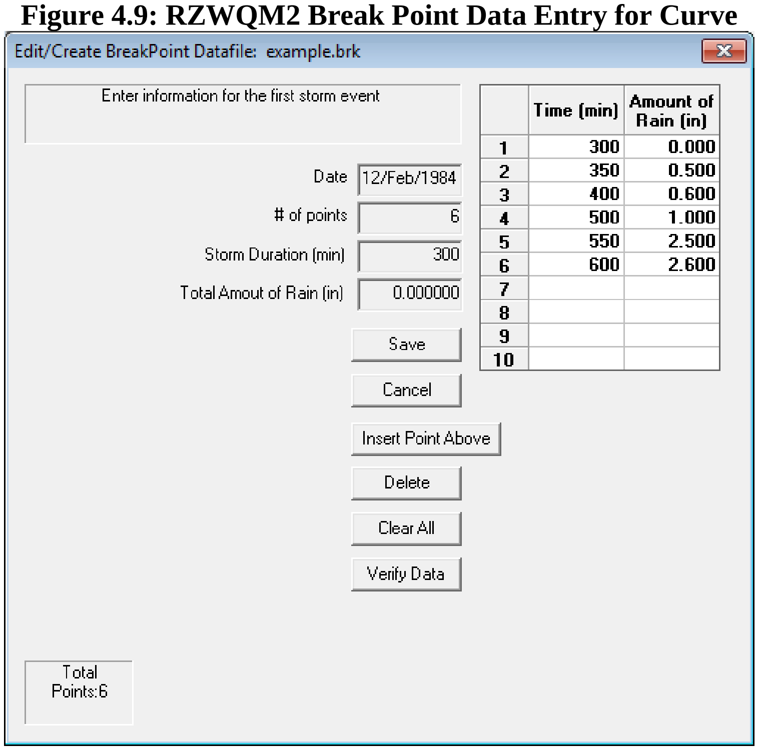


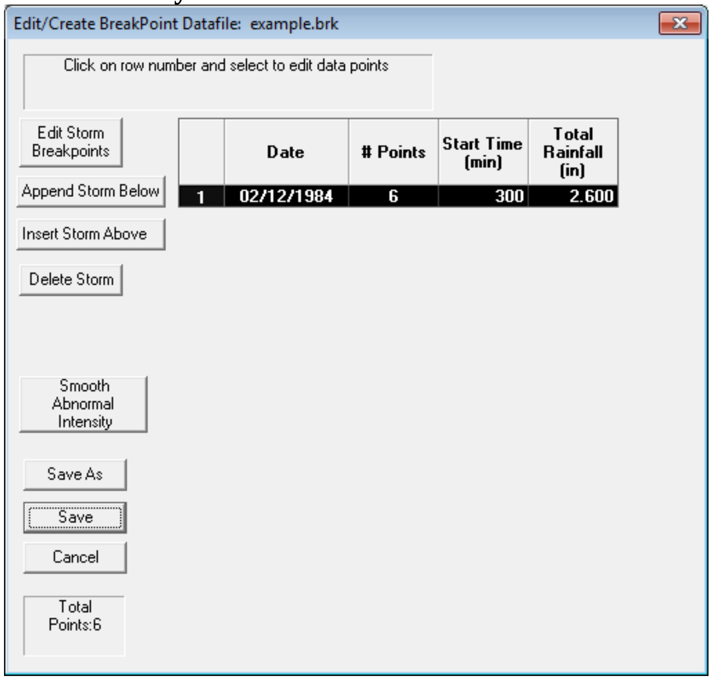
Figure 4.9: RZWQM2 Break Point Data Entry for Curve



C. ***SAVE***

D. The Storm Summary (Figure 4.10) will display and can be used to enter more storms with ***Insert*** or ***Append Storm.***

Figure 4.10: Summary of the Storm Event after Break Point Data Entry



Note: Total Rainfall in the *.BRK file is in inches! GUI entry of rainfall is in millimeters and is converted to inches for the *.BRK file.

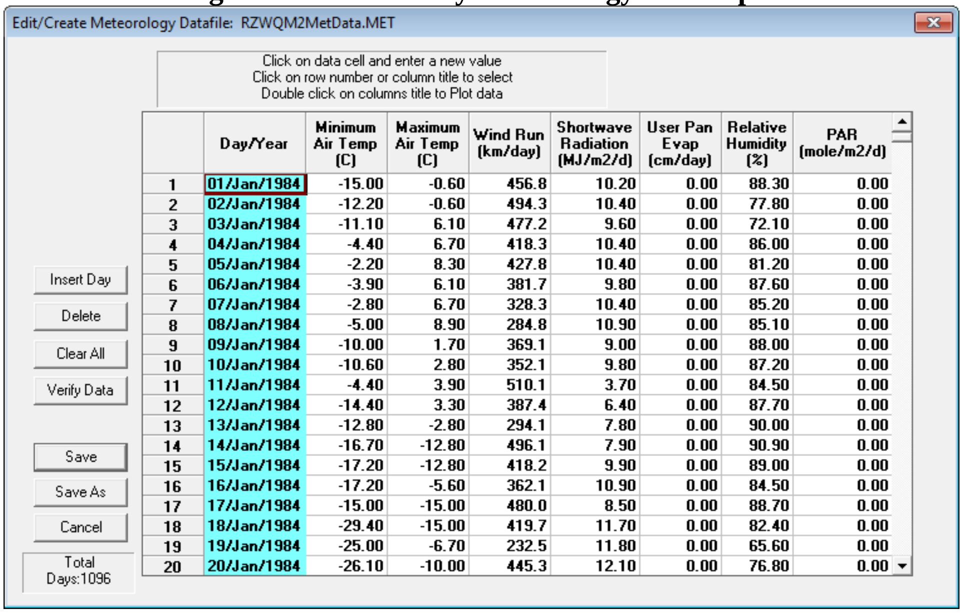
E. ***SAVE*** or ***SAVE AS*** to save the breakpoint file.

F. The snowpack dynamics file **(***.SNO) is copied from the default scenario with the FILE SAVE SCENARIO AS command. Users may modify this file by opening the scenario and under ***Input | PRMS Snow Parameters*** (See Appendix 4 for parameters).

Important Note: During storm infiltration no evapotranspiration (ET) is allowed. Extremely long storm events will affect ET and plant growth.

Note: Minimum storm duration of 5 minutes is recommended.

Figure 4.11: Edit Daily Meteorology Data Input



	Day/Year	Minimum Air Temp (C)	Maximum Air Temp (C)	Wind Run (km/day)	Shortwave Radiation (MJ/m2/d)	User Pan Evap (cm/day)	Relative Humidity (%)	PAR (mole/m2/d)
1	01/Jan/1984	-15.00	-0.60	456.8	10.20	0.00	88.30	0.00
2	02/Jan/1984	-12.20	-0.60	494.3	10.40	0.00	77.80	0.00
3	03/Jan/1984	-11.10	6.10	477.2	9.60	0.00	72.10	0.00
4	04/Jan/1984	-4.40	6.70	418.3	10.40	0.00	86.00	0.00
5	05/Jan/1984	-2.20	8.30	427.8	10.40	0.00	81.20	0.00
6	06/Jan/1984	-3.90	6.10	381.7	9.80	0.00	87.60	0.00
7	07/Jan/1984	-2.80	6.70	328.3	10.40	0.00	85.20	0.00
8	08/Jan/1984	-5.00	8.90	284.8	10.90	0.00	85.10	0.00
9	09/Jan/1984	-10.00	1.70	369.1	9.00	0.00	88.00	0.00
10	10/Jan/1984	-10.60	2.80	352.1	9.80	0.00	87.20	0.00
11	11/Jan/1984	-4.40	3.90	510.1	3.70	0.00	84.50	0.00
12	12/Jan/1984	-14.40	3.30	387.4	6.40	0.00	87.70	0.00
13	13/Jan/1984	-12.80	-2.80	294.1	7.80	0.00	90.00	0.00
14	14/Jan/1984	-16.70	-12.80	496.1	7.90	0.00	90.90	0.00
15	15/Jan/1984	-17.20	-12.80	418.2	9.90	0.00	89.00	0.00
16	16/Jan/1984	-17.20	-5.60	362.1	10.90	0.00	84.50	0.00
17	17/Jan/1984	-15.00	-15.00	480.0	8.50	0.00	88.70	0.00
18	18/Jan/1984	-29.40	-15.00	419.7	11.70	0.00	82.40	0.00
19	19/Jan/1984	-25.00	-6.70	232.5	11.80	0.00	65.60	0.00
20	20/Jan/1984	-26.10	-10.00	445.3	12.10	0.00	76.80	0.00

4.6 **How to Edit Existing Meteorology Files:**

1. Given you have created a *.MET file.
2. Click ***METEOROLOGY | EDIT*** on project menu or  on the toolbar. Existing daily meteorology (*.MET), hourly meteorology (*.HMT) or breakpoint rainfall (*.BRK) file names will be displayed, from which you can select to edit.
3. If you ***Open*** on the *.MET file type: (Figure 4.11)
   A. It can be edited with cut/paste or selecting a row and doing insert/delete.
   B. Click **VERIFY DATA.** Row numbers with data outside acceptable ranges will display in the top message box. Correct data and repeat until message reads “Data Successfully Validated”
   C. Click **SAVE** to save the data set at any time.

   Important Note: Pan evaporation is not required. If a value is entered other than 0.0 it will be used as daily potential evaporation. By default PAN is 0.0 and PET is calculated from the modified Shuttleworth-Wallace equations. Photosynthetically active radiation (PAR) has been added to meteorology input. It is entered after Relative Humidity or defaulted to 0.0. If entered the DSSAT crop growth model simulation control option “Entered PAR” can be specified as yes to use the values. Not shown here in EDIT mode is an optional final column in the *.met file for daily rain (mm). Rainfall is only available in CREATE mode so breakpoint files can be created from it. Precipitation can only come through the breakpoint file for simulation not *met or *hmt.

   Note: If you double-click a column title, a graph of that data will be displayed on the back screen. Users may also want to verify the consistency and range of meteorology data. The indicator box in the left hand corner displays pertinent election and summary information.

4. Since entering data is tedious in large files even with spreadsheet cut and paste you may wish to use an external spreadsheet or statistical package to output the MS DOS ASCII text whitespace file format of the meteorology files *.MET or *.HMT or *.BRK according to the file formats existing in the example Projects/Meteorology folder.

   Important Note: All comments in these ASCII files are denoted by an "=" in the first column and no blank lines should occur in the file if you use a spreadsheet to create weather files.

4.7 **How to Edit Existing Breakpoint Rainfall File:**

Figure 4.10 is displayed by selecting ***METEOROLOGY | EDIT*** and ***OK*** on the *.BRK file selected at the project level. Storm events can be inserted, appended or deleted. The indicator box in the lower left hand corner displays total number of storms defined. The breakpoint rainfall data file (*.BRK) can be edited by selecting a storm and **Edit Storm Breakpoints**. Figure 4.9 will display. You may enter or edit breakpoints and ***SAVE*** will re-write the *.BRK and return you to the storm summary (Figure 4.10). Total duration of rainstorm is calculated automatically.

Chapter 5:
Creating a Scenario for the Study Site

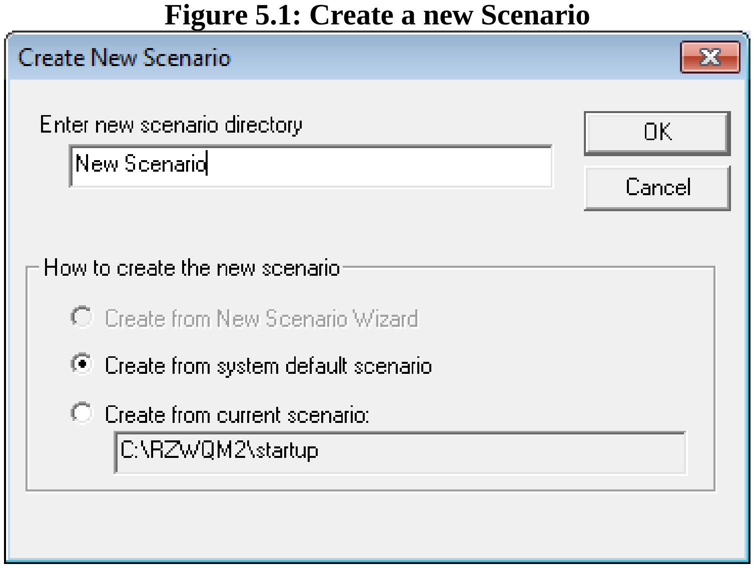
Setting up Scenarios for the study site

A scenario is where you define your system, such as soil horizons, macropores, subsurface drains, water tables, background rainfall/irrigation chemistry, potential evapotranspiration, nutrient dynamics, surface crop residue, incorporated crop residue, crop selection, management practices, pesticide selection, and initial soil status. Meteorology data files should be selected for each scenario.

To help you go through this process of parameterization there are several input wizards that provide guidance, access to databases, and tool tips that give recommended values with acceptable ranges for input. In some cases limits on input values are enforced, while others allow you to use your best judgment. Of course, you can press the ***F1*** button on your keyboard to access the help system for a more detailed description and parameter suggestions at any time. It is important to visit every data screen to enter site specific information for your study site.

5.1 **How to Create a New Scenario**

Figure 5.1: Create a new Scenario



To create a scenario on the project level:

1. With a project open, locate the mouse pointer on the scenario list window
2. Click the **right** mouse button
3. Select ***NEW SCENARIO***
4. Specify a new name, the default or the startup directory is the same scenario
5. ***OK***

6. Select and ***OPEN SCENARIO*** to work with.
Tip: Right Click on the project view will also allow you to copy a scenario from another project. Select the scenario directory, OK, browse up a directory to the projects meteorology directory, select and SAVE each meteorology file for that scenario to the current project. You may then open the scenario and work with it.

To create a scenario on the scenario level:

1. Click ***FILE | NEW SCENARIO*** or on the toolbar.
2. Specify the new name.
3. Scenarios can be created from system default scenario or current scenario open.
4. ***OK***
5. Select and ***OPEN SCENARIO*** to work with.

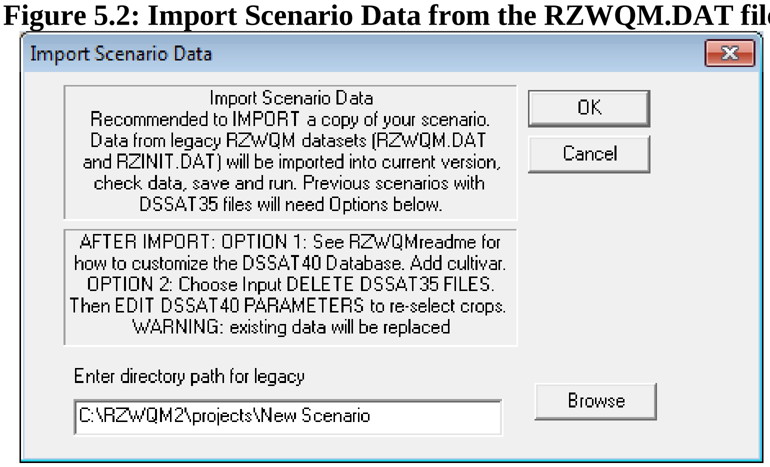
Tip: New scenarios may also be created by opening one and using ***FILE | SAVE AS*** or on the toolbar, and specifying new name and description and OK. If you modify scenario input, make sure to use ***FILE | EXIT*** and/or ***SAVE SCENARIO*** to save any key changes.

5.2 **How to Import Legacy Scenario Data Files**

RZWQM2 has the capability of importing data files from previous RZWQM versions. Making backup copies of the project is always recommended before importing scenarios to upgrade to the latest RZWQM2 interface.

1. Open the Project and Scenario to upgrade. A warning will indicate if it is a legacy scenario and advance you to **Import Scenario Data** dialog (Figure 5.2). If you cancel (i.e. you want to save to a new name first), you may then import manually with ***INPUT | IMPORT RZWQM *.DAT FILES*** on the scenario menu.
2. Click on ***OK*** on Figure 5.2 to import the current legacy scenario open.
3. For those who used RZWQM2 version 1.0 with DSSAT3.5, follow the additional directions on import and see the program readme documentation on updating to DSSAT4.0 crop cultivar database.
4. After import ***RUN*** and ***FILE | SAVE SCENARIO*** to complete scenario upgrade.

Figure 5.2: Import Scenario Data from the RZWQM.DAT file



Warning: Browse to another scenario directory will import input files from that directory to the scenario you currently have open.

5.3 **Define Your System**

Under the ***INPUT*** on the scenario menu or toolbar, there are four sub-menus that help users easily set up RZWQM2: Site Description, Initial State, Residue State, and Management Practices. Each sub-menu or icon brings up a set of tabbed dialog boxes.

Important Note: Click ***APPLY*** before leaving the dialog tab you are focused on to save the data. ***OK*** will also save, but will exit you from the sub-menu input section entirely.

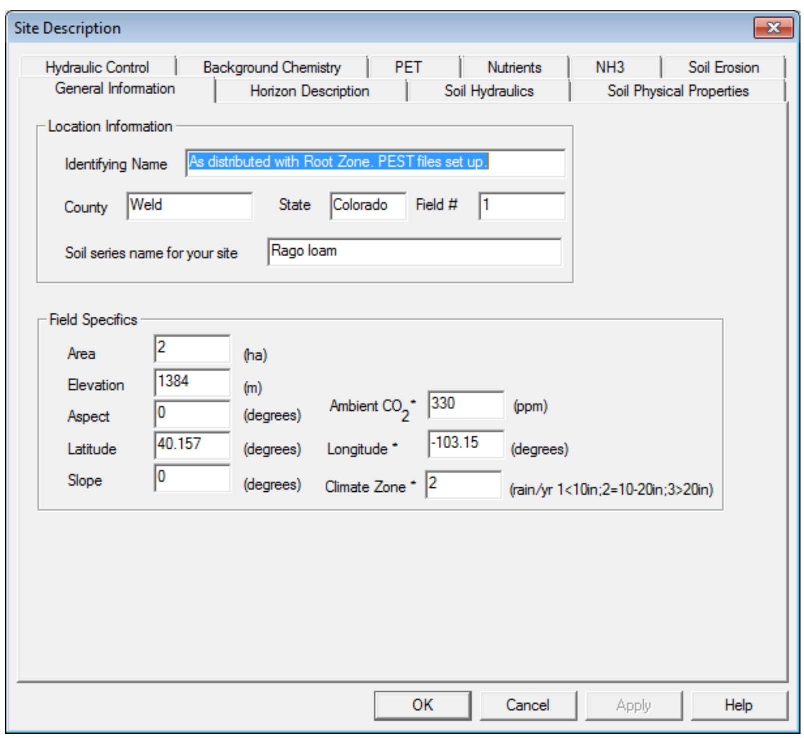
5.3.1 Site Description

INPUT | SITE DESCRIPTION or on the toolbar brings up a set of tabbed dialog screens to describe the physical, chemical and biological properties of the site.

General Information

The general information tabbed dialog includes location information and field specifics (Figure 5.3). Location information helps you recall characteristics of the plot or field in the scenario that may not fit in the scenario name itself. Field specifics determine the site's geographic and topographic attributes. Aspect is referenced in degrees to true North.

Figure 5.3 : Site Description Gen. Info. Contains Descriptive, Geographic, Topographic Info

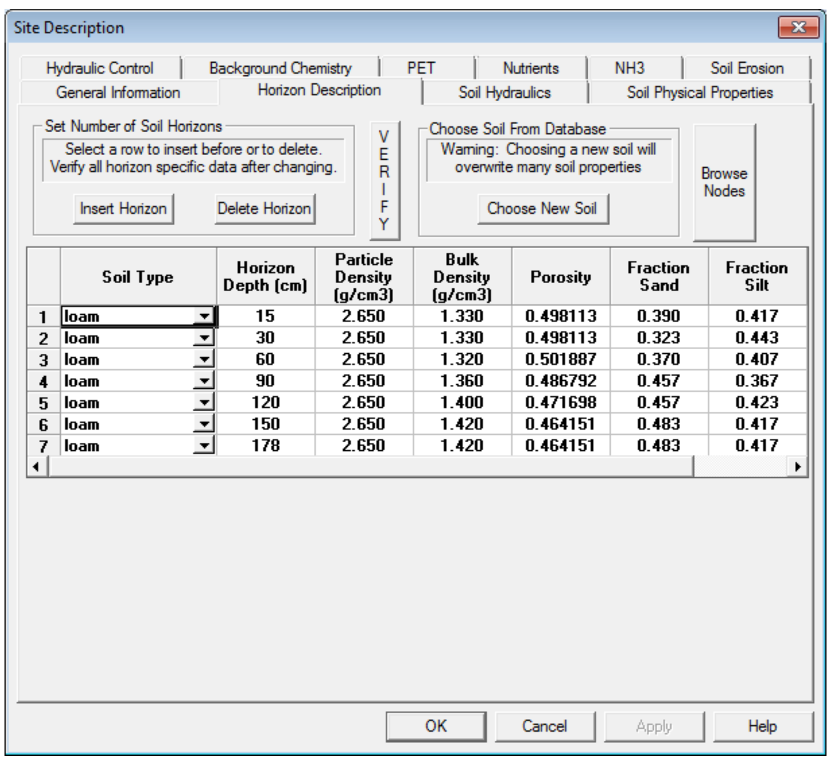


Important Note: "*" denotes new variables in Version 2.0 and 2.8. Longitude and climate zone are used in ET calculations. Science 2.0 upgrades regarding incoming radiation angle could impact results as the full effects of slope and aspect are simulated. Version 2.8 brings ambient CO2 to user access. DSSAT CSM 4.0 models assume 330 ppm which is default. Note changing the ambient ppm may require recalibration of crop parameters in the *.cul file.
Note: Field Specifics can affect the energy balance on your site.

Soil Horizon Description

Select ***Horizon Description*** to describe the soil horizons and their attributes (Figure 5.4). Spreadsheet content can be entered or chosen from the NRCS Pedon Database. Default saturated hydraulic conductivity (Ksat) and water contents not loaded from the database will be estimated from the horizon soil texture and preloaded in the Soil Hydraulics Screen; however, they can be overridden by the user when choosing a method to estimate the Brooks-Corey parameters for the RZWQM2 science model. There is a maximum of 10 soil layers for user input. The maximum profile depth simulated is 30 meters. The maximum number of numerical layers generated to accomplish profile simulation is 300; therefore, numerical layer thickness may vary depending on the depth of profile to simulate. Browse nodes show the node number at any given depth.

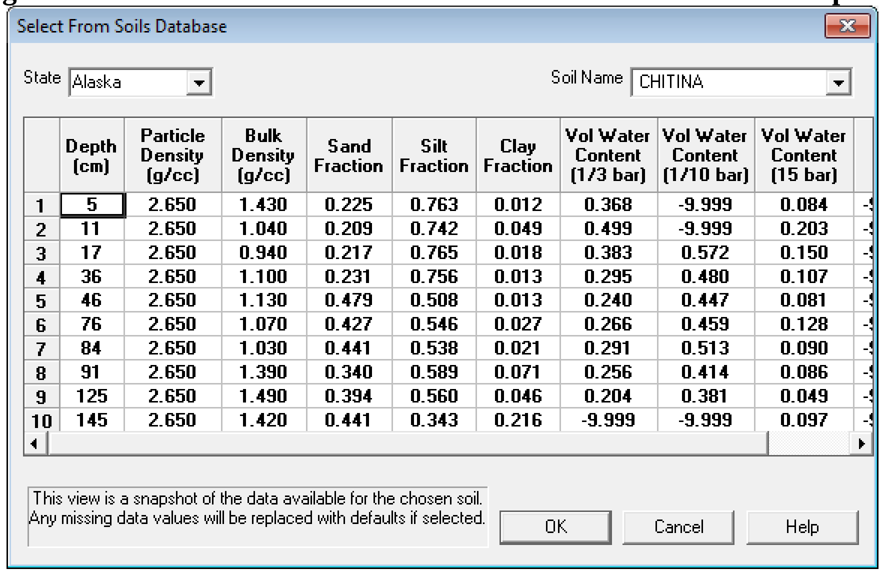
Figure 5.4: Site Description: Horizon Info. - Soil Profile Thickness, Texture, Bulk Density, Porosity, and Sand/Silt/Clay fractions.



Create a Soil with Load a New Soil from the Database:

- Click **CHOOSE NEW SOIL** and Figure 5.5 will result.

Figure 5.5: Create New Soil from Available NRCS Pedon Soil Descriptions



	Depth (cm)	Particle Density (g/cc)	Bulk Density (g/cc)	Sand Fraction	Silt Fraction	Clay Fraction	Vol Water Content (1/3 bar)	Vol Water Content (1/10 bar)	Vol Water Content (15 bar)
1	5	2.650	1.430	0.225	0.763	0.012	0.368	-9.999	0.084
2	11	2.650	1.040	0.209	0.742	0.049	0.499	-9.999	0.203
3	17	2.650	0.940	0.217	0.765	0.018	0.383	0.572	0.150
4	36	2.650	1.100	0.231	0.756	0.013	0.295	0.480	0.107
5	46	2.650	1.130	0.479	0.508	0.013	0.240	0.447	0.081
6	76	2.650	1.070	0.427	0.546	0.027	0.266	0.459	0.128
7	84	2.650	1.030	0.441	0.538	0.021	0.291	0.513	0.090
8	91	2.650	1.390	0.340	0.589	0.071	0.256	0.414	0.086
9	125	2.650	1.490	0.394	0.560	0.046	0.204	0.381	0.049
10	145	2.650	1.420	0.441	0.343	0.216	-9.999	-9.999	0.097

1. Select state of interest
2. Select soil series of interest
   Note: Series names may be repeated in the list. This means that multiple surveys have been included in the database. The soil may have been phased on different surface texture, slope or other attributes but is still classified in the same series.
3. Spreadsheet will fill with data (-9.999 is missing data), click ***OK.***
4. New horizon data will now show in horizon description (Figure 5.4) and you may want to modify these soil parameters to customize to your site.

Edit Existing Soil:

1. Set the Soil Type (Texture Class) (according to your measured sand silt clay or soil survey) and depth for each Horizon (Limit is 10 horizons) (Figure 5.4)
   Important Note: When soil type changes in the drop down box a set of default soil hydraulic parameters are loaded. Users may modify these later under **Soil Hydraulics**. Modifying the sand, silt and clay fractions may change the soil type but will not load the default soil hydraulic properties for the new soil type. Hydraulic properties only change by a re-selection in the soil type drop down or using one of the estimation methods.
2. Click **VERIFY**, this verifies whether the general layers you entered can be subdivided into the numerical nodes used in solving the Richards' equation. It may change the last horizon for you or ask you to change 1 or 2 horizons to coincide a horizon boundary with a numerical node boundary.
   Important Note: In general, all horizons must be not less than 1cm and the first horizon must be greater than 2 cm thick. Consistent horizon thickness is recommended.

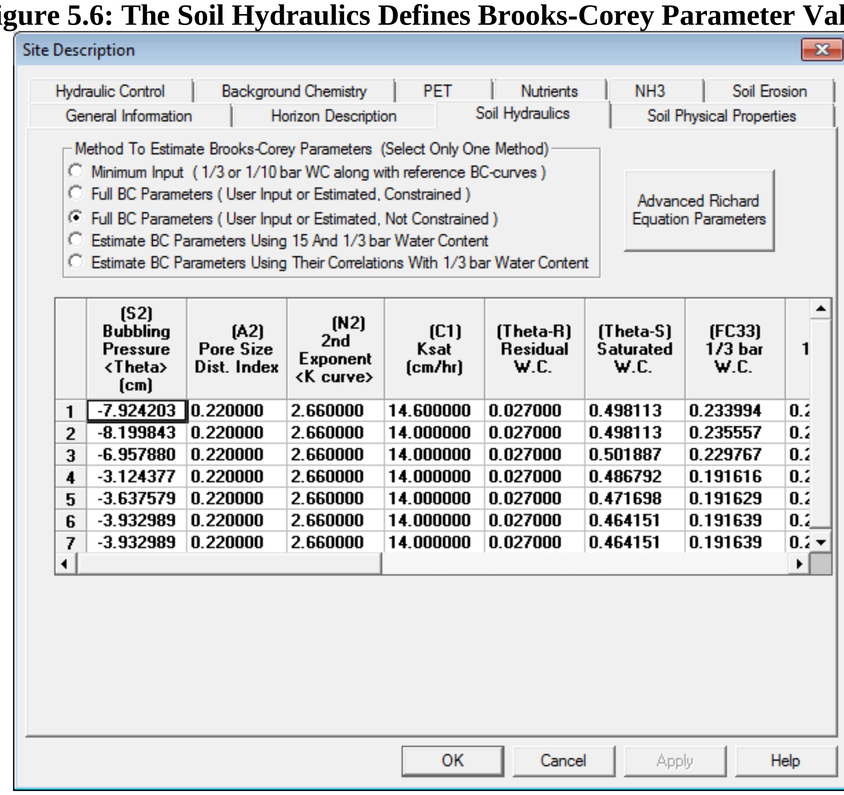
3. Once the data has been successfully verified, go to the end of the spreadsheet and enter sand, silt and clay, then enter bulk density. Let porosity default unless you have measured values.
4. ***APPLY***

Soil Hydraulics

The Full Brooks-Corey (BC) soil hydraulic property parameters required by RZWQM2 science may be entered directly in the full description (not constrained or constrained) sheet of the Soil Hydraulics screen (Figure 5.6). Other methods of estimating the BC parameters are available by choosing one of the following radial buttons: from soil texture (Minimum Input), from 1/3 and 15 bar water content, and from water content correlations with 1/3 bar water content only (One parameter model). When an "Estimate BC.." radial is chosen and O.K. from the screen will run the equation and when you return the results will be displayed under the full radial. Only Minimum or Full parameters are written for science execution. Details are found under Soil Hydraulic Property Estimation Methods section. Number of iterations and convergence criteria of the Richards' Equation can be modified using the Advanced Richard Equation Parameters button on the same page.

Note: All soil hydraulic property estimation methods use 1/3 bar water content.
Important Note: Water content is volumetric in these screen entries.

Figure 5.6: The Soil Hydraulics Defines Brooks-Corey Parameter Values



How to Operate the Soil Hydraulics Screen:

1. Decide which method you want to fill the full description sheet with estimates.
2. Choose that method by selecting ONE radial button.
3. Fill in required information (1/3, or 1/10 or 15 bar water content and residual). Ksat is estimated based on the soil texture. Retain the estimate or enter your own or put -1,-2, or -3 in the field for it to be calculated by the following equations when 1/3 bar water content is given in the method.
   Enter -1 for Ksat estimation by effective porosity Ahuja et al. (1989).
   $$K_{ep} = 764.5(\theta_s - \theta_{1/3})^{3.29}$$
   Enter -2 for KSAT estimation by Rawls et al. (1982).
   $$K_{ep} = 509.4(\theta_s - \theta_{1/3})^{3.633}$$
   Enter -3 Set the default value for the soil texture

Table 5.1: Texture Specifications

Texture	Ksat	WR	WS	1/3bar WFC	1/10bar WFC	WWP
Sand	21.000	0.020	0.437	0.0633	0.1083	0.0245
Loamy Sand	6.1100	0.035	0.437	0.1064	0.1613	0.0467
Sandy Loam	2.5900	0.041	0.453	0.1917	0.2630	0.0852
Loam	1.3200	0.027	0.463	0.2335	0.2961	0.1164
Silt Loam	0.6800	0.015	0.501	0.2855	0.3638	0.1362
Silt	0.6800	0.015	0.501	0.2855	0.3638	0.1362
Sandy Clay Loam	0.4300	0.068	0.398	0.2458	0.3082	0.1366
Clay Loam	0.2300	0.075	0.464	0.3119	0.3743	0.1882
Silty Clay Loam	0.1500	0.040	0.471	0.3433	0.4038	0.2107
Sandy Clay	0.1200	0.109	0.430	0.3222	0.3699	0.2215
Silty Clay	0.0900	0.056	0.479	0.3728	0.4251	0.2513
Clay	0.0600	0.090	0.475	0.3789	0.4283	0.2655
Default-Silt Loam	0.6800	0.015	0.501	0.2855	0.3638	0.1362

Note: WFC = Water Field Cap. WR = Water Residual WS = Water Saturated WWP = Wilting Point

4. Click ***OK*** on the method screen.
   Important Note: If bubbling pressure exceeds -100 adjust 1/3 water content and click ***OK*** again on method screen
5. You will exit out of site description. You may save your scenario noting the method you chose and run if you wish as the simulation ball is red and indicates information has changed and output is no longer up to date OR
6. If you return to **Site Description Soil Hydraulics** screen the *Full Description of the Brooks Corey (Not Constrained)* radial button is defaulted to show the

parameters estimated. ***EXCEPTION:*** If you have chosen the minimum input method the radial button is maintained, but you can manually click on the Full Description radio button (not constrained) to view the full parameters.

Important Note: Do Not Click back and forth on the radial buttons once you have completed step 6 or else it will change your BC estimations.

Tip: For any method other than Minimum Input, if you open a scenario it will show *Full Description Not Constrained* (default). It does not tell you how it was derived. Consequently results of different methods can be saved with ***FILE | SAVE SCENARIO AS*** and the method specified in the scenario name.

7. Click ***OK*** to save soil hydraulics (leaves **Site Description)**. ***APPLY*** on **Site Description Soil Hydraulics** is not available at this time.

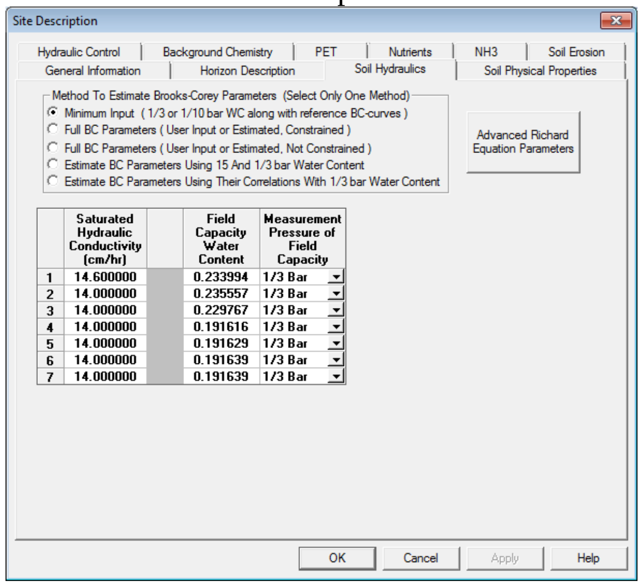
Note: Click only one radio button on **Soil Hydraulics**. The interface will accept the last radio button clicked as the method to estimate and fill in the full BC description on ***OK.***

Soil Hydraulic Property Estimation Methods:

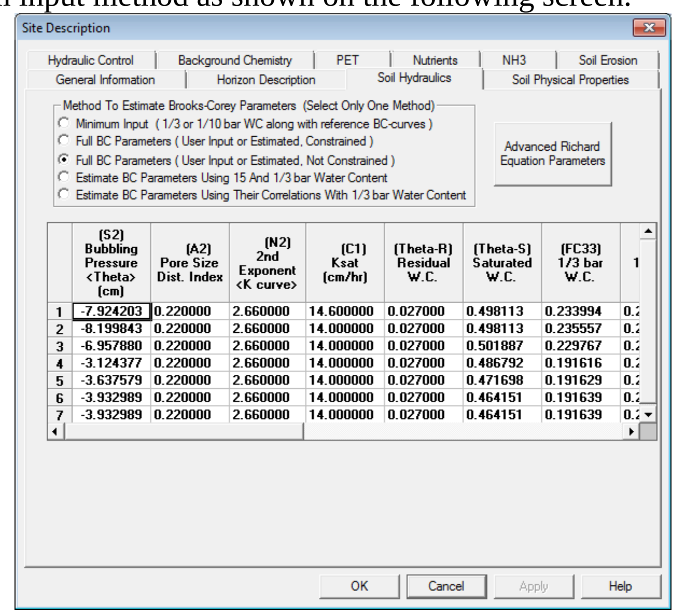
There are three methods to estimate the full description of the Brooks-Corey parameters in addition to manually entering parameters in the full description. The minimum input is based on soil texture, bulk density, 1/3 or 1/10 bar as described in Chapter 2 of the RZWQM book (Ahuja et al. 2000). The second is based on 15 and 1/3 bar soil water content, and the third is based on correlations with 1/3 bar soil water content (Williams and Ahuja 1992).

Important Note: Saturated hydraulic conductivity (cm/hr.) or Ksat can be entered or you may enter 0.0 to have it estimated for you.

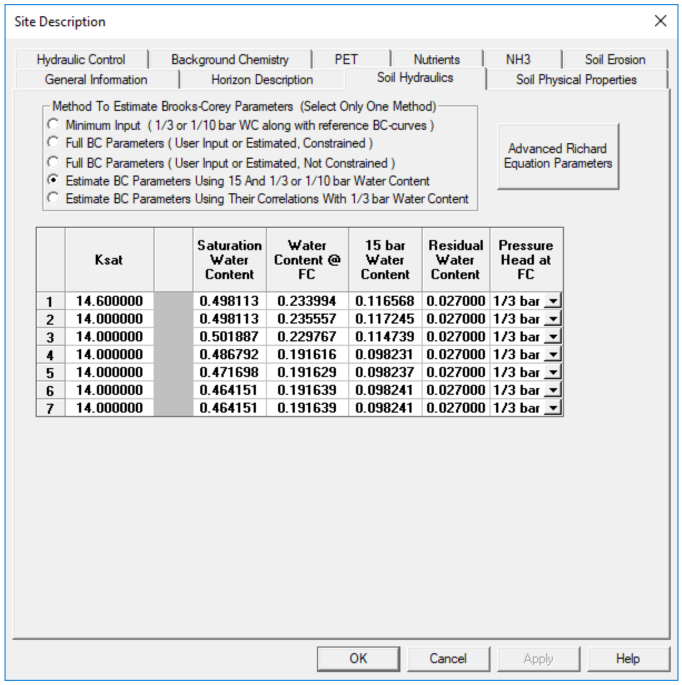
1. Estimate BC Parameters with minimum input



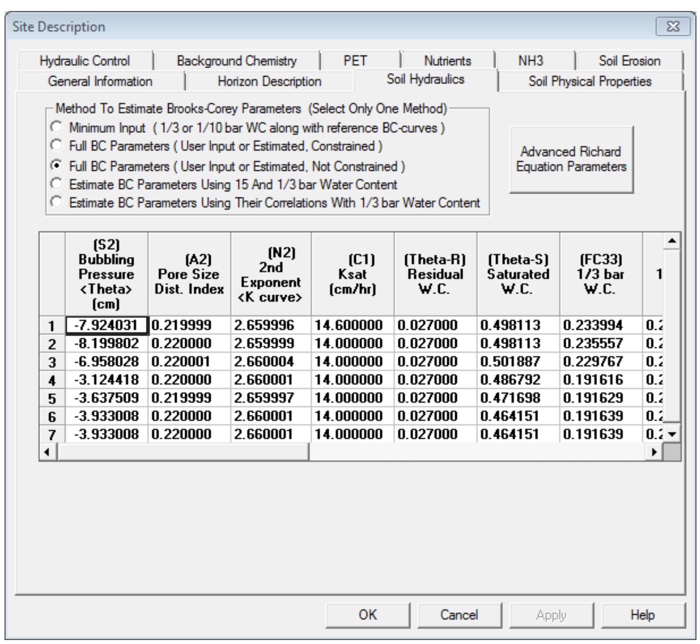
- Fields default to values based on soil type selected, but users may enter their own 1/3 bar or 1/10 bar water content and saturated hydraulic conductivity (Ksat). If Ksat is not known enter 0.0 and the model will default for you.
- ***OK*** on this screen will complete calculations and exit ***Site Description***
- If you so desire, *Re-Select* ***Site Description Soil Hydraulic Properties*** and *select* the full Brooks-Corey parameters radial button to see the estimates from the minimum input method as shown on the following screen.



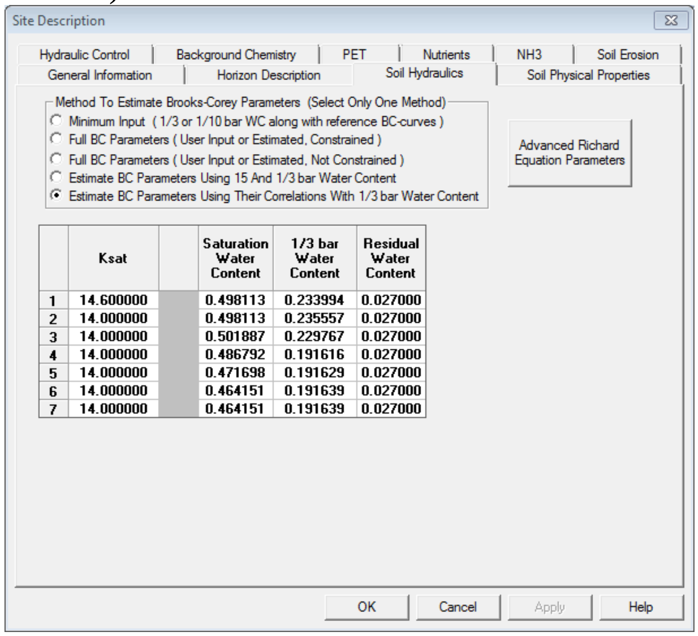
2. Estimate BC Parameters with 15 and 1/3 Bar (or 1/10 Bar) Soil Water Contents



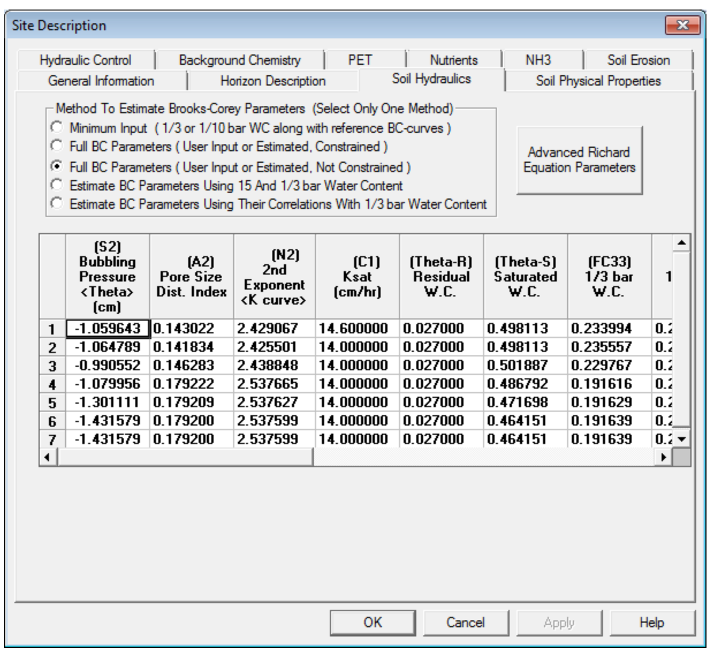
- **Enter 1/3 or 1/10 bar, 15 bar, Saturated and Residual water contents. If Ksat is not known, enter 0.0 to have the model estimate for you.**
- ***OK*** on this screen will complete calculations and exit *Site Description*
- **Re-Select** *Site Description Soil Hydraulic Properties* to see full Brooks-Corey parameters if so desired.



3. Estimate BC Parameters Using Correlations with 1/3 Bar Soil Water Content (One parameter model)



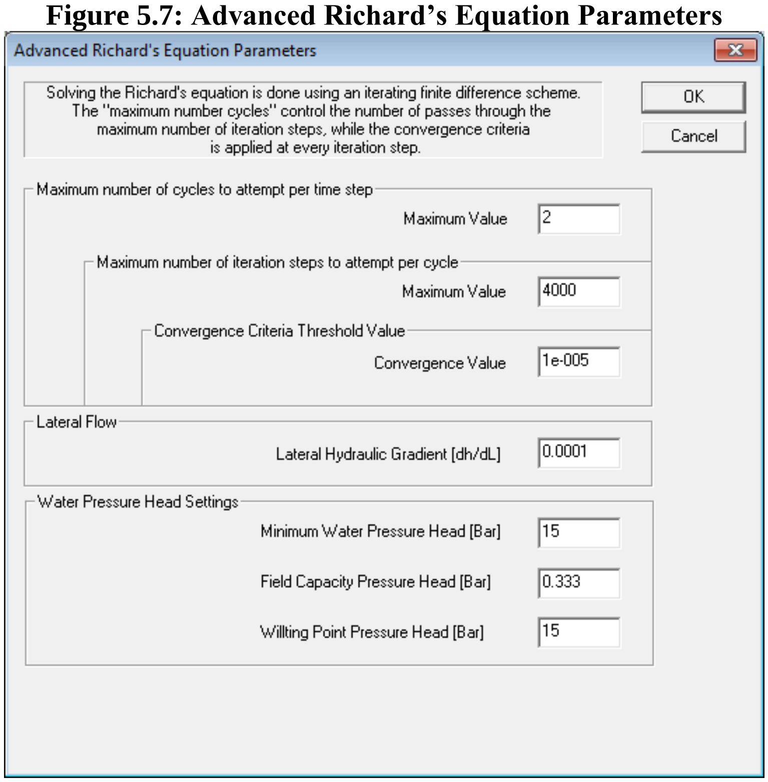
- **Enter 1/3 bar, saturated and residual soil water contents. If Ksat is not known, enter 0.0 to have the model estimate for you.**
- ***OK*** on this screen will complete calculations and exit *Site Description*
- If you so desire, Re-Select *Site Description* | *Soil Hydraulic Properties* to see results of the full Brooks-Corey parameters display as shown in the following screen



Advanced Richards Equation Parameters:

Parameters used in solving the Richards's equation and Lateral Hydraulic flow are set on the screen shown in Figure 5.7. Convergence issues can occur with or without mass balance issues. Be sure to check the MASSBAL.OUT file for the day of the year with the worst balance recorded and check to see if an input event may have caused it. If all input looks valid and is implemented correctly in the MANAGE.OUT file proceed with adjusting parameters for the Richards equation solution if a convergence warning message shows. Increase the iterations to 4000 or greater (usually <10000) to reach convergence or an increase to the convergence criterion (1e-004) may be needed. However, setting too high convergence criterion may cause a mass balance problem. Lateral flow can occur with or without tile drainage and requires a water table present. Version 2.4 allows user access to specifying the minimum pressure head required for the movement of water for the Richard's equation as well as plant water uptake. A pressure of 15 Bar will be written to the rzwqm.dat file as -15000 minimum suction required. Version 2.8 expands entry to set pressure heads for Field Capacity and Permanent Wilting Point.

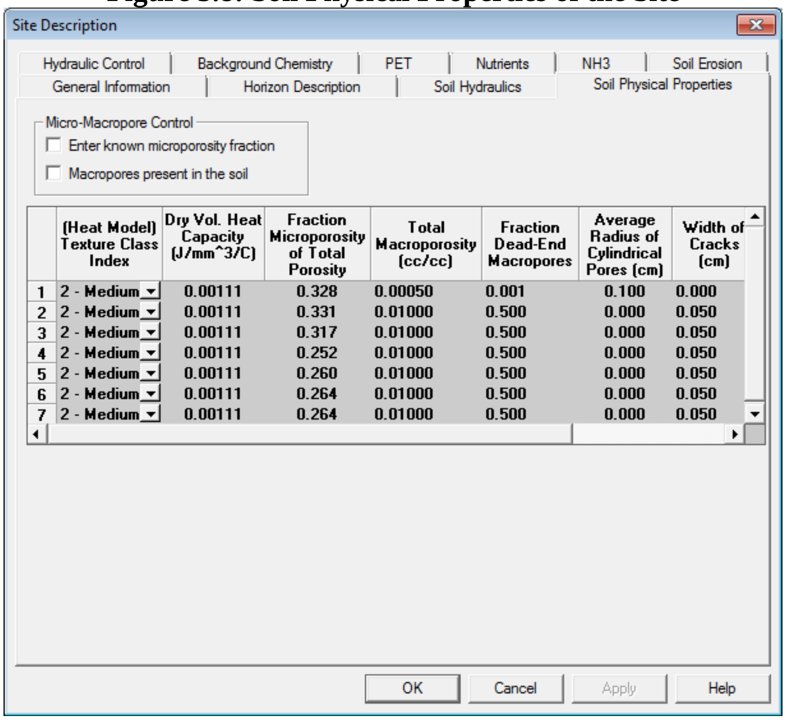
Figure 5.7: Advanced Richard's Equation Parameters



Soil Physical Properties

This screen is used to input parameters related to heat capacity, and macropore and micropore information. Users may further characterize macropores with the ***Advanced Macropore Parameters*** Button.

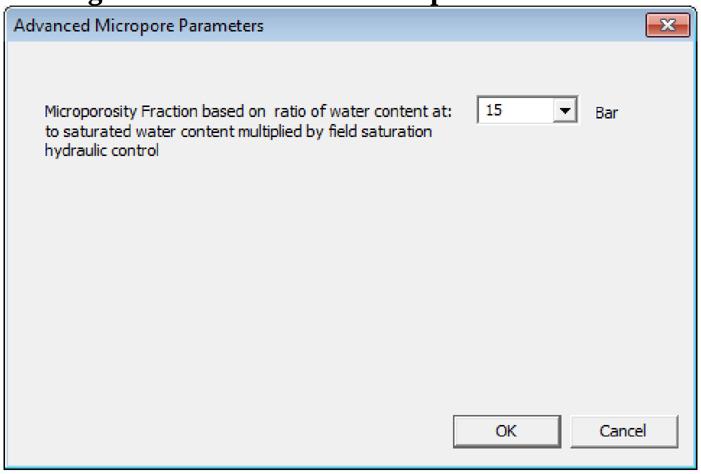
Figure 5.8: Soil Physical Properties of the Site



Important Note: If you check macropores present, the spreadsheet must have a macropore radius > 0.0 in the top horizon (O.K. will default to 0.1 if left 0.0). A macropore/tile with a large express fraction or dimensions drains the water from the soil.

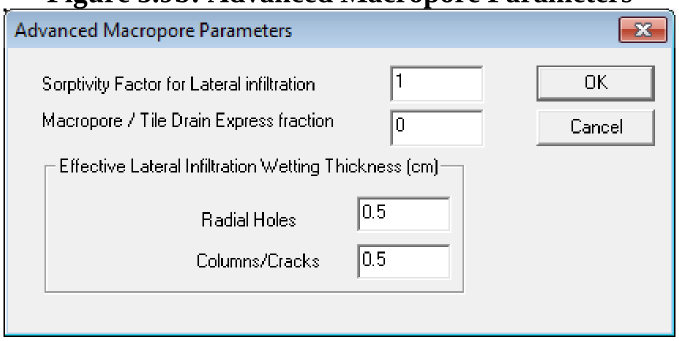
1. If you know your microporosity fraction, check the box and the grey column will become live for entry. Additionally, a new button will appear labeled "Use a Specified Microporosity". Clicking this button will open up a new screen (figure 5.9a) in which you can select your Microporosity Fraction based on the ratio of water content at a specific pressure head to saturated water content multiplied by the hydraulic control field saturation factor. This will automatically fill the spreadsheet's column. Solute in micropores is not mobile during infiltration.

Figure 5.9a: Advanced Micropore Parameters



2. If you have macropores, check box and click ***Advanced Macropore Parameters***

Figure 5.9b: Advanced Macropore Parameters



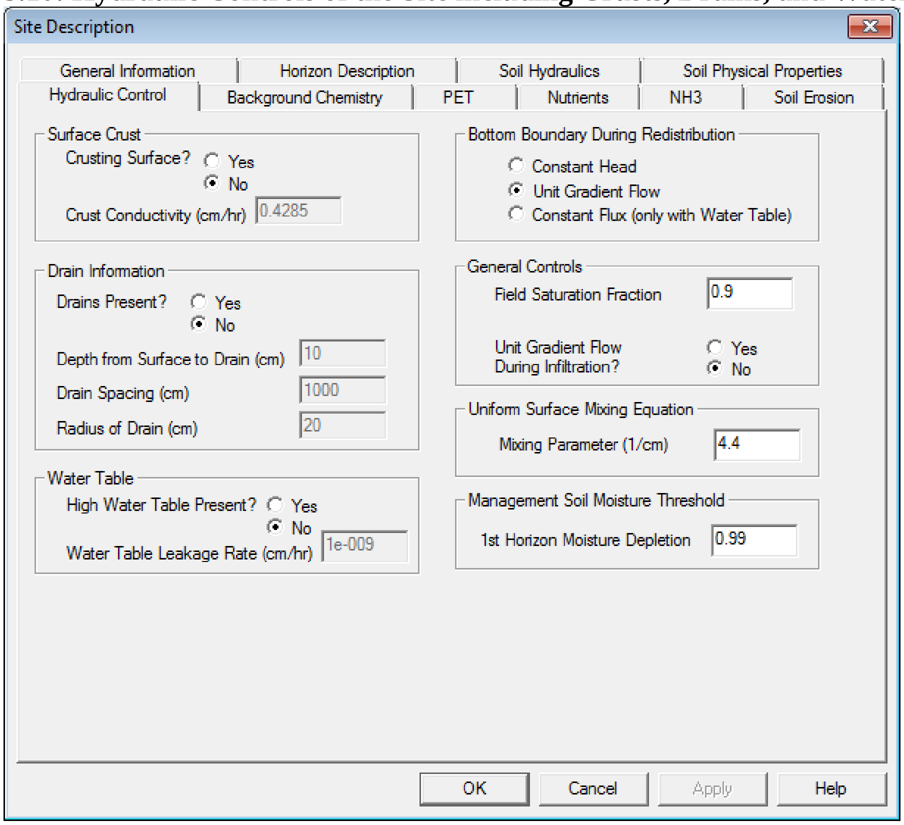
3. Fill out the Macropore Express Fraction if any. Enter other values (Figure 5.9b)
4. ***OK*** to save and exit dialog
5. Upon returning to the Soil Properties screen (Figure 5.8) be sure to enter the average pore radius, dead end pore fraction and microporosity if applicable.
6. ***APPLY*** to save on the Soil Properties screen and select the next dialog tab for data entry

Hydraulic Controls

Input information related to surface crusting, drainage, water table, boundary condition, field saturation, chemical mixing, and threshold water content for agricultural management is in this dialog tab (Figure 5.10). Some buttons and values may change automatically depending on what options the user chooses. The program default is no crust, drains, or water table within the soil profile and unit gradient flow.

Management soil moisture threshold is used to delay management implementation if soil moisture exceeds a certain fraction of saturated soil moisture. The lower the number the more chance management may not implement. If a crust exists, a default surface hydraulic conductivity of the crust is set to 1/5 of that of a no-crust soil.

Figure 5.10: Hydraulic Controls of the Site including Crusts, Drains, and Water Table

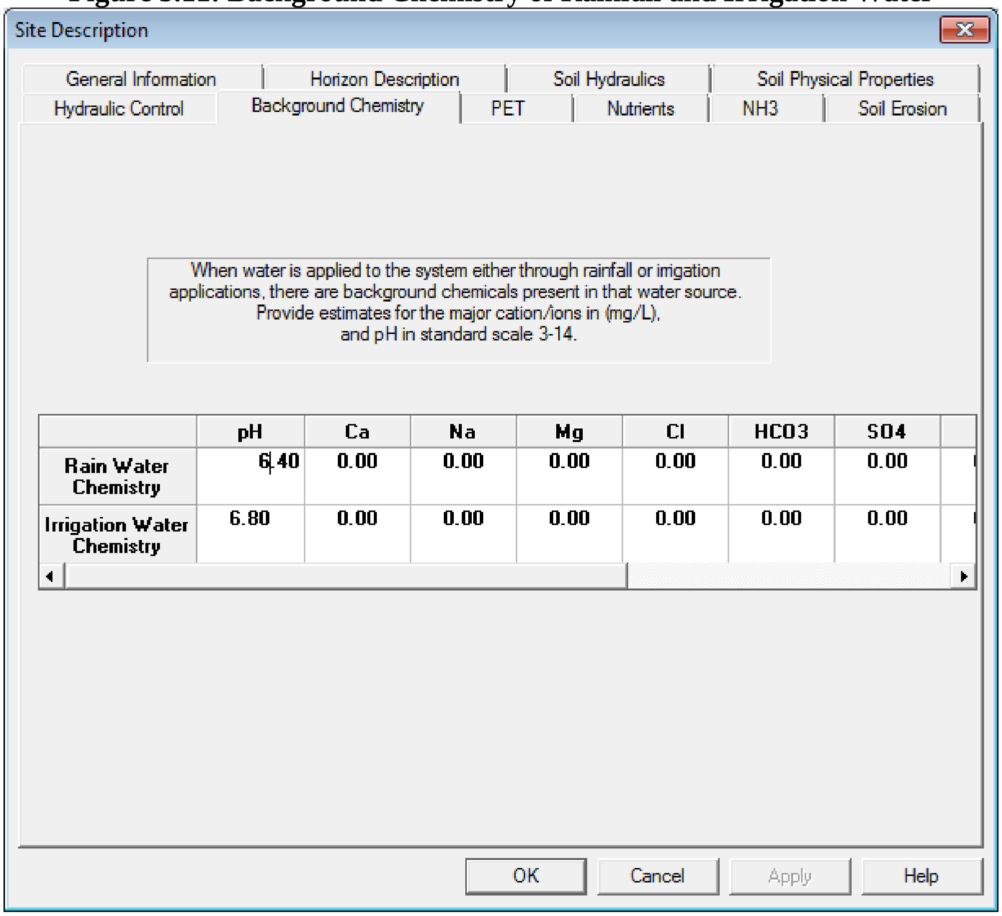


Important Note: When a water table is present, constant flux boundary condition should be used and Field Saturation Fraction should be set to 1.0. **Do not use constant flux boundary if a water table is not present.**

Background Chemistry

Input information for the chemistry of rain water or irrigation water (Figure 5.11). The pH is required; other fields are optional.

Figure 5.11: Background Chemistry of Rainfall and Irrigation Water



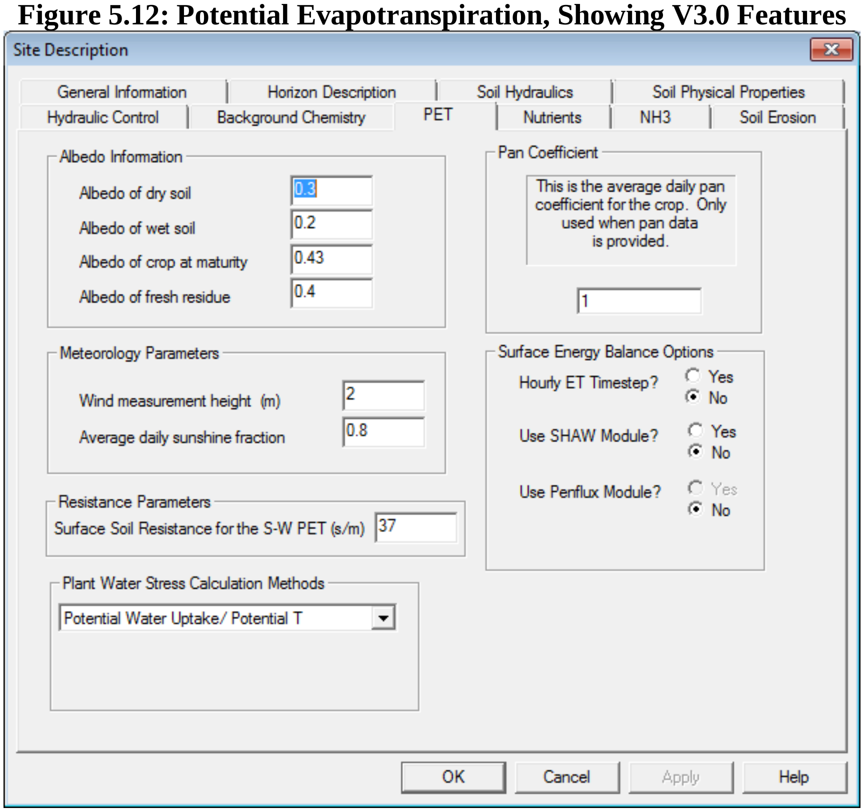
1. Enter the pH of rain water and irrigation water
2. If you have other naturally occurring chemical elements in these water sources, often NO_3-N and NH_4-N located at the end of the spread, include them here in mg/L (ppm).

Potential Evapotranspiration (PET)

Parameters used to calculate potential evapotranspiration (PET) are entered on this screen (Albedo, height of wind measurement, sunshine fraction, and pan coefficient if you provided pan data in *.MET file). New to Version 2.0 is a simulation option that can calculate ET on an hourly time step. New features will be included in version 2.5, which includes the ability to use the SHAW model or the Penflux module to calculate energy balance. These two options are mutually exclusive, and both require that the "Hourly ET Timestamp" feature is enabled to function.

If you select to use the SHAW or Penflux module, a button at the bottom of the "Surface Energy Balance Options" dialog labeled "Canopy Data" will appear. This will take you to a new dialog into which you can enter your observed plant data, which can be used for the SHAW and Penflux modules instead of an RZWQM2 simulated crop. Version 2.4 has added user access to the minimum soil surface resistance which affects potential evaporation in the Shuttleworth-Wallace ET method (see Figure 5.12). The default value is 37. Enter -1 instead of a value and RZWQM2 will calculate the resistance as a function of wind speed (Sakaguchi[1] and Zeng, 2009). To increase the potential evaporation without affecting potential transpiration and resulting yield a value of 200 is recommended. The range is -1 to 500. Alternative Plant Water Stress Calculation Methods are available with the first method defaulted as the method used by previous RZWQM versions.

Figure 5.12: Potential Evapotranspiration, Showing V3.0 Features



1. Enter soil, crop, and residue albedo (*F1* for help)
2. Enter height of wind speed measurement and average daily sunshine fraction

to account for cloud coverage.

3. Enter Pan Coefficient *only if* pan evaporation is provided in the *.MET file.
4. Hourly ET Time step optional Version 2.0 and later.
5. Use SHAW/Penflux module (optional version 2.X and later)
6. If Shaw/Penflux is selected, Enter Canopy Data.

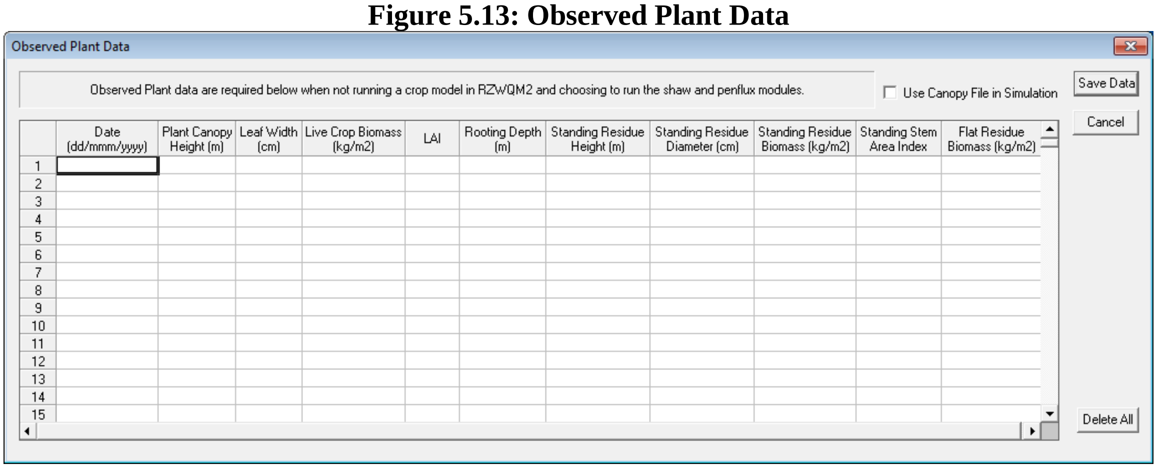
> ***Tip:*** Albedo characteristics, wind speed in the *.MET file, whether a surface crust hydraulic control is specified, and management such as tillage will affect the evaporative water loss during fallow periods

Keep In Mind: The Penflux and SHAW features are not yet in RZWQM2. This is a preview of things to come.

Observed Plant Data

This dialog contains a spreadsheet into which you will input your observed data for use by the SHAW and Penflux modules.

Figure 5.13: Observed Plant Data



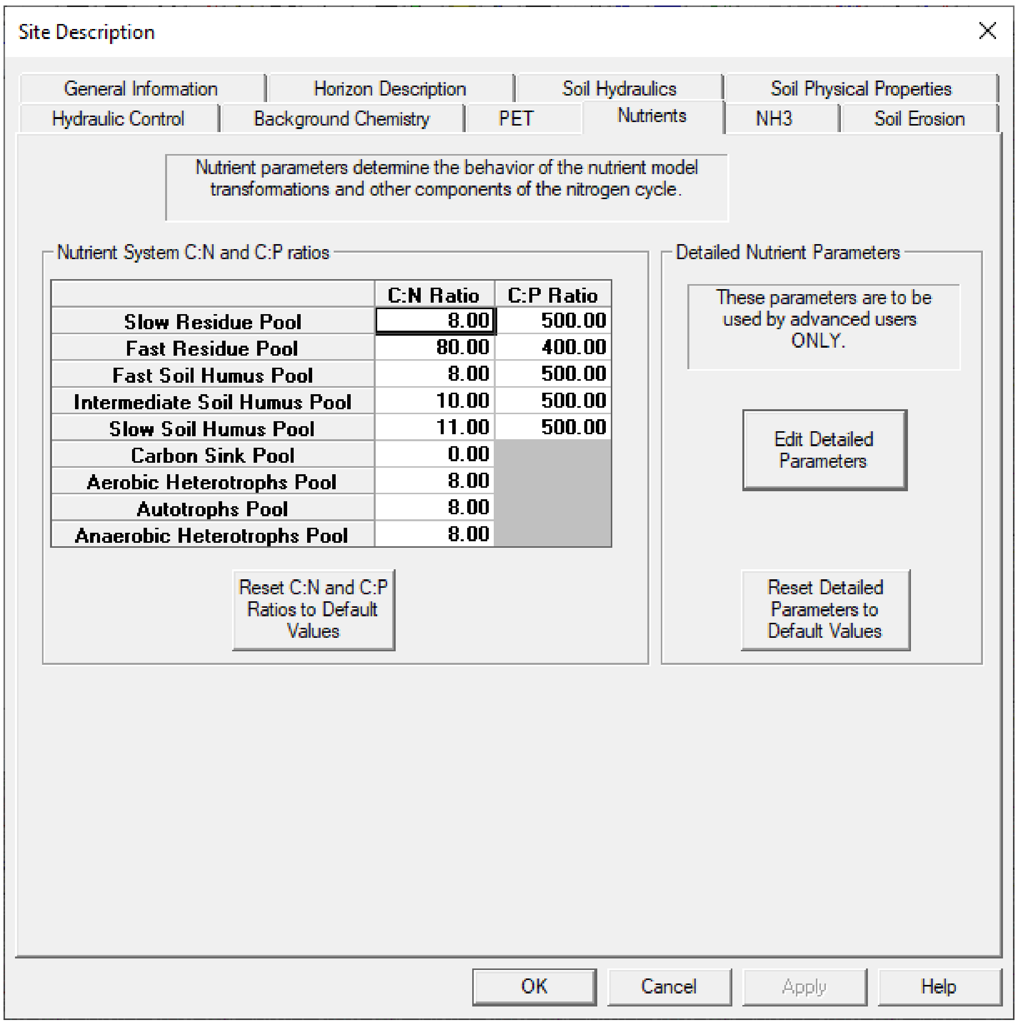
1. Enter observed plant data, row by row. The program can store 500 rows of observed plant data.
2. The “Use Canopy File in Simulation” checkbox, which defaults to checked, allows you to toggle whether or not the science will use your observations. If the box is unchecked, the data will be ignored.
3. The “Save Data” button writes the data in the Spreadsheet to the Canopy.DAT file, and returns the user to the PET tab in the Site Description dialog.
4. The “Delete Entry” button removes the currently selected row.
5. The “Delete All” button wipes out all data in the spreadsheet.

Nutrient System Parameters

Input parameters for the simulation of nutrients are entered on the screen (Figure 5.14). These parameters are defaulted and defaults can be retrieved with "*Reset…*" buttons. Users may override Carbon/nitrogen ratios of each organic pool.

Advanced users may modify the nutrient parameters including pool transfer rates, organic matter decay rates, microbial death and activation energies, and rate coefficients of carbon/nitrogen transformations by clicking ***Edit Detailed Parameters*** button. However, this is not recommended without advanced training and/or the assistance of a nutrient scientist.

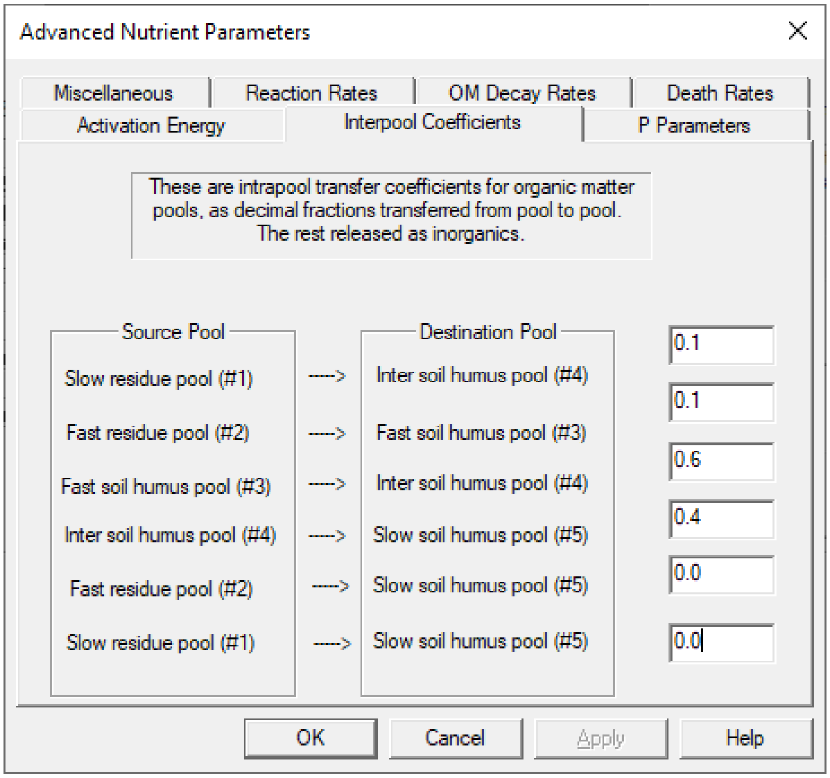
Figure 5.14: Nutrient Parameters to be used on the Site



Important Note: The slow residue pool C:N ratio may change if you apply manures under management. In this case the slow residue pool ratio is set to the minimum C:N of the manures applied.

Nutrients screen: Detailed Parameters

Figure 5.15: Interpool Transfer Coefficients



Tip: The most common parameter changed to calibrate in the nutrient section is the pool transfer coefficients. These may vary with climatic region, soil type and management history.

Figure 5.16: Organic Matter Decay Constants for the Five Humus Pools

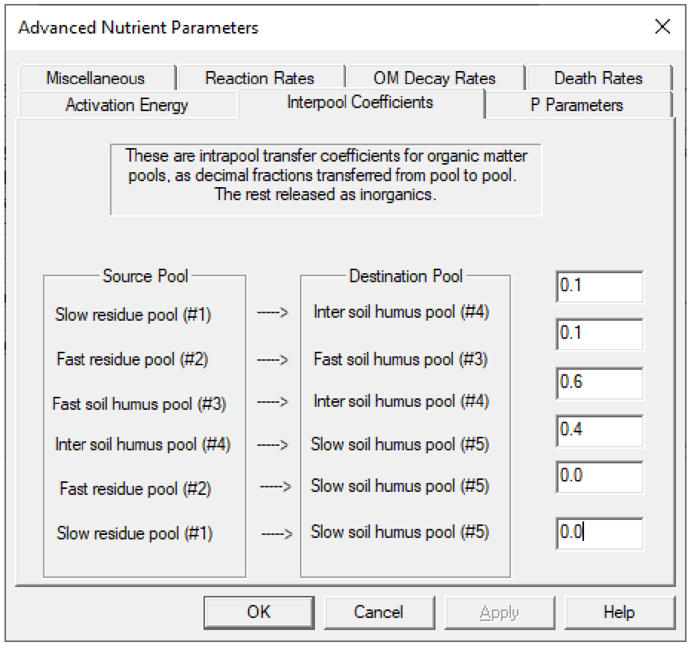
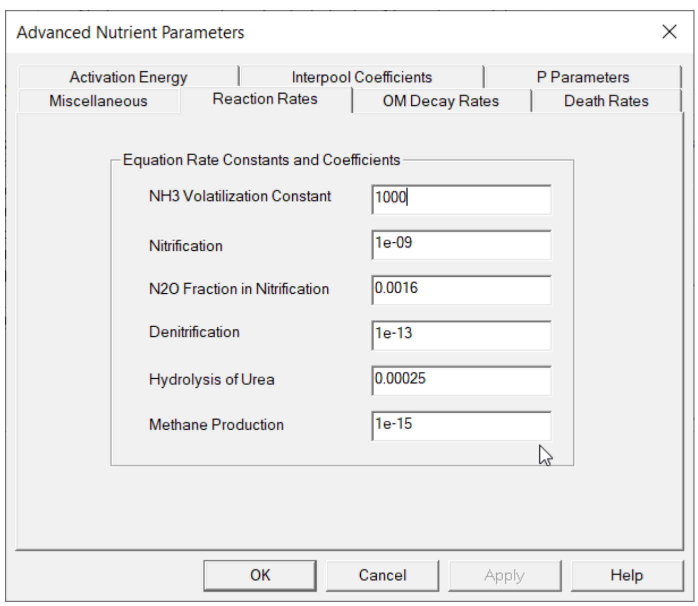


Figure 5.17: Reaction Rate Constant for Volatilization and Arrhenius Coefficients for Nitrification, Denitrification, and Hydrolysis



Important Note: User may have to adjust the denitrification rate constant after recent improvements in RZWQM2 (Version 1.8). Please view total denitrification in MBLNIT.OUT to calibrate the denitrification rate. General Organic Matter Decay Rate not needed and disabled in Version 2.0 is now replaced with Volatilization Constant in Version 2.1.

Figure 5.18: Activation Energy for Microbe Death

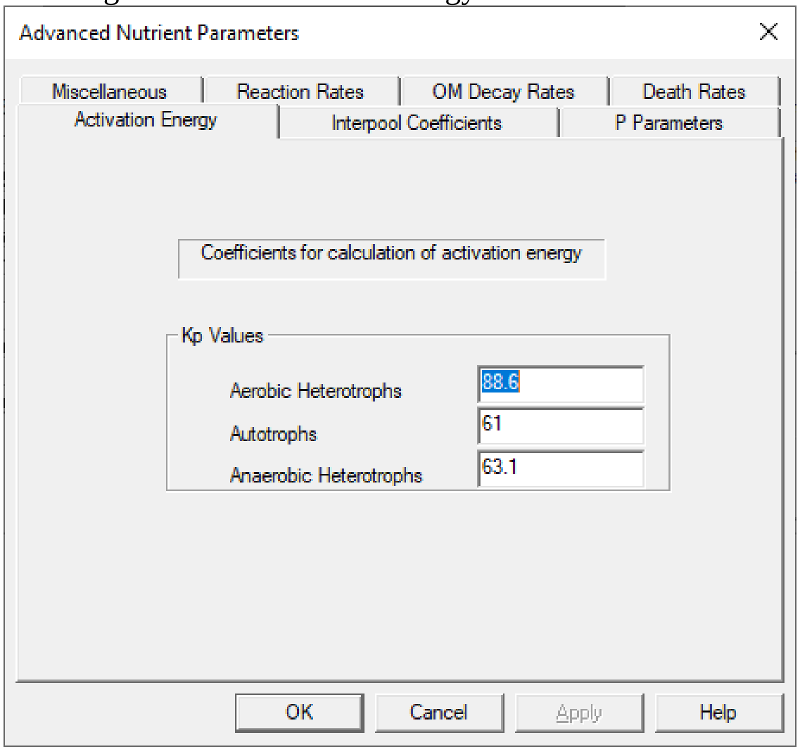


Figure 5.19: Rate Constants for Microbe Death

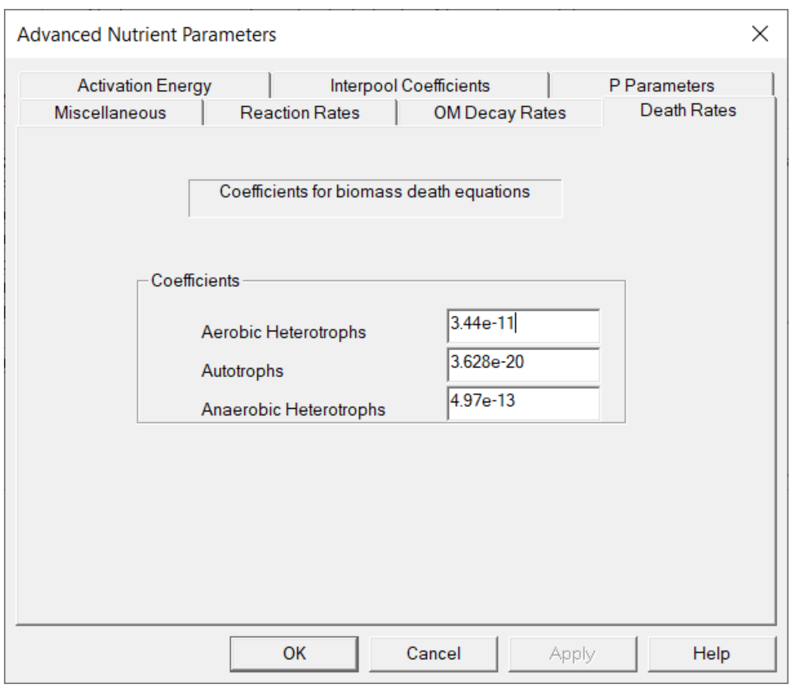


Figure 5.20: Misc. Efficiency and Microbial Population Growth Factors. Not recommended for modification.

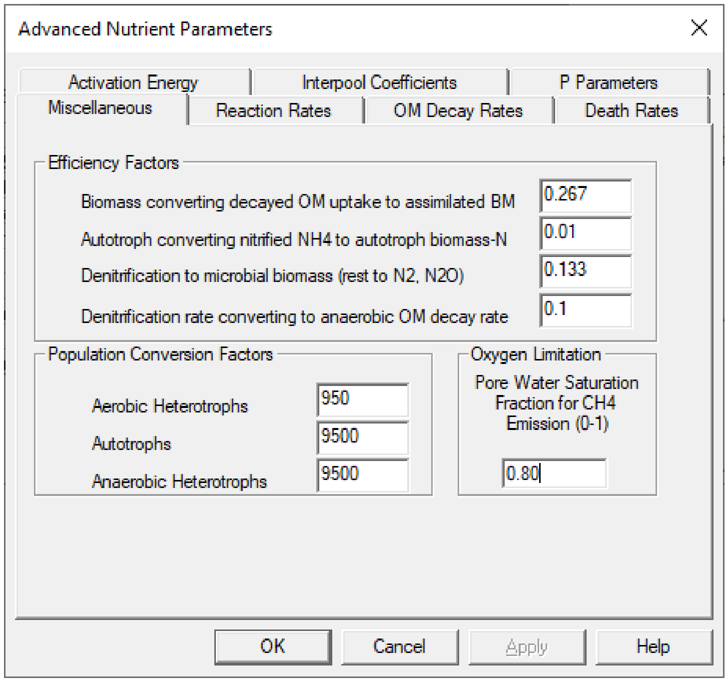
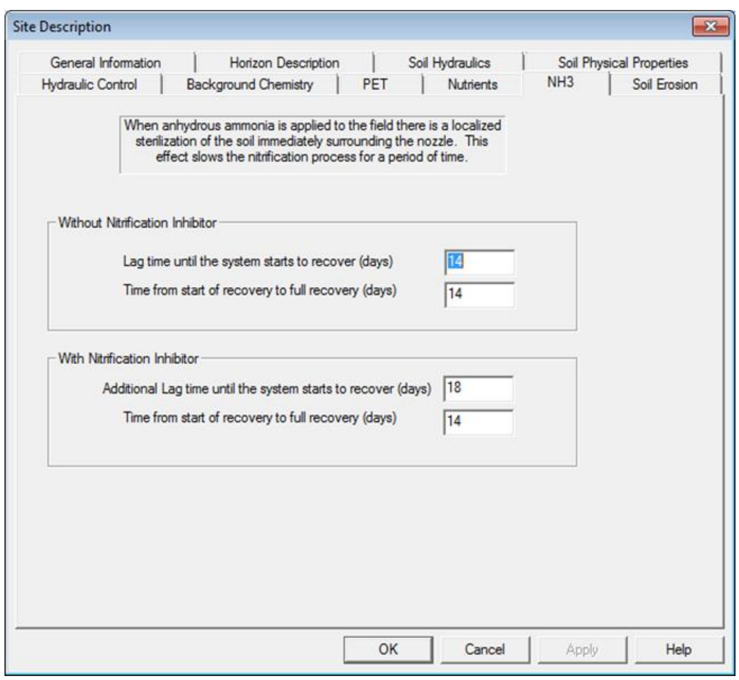


Figure 5.21: NH3 Tab, Defines Nitrification due to Anhydrous Ammonia application and Nitrification Inhibitors



Soil Erosion

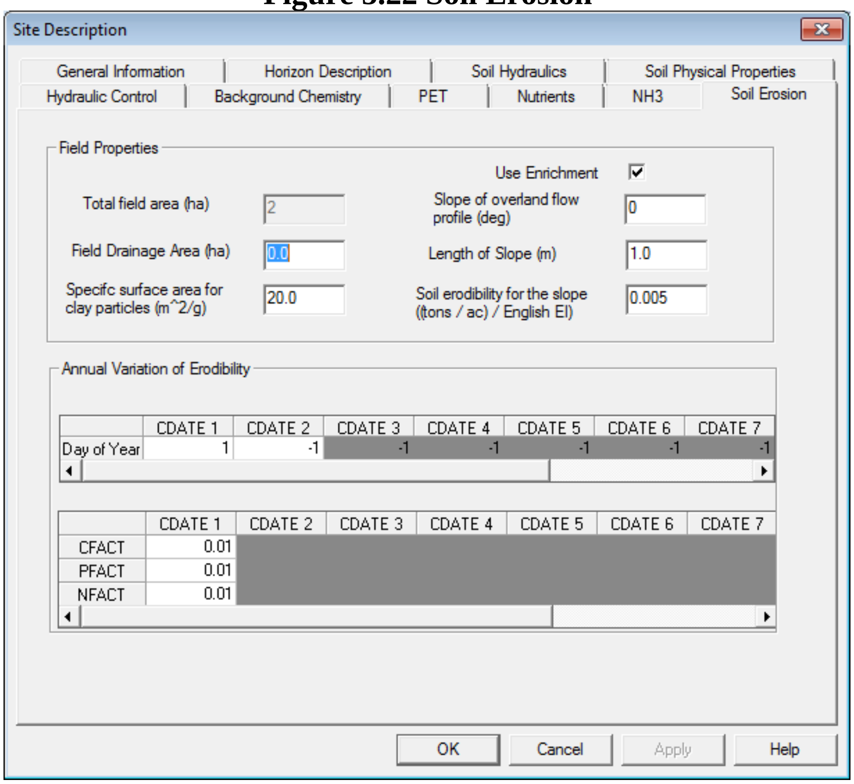
Gleams model sediment and associated chemicals on sediment loss due to overland flow erosion was added to RZWQM2 Version 3. A simplified approach assuming a single uniform slope for the field is taken. The user must specify the drainage area if not the field area, slope and slope length, and general soil erodibility. Then an erodibility index is entered for each day of year the erodibility changes. As the day of year (CDATE) is entered the table below it opens up entry for the variables CFACT (soil loss ratio for overland flow), PFACT (Contouring factor) and NFACT (Manning's N) (Figure 5.22). Suggested values are as follows:

PFACT 0.01-1.0. For any slope percentage or a slope length greater than the table value PFACT=1.0. NFACT varies between .01-.4.

Slope (percent)	PFACT	Max Length (m)
1-2	0.6	122
3-5	0.5	91
6-8	0.5	61
9-12	0.6	36
13-16	0.7	24
17-20	0.8	18
21-25	0.9	15

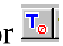
For more information on these variables see the installed GLEAMS help documentation RZWQM2\documentation\Component Models\GLEAMS Sediment Erosion\G2MAN2E.pdf

Figure 5.22 Soil Erosion



Important Note: a *Field Drainage Area* of 0.0 or a *Slope of Overland Flow* of 0 will shut off the GLEAMS module.

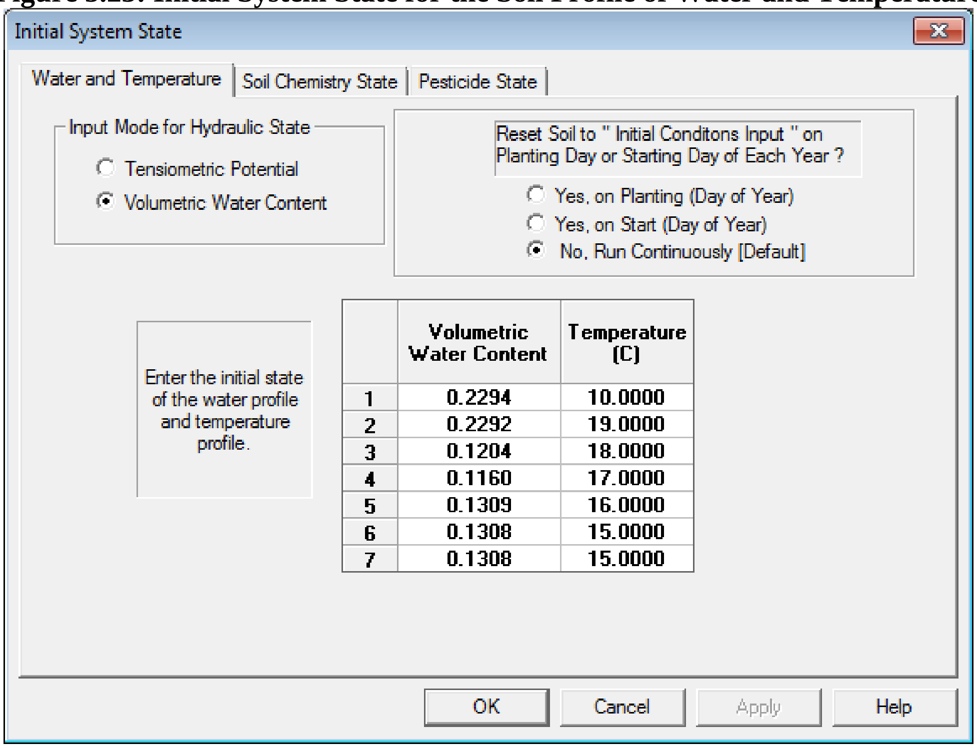
5.3.2 Initialize State of the Soil Profile

Clicking ***INPUT | INITIAL STATE*** or  on the toolbar.

Water and Temperature State

Initial water may be input as volumetric water content or tensiometric potential (Figure 5.23). Initial soil temperature in the soil profile is in ^{o} Celsius. To run RZWQM2 continuously through time, reset is defaulted to "NO."

Figure 5.23: Initial System State for the Soil Profile of Water and Temperature



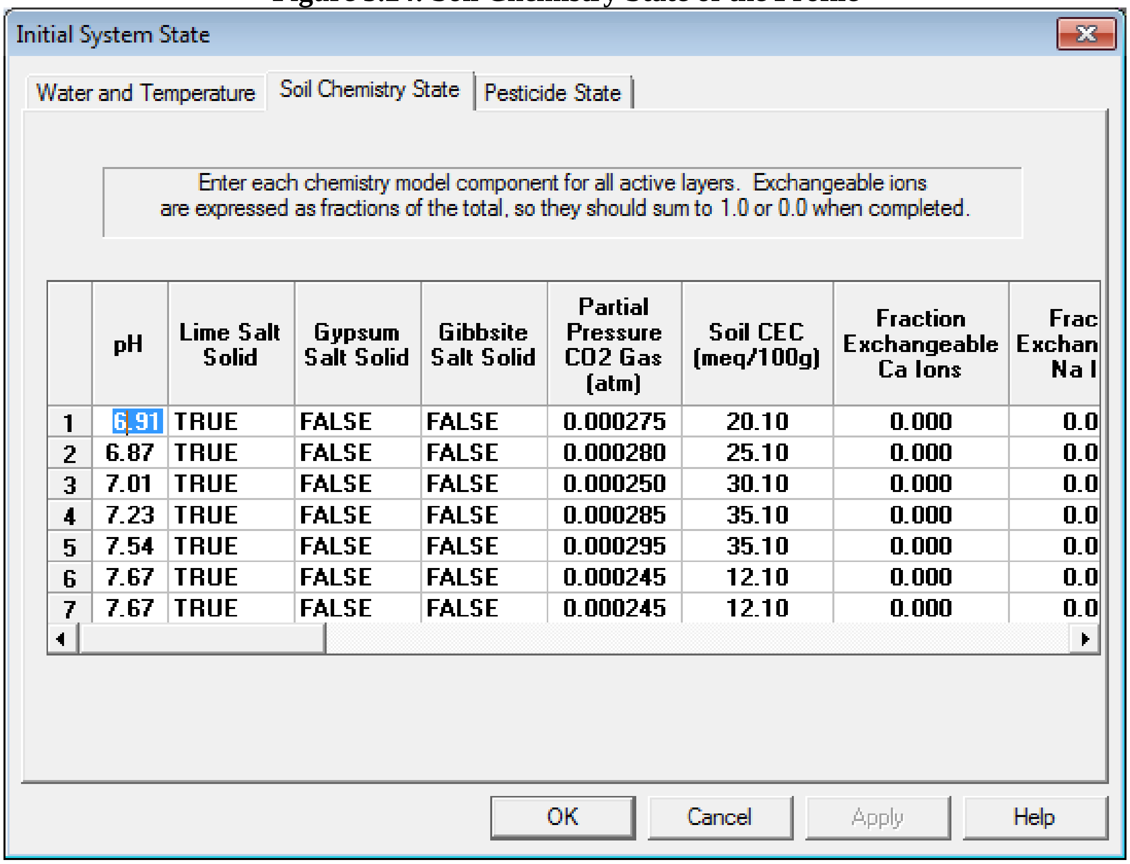
Important Note: *RZWQM2 runs continuously through time by default (No Reset).* However, the state of the system at planting or at run start can be reset to the initial state input defined by the user with the radial buttons "Yes, on Planting" or "Yes, on Start". If you use this feature it is wise to save the scenario noting this execution option.

Tip: Tensiometric potential should be used if a water table exists within the profile. Therefore, the location of the Initial water table is defined at the soil layer based on a positive potential.

Soil Chemistry State

Initialize the equilibrium chemistry state for each soil horizon with soil pH, CEC, and other cations. In addition initialize the existence of lime, gypsum, and gibbsite on the ***Soil Chemistry State*** (Figure 5.24).

Figure 5.24: Soil Chemistry State of the Profile



	pH	Lime Salt Solid	Gypsum Salt Solid	Gibbsite Salt Solid	Partial Pressure CO2 Gas (atm)	Soil CEC (meq/100g)	Fraction Exchangeable Ca Ions	Frac Exchan Na I
1	6.91	TRUE	FALSE	FALSE	0.000275	20.10	0.000	0.0
2	6.87	TRUE	FALSE	FALSE	0.000280	25.10	0.000	0.0
3	7.01	TRUE	FALSE	FALSE	0.000250	30.10	0.000	0.0
4	7.23	TRUE	FALSE	FALSE	0.000285	35.10	0.000	0.0
5	7.54	TRUE	FALSE	FALSE	0.000295	35.10	0.000	0.0
6	7.67	TRUE	FALSE	FALSE	0.000245	12.10	0.000	0.0
7	7.67	TRUE	FALSE	FALSE	0.000245	12.10	0.000	0.0

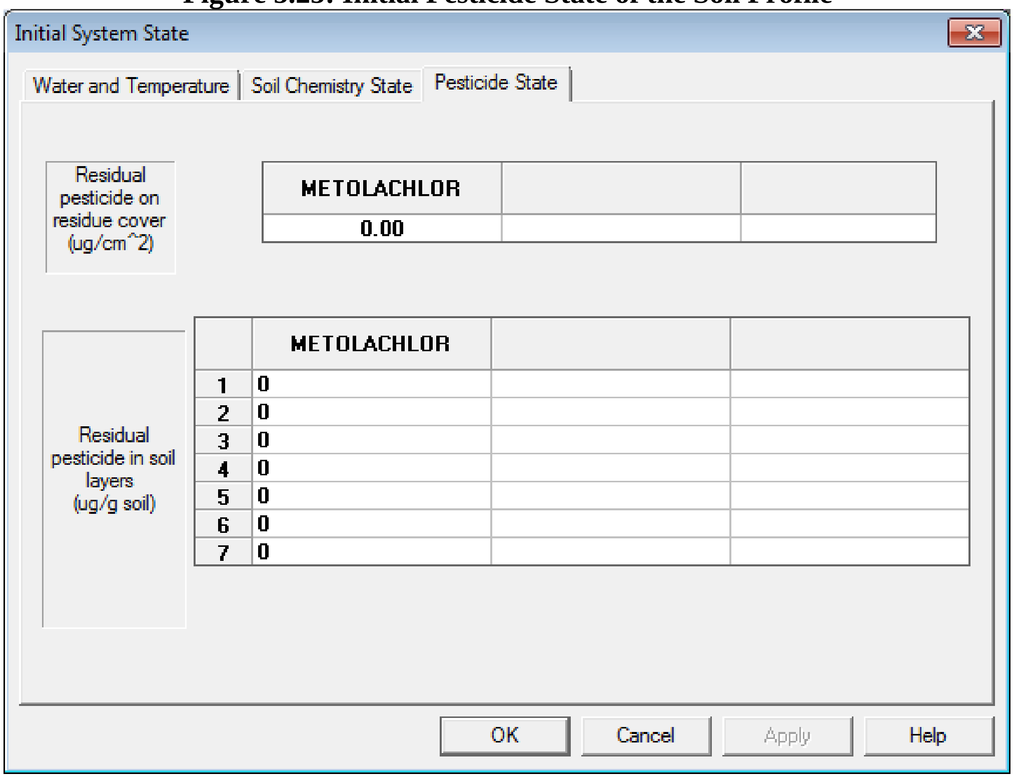
1. Input soil pH by layer if known; otherwise enter 7.0
2. Only if The Inorganic Chemistry Sub-model is active in **Simulation Controls Shutdown Module Components** are the other data required.

   Important Note: The Inorganic Chemistry Sub-model (i.e. the chemical equilibrium model) is currently disabled. Only soil pH is required.

Pesticide State

Initial total pesticide concentrations on crop residue and in the soil profile are input on this screen (Figure 5.25). Only three pesticides may be simulated at one time in the system whether it is a parent or a product of a parent. These initial total amounts are partitioned between absorbed and in solution.

Figure 5.25: Initial Pesticide State of the Soil Profile



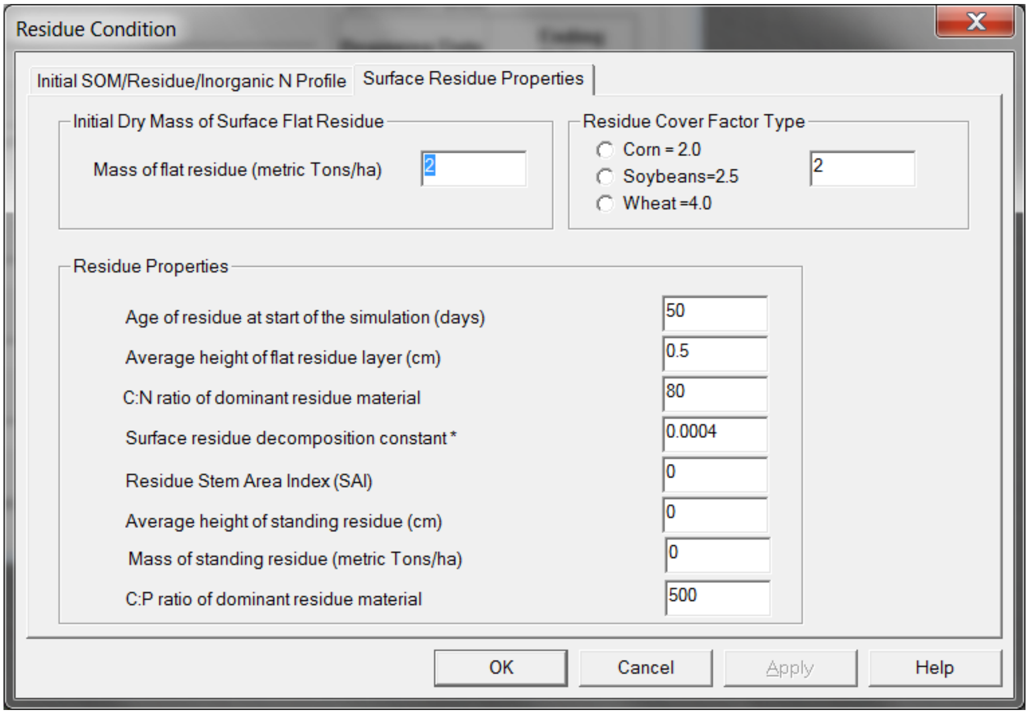
Note: Pesticides in applications are selected from the pesticide management screen under **Management Options** (Figure 5.40).

5.3.3 Initialize the State of the Soil Residue

Click ***INPUT | RESIDUE STATE*** on the menu or  on the toolbar.

Initial Surface Residue Properties screen: Users set the amount and type of residue on the surface at the start of the run. Other properties to be specified are age, height, C:N ratio, C:P ratio, fraction of natural incorporation by biologics (worms), and characteristics of standing residue (SAI, height and mass).

Figure 5.26: Initial Surface Residue



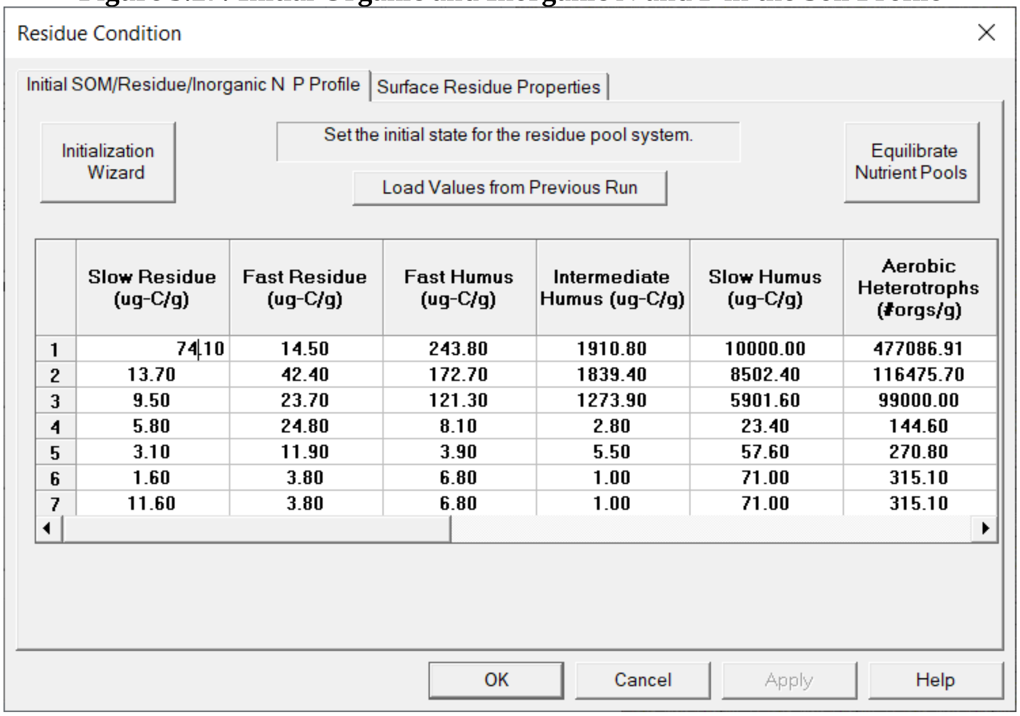
Note: Residue stem area index is not used at this time (future addition for Shaw Energy Balance Model). Residue Cover Factor Type is used to convert residue mass to % residue cover.

Tip: For crops in rotation pick the residue cover type of the most recently harvested crop.

Initial Organics, Residues and Inorganic N and P in the soil profile

All the simulated Carbon, Nitrogen and Phosphorus pools are initialized on this screen along with soil microbial pools. Users may choose to load the pools from a previous run, or follow the step-by-step initialization wizard based on the soil organic carbon measurement, or equilibrate the pools based on general weather information and agricultural practices. The last three columns of the spreadsheet require users to initialize inorganic sources of N for each layer in the profile. Appendix 3 covers in detail the methods for initializing organic matter pools.

Figure 5.27: Initial Organic and Inorganic N and P in the Soil Profile



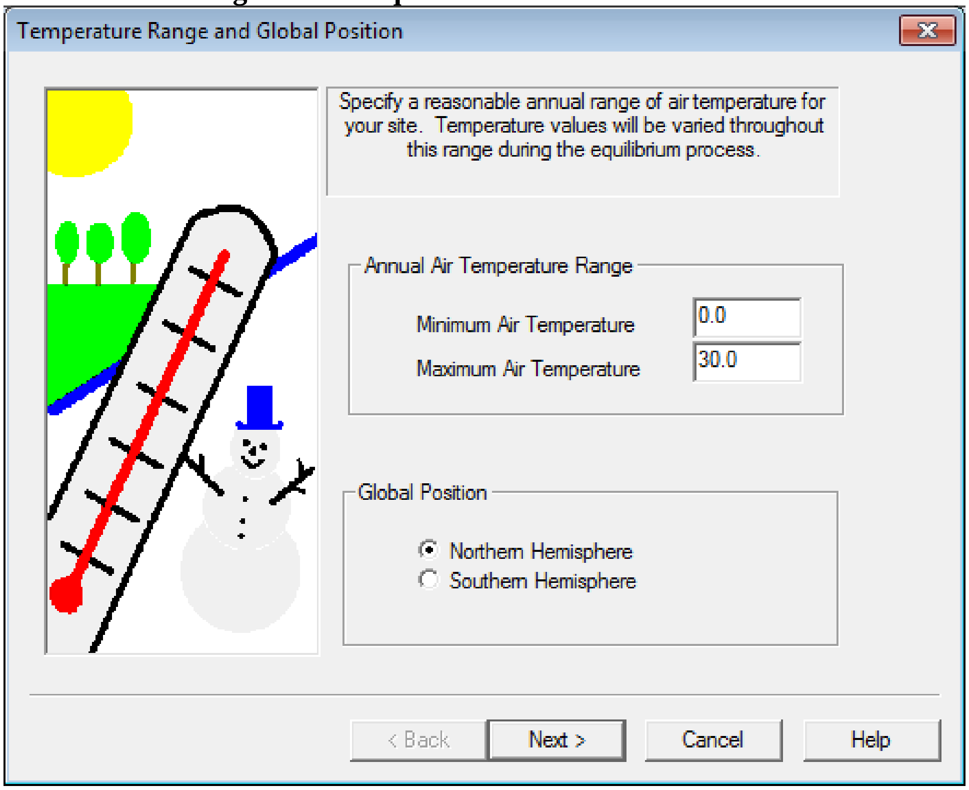
	Slow Residue (ug-C/g)	Fast Residue (ug-C/g)	Fast Humus (ug-C/g)	Intermediate Humus (ug-C/g)	Slow Humus (ug-C/g)	Aerobic Heterotrophs (#orgs/g)
1	74.10	14.50	243.80	1910.80	10000.00	477086.91
2	13.70	42.40	172.70	1839.40	8502.40	116475.70
3	9.50	23.70	121.30	1273.90	5901.60	99000.00
4	5.80	24.80	8.10	2.80	23.40	144.60
5	3.10	11.90	3.90	5.50	57.60	270.80
6	1.60	3.80	6.80	1.00	71.00	315.10
7	11.60	3.80	6.80	1.00	71.00	315.10

1. Click the ***Initialization Wizard*** and enter measured total organic carbon fraction and partition among the pools (See Appendix 3 Method 1)
2. Run iteration with a site that has 10 years of climate and management data (See Appendix 3 Method 2). If you have less than 10 years see Appendix 3 Method 3.
3. To manually run iterations of the model: Click ***Load Values from Previous Run*** (which will display after at least 1 run) and then Re-Run. **Note:** The pools are loaded from the NEWINT.OUT file created from the previous run so you may run again.

   Note: Users may choose to disable soil microbial growth under **Simulation Controls**. If that is the case default soil microbial pools should be used.

If no measurements of organic matter or carbon are available a user may also refine the estimated organic matter pools by select the ***Equilibrate Nutrient Pools*** button. Results from this wizard are obtained by taking into account general weather and cropping information available to the user.

Figure 5.28: Equilibrate Nutrient Pools Tool

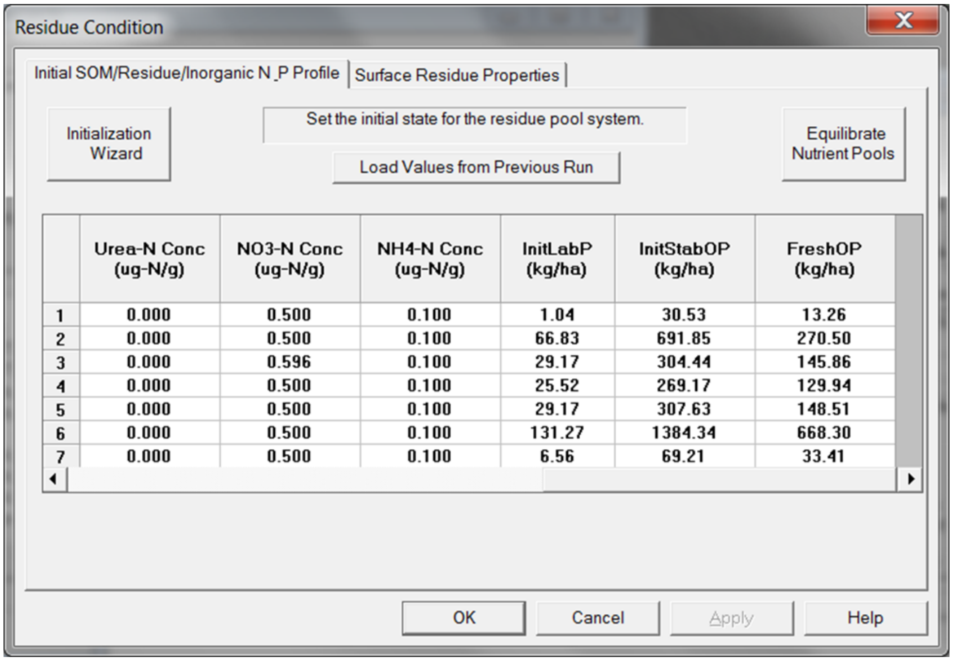


Note: This method is only recommended if you have no information or data for soil carbon in the system.

Important Note: See Appendix 3 for detail on initializing pools.

Initial Inorganic N and total P in the soil profile (last 3 columns of the spreadsheet)

Users specify the initial NO3-N, NH4-N or Urea-N, labile P, stable organic P, and fresh organic P for each soil horizon.



	Urea-N Conc (ug-N/g)	NO3-N Conc (ug-N/g)	NH4-N Conc (ug-N/g)	InitLabP (kg/ha)	InitStabOP (kg/ha)	FreshOP (kg/ha)
1	0.000	0.500	0.100	1.04	30.53	13.26
2	0.000	0.500	0.100	66.83	691.85	270.50
3	0.000	0.596	0.100	29.17	304.44	145.86
4	0.000	0.500	0.100	25.52	269.17	129.94
5	0.000	0.500	0.100	29.17	307.63	148.51
6	0.000	0.500	0.100	131.27	1384.34	668.30
7	0.000	0.500	0.100	6.56	69.21	33.41

Tip: Enter inorganic N only after you have completed the wizard or equilibrated pools.

Note: Inorganic N is not calculated by the Initialization Wizard and amounts calculated from the Equilibrate Nutrient Pools tool appear high. Please check inorganic N after organic C initialization.

5.3.4 ***Define Agricultural Management Practices***

Click ***INPUT | MANAGEMENT PRACTICES*** or  on the toolbar. Seven tabbed dialog screens are available for a full parameterization of RZWQM2 management. Management event timing can be entered as pre-plant, pre-emergence, post-emergence, lay by (after harvest) or Specific Date (DD/MM/YYYY). Fertilizer events include an additional timing option, split application.

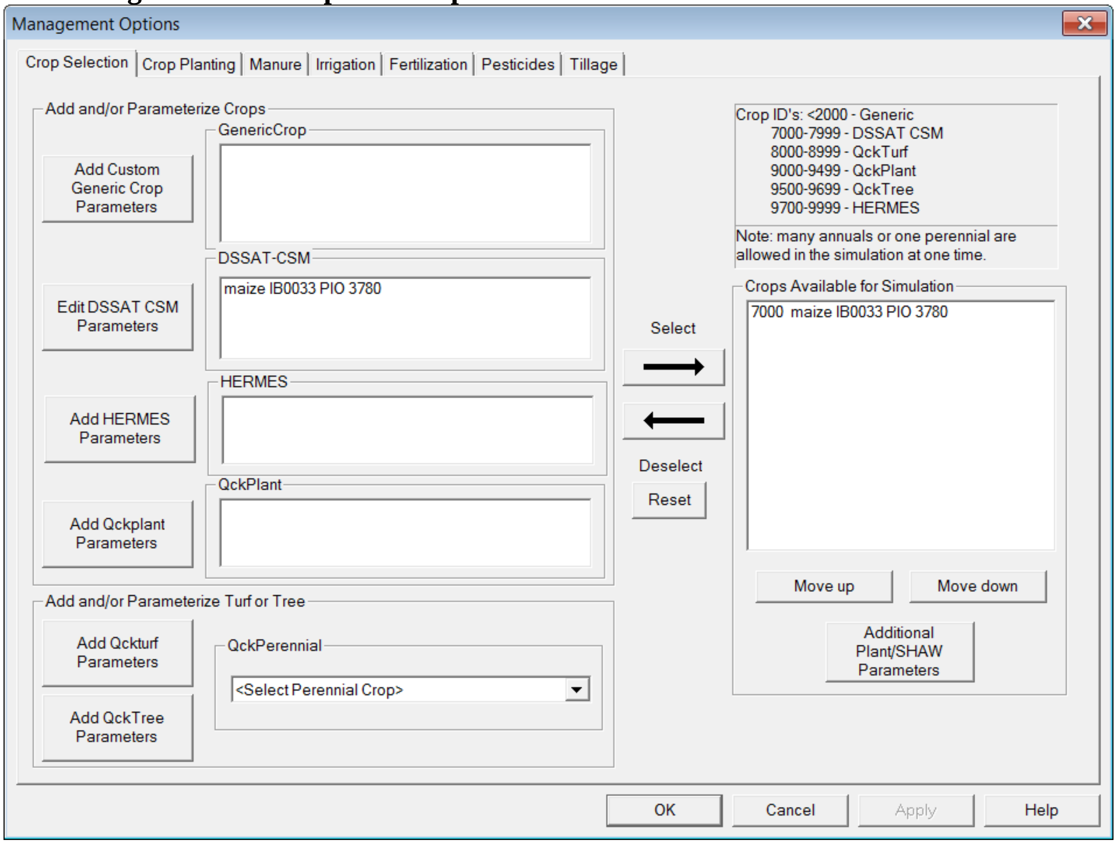
Tip: Double click on the date selection for all the management to bring up a handy calendar control.

Crop Selection

The user has the option to use the RZWQM Generic crop growth model, the DSSAT 4.0 Cropping System Models (CSM) which include CROPGRO and CERES Models, the Hermes SUCROS model or the Quick Plant, Quick Turf Models or Quick Tree Models on this screen (Figure 5.29a). The four digit Crop Identification numbers indicate which crop model is used. Numbers less Than 2000 indicate use of the Generic Crop Growth Model. The 7000 series is for DSSAT4.0 crop growth models; the 8000 series is for Quick Turf Crops; 9000-9500 is for the Quick Plant Crops, 9500-9699 is for Quick Tree Crops and 9700-9999 is for the Hermes SUCROS model.

The newest addition the Hermes SUCROS has an interface similar to the generic crop model where you may add a crop from the default database and customize it to your needs. While new and minimally tested the model within RZWQM2 has been used for alfalfa and sugarbeet production.

Figure 5.29a: Crop and Crop-Model Selection and Parameter Definitions



Important Note: Quick Plant, Quick Turf and Quick Tree are not crop growth simulation models. Quick Turf and Quick Tree cannot be used with any other crop (model) selected for simulation as they are perennial.

Selecting Crops on Crop Selection Screen (Figure 5.29a):

1. Click ***ADD*** to add to available crops list if they are not listed yet (i.e. A quick plant or turf plant , a new DSSAT cultivar)
2. Select the Crop Name from the available crop window (use the Control Key down to select a second crop from the list).
3. Click ***SELECT Arrow*** to move the selected names to the Crops Selected for Simulation list (right window).
4. Click all management tabs and enter management events for these crops.
   Tip: If a new crop variety is selected to replace an existing one, click ***APPLY*** after ***SELECT Arrow***. The old reference crop variety in all management screens (planting, tillage, fertilizer, manure, pesticide) will be replaced with the new. However irrigation management screens load from *.ss2 files so when you change crop types you must access the irrigation sub tabs for reference crop and change it manually to the new crop and O.K. to write the new *.ss2 files.
5. OK to leave management.

De-selecting Crops on the Crop Selection Screen (Figure 5.29a):

1. Select or control select from right window the crop name(s) to remove.
2. Click on DESELECT arrow
3. Your management has no crop to work on now.
4. You must return to available crops and reselect new crops.
5. Click on SELECT arrow to move the names right.
6. ***Optional*** Click **APPLY** to apply all the previous programmed management to the crops newly selected. **Important Note**: The management will be applied in the order the crops are listed in the ***crop selected for simulation*** window for planting, tillage, fertilizer, manure, pesticide). ***However, Irrigation screens must be manually edited to change the reference crop as they load from *.ss2 spreadsheet files.***
7. Manually check each tabbed dialog to see if your management is applied to the appropriate crop type (reference crop).

Generic Crop Growth Model

RZWQM2 Generic Crop Growth Model is currently calibrated for corn and soybean in several states and winter wheat in eastern Colorado. Advanced users who wish to parameterize a customized specific crop will be able to do so by accessing the ***Add Custom Generic Crop Parameters*** button in Figure 5.29a. Figure 30 shows the main generic crop entry screen. The user specifies the species, a default set of values and a name for their crop in step 3. Then on your custom crop tab appearing enter: if the crop is requires vernalization and growing degree days for each developmental stage when the growing degree day method box is checked or the number of days spent in each developmental stage. Enter plant attributes and plant optimum temperatures on the screen as well. More detailed parameters are set on subordinate screens for the custom crop with the low left button. **SAVE and EXIT** from the main generic crop screen writes the PLGEN.DAT input file for the generic crop growth model (for format See Appendix 5). However, those who want to parameterize an existing generic corn or soybean may only need to change a few model parameters by going to ***Set Site Specific Crop Parameters*** button in Figure 5.29a resulting in Figure 5.29b. DSSAT crop growth modules and QCK* models do not use parameters in **Site Specific Crop Parameters** EXCEPT for *Minimum Leaf Stomatal Resistance (all other fields are greyed)*. A default value of 200s/m is used for the minimal stomatal resistance if not defined.

Figure 5.29b: Site Specific Crop Parameters

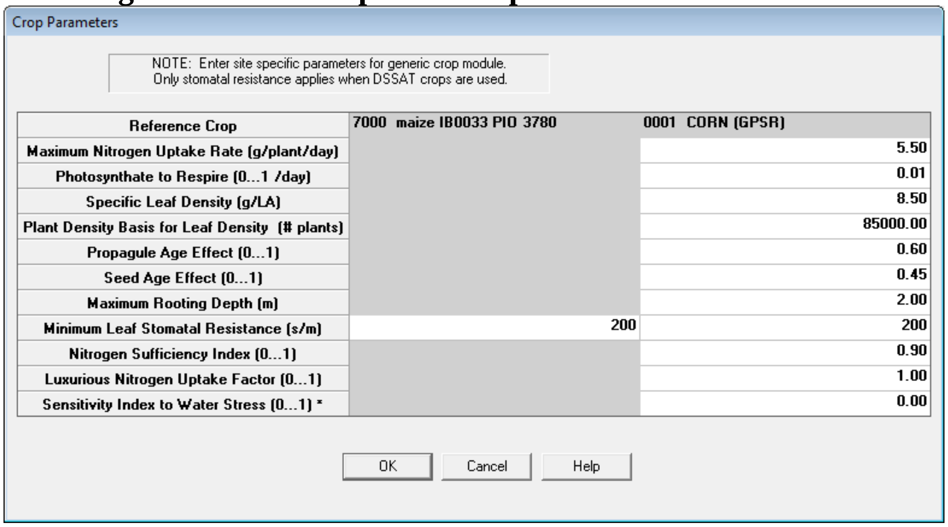
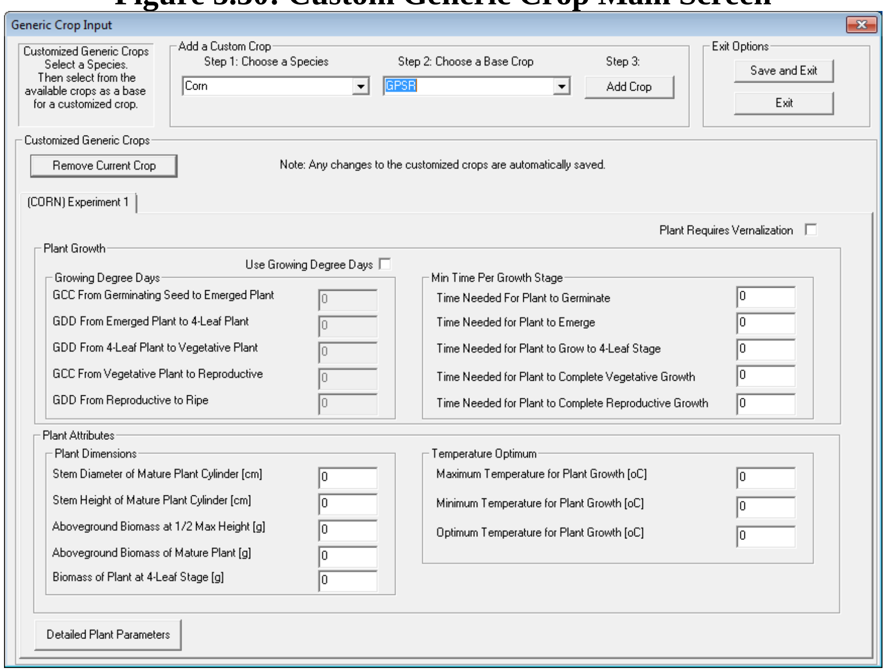


Figure 5.30: Custom Generic Crop Main Screen



DSSAT Crop Growth Models

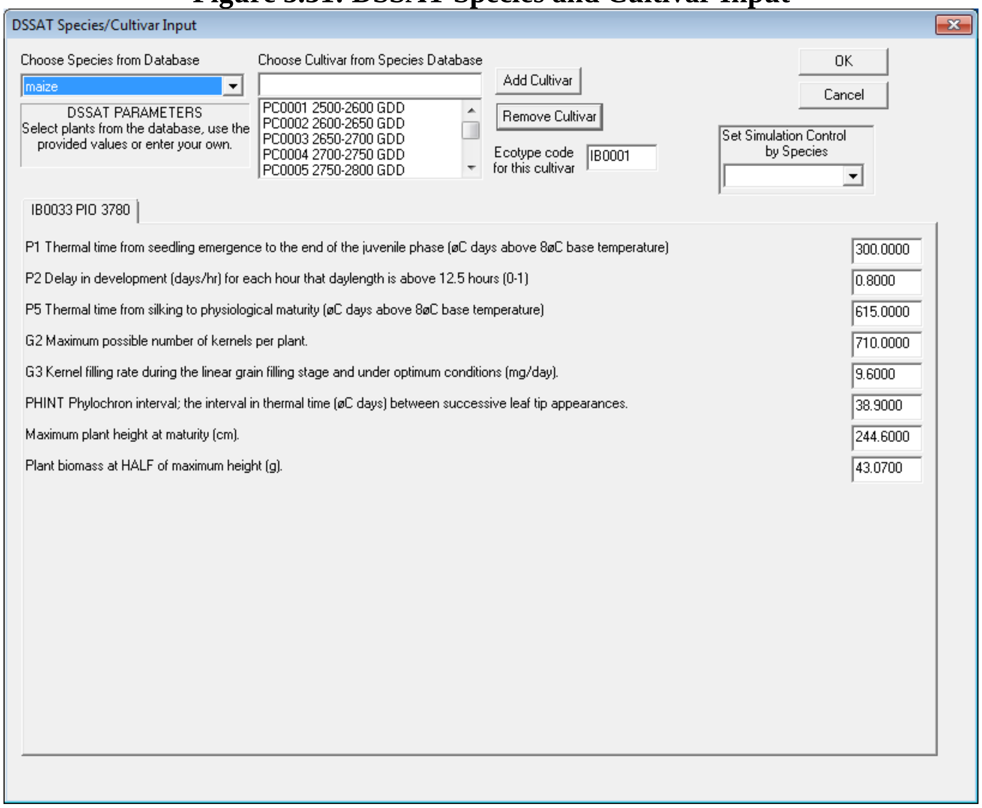
These consist of the CERES models for small grains and the CROPGRO model for legumes. Varieties already parameterized are available for selection. The user may need to try several (varieties, maturity groups) before they encounter one similar to their own. Users may calibrate the cultivar parameters also if they have crop phenology and production data. The OVERVIEW.OUT file is the DSSAT output file most useful for overall summary of crop simulation runs. RZWQM2 file

COMP2EXP.OUT is used when calibrating and evaluating observed vs. predicted results.

To Add a DSSAT crop:

1. Click ***ADD DSSAT CSM Parameters*** (if available previously it will read ***EDIT DSSAT CSM***).
   There are 2 screens of information necessary for the DSSAT crop simulation and some additional fields added to the planting management (e.g. transplant information, emergence date) applicable to DSSAT. The first screen are the cultivar parameters in the *.CUL cultivar files (Figure 5.31) and the second screen contains simulation controls for how the DSSAT crop will run and allows different soil root growth factor for each crop type (Figure 5.32).

Figure 5.31: DSSAT Species and Cultivar Input

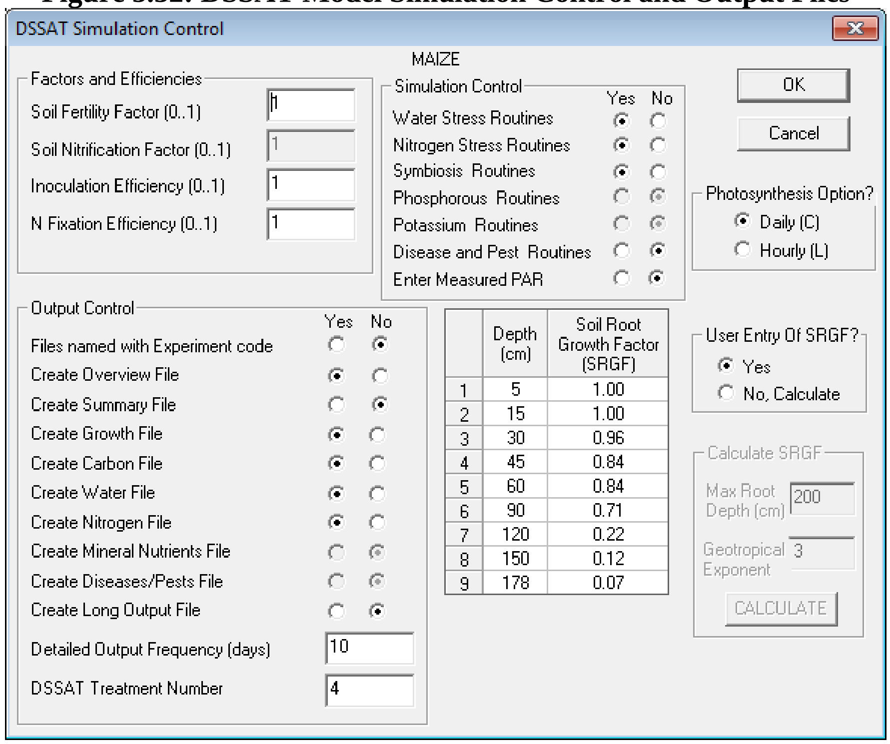


2. Select Crop Species (*.SPE file)
3. Select Cultivar from the database (*040.CUL file)
4. Click ***ADD Cultivar*** to add tabbed dialog of cultivar parameters
5. Set Simulation Control drop down to Crop Species in step 2.
6. The DSSAT simulation control screen will result (Figure 5.32).

   Note: Users can only select the cultivar names on the window. Once a cultivar is selected, they may change the parameters. The new parameters for this cultivar will be saved in the scenario folder (*.CUL file). However, if a user wants to use his/her own cultivar name to show in the window for selection it can be added in the master database directory RZWQM2\DATABASES\DSSAT\ in the *040.CUL files default parameters.

Important Note: Please backup your customized *.CUL, *.SPE, or *.ECO master databases to an alternate location as these master databases are replaced with each new RZWQM2 version installed and your copies will be overwritten.

Figure 5.32: DSSAT Model Simulation Control and Output Files



Note: If you enter PAR flag as “Yes” you must have daily Photosynthetically Active Radiation specified in the last column of your *.MET file.

7. ***OK*** to save simulation controls to the scenario’s *DSSAT.RZX file.
   Note: *DSSAT.RZX file contains database paths for DSSAT model execution. The RZWQM2 interface will update these paths to your current path upon OPEN Scenario, RUN, and saving the scenario as a different scenario. If you manually copy a project from folder to folder outside the interface, open your project and scenario and OPEN | SAVE and/or OPEN | RUN | SAVE to update these paths.
8. ***OK*** to save DSSAT parameter screen
   Note: This generates and updates the scenario’s *040.CUL file.
9. Now you may select available name to select for names selected for simulation.
10. When alfalfa is selected, users need to create a FORGMOW.DAT file using the example in the C:\RZWQM2\startup\ and copy to the current scenarios.

Quick Plant, Quick Turf and Quick Tree Models

Under management options, users who are interested in water quality only may parameterize their own crop, turf or tree parameters using measured plant parameters, such as nitrogen uptake, leaf area index, biomass, and rooting depth.

Quick Plant

Quick plant is a module to mimic plant growth by taking out water and nutrients from the soil. It does not simulate actual plant growth per se. Users need to define maximum plant height, maximum rooting depth, maximum leaf area index, and total plant N uptake (both aboveground and belowground) during a growing season. The module will grow a plant based on user defined growing period (Fig. 5.33, 5.34). It also provides a simple way for vernalization such that the plant will enter dormancy when every 5-day temperature is below zero and out of dormancy when every 10-day temperature is above zero. The module also allows user specified stover and dead root returned to the soil at harvest. If users do not specify root biomass, it will assume it to be 25% of aboveground biomass.

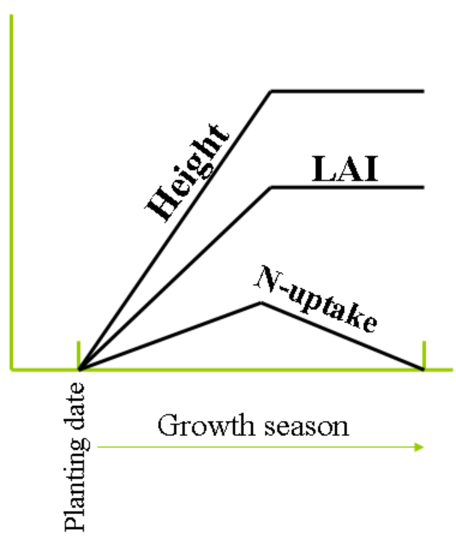


Figure 5.33: This diagram shows plant height, leaf area index, and daily N-uptake with growing season. Maximum height and LAI are reached at the beginning of the reproductive growth

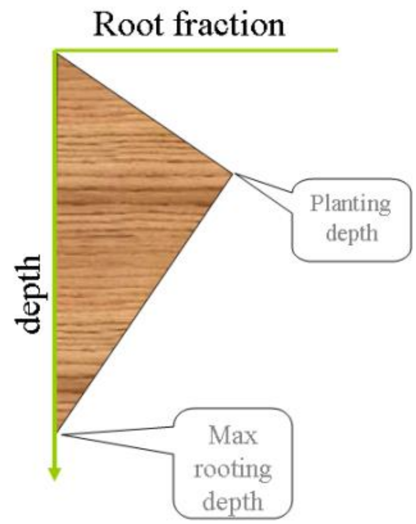


Figure 5.34: This diagram shows root distribution in the soil. The rooting depth changes with time until reaching the reproductive stage.

Quick Turf

The Quick Turf module has a constant root system that the user has to provide. LAI and height at each cut is also defined. The module assumes a uniform cut to a given LAI and height (Fig. 5.35). LAI and height will increase from one cutting to another. However, the cutting intervals can vary between cuttings. Cutting dates are in day of the year and are applicable to every year of the simulation. The interval between last cutting of a year and the first cutting of the following year is defined as the growing period for the first cutting of the following year. Turf can experience a period of dormancy at beginning of the year only. Users may define when turf is out of the dormancy; otherwise, the module will bring the plants out of dormancy when average 10-day temperature is above 0 ^{o} C. The module also allows user specified stover and dead roots returned to the soil at harvest. Dead roots will be zero if not specified. User defined plant N uptake is total uptake between cuttings. If user defined initial plant height is greater than height at cutting, the module will automatically cut it to the height after cutting.

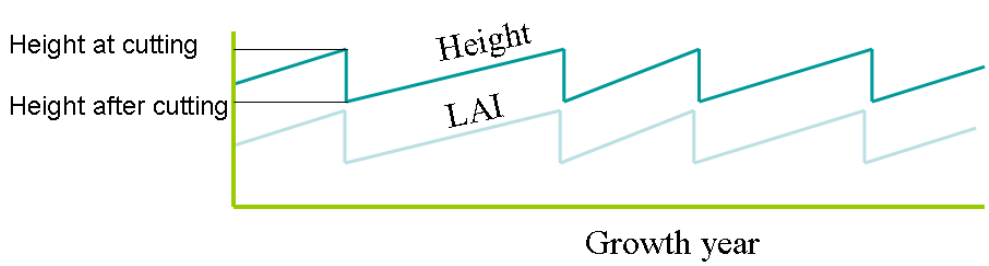


Figure 5.35: Diagram showing the height and LAI changes with each cutting.

Quick Tree

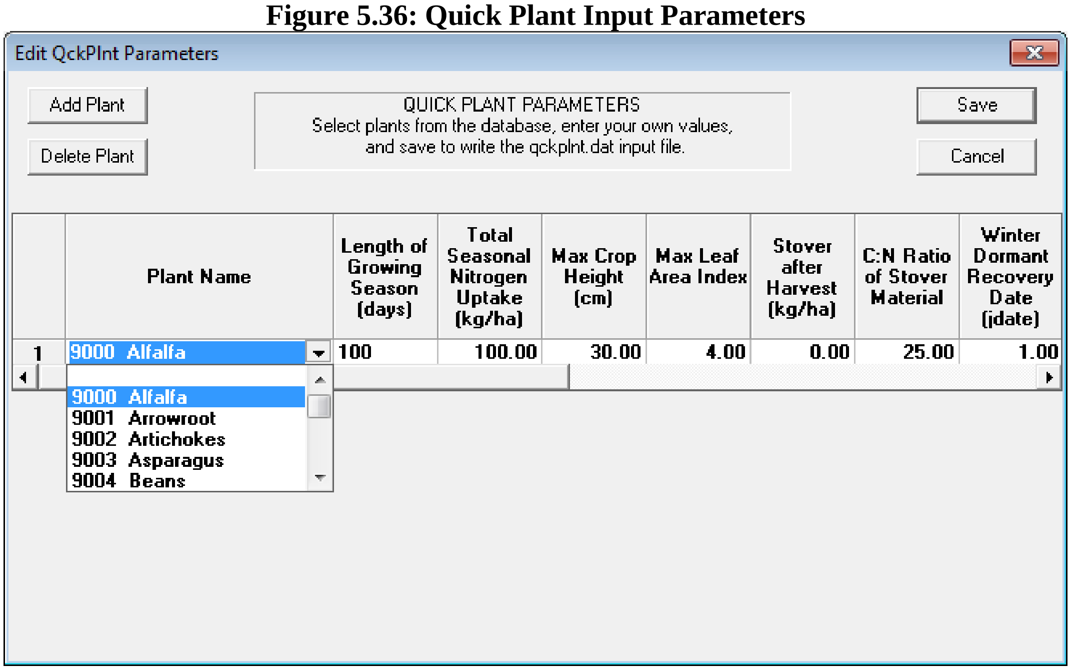
The Quick Tree module provides a way to simulate shrubs and trees where LAI may be the only variable during the simulation year. The user needs to provide constant plant height and rooting depth. LAI will change during the growing season. Users still need to ‘plant’ and ‘harvest’ a tree or shrub every year to start and end growth. The module also allows user specified dead leaves (stover) and dead roots returned to the soil at harvest. Dead roots will be zero if not specified. Plant N uptake is total amount during the growing season.

Select ***Input | Management Practices*** or  on the toolbar. Detailed information on quickplant and quickturf is available on the on-line help system *F1*

To Add a Quickplant Crop:

Click on ***ADD QCKPLNT*** on Figure 5.29a. Figure 5.36 will result.
Users are asked for total growing season length, total seasonal N uptake, maximum plant height, maximum LAI, crop residue (stover) amount and C/N ratio, dormant recovery date, and maximum rooting depth, root biomass at harvest, and C/N ratio of root biomass.

Figure 5.36: Quick Plant Input Parameters

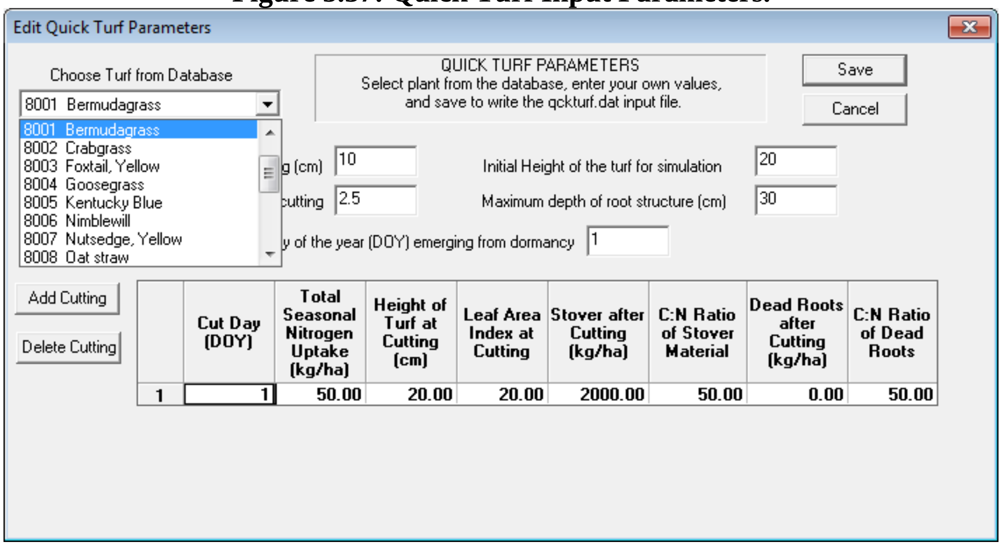


1. Select a plant from the dropdown, Click ***ADD PLANT*** to add another line to the spreadsheet for crop selection.
2. Click ***SAVE*** to add these plant identification numbers to crops available for selection on previous screen and to write the QCKPLNT.DAT file.

To Add a Quickturf plant:

Click on ***ADD QCKTURF*** on Figure 5.29a, users can parameterize a simple turf growth by providing total seasonal N uptake, cutting dates, height and LAI at cutting, maximum rooting depth, and stover amount and C/N ratio on Figure 5.37. Click ***SAVE*** write the QCKTURF.DAT file for the simulation.

Figure 5.37: Quick Turf Input Parameters.

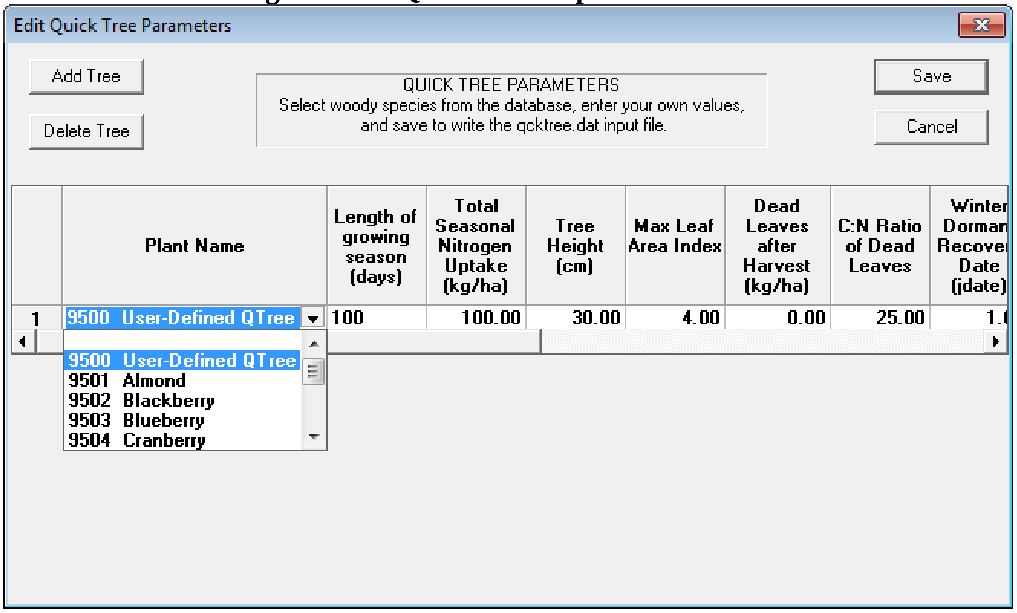


To Add a Quicktree Crop:

Click on ***ADD QCKTREE*** on Figure 5.29a.

Users are asked for total growing season length, total seasonal N uptake, plant height, maximum LAI, dead leaves after harvest to become residue amount and their C/N ratio, dormant recovery date, maximum rooting depth, dead root biomass after harvest, and C/N ratio of dead root biomass.

Figure 5.38: Quick Tree Input Parameters.



1. Select Tree or shrub from the dropdown. Click ***ADD TREE*** to add another line to the spreadsheet for crop selection.
2. Click ***SAVE*** to add these tree identification numbers to Crops Available for Selection on previous screen and with the QCKTREE.DAT file.

Important Note: Quick turf or tree may not be used in combination with other annual crops in the same scenario. The Crop Selection screen will list all of the available crops, but when choosing a perennial crop all other crops will be removed from the selected list and only the perennial will be selected.

Note 1: The Quick modules are not plant growth simulation models. For plant production models use the Generic or the DSSAT crop growth modules.

Note 2: Data for quick plant, turf and tree are written to QCKPLNT.DAT, QCKTURF.DAT and QCKTREE.DA, respectively, and stored in the scenario directory.

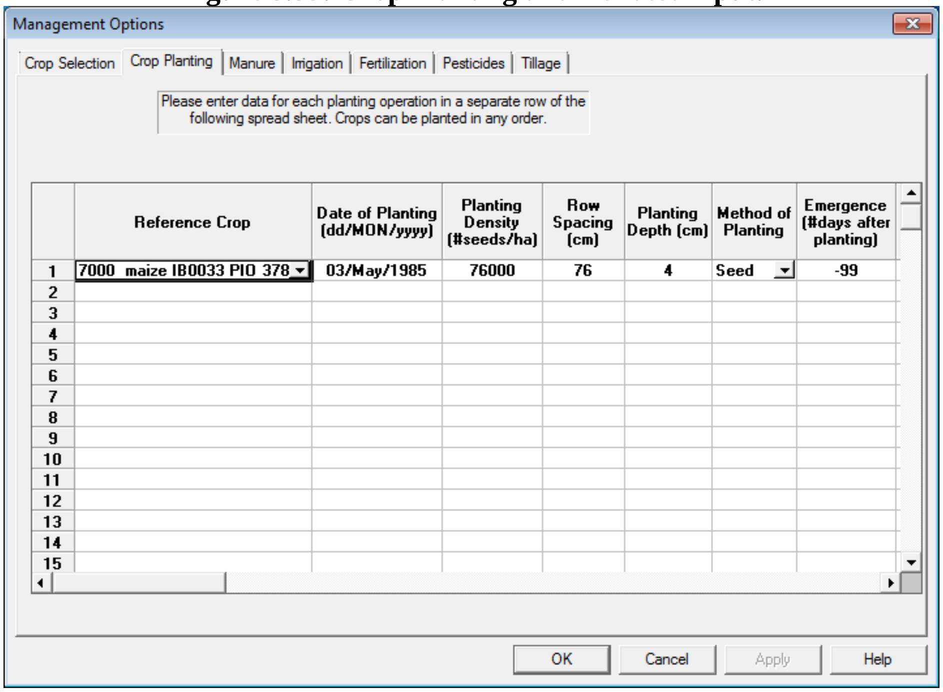
Completion of crop parameters screen will return you to the ***Crop Selection*** tab (Figure 5.29a). Select crops from the available list window and push the arrow to select them as crops for simulation. Once your crop is selected for simulation the management activities for the crop need to be defined on crop planting, manure, irrigation, fertilization, pesticides and tillage tabbed dialogs under ***Input | Management Practices***. The first field on all management asks for the "Reference Crop." This is the crop the management will be applied to.

Important Note: Reference crop is required and using the incorrect reference crop may disable a management event.

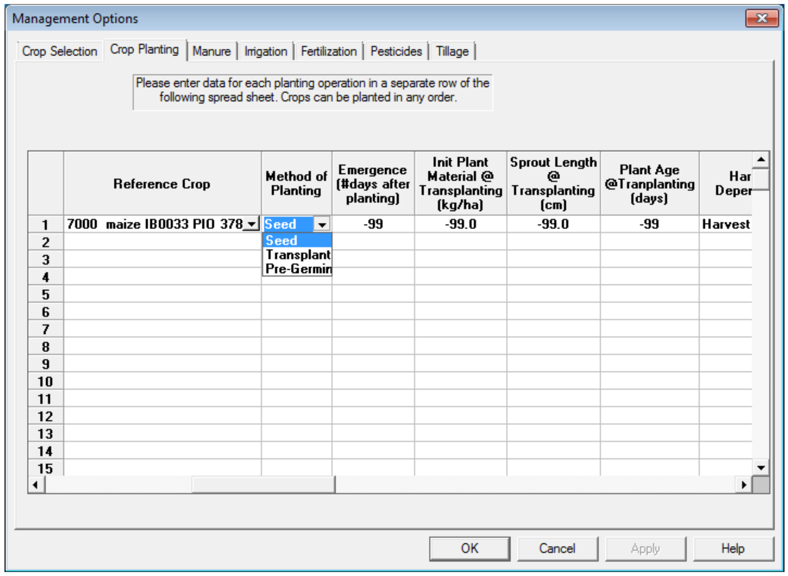
Crop planting

This screen is used to set up crop management information, such as planting (date, density, row spacing) and harvesting (date, efficiency, type). As with all the management input screens, the user can use the right mouse button to cut, copy, and paste selections for easy parameterization.

Figure 5.39: Crop Planting and Harvest Input.

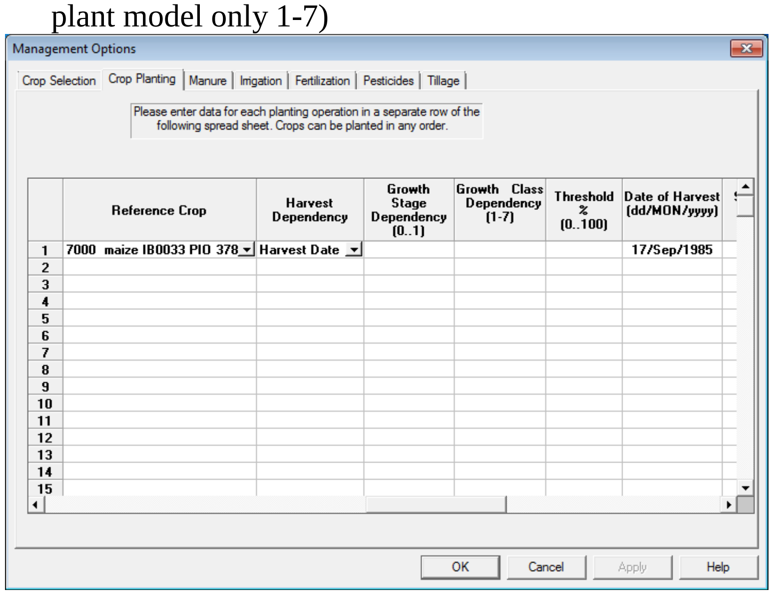


1. Select the Crop you want the management to apply to (reference crop drop down loads from the crop list you selected for simulation)
2. Double Click Date to bring up calendar to set plant date
3. Enter density, row space and planting depth
4. **Note:** Planting depth is in ***cm*** now. Previous RZWQM versions planting depth was at a numerical node.
5. The default plant method is seed. If you choose transplant for the DSSAT crops like potatoes you will need to fill in age, length and weight at transplanting.



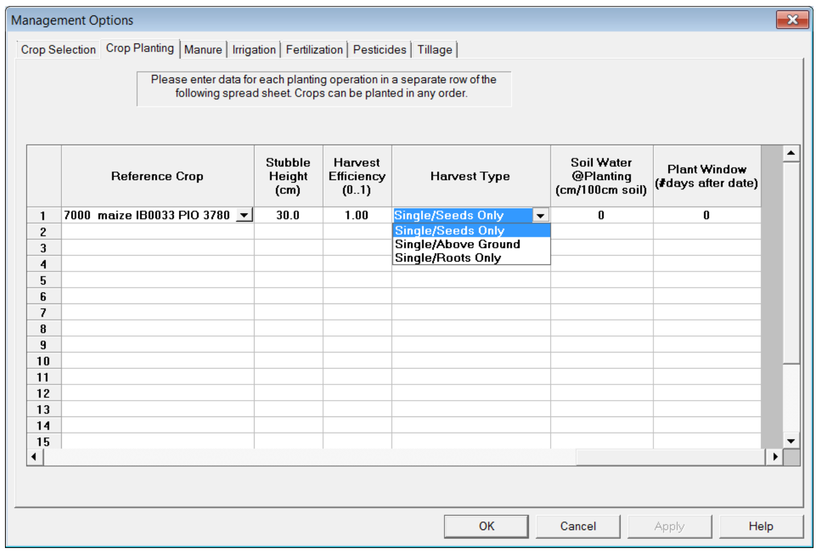
Important Note: Generally -99 denotes that DSSAT will default a number if not entered; however sprout length entry is required for potato. Emergence date is only used for the potato model.

6. The harvest dependency will dictate entry for the fields following;
   - Harvest Date requires Date of Harvest only (Calendar day)
   - Growth Stage requires Growth Stage only (0-1)
   - Growth Class requires Growth Class and Threshold % (generic plant model only 1-7)



7. Harvest efficiency and new input fields for a planting date rules finish the planting entry.
   - If seed grain harvested, efficiency is applied to seed yield.

- If aboveground biomass harvested, efficiency is applied only to the vegetative portion of the total aboveground biomass excluding seed yield.
- If roots only harvested, efficiency is applied to tuber yield.
- If a number greater than 0 is entered for cm of Soil water per 100cm of soil is entered, planting will not occur till condition is met.
- If a number greater than 0 is entered in Plant Window the Soil Water rule will be considered only that number days after the proposed plant date.

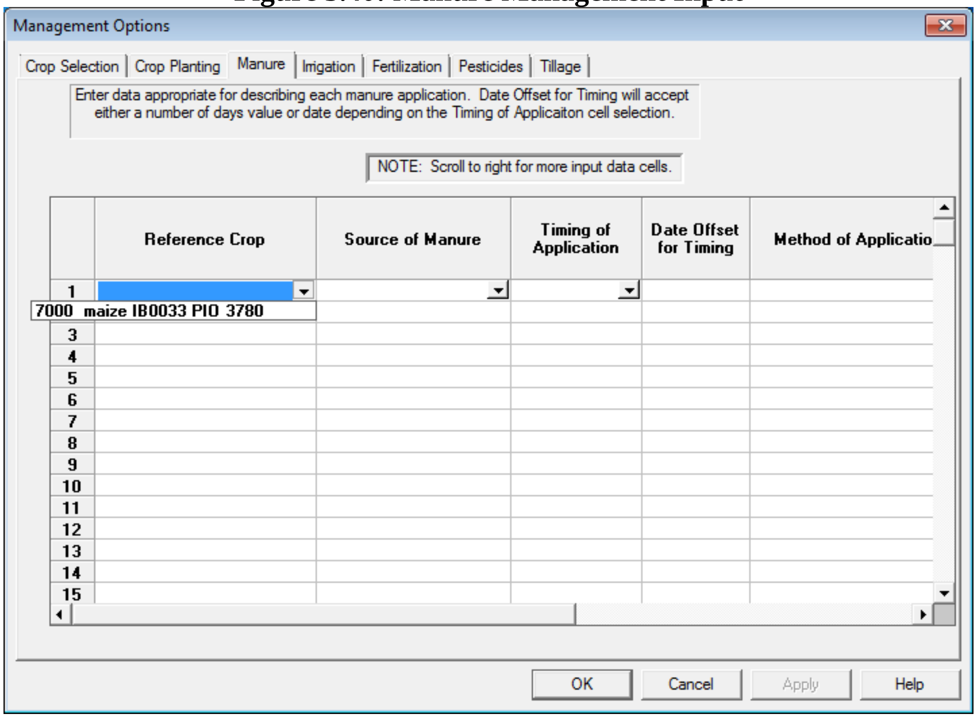


8. Click ***APPLY*** to save data then focus on another tab

Manure

This screen is designed for inputting manure management information, such as manure type, application time, application amount, fraction of carbon, C/N ratio, total phosphorus, and water extractable inorganic and organic P. A specific reference crop is chosen for the management to apply to.

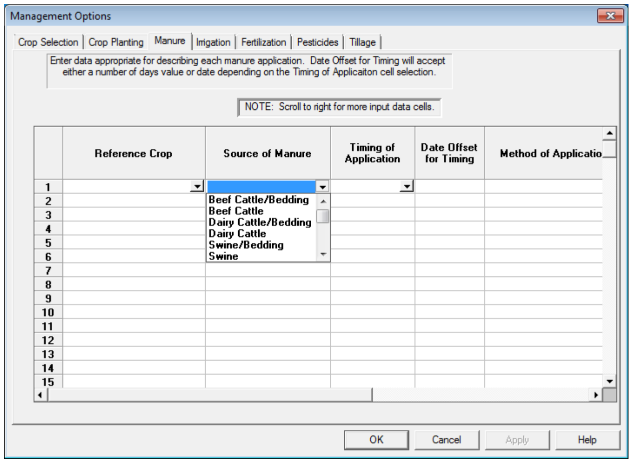
Figure 5.40: Manure Management Input



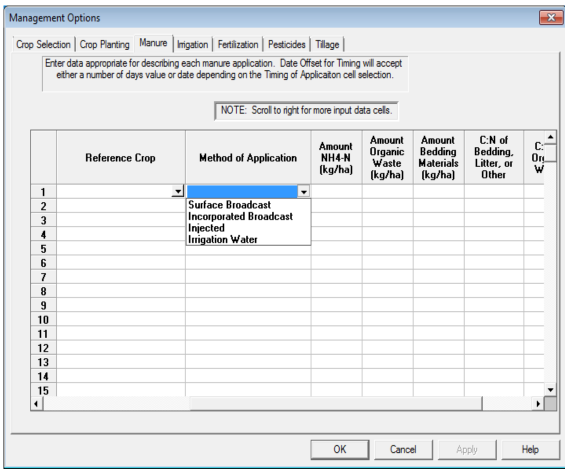
1. Enter Crop applied to
2. Select source of Manure, Timing (specific date most commonly used), offset
3. Select Application Method
4. Enter Composition and C:N Ratios
5. Click ***APPLY*** to save data and proceed to the irrigation screen

Note: Manure types will have a default C:N ratio the user should override. The user must enter C:N ratio of bedding if manure/bedding type.

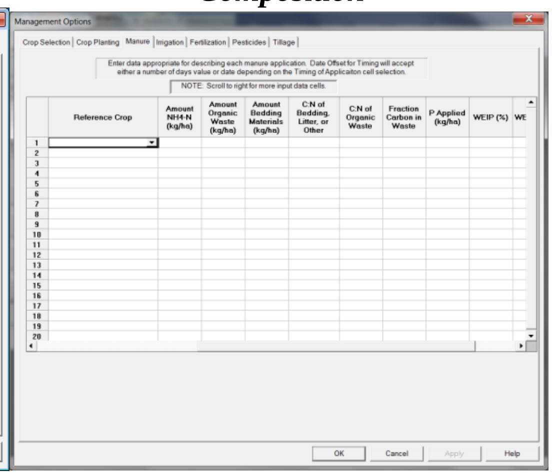
Manure Source



Method



Composition



Important Note: If manure is applied the Slow Pool C/N Ratio on Site Description Nutrients Screen may be reset if it is higher than the C/N ratio of the applied manure.

Note: RZWQM2 will default a tillage (field cultivator type) event if an incorporated method is specified and the user has not specified tillage on that date.

Irrigation

This screen is used to schedule irrigation events on specific dates, fixed intervals, or by root zone depletion. Irrigation amounts are in ***cm*** (different from rainfall) and summarized information is shown on the irrigation screen (Figure 5.41.)

Click ***ADD/EDIT IRRIGATION EVENTS*** button. You may enter events by Specific date, fixed intervals, depletion of the root zone moisture method or evapotranspiration deficit method. Each method results in a different format written to the Irrigation Management Section of the RZWQM.DAT input file. See Appendix 6 for example formats.

Under "Use?" Unchecking a box will prevent the data from being written to the RZWQM.DAT file; but, it will still be saved in a .ss2 file for later use. You may use any combination of Specific Dates, Fixed Interval and either Root Zone Depletion **or** ET Deficit. ***Note:*** You **cannot** use the Root Zone Depletion and ET Deficit methods at the same time.

Figure 5.41: Irrigation Events Summary Screen

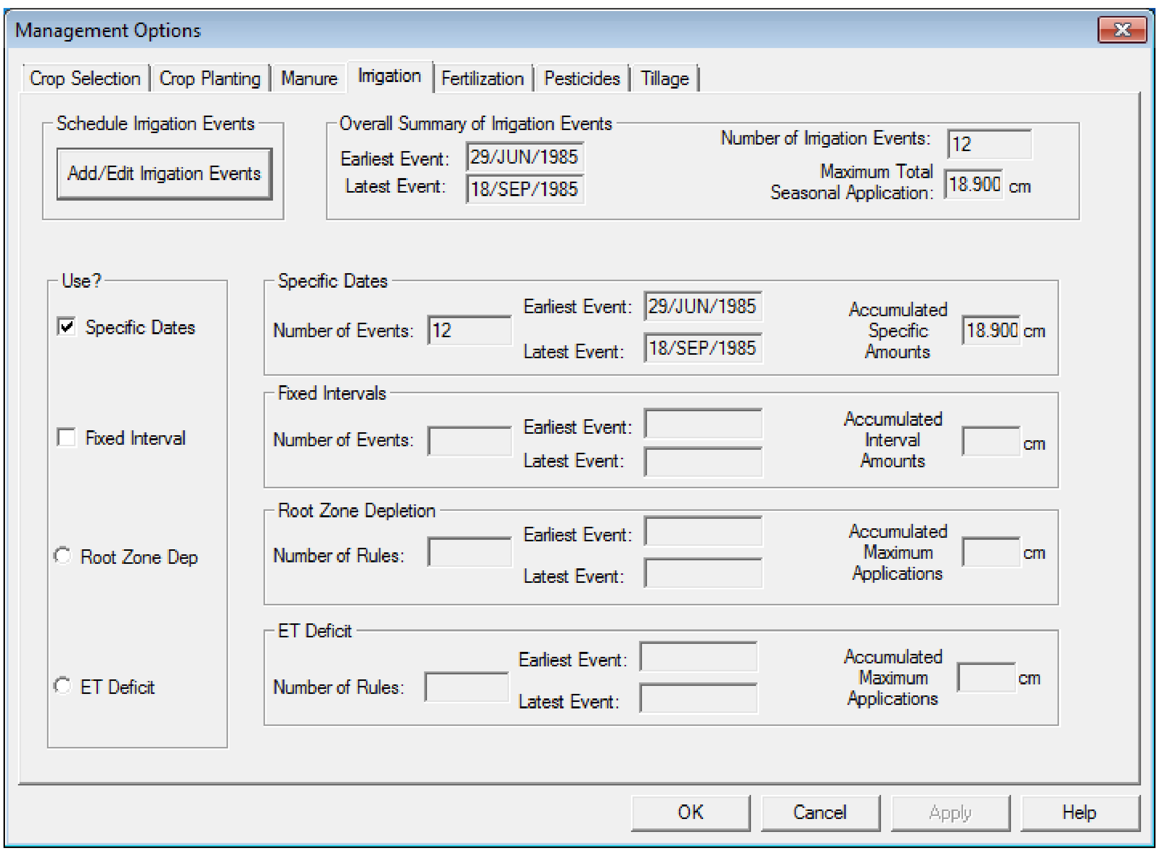
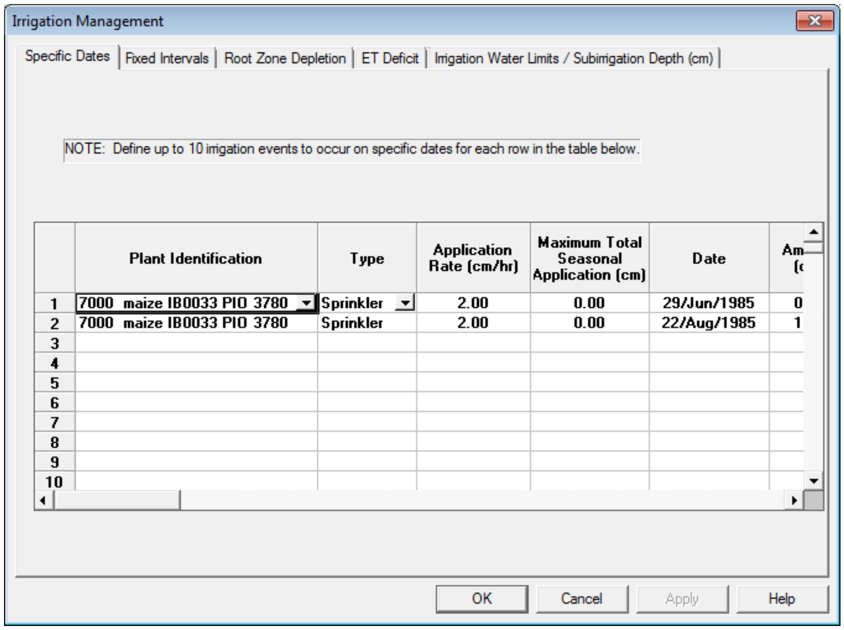
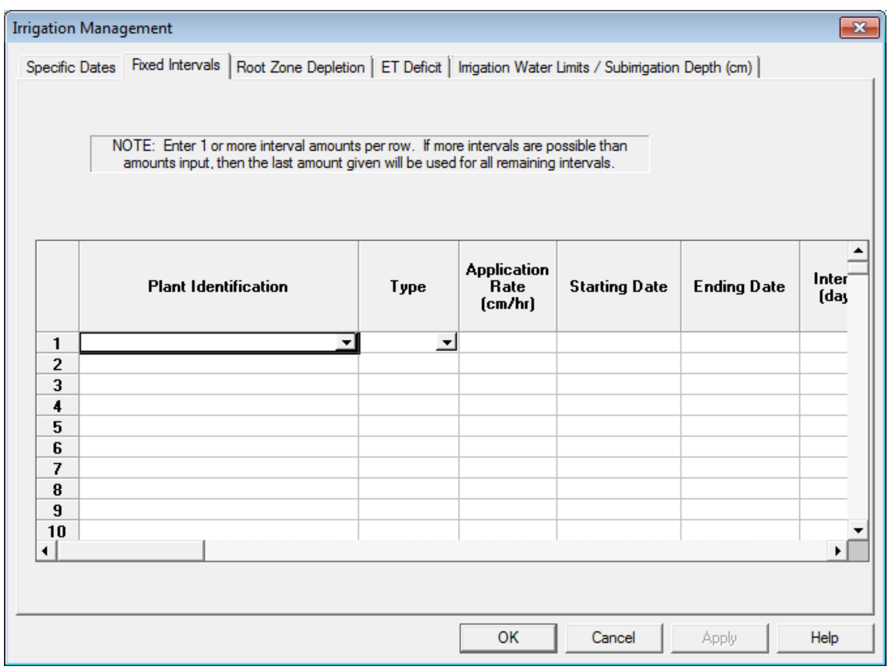


Figure 5.42: Irrigation by Specific Date



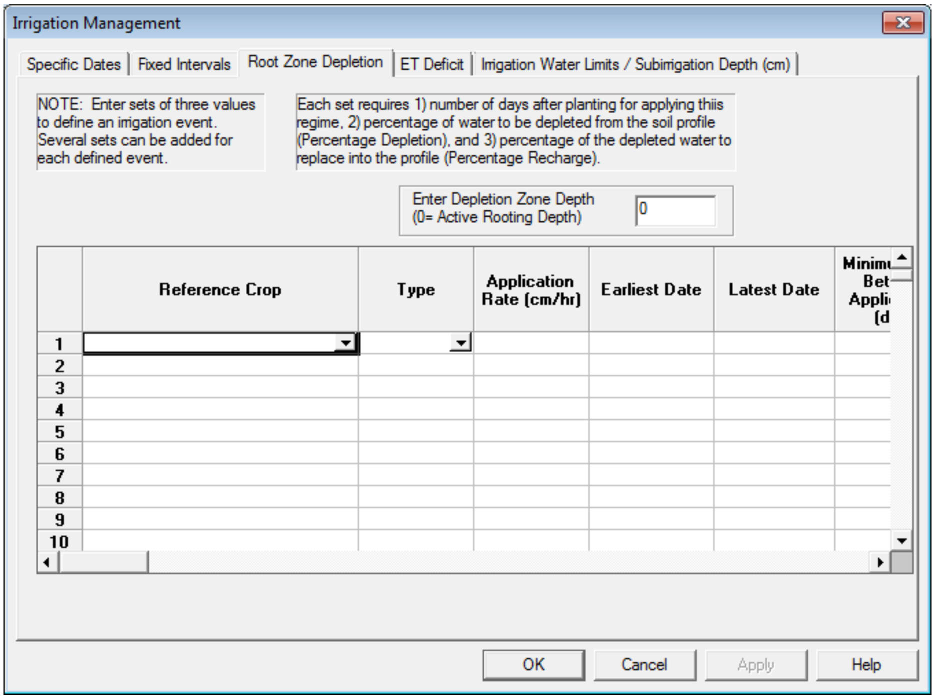
1. Enter reference crop, type, rate if sprinkler, seasonal maximum, date and amount (Figure 5.42). Continue on another line to complete all dates for that crop.
   Note: Application Rate is only applicable to sprinkler irrigation. No entry in this field for other types. A maximum of seasonal application of 0.0 means irrigation water is unlimited for the irrigation scheduling.
2. Click ***APPLY*** to save data and remain in the tabbed dialog.
3. Click **OK** when done with irrigation management

Figure 5.43: Irrigation by Fixed Interval



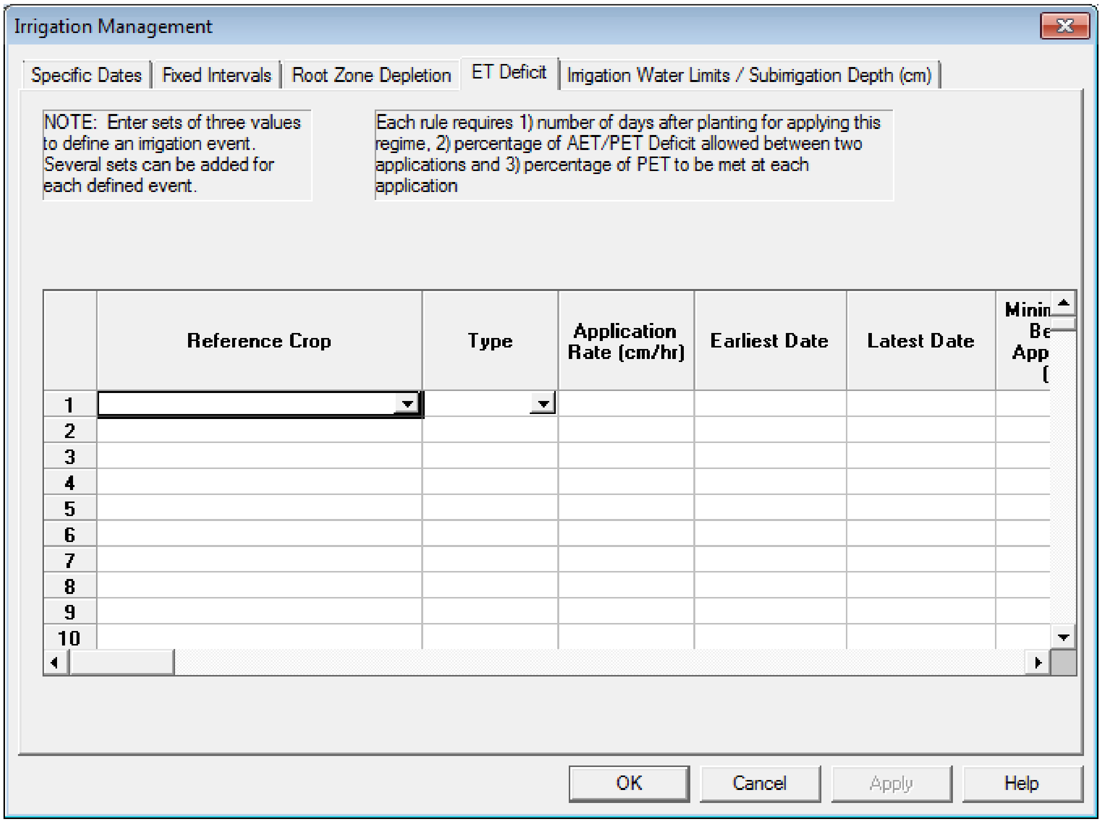
1. Enter reference crop, type, rate if sprinkler, start and end date, seasonal maximum, time interval, and amounts (Figure 5.43).
2. Click ***APPLY*** to save data and remain in the tabbed dialog.
3. Click **OK** when done with irrigation management

Figure 5.44: Irrigation by Root Zone Water Depletion Rule



1. Enter Depletion Zone (cm) or default (0) to use active rooting depth (Figure 5.44).
2. Enter Reference Crop and method (sprinkler if you want a specific application rate)
3. Enter Days after planting to apply method, depletion and recharge percentage. **Note:** The equation used for depletion and recharge are in the column titles. Depletion trigger is actual Plant Available Water / Total Plant Available Water represented as a %. Recharge trigger is % of Total Plant Available Water. Total plant available water is 1/3 bar soil water content – 15 bar soil water content for in the depth of the depletion zone considered.
4. ***APPLY*** to save and remain in dialog
5. ***OK*** to exit to Irrigation Summary

Figure 5.45: Irrigation by Meeting the ET Deficit



1. Enter reference crop and method (sprinkler if you want a specific application rate)
2. Enter days after planting to apply method, depletion and recharge percentages. **Note**: Irrigation deficit trigger refers to AET/PET expressed as a % and Irrigation recharge % refers to % of potential evapotranspiration (PET).
3. Apply to save changes.
4. Okay to exit to irrigation summary.

Important Note: A “Maximum Total Seasonal Application of 0.0 means water supply is *unlimited* for the season. A value in that field indicates the planting season is limited to that number.
Note: Application Rates are only applicable to sprinkler and subsurface irrigation. No entry in this field for other irrigation types. Rate is calculated based on soil hydraulic conductivity for drip irrigation.

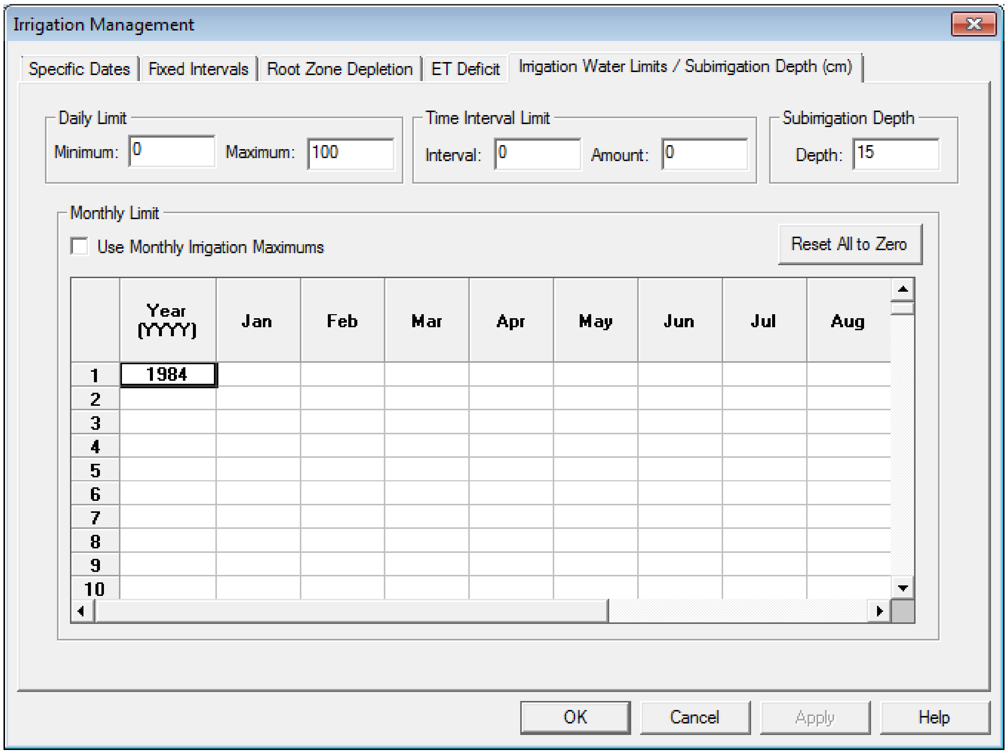
Tip 1: Use of specific date, fixed interval, and a rule is allowed in combination but not recommended. Specific date for a single irrigation say preplant is commonly used in conjunction with a rule.

Tip 2: Irrigations may not occur if you reach seasonal maximum or any other limit set on the irrigation limits tabbed dialog (Figure 5.46) . Set seasonal maximum at 0.0 to make sure all scheduled irrigations are implemented.

Tip 3: Always check the MANAGE.OUT file to verify that irrigations were applied correctly.

Warning: If you put your depletion zone as the whole profile you will get water being applied too often to try to fill the profile to your recharge specifications.

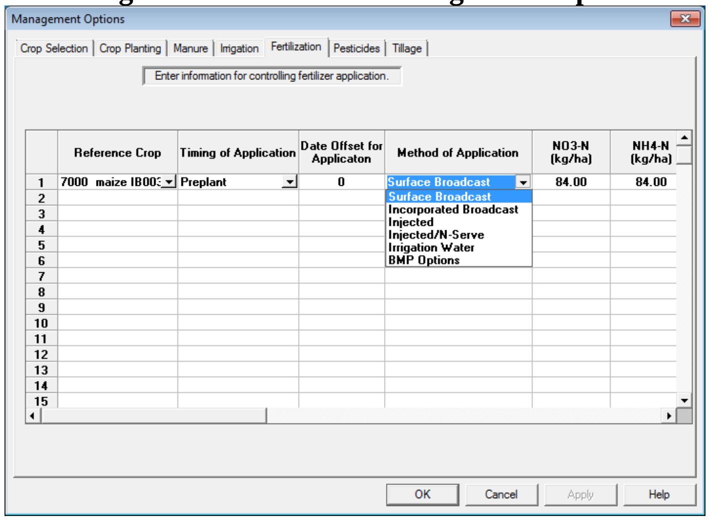
Figure 5.46: Irrigation Water Limits (Daily, Monthly, Multi-year Interval) and Subirrigation Depth



Fertilization

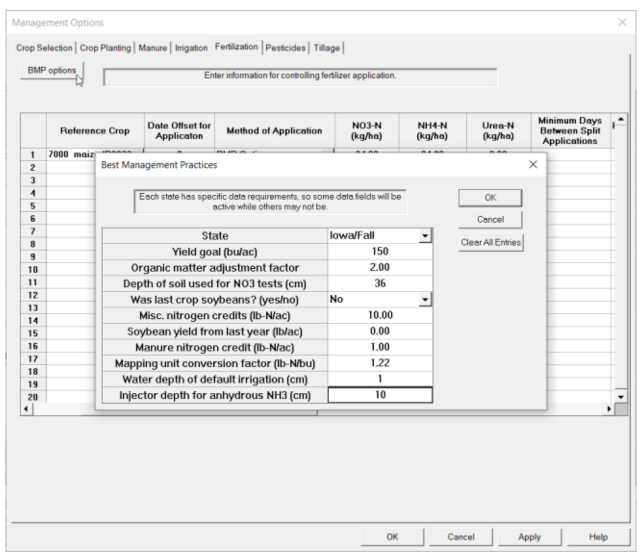
All the inorganic fertilizer applications are scheduled on this screen. It contains information on fertilizer type, application amount, application method, application time, and a Best Management Practices (BMP) option. When BMP is selected, users should go to the ***BMP OPTIONS*** button to parameterize BMP options. Two other options on the ***BMP OPTIONS*** screen are default irrigation amount (used for fertigation) and injection depth of anhydrous ammonia.

Figure 5.46: Fertilizer Management Input



Important Note: Use injected application method for "Banded". Amounts are NO_3-N, NO_4-N or Urea-N and not mass of fertilizer. The values are not needed if BMP options method is chosen.

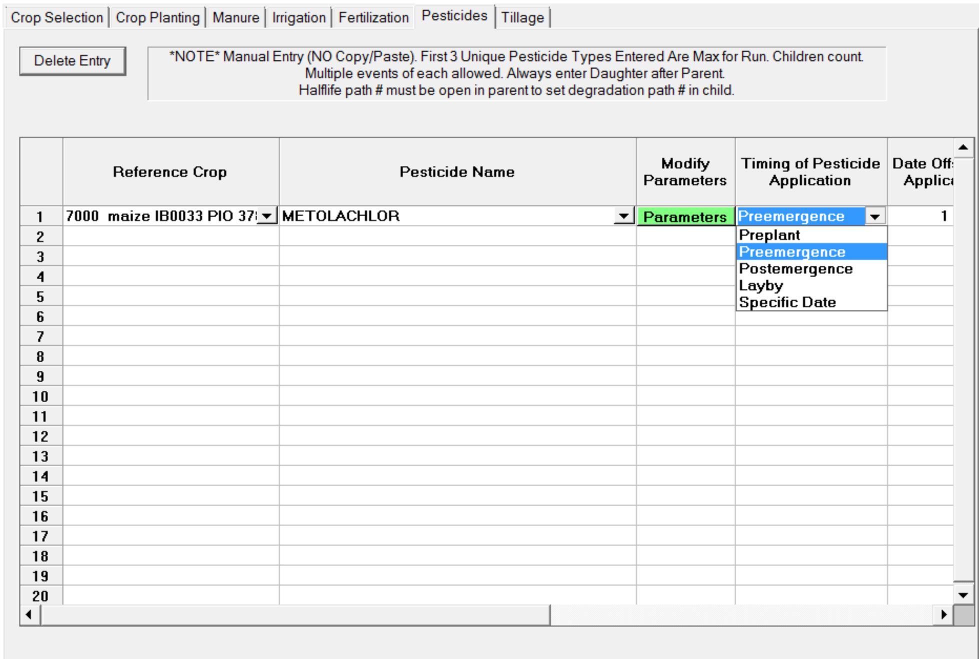
Note: "Irrigation Water" (fertigation) fertilization method defaults to a 1 cm irrigation if no irrigation event is specified for the fertilization date. When NH3 applicator is used, users may use the BMP options to define a injection depth (default = 10 cm)



Pesticides

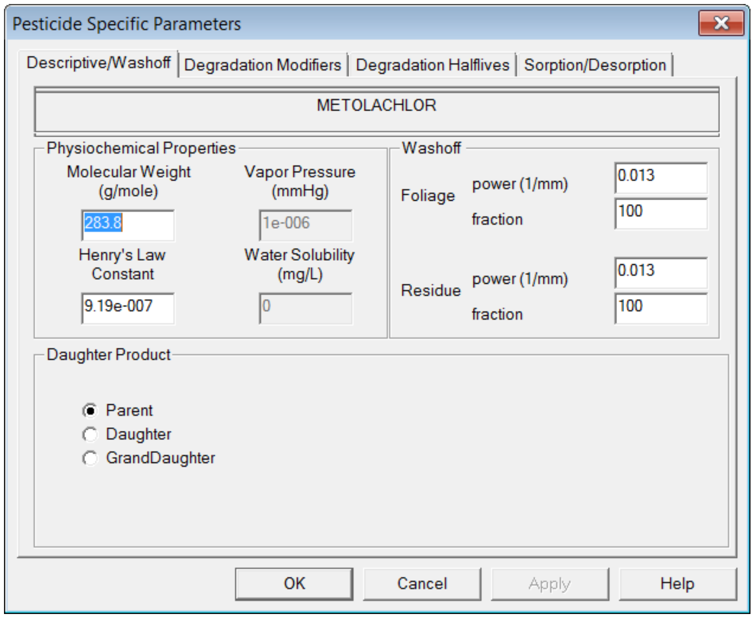
Pesticide management and pesticide parameters are input on this screen. (Figure 5.47a) It contains information on reference crop, pesticide type, application time, application amount, application method, and daily fraction of pesticide released (for slow released form).

Figure 5.47a: Pesticide Management Input



Pesticide parameters may be modified on the ***MODIFY PARAMETERS*** column. Click on the ***Parameters*** letters and Figure 5.47b will appear for specifying pesticide wash off parameters, ancestry (no products, parent, daughter granddaughter), half lives in the soil-crop system, and adsorption mechanisms. Default pesticide parameters are from Wauchope et al. (1992). Version 4.0 was updated to have only Parent, Daughter and Granddaughter as Parents without a daughter specified as the chemical following the first in figure 5.47a are equivalent to the previous ***None*** category. Choose ***Parent*** on the **Descriptive/Wash off** dialog for the first unique pesiticide a pesticide that its daughter products are not tracked by not following the chemical on 5.47a with the chemical type daughter.

Figure 5.47b: Pesticide Parameters Input



If a daughter product is tracked in the system you must first specify the pesticide as a ***Parent*** as shown in Figure 5.48a, and a degradation half-life for the process producing the daughter. For example, a user may enter 5 days as the foliar biotic degradation half-life to be used as a process for daughter formation in Figure 5.49a

Enter a daughter on the next pesticide entry line 2 (Figure 5.47a) and set it as ***Daughter.*** You may set the application timing the same as the parent with zero rates. Specify the degradation process (e.g. Foliar Biotic) for its formation and the % of the degraded material that is the daughter product (Figure 5.48b). Degradation half-lives of the daughter should be entered only if there is another degradation product from the daughter (Figure 5.49b). If the daughter is produced the same way in both the soil surface and subsurface, select #10 from the dropdown list in Figure 5.46b.

Soil surface and subsurface half-lives are distinguished to give user flexibility in defining degradation pathways. Soil surface refers to the top centimeter of the soil. Soil subsurface degradation half-lives may be adjusted for soil depth for aerobic and anaerobic conditions.

Figure 5.48a: Parent Descriptive/Washoff

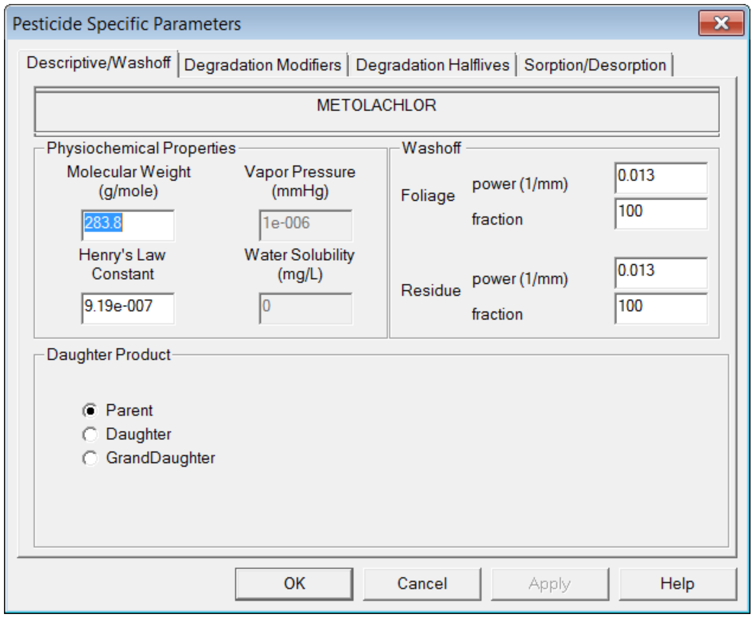
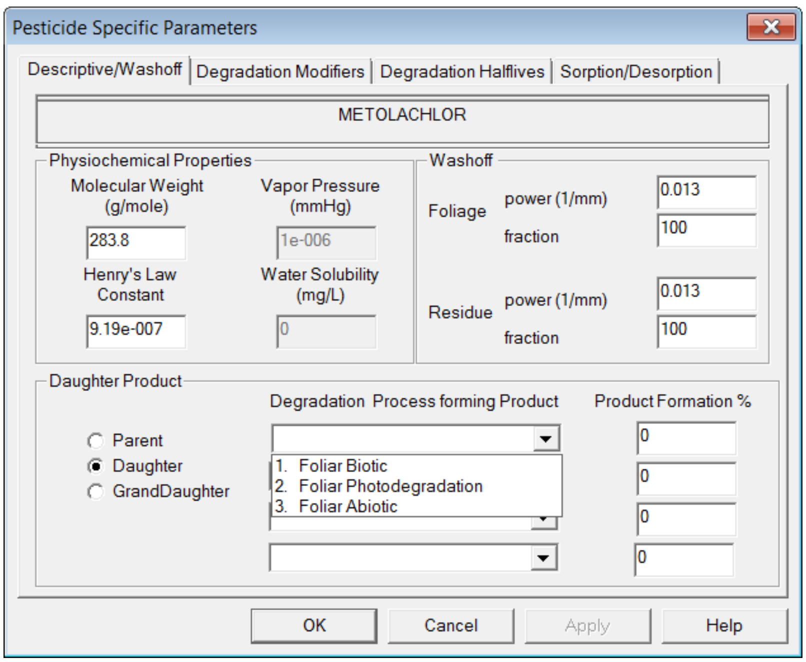
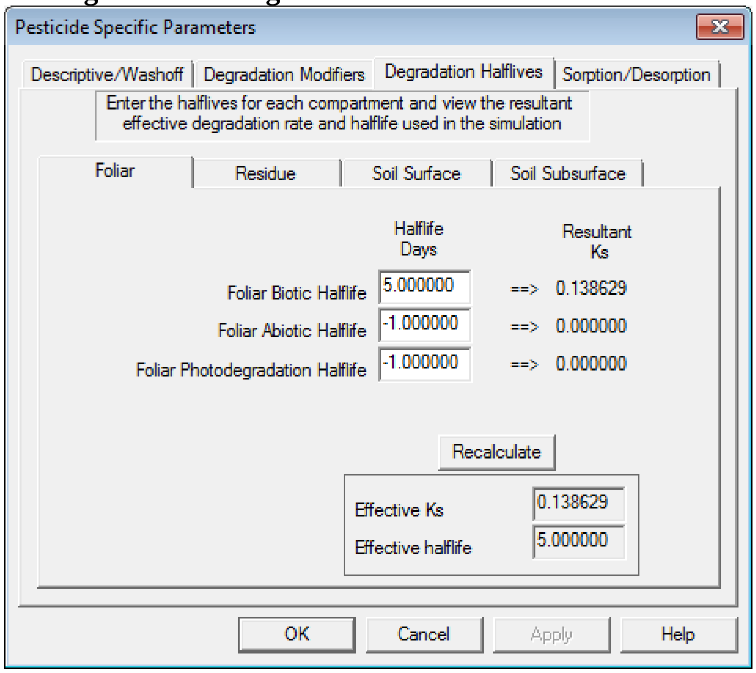


Figure 5.48b: Daughter Descriptive/Washoff and choice of Degradation Process



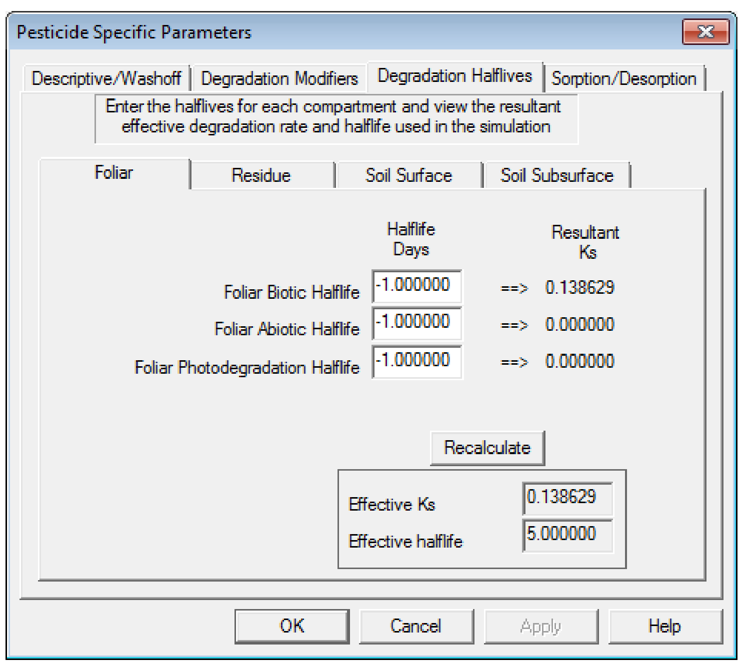
Note: Vapor pressure and water solubility are not used in the current model.

Figure 5.49a: Degradation Half-lives for the Parent



Note 1: Pesticide degradation half-lives may be adjusted for soil moisture, pH and temperature.
Note 2: Users define various degradation half-lives based on their experience. The effects of these processes are additive. Users may use a field as a surrogate if a process is not defined on the half-life list. After half-life data entry, click ***Recalculate*** to show an effective half-life.

Figure 5.49b: Degradation Half-lives for the Daughter without another Degradation Product



Note: Value of -1 for half-lives denotes process not used for simulation.

Enter the reference soil water content, temperature, pH and Walker adjustment for aerobic processes to re-calculate effective half-lives (Figure 5.50) from the reference half-lives entered under half-lives tab dialog (Figure 5.49a and 5.49b). If you would like to adjust these with depth choose radial button "yes" and set a value for the maximum level of adjustment.

Figure 5.50: Degradation Modifiers

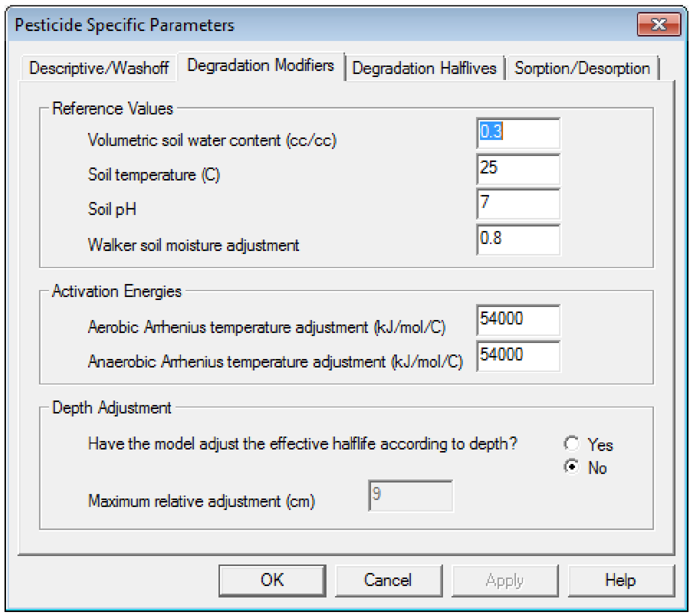
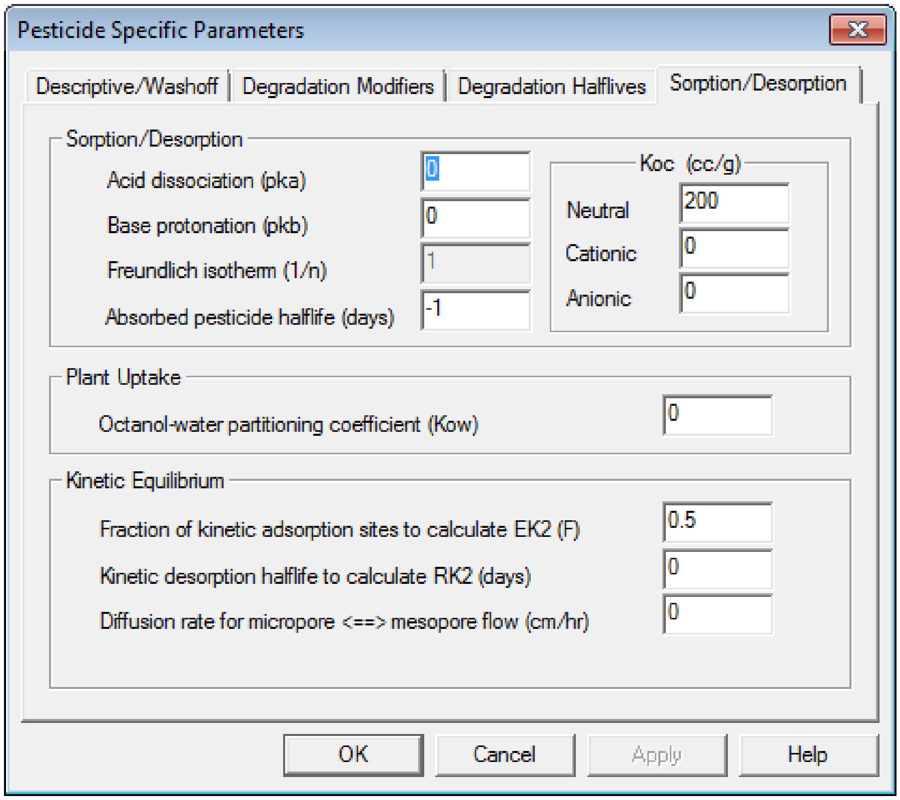


Figure 5.51: Sorption/Desorption



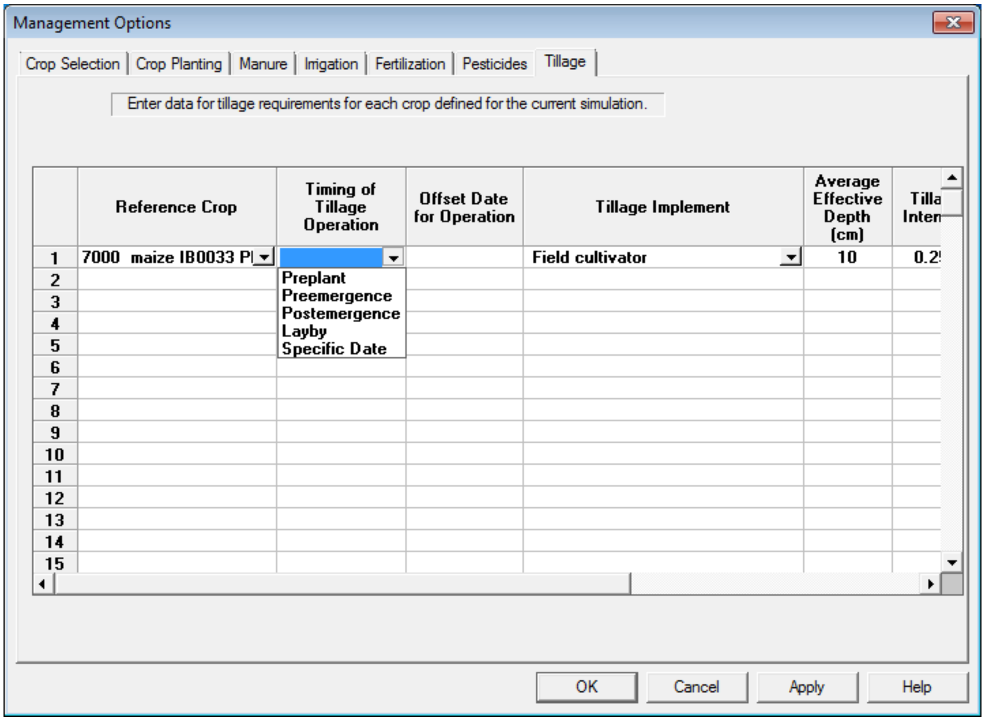
Important Note: Plant uptake of pesticide is covered on page 182 of the RZWQM book chapter by Wauchope et.al. (2000). Zero represents no uptake; a higher Kow represents lower uptake and a lower Kow represents a higher uptake of pesticide by the plant.

Note: Absorption is assumed to be linear and a fraction of absorption may be kinetic. A user can also specify diffusion between micro- and meso- pores. Greyed fields in the pesticide section are not currently used in simulation.

Tillage

Tillage practices are scheduled on this screen, the crop applied to, tillage type, tillage time, intensity, and effective depth of disturbance (Figure 5.51). A change in reference crop can change the tillage intensity. A change in tillage type can change both the tillage intensity and the effective depth of disturbance. The user may then override these defaulted values.

Figure 5.51: Tillage Management Input



1. Enter Reference Crop to apply tillage to.
2. Enter timing and offset. Timing can be prior to planting or emergence, post emergence, a number of days after harvest (Lay by) or on a specific date. The offset is the number of days to offset the timing by; except in the case of specific date, the offset is a calendar date tool.
3. Select Implement, depth and efficiency.
4. Click ***APPLY*** to save data focus on another screen
5. Click ***OK*** to exit the Management Input
6. You are done with scenario Input -- ***FILE | SAVE SCENARIO***

Note: All management events are recorded in the MANAGE.OUT file. View this file to verify all events are implemented correctly. Incorrect reference crop is one example that may disable an event. ***F1*** help on tillage screen error no wheel compaction effects simulated.

Important Note: During tillage and reconsolidation Ksat estimation by effective porosity Ahuja et al. (1989) is used: $K_{ep} = 764.5(\theta_s - \theta_{1/3})^{3.29}$

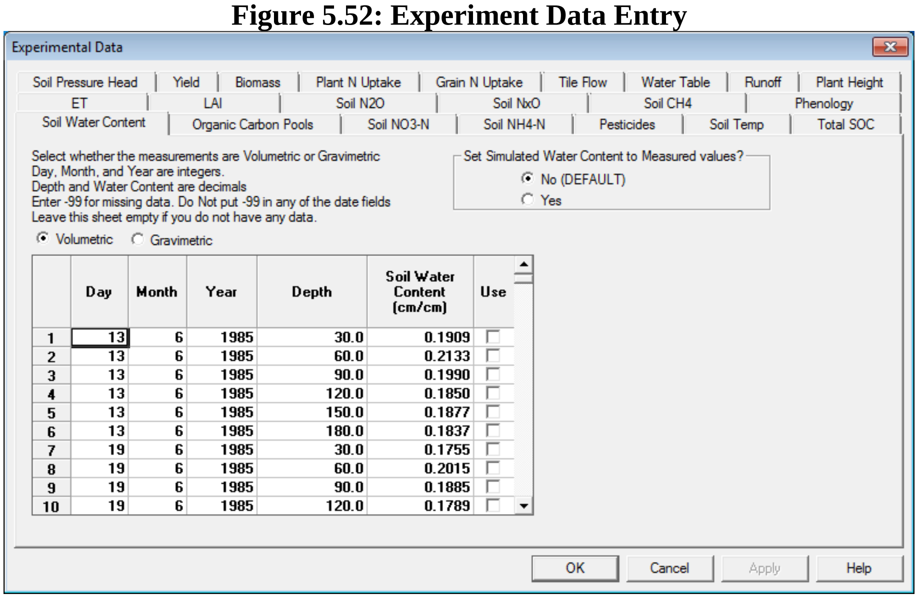
5.3.5 Define PRMS Snow Parameters

Click ***INPUT | PRMS SNOW PARAMETERS*** to customize snowpack dynamics for the PRMS snow accumulation module (Leavesley et al., 2006). See Appendix 4 for the details of this text file accessed by WordPad from the interface. The user will often customize SNOWPACK PARAMETER 1.1, fraction of snow infiltrating if losses (evaporative or runoff) from the snow pack are high.

5.3.6 Define Experiment Data Observations

Click ***INPUT | EXPERIMENT DATA*** to enter observed data from your experiment and RZWQM2 will perform comparison statistics of observed vs. predicted for variables allowed in the format. The EXPDATA.DAT, text file is edited via tabbed dialog spreadsheets from the Version 2.4 and later versions interface (See Figure 5.52). The new format is listed in Appendix 7. The RZWQM2 science application provides the dates and values compared and statistics of RMSE, R-SQUARED and T-TEST in the COMP2EXP.OUT file in Appendix 9. Please note that the Experiment Data entry may change the Output Variables you have selected in the "***SIMULATION CONTROLS | SELECT OUTPUT VARIABLES***" Screen (See Chapter 6) if needed for computation and comparison. This is done in order to insure that the Comp2Exp.out file is generated properly.

Figure 5.52: Experiment Data Entry



Observed experiment data is required to assess simulation performance in a comp2exp.out file. It is also required to use automated parameter estimation and the non-linear uncertainty analysis with P-EST utilities. A Minimal observed dataset would include: several soil water measurements for the profile throughout the season, final yield and biomass, and a maturity date (can assume harvest date - 7 to 10 days). Given experimental data the science will perform comparison statistics of observed vs. predicted for variables allowed in the EXPDATA.DAT and results written to Comp2exp.out as well as performance graphics will be available. If the Comp2exp.out is empty a problem may exist in the expdata.dat. Please check the RZWQM2.LOG file for what section may have issues.

Important Note: If you check USE to reset soil water during run you will not be able to use the Project Summary Analysis Seasonal Annual Massbal button. Please check your scenario massbal.out file instead. <u>LAI reset also adjusts leaf biomass, not leaf N.</u>
Note: OK on this the experiment data sheet SAVES and WRITES data to the EXPDATA.DAT file in a standard format for other program utilities.

Chapter 6:
Scenario Simulation Controls, Modifiers, Execution

Once scenario input is completed, Click ***Simulations Controls*** or on the toolbar to: select output variables, set up items breakthrough node monitoring, module shutdown, and debug flags for model output files. Debug flag can also be used to check new meteorology files created for proper format before allowing the scenario to run. In order to assure correct simulation results, users are recommended to run with a consistent set of input and output options and check them periodically.

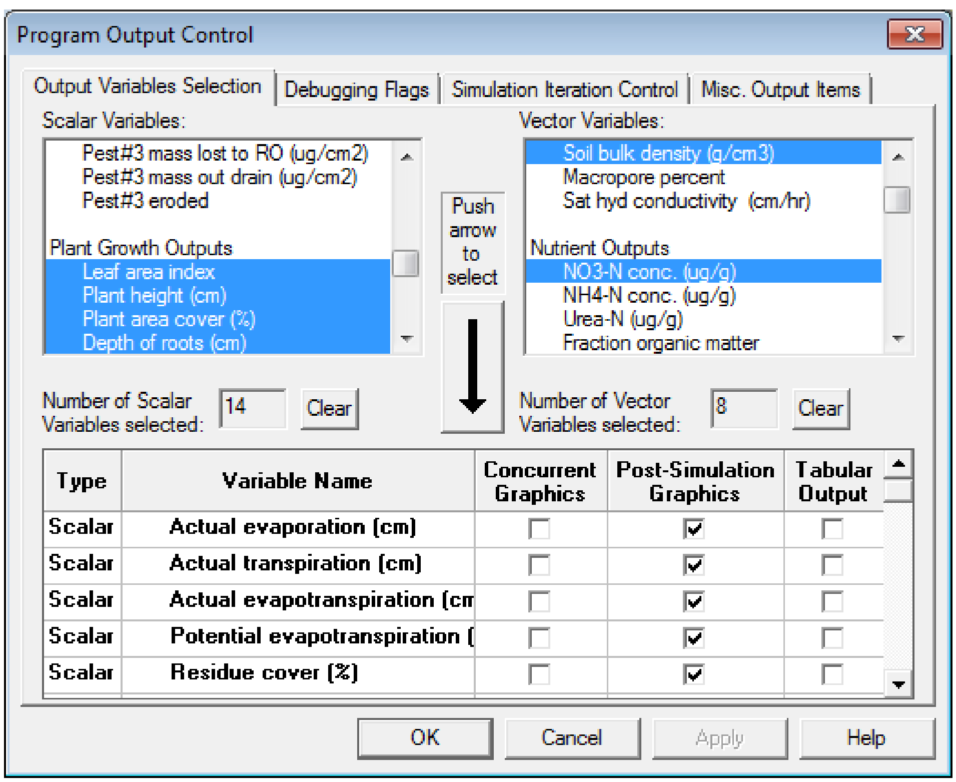
6.1 **Select output variables and controls**

The following screen may be activated by selecting any of the options on the ***OUTPUT*** menu except for the ***VIEW OUTPUT FILES*** option. Four sub-screens are available for correctly setting up output variables and controls.

6.1.1 Output variables selection screen

Users may select from 98 scalars (value-time) and 51 vector (value-horizon depth-time) output variables.

Figure 6.1: Output Variable Selection under Simulation Controls



Important Note 1: Push the downward arrow to select variables into the lower spreadsheet. Scalars are stored in DAILY.PLT and vector variables are stored in

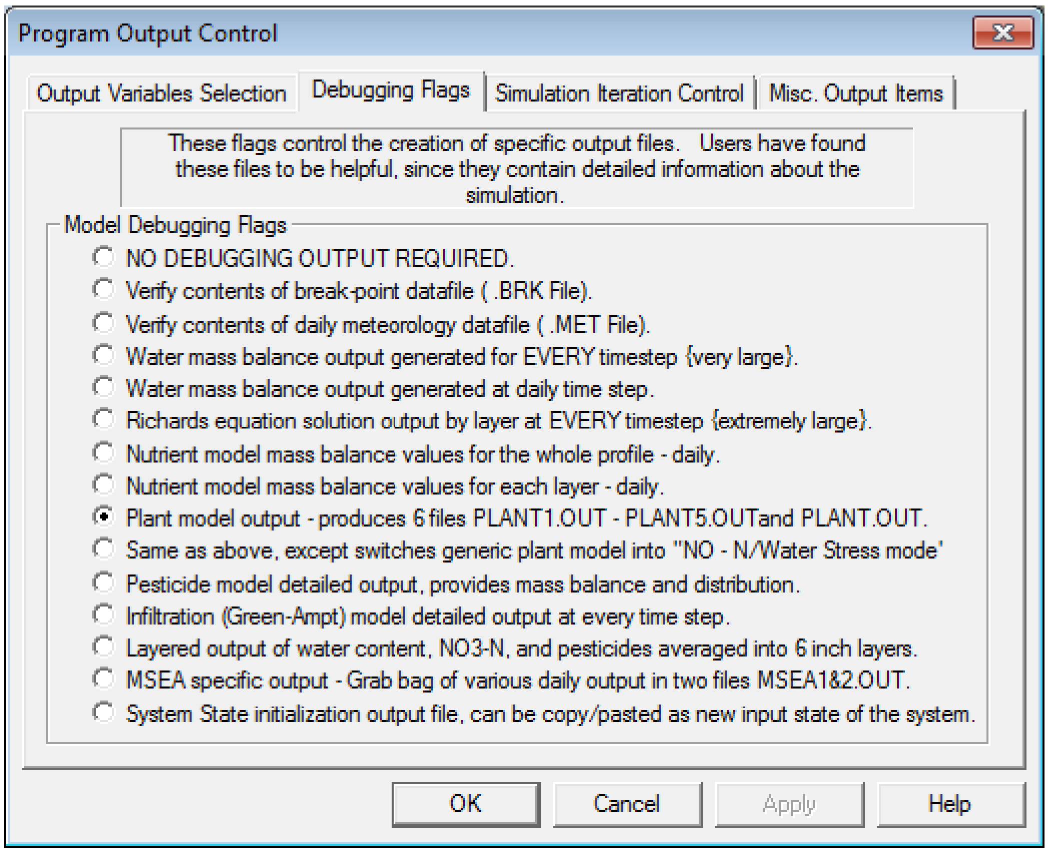
LAYER.PLT. The model will not create these plot (*.PLT) files unless at least 1 output variable is chosen.

Important Note 2: Selection of Output Variables is currently limited to 20. Removing this limit and resizing the array will require users to import their scenarios to new memory size; it will not be changed until a new science feature requires an import.

6.1.2 Debugging flags screen

Users may select the output files that are most related to their experimental Interests by selecting different debug options.

Figure 6.2: Debugging Flags under Simulation Controls



Important Note: Selecting “Verify contents of break-point data file” and “Verify contents of daily meteorology data file” only checks weather files when the model is executed with ***RUN*** (the science is not run). The 4th debug flag and below are various options for outputs (*.OUT) and the science is executed with ***RUN*** (See Appendix 8).

Tip1: Use flag 2 and 3 to do a pre-run verification of user created weather *files* ****.BRK*** *and* ****.MET***. Results of the verification will be shown in *RZWQM2.LOG*.

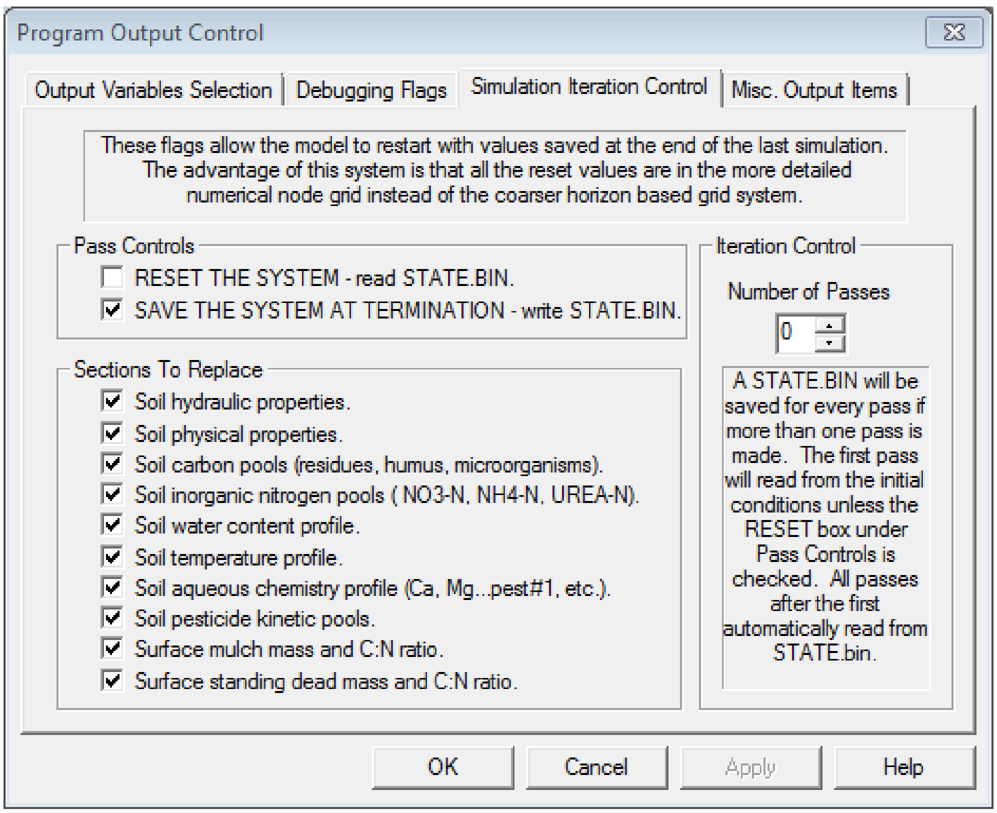
Tip2: Use flag 9 to produce output for the Generic Plant Model. DSSAT output files are specified in **the Set Simulation Control By Species** under **EDIT DSSAT Parameters.**

Tip3: Use flag 12 to obtain infiltration module output in MACRO.OUT file.

6.1.3 Simulation iteration control screen:

This screen is designed to set the number of iterations, define system status for the next simulation. Users may select ***SIMULATION ITERATION CONTROL*** on the ***SIMULATION CONTROLS*** menu to set up multiple runs. The setup screen allows users to select number of runs of the same scenario, recording of system status at end of each run, and reading in system status at the beginning of a subsequent run.

Figure 6.3: Simulation Iteration Control



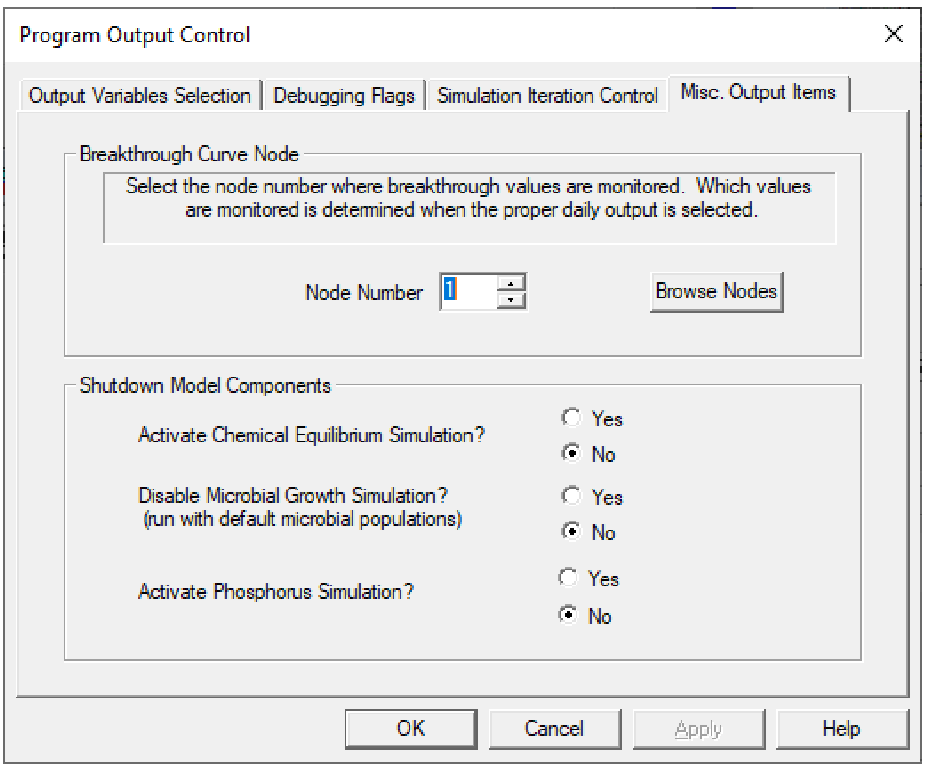
Note: This feature is useful to make multiple runs needed to stabilize the organic C pools or study long-term effect of a typical soil-weather setting (See Appendix 3).

Important Note: If the number of passes is *n* and *n>1*, the model will run for *n* times. At the beginning of the second run, the model will automatically check the "RESET THE SYSTEM" box. Users should not manually set this box with multiple passes. The second run will reset the system status from previous run with the variables specified by a check in the "Sections to Replace" boxes from the STATE.BIN file. A STATE.BIN file stores the end of the system status of a run and contains the same information as the NEWINT.OUT, except STATE.BIN saves the data at a finer resolution (numerical layers). NEWINT.OUT saves the data for the user entered soil horizons.

6.1.4 Misc. output items screen

Node to generate chemical breakthrough curves is set up on this screen. Users can also shut down the microbial growth simulation or the inorganic chemical equilibrium model on this screen. In the case of executing the phosphorus module, check "Yes" to Activate Phosphorus Simulation.

Figure 6.4: Monitoring a Specific Soil Numerical Node and Component Shutdown



6.2 **Modifiers**

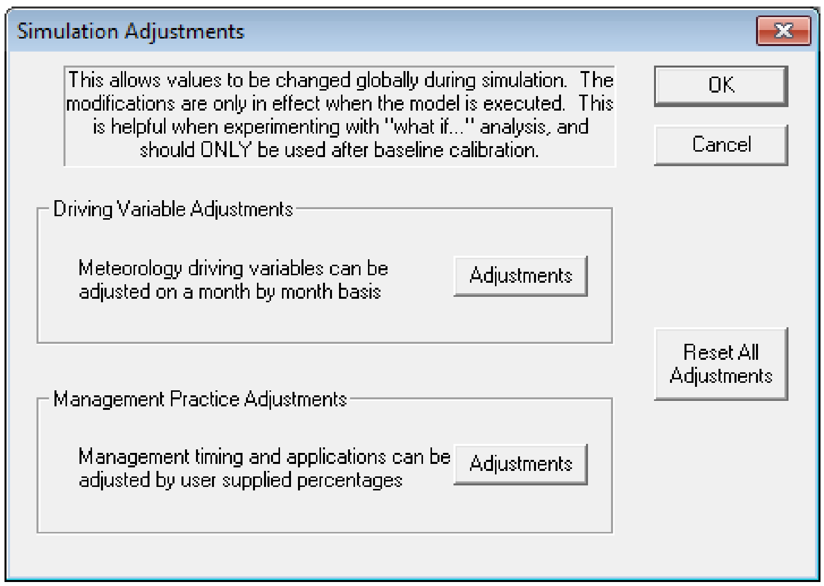
Conducting "What If?" Analysis

The interface has the capability of modifying weather information and management practices to conduct "what-if" analysis of a particular agricultural system. These are used after a baseline run is obtained for the site (See Chapter 8). ***Tip:*** Click ***FILE | SAVE AS*** New Scenario with modifiers values mentioned in the new name to help you recall this is not your baseline run and run this as a new scenario.

Click ***MODIFIERS*** menu item or the ***MODIFIERS STATE*** button on the scenario screen or  on the toolbar. Click ***ADJUSTMENTS*** for either driving variables or management (Figure 6.5).

Important Note: Modifiers are written to the IPNAMES.DAT file. This file is ***not imported*** with the input import command. Version upgrades where input or output changes and prompts the user to import will require the user to check the modifiers section to see previously set values are still set properly.

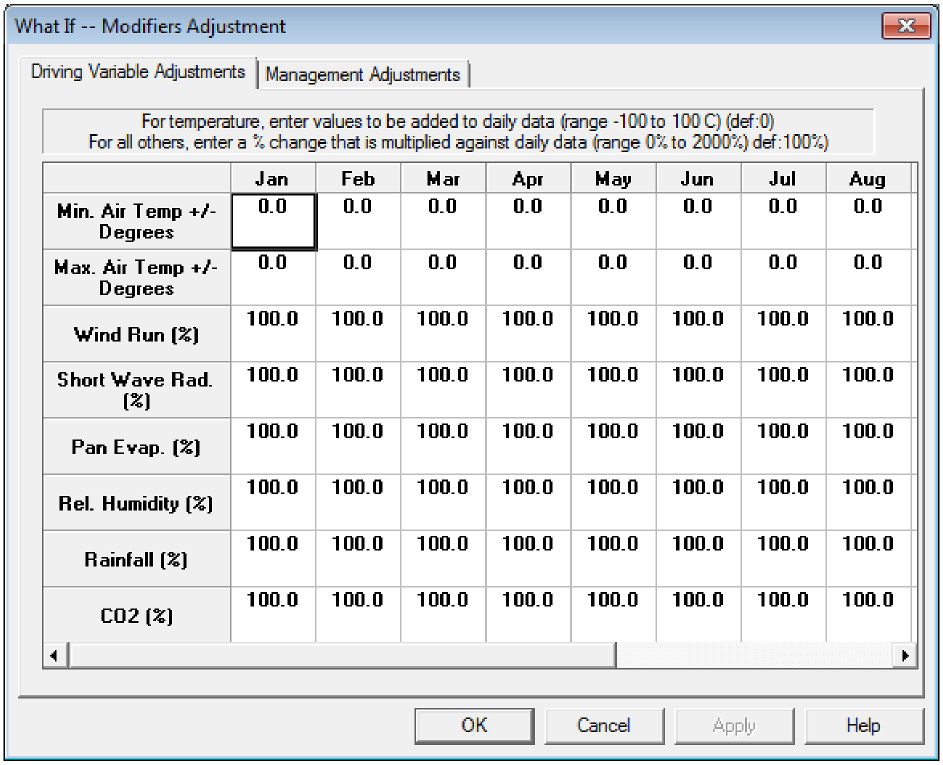
Figure 6.5: Simulation Modifiers for Climate or Management



6.2.1 Driving variable adjustments screen

Modify weather information such as air temperature, rainfall, wind speed, solar radiation, relative humidity, pan evaporation and CO_2.

Figure 6.6: Modifiers for Driving Variables (e.g. Climate)



1. Add or subtract degrees of temperature or enter a percentage multiplier for other variables.
2. Click ***OK***
3. Click ***OK*** on Simulation Adjustments
4. Click ***File Save Scenario As***
5. Rename with modifier attributes in Scenario Name to distinguish from your baseline scenario.

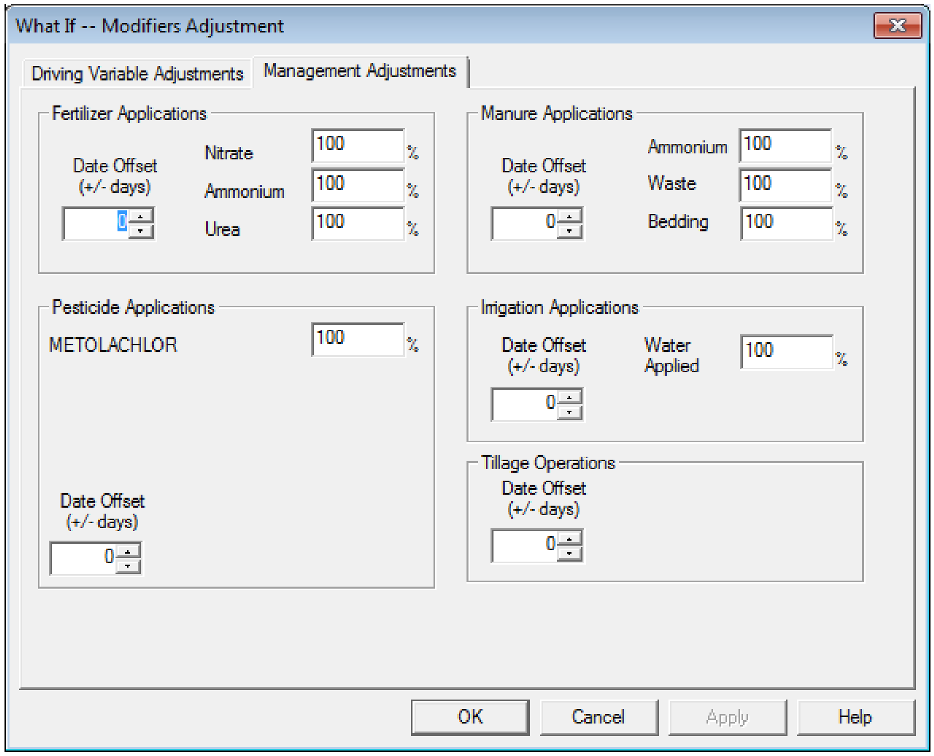
Note: Air temperature is in ^{o} C and other modifiers are in % (100% means no change)

Tip: Modifiers on climate can help assess changing climate scenarios oncrop production and the environment.

6.2.2 Management adjustments screen

This screen allows for modification of fertilizer amount and timing; manure application amount and timing; irrigation amount and timing; pesticide application amount and timing; as well as timing of tillage.

Figure 6.7: Modifiers for Management



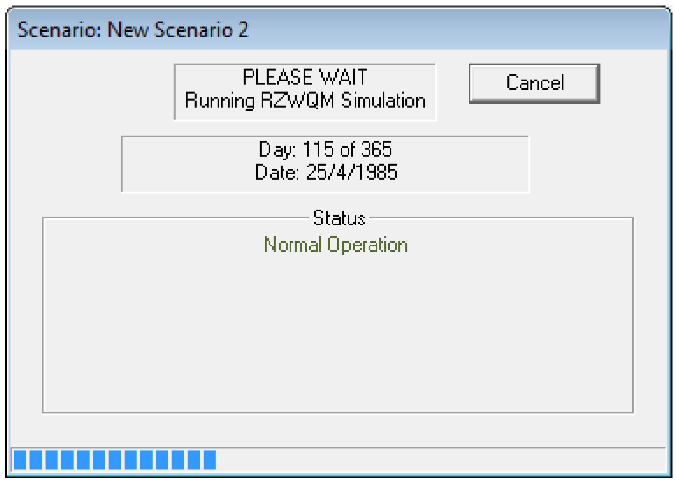
Tip: Management Modifier Scenarios can be used to alter management inputs while assessing effect on production and the environment.

6.3 **RUN The Scenario**

Click on ***RUN SCENARIO*** or icon  on the scenario toolbar to execute scenario. You should see a running progress window at the center of your monitor notifying the status of the execution at the selected time intervals. Upon completion of the run, the status of your scenario will then show with a green ball. A red ball indicates that input data was changed in the interface and the RZWQM2 *.DAT files need to be rebuilt and the output files need to be updated.

Important Note: The progress report of a run is recorded in the RZWQM2.LOG. Click ***FILES*** button under ***VIEW OUTPUT*** in Scenario Window. Select File Type *.LOG. Select File and ***VIEW*** button.

Figure 6.8: Run Scenario Process Update Screen



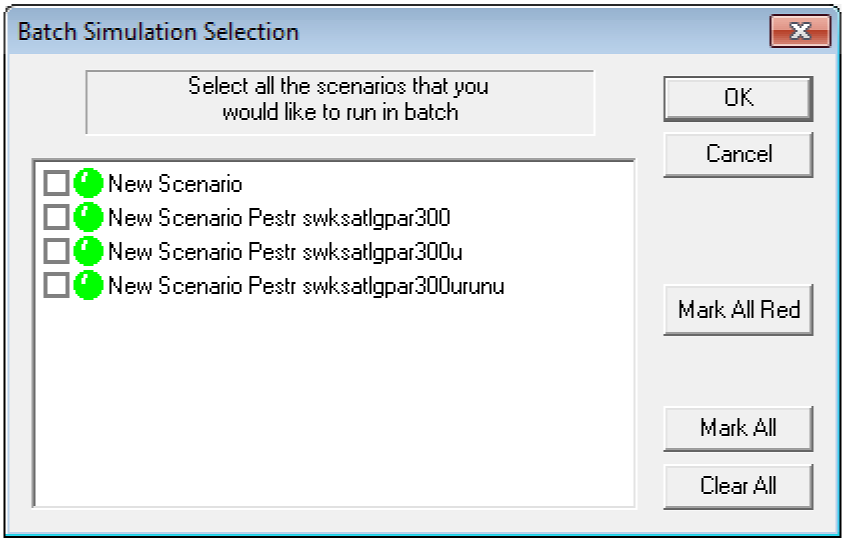
6.4 **Run Multiple Scenarios (Batch Run)**

This feature runs scenarios as a group in batch mode (one after the other). This feature can only be used with scenarios that were already created (or imported), saved and run with the current release RZWQM2 Version.

Important Note: If legacy binary files are found a message will display the scenario name and it will not execute that scenario.

1. Click on the menu ***BATCH SIMULATIONS*** or  on the toolbar.

Figure 6.9: Batch Simulation Execution Screen



2. Place check marks in the boxes to the left of the scenario name or click the ***Mark All*** button.
3. Click ***OK*** button to execute the scenarios order they are shown.

Chapter 7:
Scenario Output and Comparisons

7.1 **Output Files and File Structure**

When a project is created, RZWQM2 will automatically create two folders (ANALYSIS and METEOROLOGY) along with a default scenario folder (NEW SCENARIO). Meteorology files are saved in the METEOROLOGY folder and simulated scenario results are saved for comparisons among scenario in the ANALYSIS folder as (scenario name.ANA files). Users can create as many scenarios as they wish in one project with each scenario assigned a folder and it corresponding output file for comparison saved at the project level.

In addition, scenario output files (some default and some based on the user's selection in ***Simulation Controls***) are saved in their corresponding scenario folders with extension of *.OUT. Plotting files are saved with the extension *.PLT. Some default files include the RZWQM2.LOG, RZWQM.OUT and if DSSAT the OVERVIEW.OUT and of course the comparison file each scenario generates *.ANA in the ANALYSIS folder. See Appendix 8 for a full list of the output files.

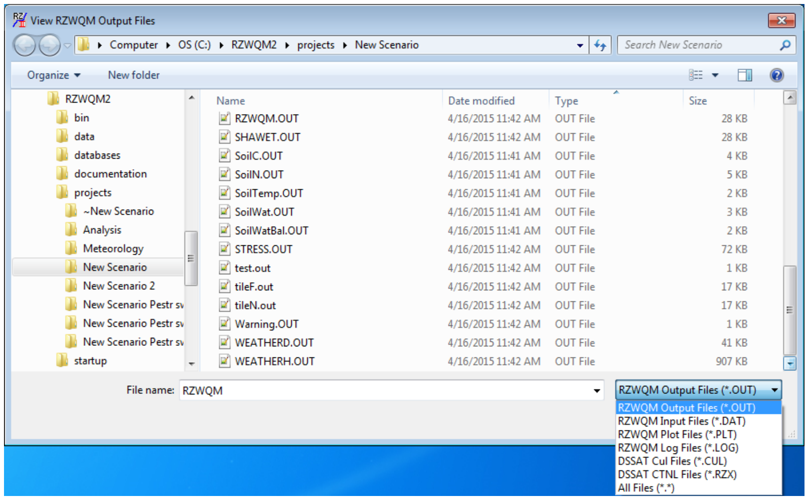
7.1.1 View Scenario Output Files by:

1. Click on the ***VIEW OUTPUT FILES*** button on the scenario level
2. OR Click menu item ***OUTPUT | VIEW OUTPUT FILES***.
3. The screen in Figure 7.1 displays. The output files are opened with the WordPad windows text editor for viewing purposes.

   Important Note: It is not recommended to change the required *.DAT files (See Appendix 1) that are managed by the RZWQM2 interface with text editors. These files are re-written at each run. If you do so you must use *Input* | *Import* to import RZWQM.DAT or RZINIT.DAT to the interface and run and save for an interface version. Other data files containing modifiers and paths are *not* imported.

4. The four types of files from the RZWQM2 interface for a simulation run are: files with extensions OUT (output results), DAT (input parameters for RZWQM2 Sciences), PLT (for graphic plotting), and LOG (daily recording of simulation run). It is recommended to view the RZWQM.LOG file to check the completion of simulation run. In addition, if you have chosen a DSSAT crop growth model you will 2 additional file types, *.CUL for cultivar plant parameters and *.RZX the simulation controls and soil root growth factor for the crop (Figure 7.2).

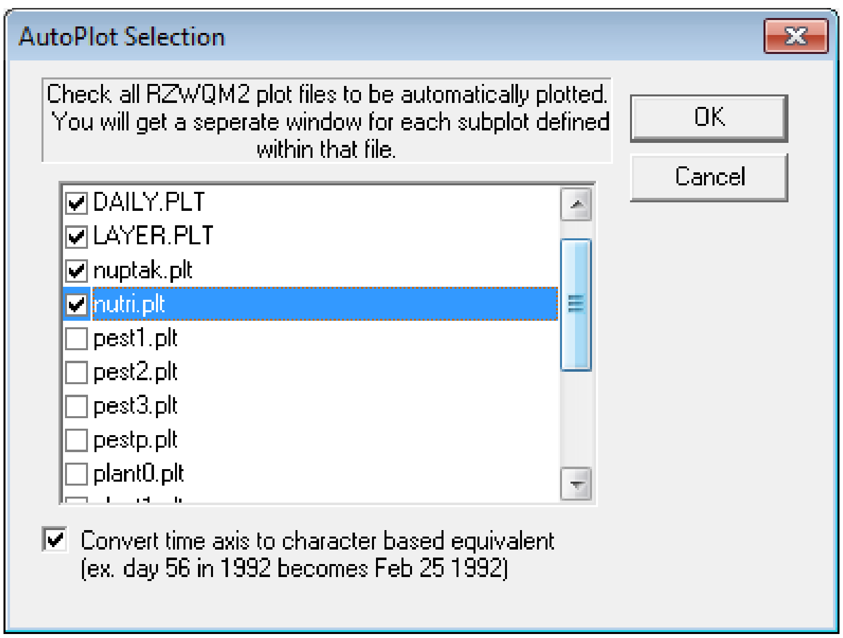
Figure 7.2: View Output Files



7.1.2 View and plot model outputs by Autoplot:

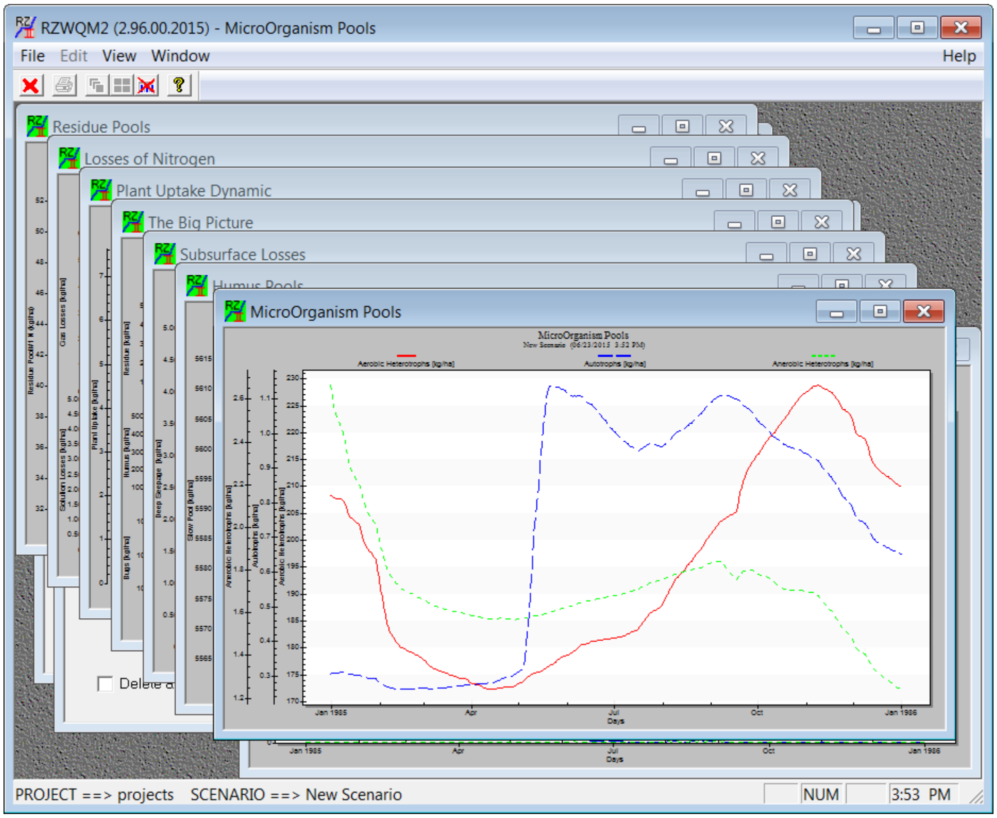
1. Click on the scenario menu ***OUTPUT | AUTOPLOT*** or  on the toolbar.

Figure 7.3: Autoplot File Selection for On-Screen Graphics



Important Note: Variables in the "PLANT*.PLT files were designed for use with the RZWQM generic crop growth model only.

Figure 7.4: Autoplot On-Screen Graphics



2. Use the red X icon farthest right to close all the plot windows and will return to the scenario view window.
3. Remember to ***FILE | EXIT SAVE SCENARIO*** before you proceed to comparing scenarios at the project level.

7.1.3 Observed vs. Predicted Graphs:

There are two types of performance graphics available if an EXPDATA.DAT was present for simulation. Performance of an observed measured experiment variable vs. its simulated result through time or a 1:1 plot of observed vs. predicted can be plotted. Variables available to graph depend on what the user has entered in the EXPDATA.DAT so the menu's are loaded dynamically (Figure 7.5). Select the main category of variables in the drop down. The variables for that category appear in the left window. Multi-select the variable and ADD to the right window and O.K. Figure 7.6 is an example of biomass simulation over time with red points for observed data.

Figure 7.5: Performance Graphics Time Series Variables

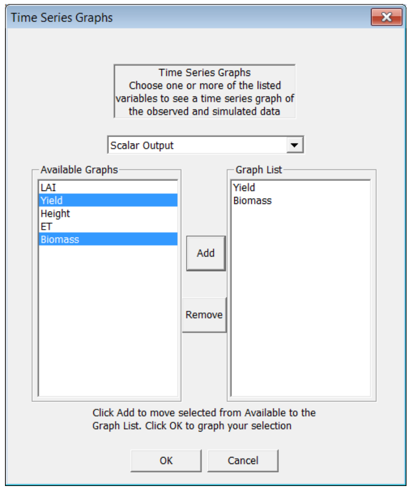
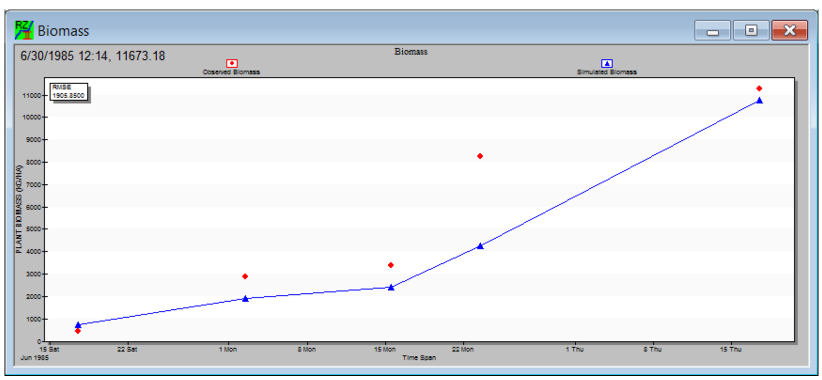


Figure 7.6: Performance Graphics Time Series Plot



7.1.4 General Steps in Scenario Execution and Output:

1. You may wish to do a pre-run with ***Simulation Controls*** debug flags on to check weather files (*.BRK and *.MET) to see they are in correct format.
2. Reset ***Simulation Controls*** debug flags to put out daily mass balance and other output of interest
3. Run the Scenario
4. Select and View the RZWQM2.LOG – This tells you of any errors during the run
5. Select and View MANAGE.OUT – This tells you if management occurred as planned
6. Select and view MASSBAL.OUT for mass balance problems.
7. Select and View Water (MBLWAT.OUT) and Nitrogen (MBLNIT.OUT), and Carbon balance files (MBLCARBON.OUT).

8. Select and View your crop production in OVERVIEW.OUT in DSSAT and begin to look at whether growth stages were hit properly. If not, adjustment of DSSAT cultivar parameters off the "**Edit DSSAT Parameters"** button will be necessary.
9. Select and View COMP2EXP.OUT if you have supplied observed experiment data in EXPDATA.DAT to see statistics on observed versus predicted values. (See Appendix 9)

7.2 **How to Make Scenario Comparisons**

Scenario Comparisons are done at the project level and may also be referred to as analyzing project results. Each successful RZWQM2 scenario run produces a standard output file for analysis (*.ANA) in the project's "ANALYSIS" subdirectory. RZWQM2 allows for simple statistical analysis of output variables within or across scenarios (if the simulations have the same simulation period) (Figure 7.7).

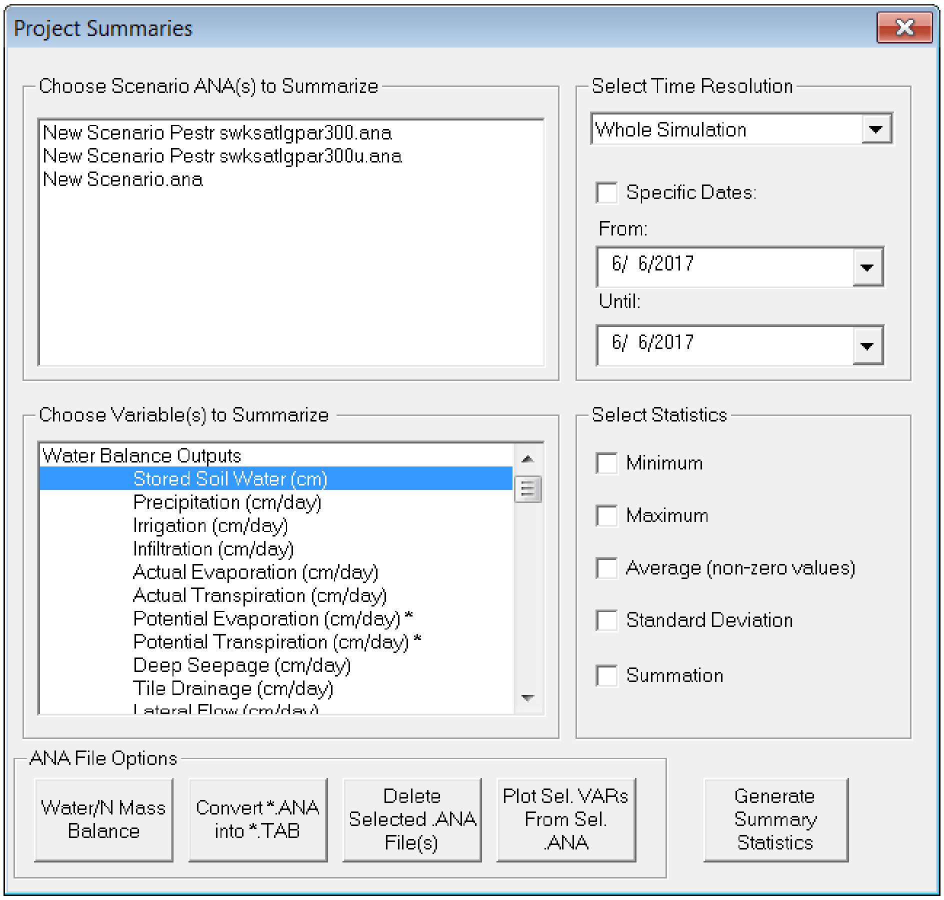
As of version 2.4, there is a new button on this dialog: "Convert .ANA to .TAB". This button coverts all selected *.ANA files into *.TAB files of the same name. A .TAB file is a tab-delimited text file that most spreadsheet programs (Such as Microsoft Excel or Open Office) recognize and can open directly, allowing the user to quickly convert into a format usable for more complicated manipulations and comparisons. The new .TAB files will be located in the Analysis directory.

1. If a scenario is open, ***FILE EXIT*** and ***SAVE SCENARIO.***
2. Click ***PROJECT SUMMARY*** on the project menu or  on the toolbar.
3. Select one scenario and multiple variable comparisons can be made by using the control key with selection.
4. Select the summary's time resolution: whole simulation period, yearly, month-by-month, and month-by-year. If you wish, you may restrict the dates between which the comparison will occur. The boxes default to the maximum and minimum dates common to all selected scenarios.
5. Select the check boxes for the summary statistic (max, min, average, sum)
6. Click ***GENERATE SUMMARY*** button and a summary sheet results. It can be printed or copied to other Windows software then ***CLOSE.***

   Note: *if you select multiple scenarios that share no dates in common, you will* ***only*** *be able to compare them across the "Whole Simulation" time resolution. At all times, you are only able to compare different scenarios across a time period common to both scenarios (i.e. you must use Specific Dates).*

Version 3.0 release and later include a mass balance button which executes analysis.exe from the RZWQM2/bin. For all scenarios selected all water and nitrogen components of mass balance are output to the project directory/ANALYSIS/ SEASONAL_MASSBL.OUT and ANNUAL_MASSBL.OUT.

Figure 7.7: Project Summary at the Project Level



Important Note: "*" next to a variable in the interface indicates the variable was added replacing an existing variable so the list definition has changed. In Version 2.0. ANA variables Actual ET and Potential ET variables have been replaced with Potential Evaporation and Potential Transpiration. There was no change to the previous listed Actual Evaporation or Actual Transpiration. In Version 3.0 ANA variable Total Residue C has replaces the list slot for Total Soil Humus N (Versions <=2.94).

Note 1: Water/N Mass Balance may not work if you reset to observed soil water data during simulation.

Note 2: <PROJECTDIR>\ANALYSIS*.ANA files are large and get created for each scenario run. Deleting a scenario will prompt the user to remove them from the Analysis directory. "Delete Selected Analysis files" here may be deleted from will only delete the scenario master output not the scenario itself.

Chapter 8:
Getting a Baseline Run

How to Get a Baseline Run of a Study Site

1. Choose the most unstressed data for your top producing yield to calibrate. Choose a stressed year data for your low end to monitor. One set of calibrated parameters should be able to perform within 2 standards deviations of your measured data for both situations. Save the rest of your data for validation. Set up the daily meteorology (met file) and the storm data (breakpoint data file).
   - You can use a generated climate from the station nearest to your site **until you have the observed experiment climate data ready from the weather station on site**. Climate generators randomly generate daily weather data reflecting the distribution of the weather station's observed 30-year average weather statistics.
   - After you have observed data .MET and .BRK, run RZWQM2 with "Verify" debug flags to do a check on them.

2. Set up the **Site Description,** which includes your site resources (soil attributes, PET, nutrient C/N, NH3 info). For Soil:
   - Enter your soil textures and depths for all horizons.
   - Enter soil bulk density. Leave the default values if you don't have any to override with.
   - Select a method to estimate soil hydraulic properties if not measured.

   Tip: present *the simplest description* of your soil as possible, without a complex features like macropores, drains, etc., which can be added later.

3. Initial State Setup: Enter initial state of the soil profile for moisture, temperature, inorganic chemicals or pesticides. If you have measured values of the soil moisture profile at the start of your study simulation you may use them here.

4. Initializing the Residue State: Enter you soil surface residues and use the initialization wizard if you have some measured organic matter in the soil profile. Initializing the state of your organic and inorganic N in the soil as in Appendix 4.

5. Setup the Management: Crop and crop module selection, planting, harvest, fertilizer, manure, irrigation, fertilizer and pesticide applications, and tillage practices need to be specified.

6. Enter in the EXPDATA.DAT the observed data you have available for the site. Work to reduce your RMSE reported in COMP2EXP.OUT, or increase the R-squared but remember plotting variables observed vs. predicted over time gives the best idea of model performance.
7. Set Up Simulation Control: Check debug flags 4 or below to get additional output files. Select the output variables for files. Choose at least one scalar and one vector.

8. Review your output:
   - View the .LOG to see it there is normal termination
   - View MANAGE.OUT to make sure all management events were recorded correctly. If not return to that section and review input.
   - View MASSBAL.OUT, MBLWAT, MBLNIT, MBLCARBON.OUT files for mass balances. If there is water balance problem, check your storm level data with smoothing and or access ***Advance Richards Equation Parameters*** to increase iterations or convergence criterion.

     Important Note: If the temperature is below 0 and you have snow pack you will not see precipitation reported in the ACCWAT.OUT or MASSBAL.OUT file.
   - Compare your simulated N and water budget with your experiment data or experience to make sure that gains and losses are reasonable.
   - Review soil water content simulation for periods without crop. You may need to add macropores, crusts etc. Review your runoff. Is it too much? Is it because the storm input intensity is too high or are the hydraulic controls in the soil set too low (e.g. Decrease 1/3 bar and increase Ksat)? Revisit soil properties and soil hydraulic properties if needed.
   - Review N balance to see if annual mineralization, annual denitrification, annual volatilization, annual runoff N, N fixation, and plant N uptake are reasonable. If not, some of the nutrient parameters may need to be modified.
   - View OVERVIEW.OUT (for DSSAT) to see if crop development stages are correct. If not, the DSSAT4.0 CSM in RZ2 have a variety of cultivars available and the user may choose one closest to theirs. Try different maturity groups and cultivars and re run to get the one closest to your own phenology and review the general performance of the crop, soil water and N system.
   - View COMP2EXP.OUT performance statistics

9. Crop parameters may need to be adjusted properly if defaults are not adequate for your cultivar. Some advanced training is needed for this. Calibrate crop parameters to meet crop phenology first then production.

10. Re-Run with your best set of parameters from the calibration (the process of changing a small set of crop parameters to get proper development and growth) for the crop and review the general crop, soil water and N performance again with and without a crop.

11. Is this run good enough? Re-examine COMP2EXP.OUT. The user makes this decision. Keep in mind all models have errors (error from uncertain input in the model and error in the knowledge gaps of the model itself); but is the error within an acceptable range considering model error itself and observed measurement error (e.g. d statistic).
12. If it is adequate, run the rest of your datasets with the calibrated soil and crop parameters.

Recommended parameterization and calibration publications are located in the RZWQM2/DOCUMENTATION directory. Additional publications are: Current Issues and techniques (Ahuja and Ma, 2002), Concepts for calibrating crop growth models (Boote, 1999), and Calibrating the Root Zone Water Quality Model (Hanson et.al., 1999).

Important Note: Parameters for crop models are meant to reflect the genetic potential for that crop variety under no-stress conditions. Calibrating a model too closely to match a stressed field condition could result in model under-prediction in wet years.

Chapter 9:
Automated Parameter Estimation and Optimization

The parameter estimation model (PEST) developed by Dr. John Doherty and the parallel processing version (BEOPEST) developed by Dr. Willem Schrueder PEST is a parameter optimizer that is model independent. It has been added to RZWQM2 to provide an automated calibration tool where input parameter estimation sets are proposed to minimize the differences (the Phi Value) between observed and simulated values defined in an objective function according to their weighted importance. In addition the tool utilities can be used to run non-linear uncertainty analysis. Upon completion of the analysis you can then Run RZWQM2 with uncertainty which will be covered in Chapter 10. See Appendix 10 for critical input and output files of the RZWQM2 PEST linkage. For further information on PEST and BEOPEST their help documentation can be installed in RZWQM2\bin\PEST\doc directory.

9.1 **Prerequisites to executing PEST**

You must have the following in order for PEST RUN to execute.

1. Observed data from your experiment entered into the program so a valid EXPDATA.DAT file is present in the scenario.
2. A successful RZWQM2 run with observed and simulated data so that a filled COMP2EXP.OUT file is produced and resides in the scenario directory you will use pest on.
3. A run without obvious mass balance or convergence errors.
4. Scenario View Menu PEST INPUT WEIGHTS/PARAMETERS tabbed input must completed.

Important Note: Keep a copy of your previous scenario and when you “SAVE AS” for a scenario with PEST include estimation parameters you will estimate (optimize) and pest execution options like Regularization or not in the scenario name. Execute the save as copy as a regular RZWQM RUN before proceeding with Run PEST or BEOPEST.

9.2 **Operating PEST For Parameter Estimation and Optimizing**

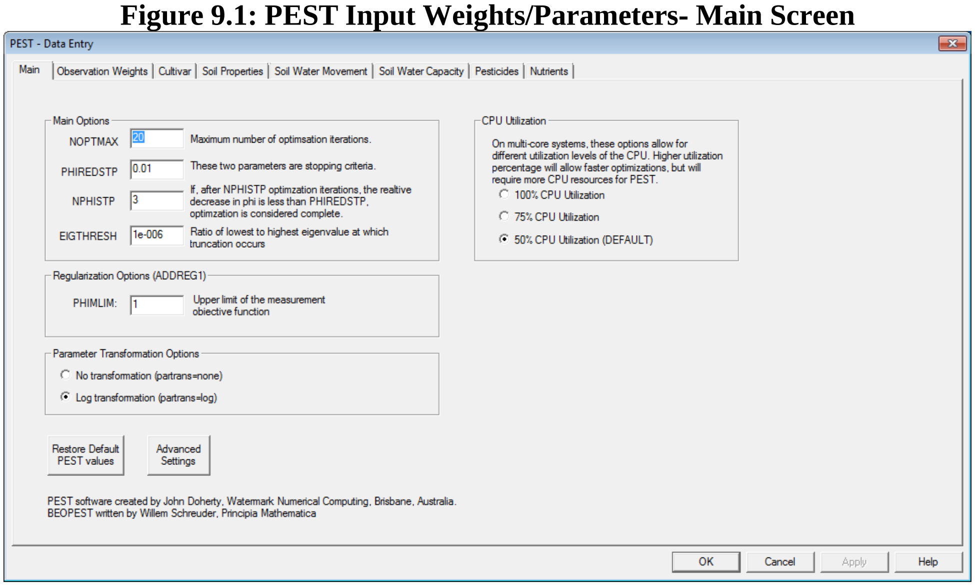
Choose on the main scenario menu ***P-EST*** (input parameter estimation) located after ***OUTPUT***. The sub menus from this are ***Input Weights/Parameters*** or ***Run Optimization/Uncertainty.***

9.2.1 PEST Input Weights/Parameters

The P-EST input has tabbed entry screens with the first tab containing PEST model input, the second containing weights for the observed data importance and the third through eighth tabs the RZWQM2 input parameters available for PEST to access and auto-calibrate within user specified rules and parameter ranges (specified with 95% Confidence).

The ***Main*** input screen sets input variables for the PEST model and values on this screen are used in the creation of the main.pst file for PEST runs (Figure 9.1). All values are defaulted for ease of use by the user. Of critical importance is NOPTMAX, the number of optimization runs attempted. Several hundred model runs occur during a single optimization run. A value of "1" or more is required for optimization to occur. A value of "0" means observation weights can be determined but not optimization occurs. A value of "-1" is required to generate a Jacobian Matrix after optimization. Observed experimental data is your "prior information". It is recommended to log transform parameters but the user has an option to use them directly. If you are running PEST with "Regularization" checked on it uses the PHILIM field to suspend optimizing when a certain Phi is reached. While other variables are available NOPTMAX and PHILIM are the most likely to be changed by a user. Advanced Settings may only be needed by the most advanced PEST users. By default only half the cores will be used in optimization and uncertainty to allow the user responsiveness when using other applications; however, the user may choose higher CPU usage to accomplish runs faster.

Figure 9.1: PEST Input Weights/Parameters- Main Screen



Important Note: PEST input and output files are located in a PEST folder under the scenario. Some pest files like *MAIN.PST* have a file extension used by **MS OUTLOOK** please use ***Open with WordPad*** when viewing these files outside the interface.

Observation Weights

The Figure 9.2 input screen allows the user to weight the importance of measured data that was entered in the expdata.dat. Click on the lower left button *Generate Default Weights* and a 1 will default for every data category present in the expdata.dat. Each observation will have the same weight to start and you will ask PEST to balance weights to come up with a balanced contribution of each category of observations to the objective function. The user can also set weights of observations manually for the objective function and not use PEST to determine them. A weight of “0” means the observation group will not be used in the objective function and will not contribute to the Phi value or be considered in optimization.

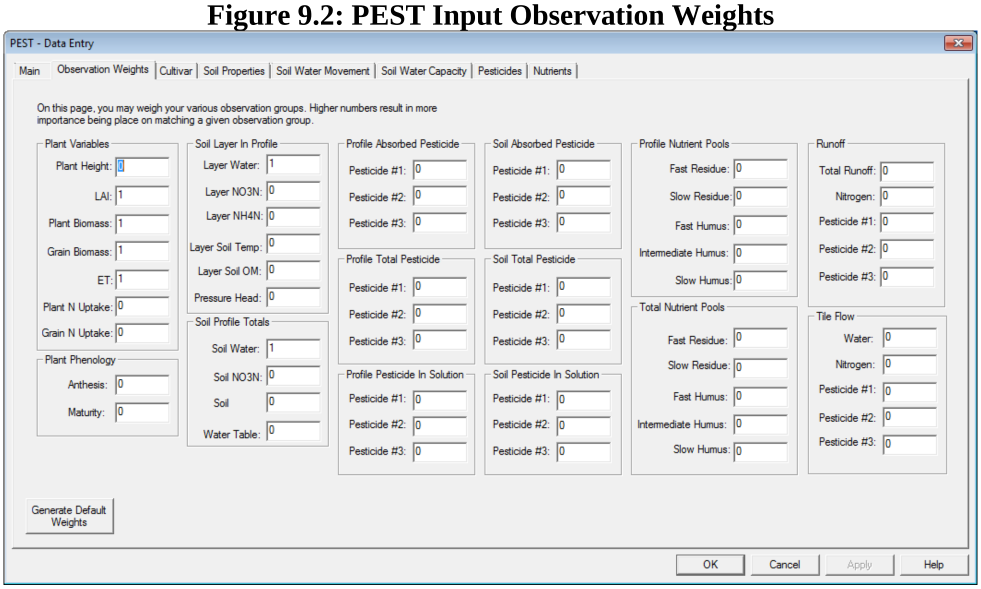
Objective Function

The sum of the weighted, squared, model to measurement discrepancies is referred to as the “objective function” (Doherty 2004). The objective in parameter estimation is to adjust parameters such that the differences between measurement and model prediction for that measurement are minimized. This is expressed in the following equation for Phi.

$$Phi=(weightobsgrp1)^2 *(RMSEgrp1)^2 * Numberobsgrp1 + (weightobsgrp2)^2 *(RMSEgrp2)^2 * Numberobsgrp2 + \ldots.$$

A parameter set yielding the lowest Phi may not be desired if the values hit many minimum or maximum limits of the parameter. In this case over fitting may occur to avoid this regularization and a PHILIM on the Main input screen may need to be defined.

Figure 9.2: PEST Input Observation Weights



Parameters to be Adjusted and the Solution Space Explored

The remaining tabs of ***Cultivar, Soil Properties, Soil Water Movement, Soil Water Capacity, Pesticides and Nutrients*** include all the RZWQM2 input parameters that we have allowed PEST to access to calibrate the model predictions to measured data. These variables were identified by previous RZWQM2 users who used PEST independently of RZWQM2 and identified them as the most useful to calibrate.

These screens have essentially two required parts the lower left button sets a default minimum and maximum value of the parameter to be explored and the check box that will indicate you want to fit the variable. The default min/max values are just a % in relation to your initial value. However, the user should set these the minimum and maximum values based on their expert knowledge of the parameter. The range of the parameter defines the solution space explored. It is important to set the range so 95% of the values for that parameter occur between the maximum and minimum values. With this assumption then the range divided by 4 can be used to set the standard deviation of the parameter to be used in parameter uncertainty.

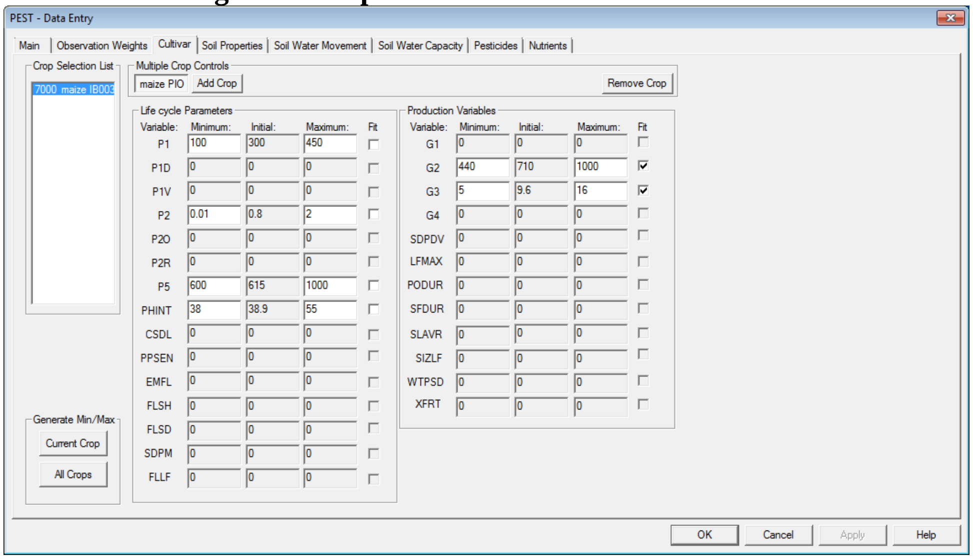
Fit variables you have no measurements for or you are unsure of the robustness of the measurement and in which variability could occur. Any variable not "Fitted" is "Fixed" to a value in PEST. The fixed value the one initially set or in special cases the RZWQM2-PEST linkage allows a calculated one.

For the ***Cultivar*** input parameter estimates you must:

1. Highlight the crop in the left drop down you want to address. Variables associated with that crop will display as Crop1 in the crop controls. If you have a rotation a maximum of 5 crop or crop varieties can be optimized in a run.
2. Choose Lower Left Button to Generate Min/Max for the parameters for each crop.
3. Check the box next to the parameter you want fit.
4. Review the ranges for your parameter you want to fit.
5. Click OK when done with this tab or others to save changes to binary data file for PEST rzzzzz.005 as with the rest of RZWQM2 O.K. functions. Cancel will not save changes.

   Important Note: The DSSAT CSM model is calibrated by obtaining phenology related variables first (P* variables) and then production (G* variables) ones. If you ask PEST to fit all parameters your EXPDATA.DAT, you should have observed phenology. If you have no estimates or only a general knowledge of phenology (maturity date) you may manually pick a cultivar for the maturity date and then just use PEST to fine tune production like the NEW SCENARIO example distributed with PEST example.

Figure 9.3: Input Parameters for CSM Cultivar



Figures 9.4 through 9.6 are input screens for parameters affecting soil properties, soil water movement and soil water retained with soil water capacity. Unique features found on these screens are the radial buttons that can give calculated variables from another variable that is fitted and the tied horizon button. If you fit a variable with pest the calculated options are not available. If you do not fit, the default radial is to use the initial value that was entered in your run.

For Bulk Density and vertical Ksat ***Soil Properties*** to be estimated you must:

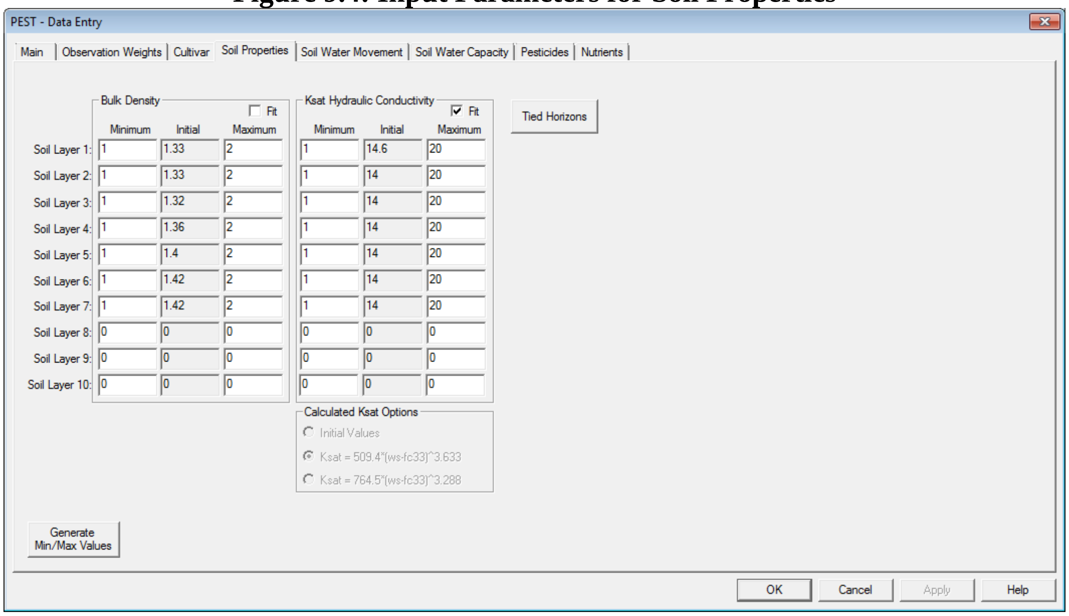
1. Choose Lower Left Button to Generate Min/Max for the parameter.
2. Check the box next to the parameter you want fit.
3. Review the ranges for your parameter you want to fit.
   Ksat is most often a fit parameter as it is rarely measured but is important to the vertical movement of water in the profile. Ksat is highly variable with soil texture so wide ranges are suggested for fitting heterogeneous profiles.

   If not fitting Ksat a radial choice must be used. Initial values are useful if you have measured them or have confidence in the exact textural classification of the soil layers. If fitting a 1/3 bar water content on the ***Soil Water Capacity*** screen you should choose a radial whose calculation involves the fitted parameter if not fitting Ksat.
4. Review the calculation chosen if not fitted.
5. Optional step to Tie Horizons
   Tied horizons will tie (make them the same) parameters of 1 soil layer to another soil layer for the parameter specified. If you have sections of the soil profile that behave the same you can cut down optimizing time by allowing more layers to have the same parameters set. Instead of optimizing 7 layers you may only need 3 sections, top, middle bottom. The same dialog appears from any tied horizon button.
   To Tie:
   - Click ***Tied Horizons***

- Check Parameter(s) to be the same among horizons
- Number the horizons with the same number that you would like the same. For example, layer1- 1, layer2 - 1, layer3 - 2, layer4 - 2, layer5 – 3.
- Click OK to save tied data entry

6. Click OK to save

Figure 9.4: Input Parameters for Soil Properties



By using calculations of parameters not fitted, tying horizons, and setting the minimum and maximum ranges for parameters you are constraining the parameter ranges tested and the potential amount of outcomes resulting in a smaller solution space. Optimization may be quicker but you can risk not exploring all possible solution space. However, constraining parameter ranges tested is necessary to insure valid parameter sets are tested.

Soil Water Movement

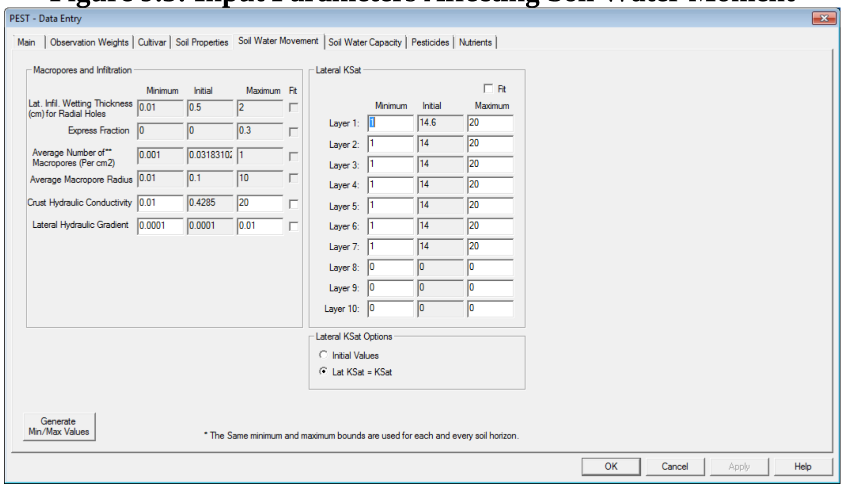
The Figure 9.5 shows the Soil Water Movement screen and the associated variables variables affecting infiltration and water movement through the soil macropores. Crust hydraulic conductivity directly affects infiltration at the soil surface and lateral hydraulic gradient controls water moving laterally within the soil and not available for vertical movement. If you did not specify macropores in your scenario, these fields will be unavailable. For scenarios with a macropore feature specified parameters are often not measured and all critical are available to be estimated. An "*" for macropores indicates the parameters estimated apply for all soil layers. Lateral Ksat for each layer also affects water retained in a layer and not available for movement. If you do not fit lateral Ksat but fit vertical Ksat, it is best to default lateral Ksat to equal vertical Ksat for lack of better information.

For Soil Water Movement parameters to be estimated you must:

1. Choose Lower Left Button to Generate Min/Max for the parameters.

2. Check the box next to the parameter you want fit.
3. Review and modify the ranges for your parameter you want to fit.
4. If not fitting lateral Ksat choose to keep fixed initial value or set equal to the vertical Ksat.
5. O.K. to save

Figure 9.5: Input Parameters Affecting Soil Water Moment



Soil Water Capacity

Soil water Capacity is shown in figure 9.6. This screen will have live fields depending on which soil hydraulic property method you had used in the scenario you are currently in setting the PEST input for. If you chose the Minimum Input radial on ***Site Description Soil Hydraulics*** for determining Brooks Corey parameters only the water content at 1/3 will fields will be active. If you chose any other radial that resulted in the full Brooks Corey set of parameters written to RZWQM.DAT for science all fields will be available.

For Soil Water Capacity at 1/3 and 15 bar input parameters to be estimated you must:

1. Choose Lower Left Button to Generate Min/Max for the parameters.
2. Check the box next to the parameter you want fit.

If you choose to fit only 1/3 bar, when both parameters are available you will need to select from the 15bar Options to fix the value (not recommended as discontinuity of the BC curve may occur giving convergence issues) or set 15 bar as a percentage of the 1/3 bar.

3. Review and modify the ranges for the parameters you want to fit.

Important Note: The parameter range when fitting 1/3 bar *CAN NOT* overlap the parameter range for 15 bars for any given layer.

4. Optional Tie horizons are shown in figure 9.7 where both water capacities are fit and the first three layers will be one set of parameters and the last 4 horizons will be the second set of parameters. All the layers after the defined soil profile may have any number as they are not present.

Important Note*:* A change in number means a new parameter set will apply. Putting a “1” on layer 7 does not mean it will have parameters like layer 1 it will have a unique set of parameters.

5. OK to save tied data entry
6. OK to save

Figure 9.6: Input Parameters for Soil Water Capacity at 1/3 and 15 bar

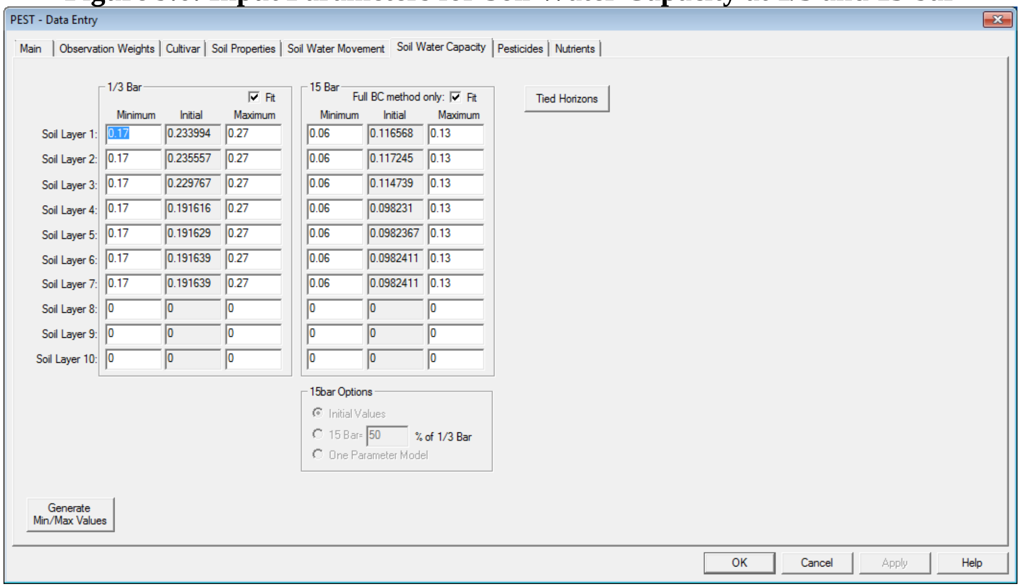
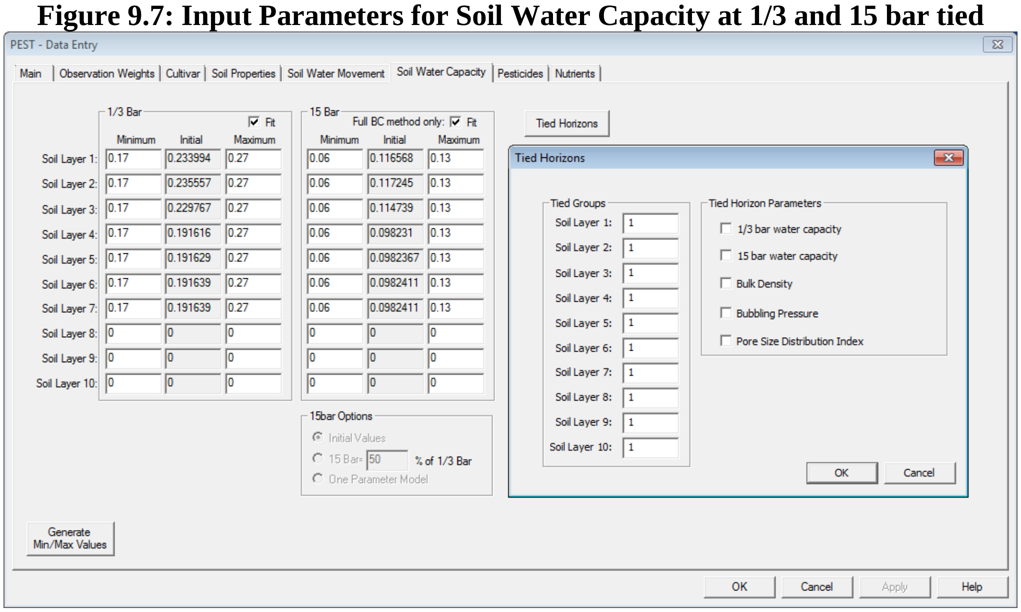


Figure 9.7: Input Parameters for Soil Water Capacity at 1/3 and 15 bar tied

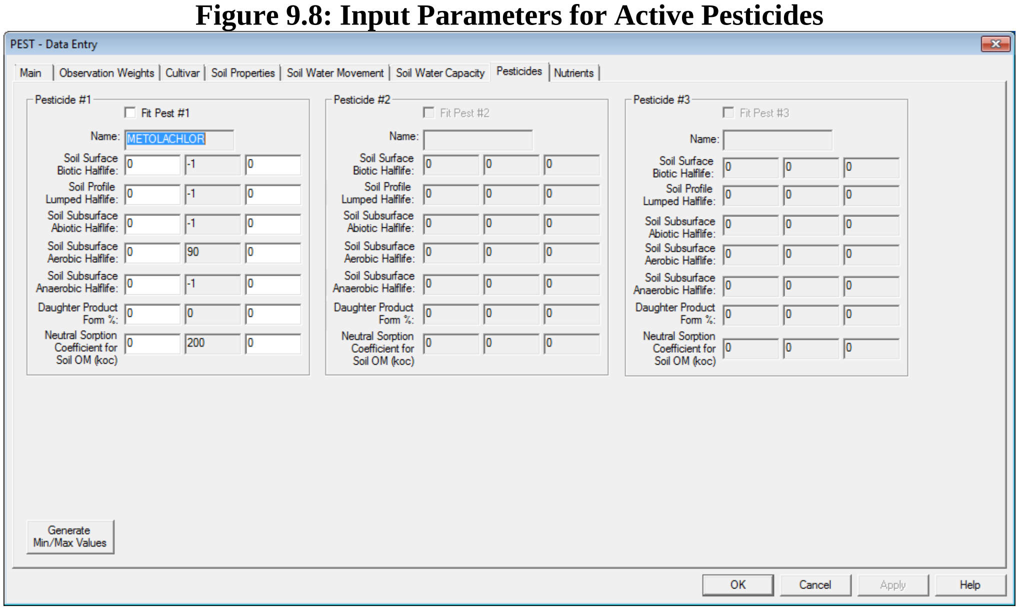


Pesticides screen shown in figure 9.8 will have live fields depending on how many pesticides are in your scenario (maximum of 3). One pesticide is present in this example and the soil subsurface aerobic pathway half-life is defined by 90, a number greater than -1; however, the expdata.dat has no observations of pesticide concentration in the soil even though. In this case optimizing this parameter would have no effect on the objective function as soil pesticide concentration observations are not included in it. This may not be true of other variables that are indirectly correlated. For example, cultivar parameters influence yield directly; but, soil properties, infiltration and soil water also influence yield indirectly. Soil water may not be in the expdata and may still be a fitted parameter even though it may not directly be represented by an observation in the objective function.

For Pesticide parameters to be estimated you must:

1. Choose Lower Left Button to Generate Min/Max for the parameters.
2. Check the box next to the parameter you want fit.
   If the parameter has loaded a negative 1 from your original scenario the pathway is not defined for it. Fit the pesticide will fit the halflife of the defined pathways. In general, fit variables that are not specifically known but likely to have impact on the simulated output that will be compared in your objective function.
3. Review and modify the ranges for the parameters you want to fit.
4. Click OK to save

Figure 9.8: Input Parameters for Active Pesticides



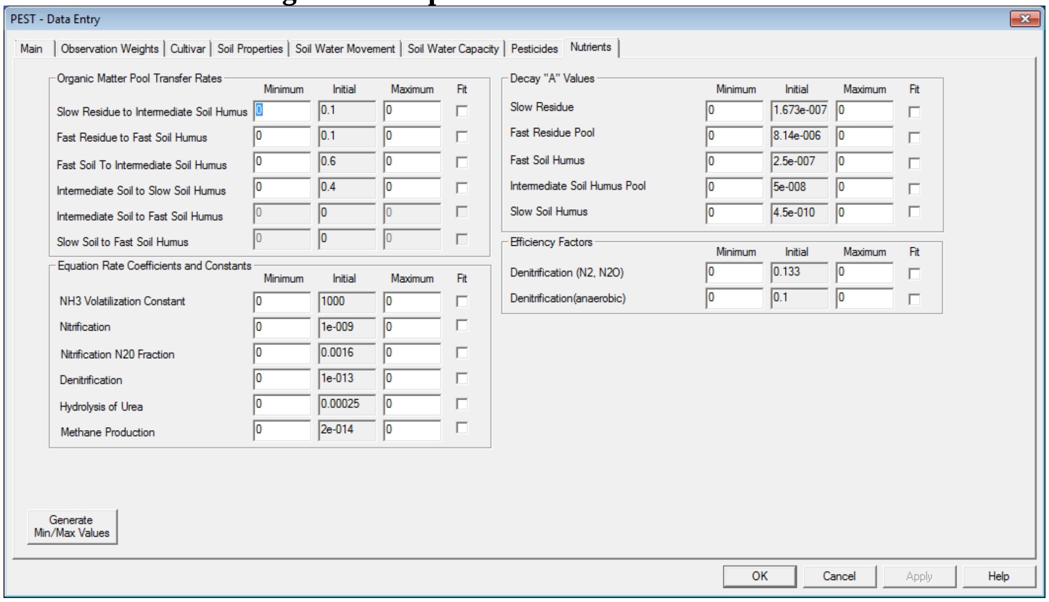
The Nutrients screen contains the most common nutrient input parameters influencing Organic and Inorganic Carbon and Nitrogen status. These parameters are rarely measured but their impact the residual C, NO3-N and NH4-N are measured in the soil layer and included in the expdata.dat file. Most commonly optimized are nutrient pool transfer coefficients and efficiency factors; however other reaction coefficients and pool decay rates are also available (Figure 9.9).

For Nutrient input parameters to be estimated you must:

1. Choose Lower Left Button to Generate Min/Max for the parameters.
2. Check the box next to the parameter you want fit.
3. Review and modify the ranges for the parameters you want to fit.
   Important Note*:* The maximum value for a transfer rate, coefficient, decay rate or efficiency factor is 1. The only exception on this screen is the NH3 volatilization constant which can be from 100 to 4000.

Click OK to save, cancel to exit not save

Figure 9.9: Input Parameters for Nutrients



9.2.2 PEST Run Optimization

OK from any PEST input screen will return you to the scenario view.
Choose ***PEST*** and the second choice ***Run Optimization/Uncertainty.***
The ***PEST Run Control*** dialog will appear with the left side of the screen reporting results from pest check step and the right side of the screen requiring user input on whether to Balance Weights, add Regularization or use the parallel processing form of PEST, BEOPEST, to speed the testing of input parameters sets (Figure 9.10). The use of BEOPEST with a multi-core machines to is highly recommended if not required in some cases. A very large scenario could take a week to optimize without it. Single core machines should only use PEST.

Important Note*:* To use BEOPEST you must have administrator privileges on your machine. A message may appear asking you to allow it to access the firewall. It is actually testing a port to allow parallel runs. BEOPEST is only given the IP address of the current machine when running from RZWQM2 so this is not a security issue. It will not access other computers to do parallel runs through this port.

Optimization can involve selecting different weights of observation importance, different parameters to optimize, and different checkboxes (balance weights, add regularization, use beopest) before the run. Due to the many options it is useful to open the original scenario and SAVE AS a name that includes options used. This requires forethought of which parameters you want to optimize and whether you want to regularize or not. The following instruction sequence was performed on the program Distributed New Scenario.

9.2.3 Running PEST

PEST INPUT

1. Open the original New Scenario and SAVE AS “New Scenario Pestr swksatlgpar300”
2. Run RZWQM2 so that it will update all relevant files, and produce the comp2exp file required for optimization.
3. Go to P-Est->Input Weights/observations
4. On the main screen:
    a. Set the PHILIM field to 300 (previous runs to lower phi had solution hitting the limits on multiple soil layers which is undesirable could indicate overfitting).
    b. Adjust the CPU utilization based on how much processing power you wish to provide to BEOPEST (PEST does not use this option).
    c. Keep the rest of the values at default.
5. On the Observation Weights Screen:
    a. Generate default weights.
    b. Set Plant height weight to 0 (not wanted in this objective function)
6. Parameters that should be optimized:
    a. On the cultivar screen: fit g2 and g3 parameters, do not modify the ranges provided.
    b. On the soil properties screen: fit Ksat, do not modify the ranges provided.
    c. On the soil water movement screen: Set lateral Ksat = Ksat.
    d. On the soil water capacity screen: fit 1/3 and 15 water content, do not modify the ranges provided.

PEST RUN for Optimization

7. Go to P-Est->Run Optimization/Uncertainty (Figure 9.10)
8. Check PEST to balance weights, Regularization, BEOPEST.
9. Perform Step 1 (writes *.pst,*.in, *.tpl, *.ins to PEST directory)
10. Perform Step 2 (runs pstchk, produces pstchk.txt in pest directory and displays on left side of screen to tell you if errors or warnings)
11. If there are errors: return to P-Est->Input Weights/Parameters to correct any listed issues.
    - For example it will say 0 in a max min value etc..
    - If warnings it will usually not stop a PEST optimization
12. Rerun Step 2 to make sure no errors
13. Perform Step 3 (This step takes less than one hour using BEOPEST and a machine with a 4 core, hyper-threaded processor using ¾ of the processing power. Your time and performance will vary depending on your machine and the choice of using BEOPEST or PEST)
14. Perform Step4: View run record main.rec
    - Locate the beginning and ending phi values.
        - Check that the ending phi was lower than the initial (starting) phi.
    - If there is no section titled “Optimisation Results” after the last listed optimization iteration, there was a problem with the run.

  - This section lists the ending parameter set, observations, and ending phi value.
    Note: The listed optimized parameter set may be your initial set of values, this means the run did not find another optimal set, or the run was not used for optimization.
- Always check the ending parameter set for sanity of the parameters.
  - Pest can over fit parameters. This produces a low phi value, but the parameter may be nonsense.

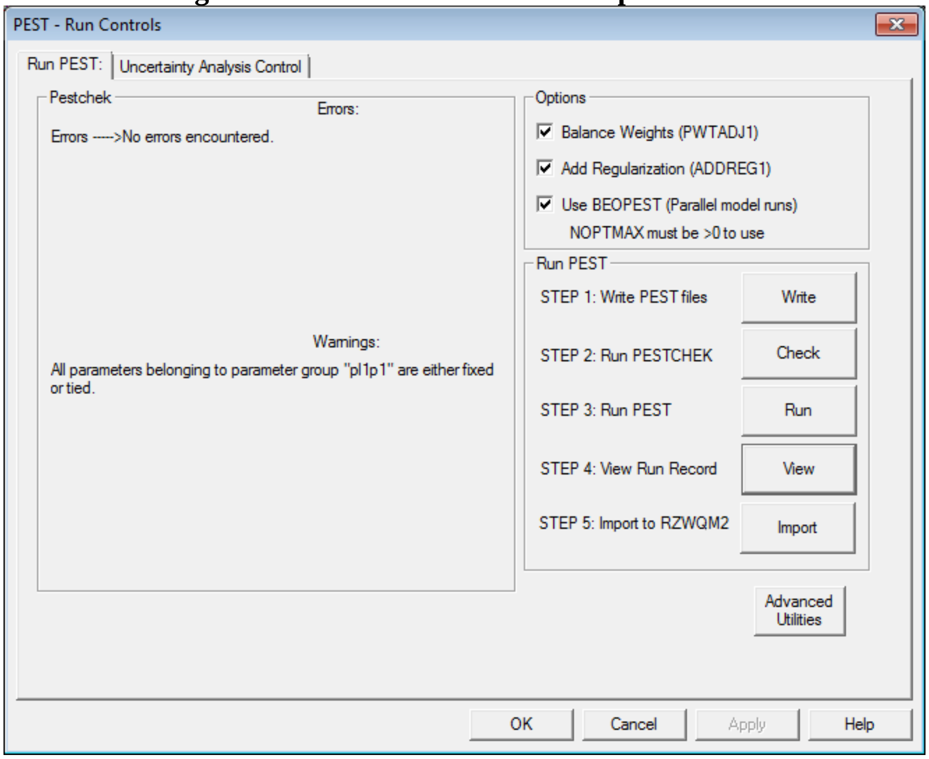
15. If you want to accept the optimized parameters:
    - Click STEP 5 Import
    - OK to return to the Scenario View
    - RUN Scenario from Scenario View
    - View Comp2exp.out and RMSE at file end
    - If the results are acceptable -> SAVE the scenario, or EXIT and YES to SAVE.

    - Alternately: You may O.K. from the Screen and return to the Scenario View to
      - INPUT IMPORT
      - RUN Scenario
      - View Comp2exp.out and RMSE at file end
      - If the results are acceptable -> SAVE the scenario, or EXIT and YES to SAVE.
16. If you DO NOT want the optimized parameters click OK to get back to the Scenario View.

    Options for changing the optimization:
    Head back to the PEST data entry screen
    (Click P-EST -> Input Weights/Parameters)
    - View the weights calculated for the observations, you may wish to manually adjust them.
      Important Note*:* If you manually set weights to use DO NOT use “Balance Weights” when you run the optimization (PEST/BEOPEST).
    - Check the parameters that you are fitting. You can:
      - Modify their ranges.
      - Either check or uncheck them to be optimized.
    - Modify the algorithms used for some parameters:
      - KSAT and 15 Bar can be calculated used different algorithms if they are not being used in the optimization process.

    Advance utilities are available to the advanced PEST user who may want to revise generally recommended variables values.

Figure 9.10: PEST Run PEST for Optimization



9.2.4 Optimization Differences between PEST and BEOPEST

PEST and BEOPEST can and often will produce different optimal parameter solutions. These changes are due to several differences between the programs and how they were compiled.

BEOPEST was designed to do more than PEST. Not only does BEOPEST parallelize the Jacobian matrix generation steps, it also partially parallelizes the lambda search. Partially parallelizing the lambda search allows BEOPEST to explore more solutions before being shut down by other criteria checks. PEST running in serial will exit optimization iteration due to the criteria being met. Consequently, BEOPEST may look a few more lambdas further and come up with a different, more optimal parameter set.

Also of note is the fact that PEST and BEOPEST were built using different compilers. Normally this would not produce too many issues, but it appears that in certain parts of the calculations it does have an impact. This is most likely due to the differences in the mathematical results these different compilers produce. Unfortunately, the differences between the two compilers can cause results to not match between the two programs.

The RZWMQ2 team was able to recompile both programs using the same compiler, and when run similarly produce the same solution. A similar setup does not the produce the same solution between the publicly distributed PEST and BEOPEST versions. We have opted to release RZWQM2 with the publicly distributed version of PEST and BEOPEST as to not increase the complexity of this issue with multiple different versions.

Chapter 10:
Non-Linear Uncertainty Analysis, Run with Uncertainty

Uncertainty analysis is a process required before the RZWQM2 Run with Uncertainty can be executed from the main **Scenario View | Run** menu.

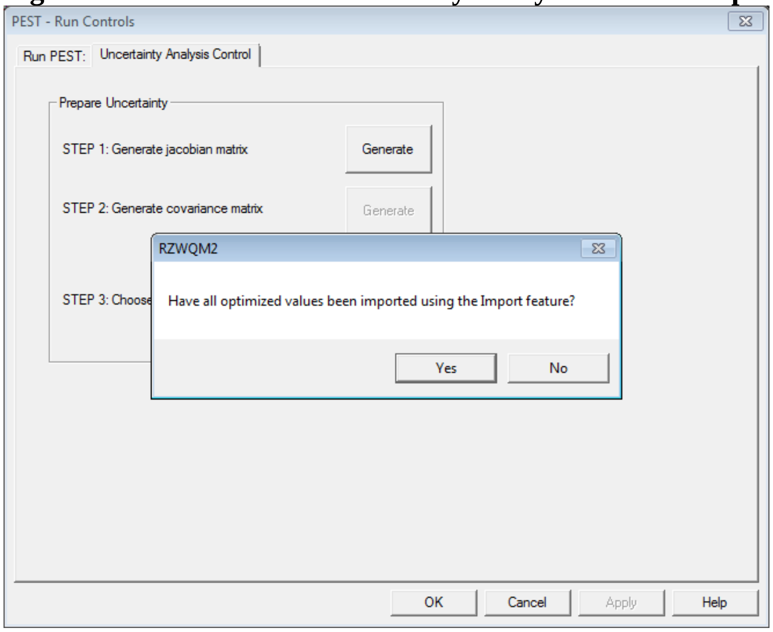
10.1 **Prerequisites for Non-Linear Uncertainty Analysis**

1. Meet all prerequisites of running PEST for Optimization
2. Run Optimization and view the record to determine if reasonable and acceptable.
3. Select the solution by using Input Import the best case written *.dat files are imported to the interface binary file and Run the scenario.
4. Proceed to the Uncertainty Analysis Control Tab

10.2 **Non-Linear Uncertainty Analysis Control Tab**

There are three parts to the control before proceeding to the Uncertainty analysis screen to generate cases of input parameters that may give a range of outcomes. The first step is Generating a posterior Jacobian matrix. Posterior means after optimization so you are prompted if you have imported the optimized parameters, if you have not you may not continue to step 2 with “No” (Figure 10.1). Both Step 1 and Step 2 will run PEST utilities in a DOS command window. Upon completion of the step writing to the window you will be left and a command prompt type “exit” or X the window. Step 1 will take a long time, step 2 will be almost instantaneous. The final step is to Click the radial to continue with non-linear uncertainty analysis and dialog will come available.

Figure 10.1: Non-Linear Uncertainty Analysis Control Step 1



10.2.1 Non-Linear Uncertainty Analysis Control Sequence

1. Generate Jacobian Matrix having imported optimized parameters
2. Exit at the DOS command prompt
3. Generate the covariance matrix
4. Exit at the DOS command prompt
5. Choose Non-Linear Uncertainty Analysis Radial
6. The Non-Linear Uncertainty tab will now display

   Note: Non-Linear Uncertainty is most applicable for the RZWQM2 agricultural system modeled. Linear option is not available at this time

Important Note*:* ALWAYS "EXIT" or close the PEST DOS command windows using the "x" button when execution is completed. Failure to exit before proceeding in the program may cause it to be hidden. Upon exit of the RZWQM2 scenario a general check is made for open windows and you will be prompted to close.

10.3 **Non-Linear Uncertainty Analysis Tab**

Uncertainty Analysis involves generating potential parameter sets STEP1 (either with RANDPAR or LHS) based on the relationships found in the optimized solution. Then, within your acceptable limits PHITHRESH (user entered or default 5% higher than optimized phi), select sets STEP 2 which you feel meet your tolerance for phi error to be run an additional optimization (max of 3) to produce the most refined sets for running RZWQM2 with uncertainty of input to produce uncertainty of output. Optimization up to 3 will not run on sets where the phi is below the threshold set.

10.3.1 Non-Linear Uncertainty Analysis Sequence

1. Specify the number of input parameter sets to generate
2. Set a PHISTOP threshold (default is the optimized phi displayed+5%)

Sets meeting the threshold criteria will have a red check in the Met Threshold column

3. Choose the random or LHS method to generate parameters
4. Recommended to choose Refine YES which will run 1 optimization run of each of generated set to make sure phi is not outrageous to display in table. Refine NO is faster but high phi values can be produced.
5. Recommended to let PEST set the internal seed with NO to custom seed. You can manually set or generate a seed when you answer YES to use custom seed.
6. Choose *Step 1* to generate.
7. Cases will appear
8. Select Cases you would like to proceed with as uncertainty cases
    a) Manually check the left column
    b) Enter a number in the field to the right of the spread and push button *Select Best.* It will select that number of lowest phi cases
9. Choose *Step 2* to generate the cases and run with up to 3 optimizations allowed to improve answers.
10. The resulting cases, phi and RMSE of observed versus predicted variables are display in *RMSE RESULTS* figure 10.3.
11. EXIT, YES to save scenario.

Figure 10.2: Non-linear Uncertainty Analysis Generate Parameter Sets

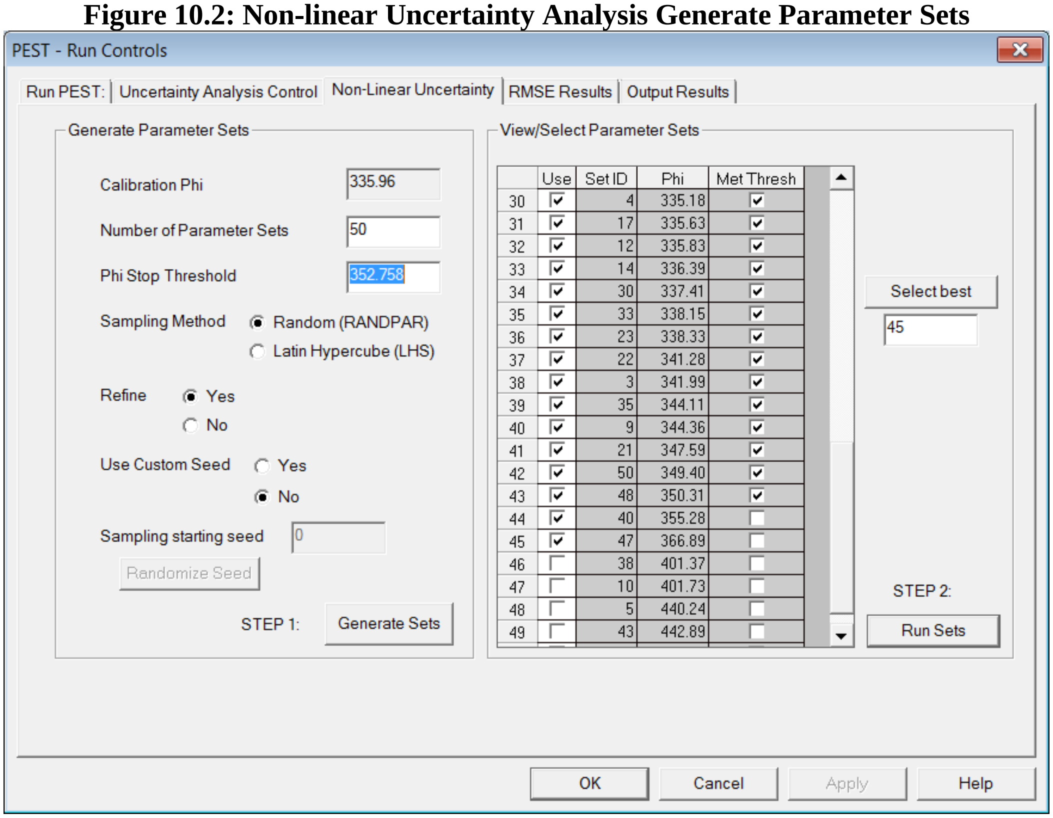
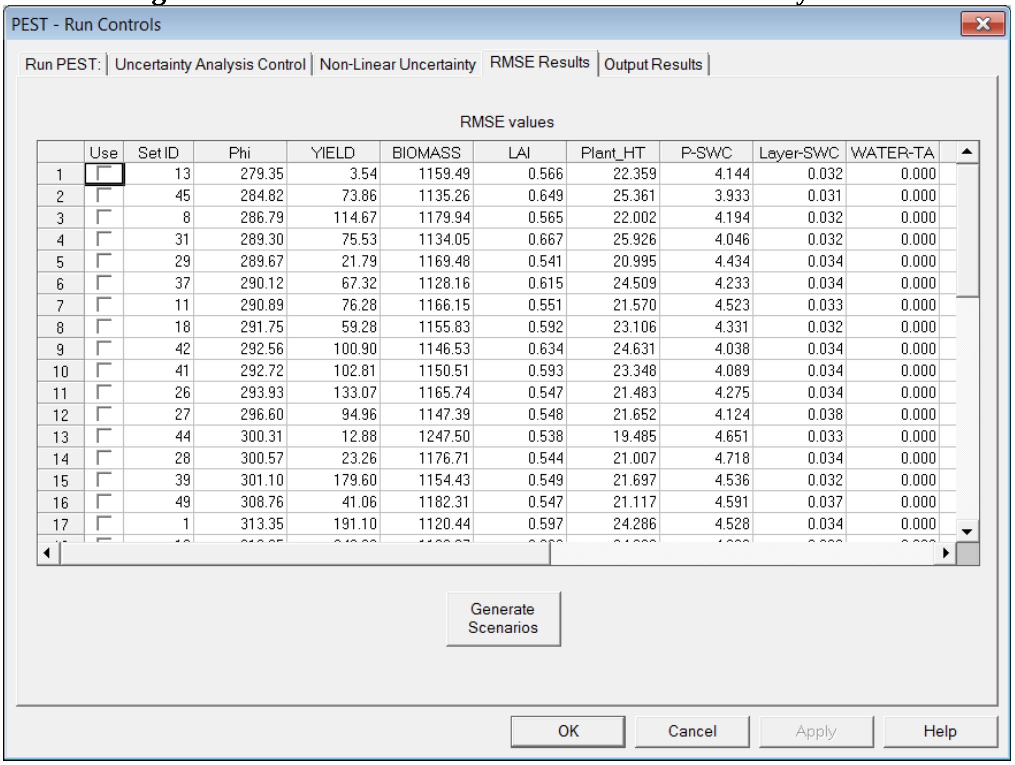
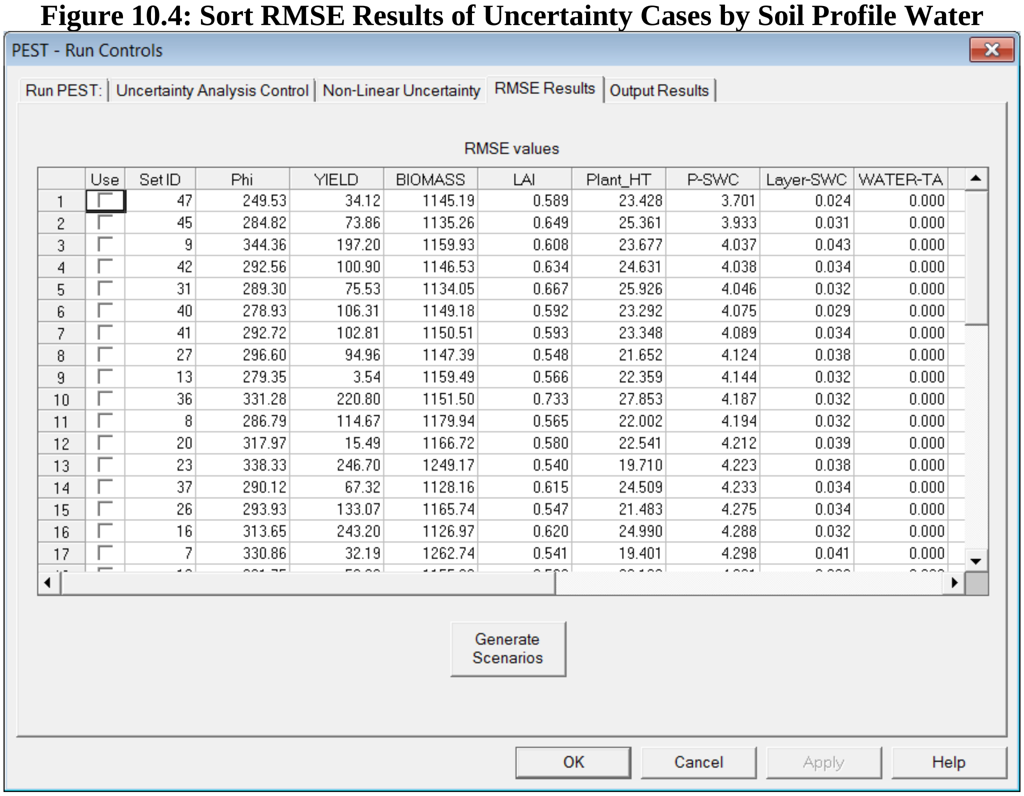


Figure 10.3: RMSE Results of Non-Linear Uncertainty Cases



	Use	Set ID	Phi	YIELD	BIOMASS	LAI	Plant_HT	P-SWC	Layer-SWC	WATER-TA
1		13	279.35	3.54	1159.49	0.566	22.359	4.144	0.032	0.000
2		45	284.82	73.86	1135.26	0.649	25.361	3.933	0.031	0.000
3		8	286.79	114.67	1179.94	0.565	22.002	4.194	0.032	0.000
4		31	289.30	75.53	1134.05	0.667	25.926	4.046	0.032	0.000
5		29	289.67	21.79	1169.48	0.541	20.995	4.434	0.034	0.000
6		37	290.12	67.32	1128.16	0.615	24.509	4.233	0.034	0.000
7		11	290.89	76.28	1166.15	0.551	21.570	4.523	0.033	0.000
8		18	291.75	59.28	1155.83	0.592	23.106	4.331	0.032	0.000
9		42	292.56	100.90	1146.53	0.634	24.631	4.038	0.034	0.000
10		41	292.72	102.81	1150.51	0.593	23.348	4.089	0.034	0.000
11		26	293.93	133.07	1165.74	0.547	21.483	4.275	0.034	0.000
12		27	296.60	94.96	1147.39	0.548	21.652	4.124	0.038	0.000
13		44	300.31	12.88	1247.50	0.538	19.485	4.651	0.033	0.000
14		28	300.57	23.26	1176.71	0.544	21.007	4.718	0.034	0.000
15		39	301.10	179.60	1154.43	0.549	21.697	4.536	0.032	0.000
16		49	308.76	41.06	1182.31	0.547	21.117	4.591	0.037	0.000
17		1	313.35	191.10	1120.44	0.597	24.286	4.528	0.034	0.000

Figure 10.4: Sort RMSE Results of Uncertainty Cases by Soil Profile Water



	Use	Set ID	Phi	YIELD	BIOMASS	LAI	Plant_HT	P-SWC	Layer-SWC	WATER-TA
1		47	249.53	34.12	1145.19	0.589	23.428	3.701	0.024	0.000
2		45	284.82	73.86	1135.26	0.649	25.361	3.933	0.031	0.000
3		9	344.36	197.20	1159.93	0.608	23.677	4.037	0.043	0.000
4		42	292.56	100.90	1146.53	0.634	24.631	4.038	0.034	0.000
5		31	289.30	75.53	1134.05	0.667	25.926	4.046	0.032	0.000
6		40	278.93	106.31	1149.18	0.592	23.292	4.075	0.029	0.000
7		41	292.72	102.81	1150.51	0.593	23.348	4.089	0.034	0.000
8		27	296.60	94.96	1147.39	0.548	21.652	4.124	0.038	0.000
9		13	279.35	3.54	1159.49	0.566	22.359	4.144	0.032	0.000
10		36	331.28	220.80	1151.50	0.733	27.853	4.187	0.032	0.000
11		8	286.79	114.67	1179.94	0.565	22.002	4.194	0.032	0.000
12		20	317.97	15.49	1166.72	0.580	22.541	4.212	0.039	0.000
13		23	338.33	246.70	1249.17	0.540	19.710	4.223	0.038	0.000
14		37	290.12	67.32	1128.16	0.615	24.509	4.233	0.034	0.000
15		26	293.93	133.07	1165.74	0.547	21.483	4.275	0.034	0.000
16		16	313.65	243.20	1126.97	0.620	24.990	4.288	0.032	0.000
17		7	330.86	32.19	1262.74	0.541	19.401	4.298	0.041	0.000

10.4 **RMSE Results**

The resulting cases, phi and RMSE of observed versus predicted variables are display in *RMSE RESULTS* figure 10.3 can be found under the PEST directory in your RZWQM2 scenario. Double click on any title heading and the parameters cases will be sorted by that RMSE (Figure 10.4). Try different columns to note which parameter set numbers always occur in the top 5. You may want to then check parameters set you would like a full RZWQM2 scenario to look at.

10.4.1 Generate Individual RZWQM2 Scenarios

If you check USE next to the RMSE results that parameter set will be used when you choose *GENERATE* Scenarios. The scenario generated will appear in the project window when you exit and save the scenario. It will have a yellow execution ball next to it to denote it was generated and you have not opened and ran it yet. This is a small side feature to allow you to build specific RZWQM2 scenarios with parameter set you like. It IS NOT the Run with Uncertainty. This can be found on the main scenario view menu under RUN, RUN with Uncertainty. The scenario view menu item will become live when you have completed uncertainty analysis through the RMSE table results. All cases of parameter sets in the RMSE Results table will be run on a RUN with Uncertainty run.

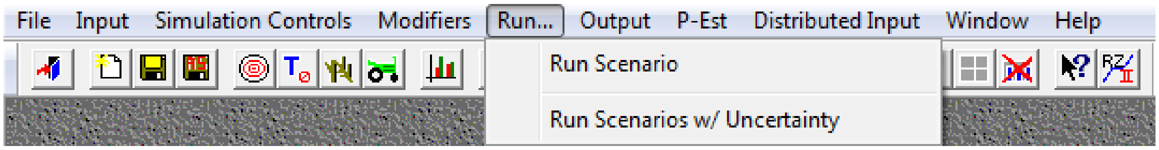
10.5 **Output Results**

The *Output Results* screen shows several variables related to the results from the Uncertainty run. This screen allows you to select an input parameter or simulated output and generates a histogram of occurrences among the previously run uncertainty cases.

10.6 **RZWQM2 Run with Uncertainty**

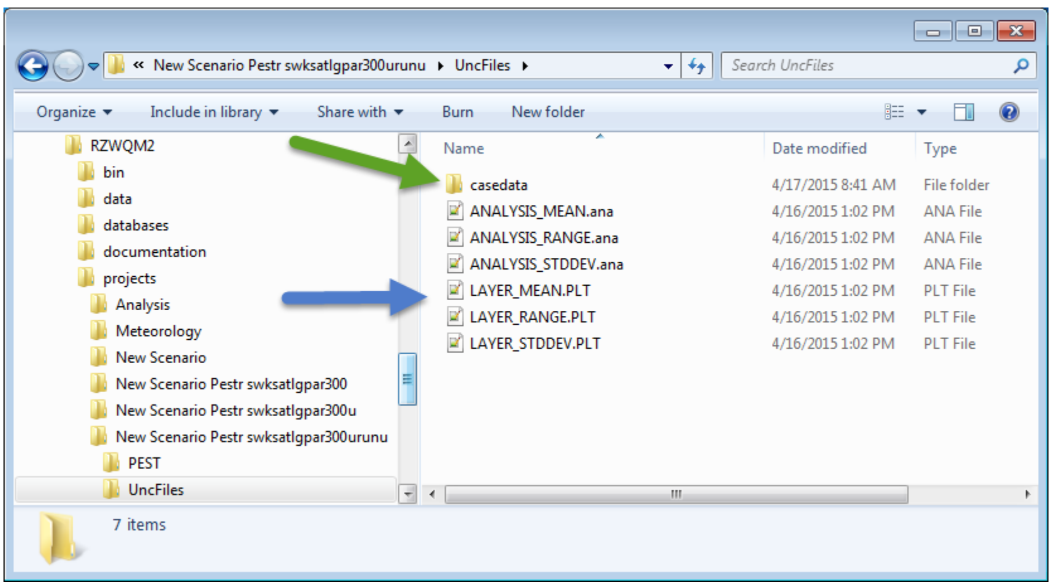
This feature is only available after Parameter Optimization, generation of the posterior uncertainty (Jacobian matrix and covariance matrix from the optimized parameter set) and Uncertainty Analysis is completed. The generated parameter cases and params.unc file must exist in the pest directory of your scenario for the “Run Scenarios w/ Uncertainty” to un-grey from the scenario view RUN menu.

Execute Run with Uncertainty from the scenario view RUN menu. A question will appear to save the individual parameter cases’ output. The analysis files which are the standard daily output of RZWQM2 are quite large so you may choose to not save individual case output files if you are limited to space. Statistics are calculated for all the possible output variables in these files before the cases are deleted if you choose not to save.



An UncFiles directory will be generated in your scenario directory. This directory will have the statistical summary files: mean, standard deviation, and range files for all output variables in the analysis and layer.plt files for all cases appearing in the RMSE Results from your non-linear uncertainty analysis. Import these files to your preferred plotting packages to generate your confidence bands of output.

The UncFiles directory will also have a subdirectory called "casedata". The output files from the cases labeled case#.PLT or case#.ana. Currently ana and layer files for each case are saved under case data.



Chapter 11:
Distributed Input

The option to run a RZWQM2 scenario with a range of soil property and soil water attribute input was developed to quantify the uncertainty associated with ranges in soil property field measurements. The input and run templates were taken from the Optimization and Uncertainty Section as some PEST pest utilities like the random number generator or latin hypercube generator are used to generate cases of input parameters.

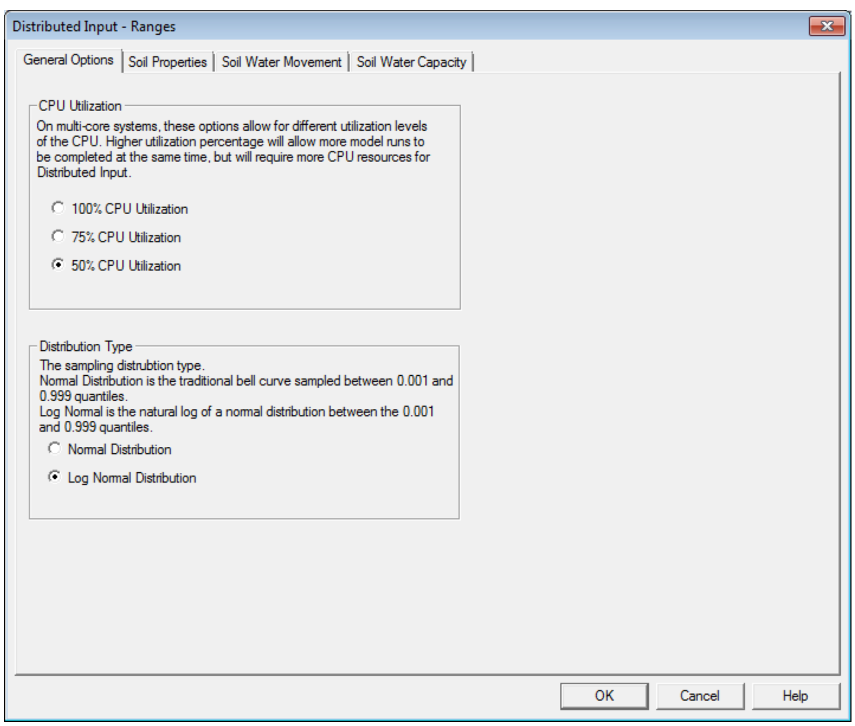
11.1 **Prerequisites for Distributed Input**

This option does not require observed experimental data (expdata.dat) as it is not fitting the input parameter to an RMSE performance of a simulated vs. observed output variable. The minimum and maximum parameter range entered does not reflect as statistical confidence of 95% as does the fit parameters for Optimization and Uncertainty process. The range to vary is subject to the user's natural occurring variability of the soil properties measured in the field and is set to test data sets within the range.

11.2 **Input for Distributed Input**

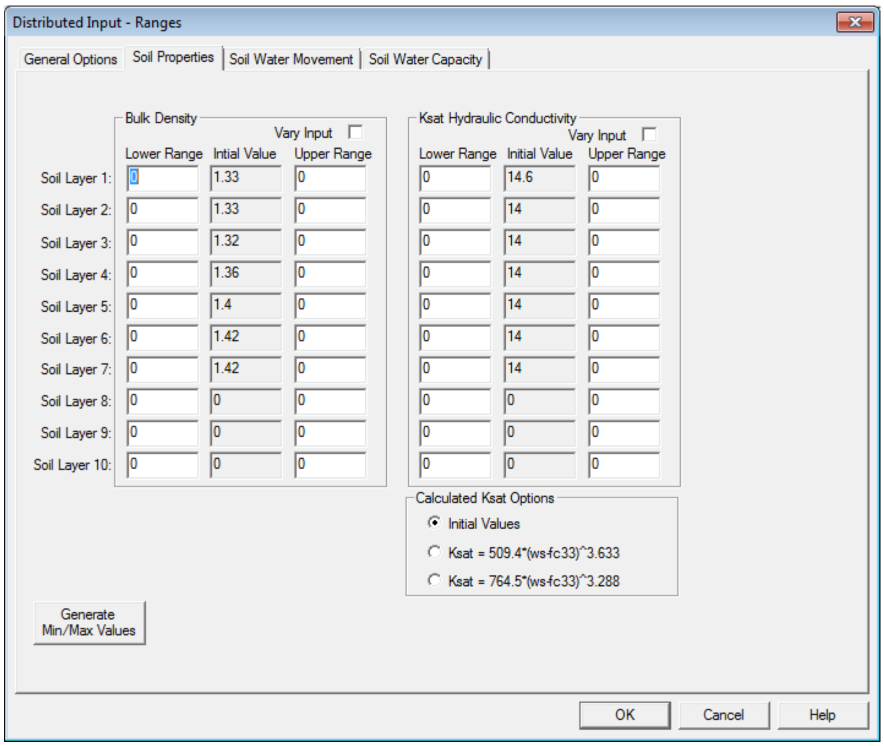
The input design is taken from P-Est input as parameters can be sampled from a log or normal distribution and portion of CPU can be allocated for execution speed of cases (Figure 11.1). The input parameters available to vary focus on soil properties, soil water movement attributes or soil water content at 1/3 or 15 bar.

Figure 11.1 Main input for distributed input



As with PEST fitting options on soil water content the water content to vary at 1/3 or 15 bar availability depends on the method used for hydraulic properties in the original scenario. If only 1/3 bar was available from the original scenario it will only be available to vary. If you are fitting 1/3 bar you may use it to calculate KSAT with the same equations used under the PEST Input Screens (Figure 11.2). If you vary input you must provide a lower and upper range around your initial values.

Figure 11.2 Vary Soil Properties in Distributed input

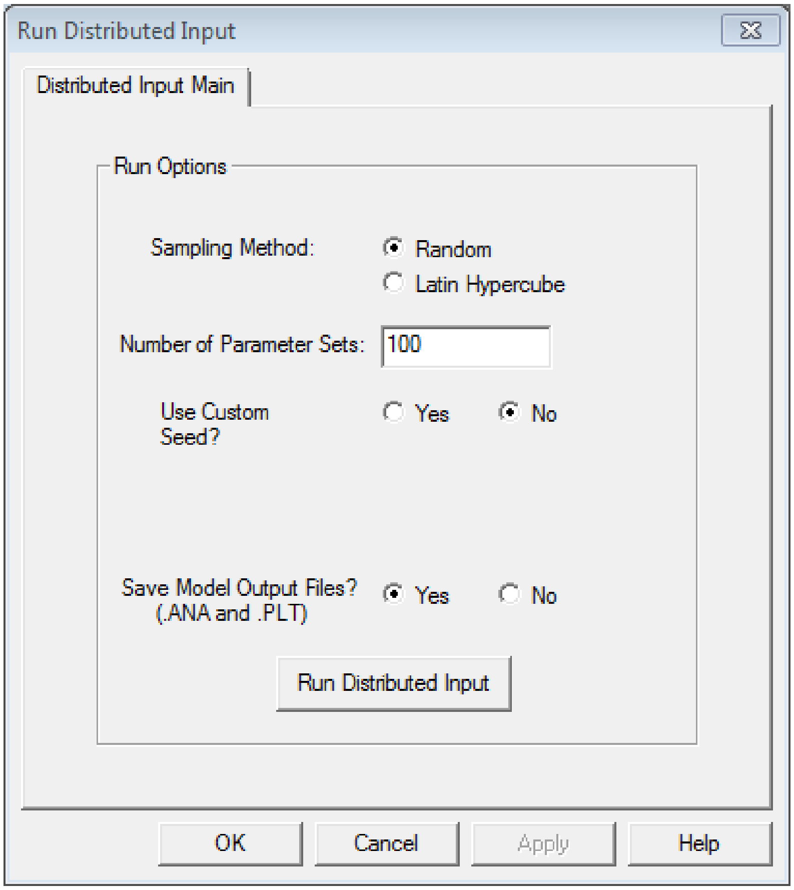


11.3 **RUN for Distributed Input**

Parameter sets are generated in this section using the same PEST utilities use by P-Est (Figure 11.3).

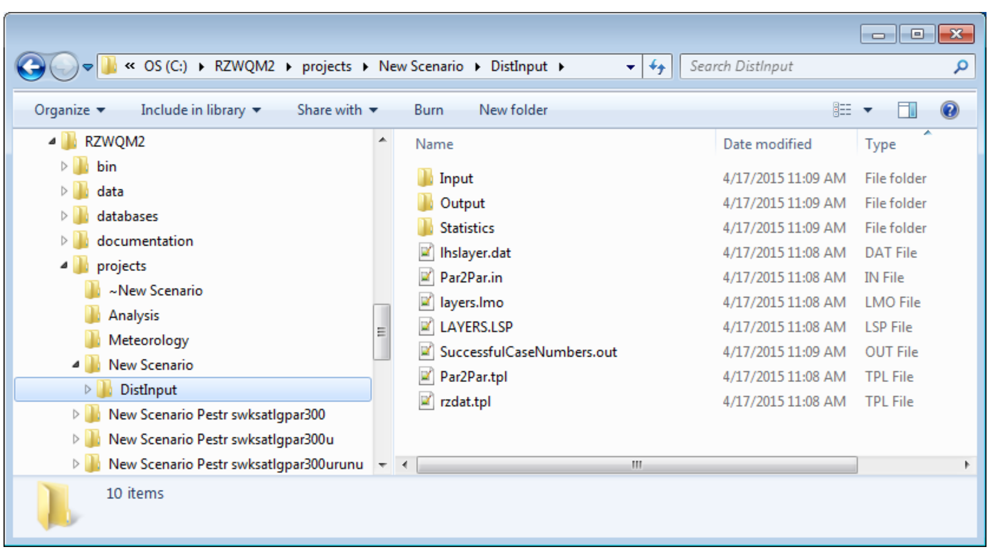
1. Select Random (randpar.exe) or Latin Hypercube (Pest2lhs.exe).
2. Set the number of sets to generate.
3. Use default seed or set your own.
4. Keep or delete the Casedata
5. Select Run

Figure 11.3 Run Distributed Input



Go to <SCENARIO DIR>\DISTINPUT to see ANA and LAYER files in output or mean, range and standard deviation of outputs in the STATISTICS directory.

Figure 11.4 Distributed Input directory



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Appendix 1:
RZWQM2 Science Scenario Input Files

File name	Description
CNTRL.DAT	Selected output variables and files. Totally managed by the RZWQM2 interface and the file is rewritten at each simulation.
IPNAMES.DAT	Direct the folders of model input and output. Totally managed by RZWQM2 Interface.
PLGEN.DAT	Parameters related to the GENERIC plant growth model. Users may access it from the interface *Edit Generic Parameters*.
RZINIT.DAT	Initial conditions for RZWQM simulation. Totally managed by RZWQM2 interface.
RZWQM.DAT	Parameters needed for RZWQM science simulation. It contains information on soils, nutrient, and management. Totally managed by RZWQM2 interface. *It is rewritten on Run for each science simulation.*
EXPDATA.DAT	OPTIONAL – Observed experiment data file
GLEAMS.DAT	Input File for GLEAMS overland flow sediment erosion module
*. CUL	Cultivar specific parameters for DSSAT customized for the scenario
*.SPE	Species specific parameters for DSSAT located in <RZWQM2 INSTALLDIR>/databases/DSSAT
*.ECO	Ecotype file for DSSAT in <RZWQM2 INSTALLDIR>/databases/DSSAT
EXPDSSAT.*A	Optional not used by RZWQM2: DSSAT can accept non-time series observed experiment data in this file (e.g. yield, maximum lai…)
EXPDSSAT.*T	Optional not used by RZWQM2: DSSAT can accept time-series observed experiment data in the this file (e.g. biomass, lai, soil moisture)

Appendix 2:
Data Requirements

Data you need to run RZWQM2

- Weather data: daily solar radiation, maximum and minimum air temperature, relative humidity, rainfall, and wind speed. Precipitation event data (sub-hourly).

- Site Description: slope and aspect

- Initial conditions: initial surface crop residue, soil temperature, soil water, and inorganic soil N.

- Soil data: soil horizons depth, soil texture, soil bulk density, soil organic C, soil pH, soil hydraulic property (if known)

- Management data: planting depth and date, variety (help in identifying the cultivar parameters from the available database), initial planting material weight (for potato only), sprout length at planting (for potato only), emergence date (required for potato only), row spacing, harvest dates, fertilizer and manure applications, irrigation, tillage, and pesticide applications.

* RZWQM2 can estimate soil hydraulic properties from soil texture. If macropore flow is a concern, users should input macroporosity.

Data needed to effectively evaluate RZWQM2 performance**

- Plant data: crop yield, crop biomass and LAI (if possible, a few measurements during the growing season), phenology observations.
- H_2O data: water balance measurements, e.g., soil water contents, leaching, runoff, and ET.
- N data: N balance measurements, e.g., residual soil N, N leaching, and N uptake.
- C data: if soil C sequestration is of interest, soil organic C should be measured before and after the experiments.
- T/energy balance: if energy balance is of interest, temperature measurements in the canopy and soil are needed. Heat fluxes should also be measured.

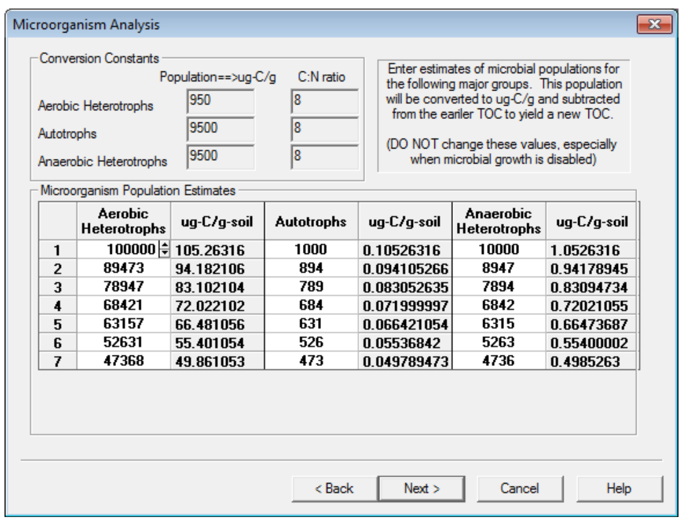
** Not all the data are essential. A balanced data collection for all the categories is recommended.

Appendix 3:
Initializing Soil Carbon and Nitrogen Organics/Inorganics

First, provide an initial estimate of soil organic Carbon (from measured or soil survey estimates) for each horizon using *Residues | Initialization Wizard* and partition pools (based on management fast recycling manures partition more to fast pool) (Method 1). The "Equilibrate Nutrient Pools" button is another tool available to get an initial estimate, but is not recommended if the user has some idea of their initial organic carbon and inorganic N in the soil profile. Second, run for a time period and use the result to initialize organic pools and microbes (Method 2, 3 or 4) for the baseline run.

Method 1: Use of the Residues Initialization Wizard

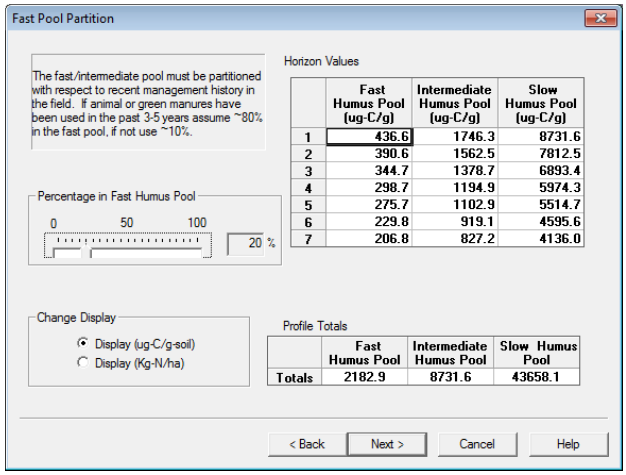
1. Go to Residues, **Initial Residue/Inorganic N Profile** dialog and click ***Initialization Wizard.***
2. Enter your soil layer fraction of organic matter OR set the conversion organic matter to carbon coefficient to 1 and enter measured organic C directly.
3. Click ***Next***, default microbial populations are loaded and distributed in the soil profile according to soil carbon in each layer. It is not recommended to modify these values.



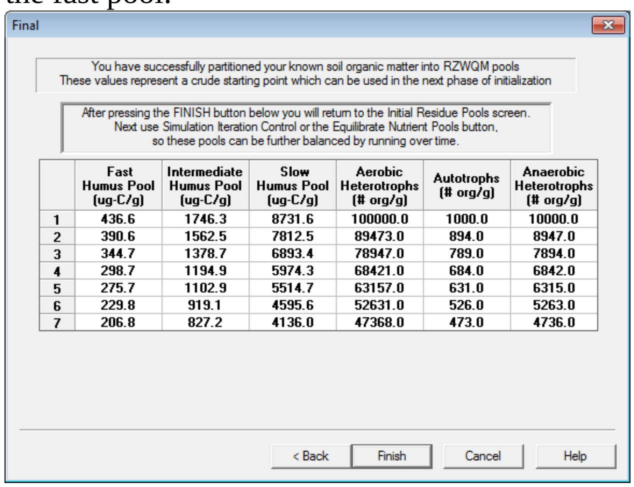
	Aerobic Heterotrophs	ug-C/g-soil	Autotrophs	ug-C/g-soil	Anaerobic Heterotrophs	ug-C/g-soil
1	100000	105.26316	1000	0.10526316	10000	1.0526316
2	89473	94.182106	894	0.094105266	8947	0.94178945
3	78947	83.102104	789	0.083052635	7894	0.83094734
4	68421	72.022102	684	0.071999997	6842	0.72021055
5	63157	66.481056	631	0.066421054	6315	0.66473687
6	52631	55.401054	526	0.05536842	5263	0.55400002
7	47368	49.861053	473	0.049789473	4736	0.4985263

4. Click ***Next*** to partition the TOC between fast/intermediate carbon and the slow carbon. Use the left and right arrow on the keyboard to adjust partition down and up by 1%, respectively.

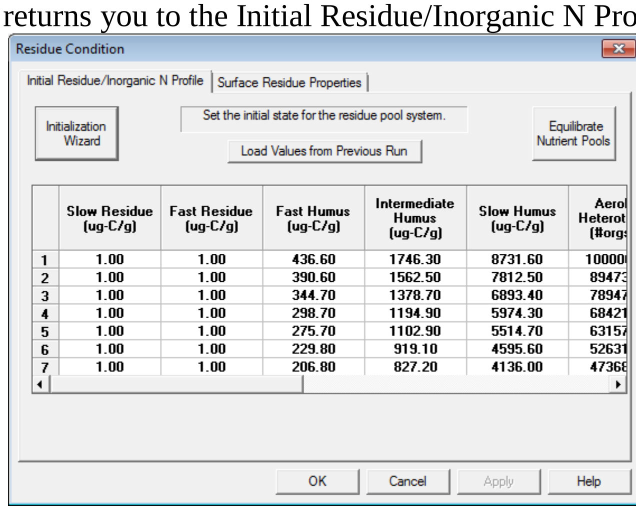
5. Click ***Next*** to partition the fast carbon from the intermediate.



Tip: Partition numbers vary by ecosystem and management. Management with animal or green manures in the past 5 years or new cultivated land requires higher percentages in the fast pool.



6. Press ***Finish.*** It returns you to the Initial Residue/Inorganic N Profile dialog.



7. Scroll to the last three columns in the spread and enter residual Soil NO_3-N and NH_4-N for each layer. If you have Urea you may enter it separately and RZWQM2 will hydrolyze it to the NH_4-N form, instead of entering it as NH_4-N.

8. ***Apply*** or O.K**.**

Method 2: Use of Historical Weather and Management in a Scenario (10 or more years)

1. Set up an RZWQM2 Run for a time period of ten years or more.
2. Use Method 1 to partition the OM into C pools by horizon.
   (Simulation Controls Default is: 0 iteration, unchecked RESET, checked Save, check all boxes for Replace)
3. Run the model.
4. View the NEWINT.OUT file under the scenario directory.
5. Use the carbon status of the organic matter and microbial pools in NEWINT.OUT to initialize your next run by using ***Load Values from Previous Run*** button.
   Note: You may also run iterations of your 10 year runs.

Method 3: Historical Weather and Management Data Less Than Ten Years.

1. Start and end your simulation on the same month and day (or end day-1) for as many years that you have data for and include all management during that time
2. On the **Simulation Iteration Control** screen *check* the box *Save the System at Termination*. *Check* under *Sections to Replace* the Soil Carbon and Soil Inorganic Nitrogen, Surface Mulch, and Surface Standing mass. You may also want to replace other items that you do not have measured initial values for.
3. Increase your run iterations to 10 or more.
4. Look in NEWINT.OUT file will give you C in organic matter pools and microbes at the end of the iteration.
5. Load these carbon values as initial values on your scenario (using ***Load values from Previous Ru*n**)

   Note: When iteration control is greater than 1, the RESET THE SYSTEM box will automatically be checked at the end of the first iteration. No action is required by the user.

Method 4: Manual Calibration without microbial growth:

Users may use the no-microbial growth option (Figure 6.3) to manually partition the soil C pools.

1. Based on the best knowledge of the user on the soil, enter fraction of the soil C pools using the **Initialization Wizard (Method 1)**, leaving the microbial population as default.
2. Run the model for a period of time.
3. Check the MBLNIT.OUT for N budget, especially annual N mineralization rate to make sure that this rate is reasonable for the soil and climate conditions.
4. If the annual mineralization rate is too high, partition more toward the slow pool. If the rate is too low, partition more toward the fast/intermediate pool.
5. Repeat steps 1-4 until the annual mineralization rate is reasonable.

Organic Matter Pool Stability

In the model, we assume the slow pool to be stable within 200-2000yrs, the intermediate pool within 20-100 yrs and the fast pool within 1-5yrs. RZWQM2 will have NUTRI1.PLT, NUTRI2.PLT..NUTRI*.PLT output files numbered for each iteration. The plot files can be appended and used in a spreadsheet tool to plot organic matter pool activity over time. While seasonal fluctuation is normal, you want to run iterations until there are no overriding upward or downward trends from year to year and then use the ending state as a near stable pool to initialize with.

Appendix 4:
Snow Dynamics File - *.SNO

The snow dynamics file is an ASCII text file that can be accessed from ***INPUT | PRMS Snow Parameters***. The format is contained in the file and the default file is listed below (<RZWQM install>\Startup*.SNO). Water infiltrated from the snowpack can be controlled by the SNOWMELT PARAMETER 1.1, fraction of snowmelt that will infiltrate, the rest may be lost to evaporation or runoff.

The "=" sign in the first column indicates the line will be a comment. A blank or a number will indicate data to read. No blank lines should be in this file (it will read as 0 data). There is no missing data value for this file all numbers should be defaulted as below.

```
========================================================================
==
==            P R M S   S N O W   P A R A M E T E R S
==
==
==
==              WATERSHED:  Lucerne, CO 1994-95
==
==
========================================================================
=
=     GENERAL SNOWPACK DYNAMICS
=
= REC.NUM   DESCRIPTION
= --------  -----------
=   1.1    AVERAGE MAXIMUM SNOWPACK DENSITY [GM/CM3]
=   1.2    INITIAL DENSITY OF NEW-FALLEN SNOW IN [GM/CM3]
=   1.3    FREE-WATER HOLDING CAPACITY OF SNOWPACK
=   1.4    SNOWPACK SETTLEMENT TIME CONSTANT
=   1.5    PRECIP ALL SNOW IF MAX TEMPERATURE BELOW THIS VALUE [F]
========================================================================
0.6  0.1  0.05  0.1  32.0
========================================================================
=
=     ALBEDO ADJUSTMENTS
=
= REC.NUM   DESCRIPTION
= --------  -----------
=   1.1    ALBEDO RESET - RAIN, MELT STAGE
=   1.2    ALBEDO RESET - RAIN, ACCUMULATION STAGE
=   1.3    ALBEDO RESET - SNOW, MELT STAGE
=   1.4    ALBEDO RESET - SNOW, ACCUMULATION STAGE
========================================================================
0.6  0.8  0.2  0.05
========================================================================
=
=     SURFACE COVER ADJUSTMENTS
=
```

```
= REC.NUM   DESCRIPTION
```

Appendix 4 continued...

```
= --------  -----------
=   1.1    COVER TYPE DESIGNATION FOR HRU
=   1.2    SUMMER VEGETATION COVER DENSITY FOR MAJOR VEGETATION TYPE
=   1.3    WINTER VEGETATION COVER DENSITY FOR MAJOR VEGETATION TYPE
======================================================================
1  0.5  0.5
======================================================================
=
=     GENERAL ATMOSPHERIC PARAMETERS
=
= REC.NUM   DESCRIPTION
= --------  -----------
=   1.1    SOLAR RADIATION TRANSMISSION COEFFICIENT
=   1.2    EMISSIVITY OF AIR ON DAYS WITHOUT PRECIPITATION
=   1.3    PROPORTION OF POTENTIAL ET THAT IS SUBLIMATED FROM SNOW
=   2      CONVECTION CONDENSATION ENERGY COEFFICIENT
=   3      SET TO 1 IF THUNDERSTORMS PREVALENT DURING MONTH
======================================================================
0.5  0.757  0.5
5.0  5.0  5.0  5.0  5.0  5.0  5.0  5.0  5.0  5.0  5.0  5.0
0  0  0  0  0  0  0  0  0  0  0  0
======================================================================
=
=     SNOWMELT PARAMETERS
=
= REC.NUM   DESCRIPTION
= --------  -----------
=   1.1    FRACTION OF SNOWMELT THAT WILL INFILTRATED, THE REMAINDER
=          WILL BE RUNOFF.                                      [0..1]
=   1.2    JULIAN DATE TO START LOOKING FOR SPRING SNOWMELT    [1..365]
=   1.3    JULIAN DATE TO FORCE SNOWPACK TO SPRING SNOWMELT STA[1..365]
=   1.4    MAXIMUM THRESHHOLD WATER EQUIVALENT FOR SNOW DEPLETION
=   1.5    INDEX NUMBER SPECIFING WHICH SNOWPACK AREAL DEPLETION
=          CURVE TO USE FOR SNOWMELT                            [1..5]
=   1.6    NUMBER OF SNOWPACK DEPLETION CURVES INPUT            [1..5]
=   2...   SNOWPACK AREAL DEPLETION CURVE VALUES (11 VALUES/CURVE[0..1]
======================================================================
0.8  90  90  50.0  1  1
0.05  0.24  0.40  0.53  0.65  0.75  0.82  0.88  0.93  0.99  1.0
```

Note: *.SNO is accessed via a shell to WordPad from RZWQM2. WordPad File commands will save data or cancel in this file; it is not controlled by RZWQM2 File Exit and Save or Do Not Save commands.

Appendix 5:
Generic Plant Growth File – PLGEN.DAT

The default PLGEN.DAT is located in the <RZWQM2 INSTALLDIR>/DATABASES. Each new scenario has a copy of this database in the scenario directory. ADD Custom Generic Plant will create a PLGEN.DAT for the scenario and user may customize and save to the scenario directory. Each crop has an entry in each section of the file. Advanced training is recommended to customize these parameters.

The "=" sign in the first column indicates the line will be a comment. A blank or a number will indicate data to read. No blank lines should be in this file (it will read as 0 data). There is no missing data value for this file all numbers should be defaulted as below.

```
==============================================================
==                                                          ==
==          G E N E R I C   P L A N T   G R O W T H         ==
==                    MODEL PARAMETERS                      ==
==                                                          ==
==============================================================
==        Text Headers Last Modified:  7.05.2017 (V4.00.00) ==
==============================================================
=
=    DATA BASE HEADER
=
= ITEM NO.   DESCRIPTION
= --------   -----------
=     1      NUMBER OF PARAMETERIZED SPECIES
=     2      VARIETY INDEX NUMBER
=     3      DESCRIPTION OF VARIETY (30 CHARACTERS, MAXIMUM)
=
=   NOTE:  ENTER NUMBER OF SPECIES PARAMERIZED THEN
=   ENTER ONE LINE OF INFORMATION FOR EACH PARAMETER IN THE FILE
==============================================================
14
=    ++++++++++++++++++++++++++++++
0001  CORN (GPSR)
0002  CORN (NEBRASKA VARIETY)
0003  CORN (EASTERN COLORADO DRYLAND
0004  CORN (OHIO)
0005  CORN (MINNESOTA)
0006  CORN (MISSOURI)
0007  CORN (EASTERN COLORADO IRRIG)
0008  CORN (IOWA)
0101  SOYBEAN (GPSR)
0102  SOYBEAN (MISSOURI)
```

Appendix 5 continued...

```
0103  SOYBEAN (MINNESOTA)
0104  SOYBEAN (OHIO)
0201  WHEAT (WINTER AKRON)
0301  ALFALFA (IOWA)
=====================================================================
=
=   THE PARAMETERS BELOW ARE FOR THE GENERIC PLANT GROWTH
MODEL ONLY.
=
=        ..THERE MUST BE AT LEAST ONE PLANT PARAMETERIZED
=            EVEN IF THE GENERIC MODEL ISN'T USED.
=
=        ...REPEAT THE GROWTH PARAMETER RECORDS BELOW FOR EACH
=                   PLANT USING THE GENERIC PLANT GROWTH MODEL
=
====================================================================
=    PLANT DIMENSIONS  (ASSUMING A CYLINDERICAL SHAPE)
=
= ITEM NO.   DESCRIPTION
= --------   -----------
=    1    STEM DIAMETER OF THE MATURE PLANT CYLINDER  [CM]
=    2    STEM HEIGHT OF THE MATURE PLANT CYLINDER    [CM]
=    3    ABOVEGROUND BIOMASS AT 1/2 MAX  HEIGHT [GM/PLANT(<=100)
OR KG/HA (>100)]
=    4     ABOVEGROUND BIOMASS OF A MATURE PLANT   [GM/PLANT OR
KG/HA]
=    5      BIOMASS OF PLANT AT 4-LEAF STAGE     [GM/PLANT OR KG/HA]
=
=====================================================================
80.0  250.0  60.0   215.0  20.0
70.0  250.0  30.0    70.0  10.0
…
====================================================================
=    NITROGEN MANAGEMENT
=
= ITEM NO.   DESCRIPTION
= --------   -----------
=    1  MAXIMUM WHOLE-PLANT N CONTENT                              [PROP]
=    2  NITROGEN CONTENT WHEN PLANT IS AT GROWTH STAGE 1.0  [PROP]
=    3  MINIMUM SHOOT NITROGEN NEEDED FOR PLANT GROWTH     [PROP]
=    4  NITROGEN UPTAKE EFFICIENCY COEFFICIENT (CNUP2)            [PPM]
=    5  NITROGEN USE EFFICIENCY COEFFICIENT                          [0..1]
```

Appendix 5 continued...

```
=    6  TYPE OF PLANT BEING MODELED  [1=NITROGEN FIXER, 2=NONFIXER]
=    7  PLANT REQUIRES VERNALIZATION               [1=YES, 0=NO]
=    8  NITROGEN N FIXING NODULE SIMULATION          [1=YES, 0=NO]
=================================================================
0.026  0.016   0.005  1.5  160.0  2  0 0
0.026  0.016   0.005  1.5  160.0  2  0 0
…
=================================================================
=    NITROGEN CONTENT LIMITS
=
= ITEM NO.   DESCRIPTION
= --------   -----------
=    1     MINIMUM N CONTENT FOR LEAVES
=    2     MAXIMUM N CONTENT FOR LEAVES
=    3     MINIMUM N CONTENT FOR STEMS
=    4     MAXIMUM N CONTENT FOR STEMS
=    5     MINIMUM N CONTENT FOR ROOTS
=    6     MINIMUM N CONTENT FOR PROPAGULES
=    7     MAXIMUM N CONTENT FOR SEEDS
=================================================================
0.02  0.03    0.01    0.03    0.01  0.02      0.02
0.01  0.03    0.01    0.03    0.01  0.0075    0.025
…
=================================================================
=
=      PHOTOSYNTHESIS
…
```

Important Note : Version 4.0 and later Plant Biomass at ½ Max Height can be g/plant if <=100 or considered as kg/ha input if input >100. Plant Dimension Item 3 entry will dictate the unit entry required for item 4 and 5. If Item 3 <=100 All entry is g/plant (legacy RZWQM generic plant model function) and if Item 3 is >100 all input units are kg/ha (new accommodation).

Note: PLGEN.DAT is created from the Interface Add Custom Generic Crop, a default can be loaded from the RZWQM2/DATABASES/PLGEN.DAT. *SAVE and Exit* from Add Generic Custom Crop will write customized plgen.dat to scenario directory.

Appendix 6:
Irrigation Scheduling Format in RZWQM.DAT

This section describes the irrigation management portion of the RZWQM.DAT input file to the science simulation. There are three timing controls and three application controls from which to choose from. There are also simulation period controls which constrain the irrigation practices for the season. It's important to note that the water applied as irrigation is the amount that actually goes to the soil surface.

The format for lines 1 and 2 is standard as in the comments from the RZWQM.DAT listed below. However the format for record sections 3 and 4 depend on whether Specific Date, Fixed Interval, or Rule based (via Root Zone Depletion or ET Deficit) scheduling method (Item 2.3 and 2.4) is used.

```
========================================================================
==                                                                    ==
==          I R R I G A T I O N   M A N A G E M E N T                 ==
==               MODEL VARIABLES AND PARAMETERS                       ==
==                                                                    ==
========================================================================
=     Irrigation control parameters
=
= ITEM NO.   DESCRIPTION
= --------   -----------
=   1.1      number of irrigation operations
=   1.2      depth of the rooting zone for depletion calculation,
=            (DEFAULT = 0 use active rooting zone) [0...3000 cm]
=   1.3      flag to use maximum monthly irrigation amounts [0=NO,1=YES]
=   1.4      minimum daily irrigation amount                        [cm]
=   1.5      maximum daily irrigation amount                        [cm]
=   1.6      subirrigation depth (default=15 cm)                    [cm]
=   1.7      irrigation interval limit amount(default=0)            [cm]
=   1.8      irrigation interval time in years(default=0)           [yrs]
=   2.1      plant reference number
=   2.2      type of irrigation =
=[1=sprinkler,2=flood,3=furrow,4=drip,5=subirrigation]
=   2.3      timing for irrigation practice (interval,specific,root zone
= deplete, ET deficit)[1..4]
=    2.4      methodology for irrigation practice                  [1..4]
=   2.5-7    earliest date for irrigation                  [dd mm yyyy]
=   2.8-10   latest date for irrigation (D/M/Y)            [dd mm yyyy]
=   2.11     minimum number of days between irrigations            [day]
=            note (if item 2.3 equals 1 or 2, set to zero)
=   2.12     maximum depth of irrigation to be applied during the
=              season (if = 0.0, water assumed unlimited)           [cm]
=   2.13     rate at which sprinkler irrigation is applied       [cm/hr]
=   3        timing of irrigation practice
=              see the user manual for details about this line
=   4        methodology of irrigation practice
=              see the user manual for details about this data
```

```
=   . . .  repeat Rec 2-4 for each operation
=================================================================
2  0
1  1  2  2  29  6  1985  20  8  1985  0  0.0  2.0
10
29  6  1985
30  7  1985
8  8  1985
9  8  1985
12  8  1985
14  8  1985
15  8  1985
17  8  1985
19  8  1985
20  8  1985
10
0.5  7.6  0.9  0.2  2.5  1.0  1.7  0.9  1.2  1.2
1  1  2  2  22  8  1985  18  9  1985  0  0.0  2.0
2
22  8  1985
18  9  1985
2
1.2  0.0
```

Record 1 (Item 1.1) above indicates the "New Scenario" had specific date irrigations entered in 2 rows of the spreadsheet. Item 1.2 is a 0 default to use the active root zone if irrigation depletion is chosen. Item 2.1 specifies the crop ID number receiving the irrigation water. Item 2.2 denotes 1 for Sprinkler irrigation. The types of irrigation to choose from: 1) sprinkler , 2) furrow, 3) flood, 4) drip and 5) subirrigation. Sprinkler irrigation requires application rate as input and is treated as a rain storm. Drip irrigation is treated as a low intensity rain storm where intensity < Ksat.

Irrigation timing options (Item 2.3) are fixed interval (1), specified dates (2), root zone water depletion (3) and ET deficit (4). Methodology choices (Item 2.4) are fixed amount for all irrigations (1), varying amount (2), and amount to restore soil water to a percentage of plant available water (3) or % of PET. In New Scenario, specified dates are used with varying amounts; where, item 2.3 is 2 and item 2.4 is 2, respectively.

Constraints on irrigation amount and timing can be entered to help customize the operation of the irrigation submodel. The earliest and latest dates (Item 2.5-2.10) of irrigation are entered, along with the minimum number of days between irrigations (Item 2.11). A cap (Item 2.12) on irrigation amount is entered next as the maximum amount (cm) of irrigation applied during the growing season. If this cap is set to 0.0, water availability is assumed unlimited.

Finally, the rate at which the sprinkler irrigation is applied (Item 2.13) is entered (cm/hr). If sprinkler irrigation is selected the irrigation is treated as a rain storm event, so that the amount of water is divided by the application rate to obtain the amount of time required to apply the water. The length of the application **cannot** be longer than 12 hours or the program will abort the application. This constraint allows the simulation program enough time to stabilize after the influx of new water and chemicals.

For Record 3, timing of irrigation, the format varies according to the user's timing choice selected for item 2.3 :

- Fixed interval (option 1) requires that you enter a single number that is the day step size within the period stated in item 2.5-7 and 2.8-10.

Example: 7 [irrigation would be triggered every seven days]

-Specific dates (option 2) as in the New Scenario example above requires that you enter the number of irrigation dates followed by each date (dd/mm/yyyy). Each date should be on a separate line (maximum number of dates is 100) resulting in 11 records for the formatted Record 3 section.

Example:

```
10
29  6  1985
30  7  1985
8  8  1985
9  8  1985
12  8  1985
14  8  1985
15  8  1985
17  8  1985
19  8  1985
20  8  1985
```

- If Root Zone Soil Water Depletion Rule is chosen threshold values are entered for as an automatic irrigation mechanism (option 3). In this mode the simulation program monitors the total plant available (1/3 bar - 15 bar) water (TPAW) in the depletion zone specified and compares this value with the user's depletion threshold value. When available water falls below this threshold, it triggers irrigation. Record 3 section values are entered in pairs consisting of the number of days from planting and the threshold value. The pairs should be arranged chronologically to set up a cascading switch from one threshold to the next. First enter the number of threshold pairs, then enter each pair (up to 10 pairs can be entered). This option can only be used with method option 3. A maximum of 20 sets can be entered.

Example:

```
1  0
1  1  3  3  29  6  1985  18  9  1985  7  0.0  2.0
2 [record section 3, 2 lines]
  7 0.6    30 0.7
2 [record section 4, 2 lines]
 7  .8   30  1.0
```

On record 3, 2 pairs are entered where the first 7 days after planting, irrigation would be triggered if the root zone water content is depleted below 60% of TPAW. At 30 days after planting irrigation would be triggered at 70% of TPAW. If ET Deficit Irrigation Rule instead of the Root Zone Soil Water Depletion the same format is used above but the trigger is AET/PET for record 3.

Example:

```
2
  7 1.0 [7 days after planting if AET/PET is less than 1 irrigate]
```

For Record 4**,** methodology of irrigation practice, the user inputs data depending on the methodology choice selected for Item 2.4:

- Fixed amount of irrigation per event (option 1) is the most straight forward method.

Just enter the amount of water (cm) to apply every time irrigation is triggered.
Example:

2.1

-If Varying amount of irrigation (option 2) as in Specific Date or Fixed interval Irrigation entry, the user specifies how much is applied every time irrigation is triggered. Enter the number of irrigation events, followed by the amount(s) (cm) to be applied for every irrigation event. If the number of amounts listed is less than the number of irrigation events, the last value is used for the rest of the events.

Example:
```
     10
      0.5  7.6  0.9  0.2  2.5  1.0  1.7  0.9  1.2  1.2
```

-If Recharging the soil water in the depletion zone to a percentage of Total Plant Available water (option 3) is used only with timing Root Zone Depletion option 3 above. As with the timing option the user has control over the distribution of the replacement of water throughout the growing season. Enter the number of replacement pairs, which consists of the period after planting and the percentage of Total Plant Available Water to irrigate to. A maximum of 20 sets can be entered.
Example:
```
2 [record 4, 2 lines]
 7  .8   30  1.0
```

The first 7 days after planting irrigation will be applied to reach 80% TPAW if under the depletion threshold. At 30 days after planting irrigation will be applied to reach 100% TPAW if under the depletion threshold.

-If Recharging the ET Deficit rule the amount of recharge 80% of PET input to the interface is expressed as a fraction in the RZWQM.DAT similar to the depletion rule
```
2 [record 4, 2 lines]
 7  .8
```

The irrigation management section from the RZWQM.DAT shown above for New Scenario completes the fourth record section and then repeats record sections 2-4 for the next irrigation operation the user entered on row 2 of the specific date spreadsheet.
Example:
```
1  1  2  2  22  8  1985  18  9  1985  0  0.0  2.0
2
22  8  1985
18  9  1985
2
1.2  0.0
```

Important Note*:* Please choose which irrigation to execute for simulation off the Irrigation Summary Screen. Only one rule method can be chosen. Fixed Interval and Specific Date irrigation can run simultaneously with a rule; however, it is not recommended as conflicting irrigation events may be implemented.

Appendix 7:
Experiment Data File – EXPDATA.DAT

The Scenario Menu **INPUT | Experiment Data** gives access to entering experiment data. Previous WordPad connection to EXPDATA.DAT file has been replaced with dialogs containing spreadsheet entry in Versions 2.4+. Previous new scenario expdata.dat will load into spreadsheets. If not, cut and paste data into appropriate tabbed dialog spreadsheets. Click OK in the master dialog to write the new EXPDATA.DAT format. RZWQM2 will perform comparison statistics of observed vs. predicted for variables allowed in the format. Version 3.1 Adds a section to the top of the expdata.dat. This section allows the user to customize horizon depths for simulated soil layer value reporting (HORIZON.OUT) for all soil layer variables measured in the expdata.dat. The horizon depths are defaulted to the depths shown under Site Description Horizon Description.

There are 2 variable control sections in the file. The first control section has 1 (=0 if no data provided) specified for each variable you will provide data on for the soil profile (vector of data). The second control section has 1 (=0 if no data provided) for scalar values that you will include data on such as plant attributes, pesticides in the tile drain, or ET. Following each control section is the data to be read for the variable. The first line of data defines how many data points to read. For a vector a data point is defined as a measurement for the whole soil profile on one data. Data point limitations are listed in the comments and vary by variable.

The "=" sign in the first column indicates the line will be a comment. A blank or a number will indicate data to read. No blank lines should be in this file (it will read as 0 data). If you have missing data but would like to include the line, enter -99 for a missing value.

Horizon.out Depths(cm)

```
================================================================
15  30  60  90  120  150  178
```

The Vector Control Section...

```
=================================================================
= C:\RZWQM2\projects\New Scenario =
= DO NOT EDIT OR ADD COMMENTS. IT MAY PREVENT THE FILE FROM
FUNCTIONING.                =
=   DATA INPUT TO CREATE COMP2EXP.OUT FILE FOR MODEL COMPARISON
=
= MEASURED SOIL PROPERTIES BY SOIL LAYER (LIMIT TO 1200 DATA
POINTS PER VARIABLE)         =
= DAY
= DEPTH (cm)
= SWC, SOIL WATER CONTENT (cm3/cm3 or g/g)
= SRP, SLOW RESIDUE POOL (g-C/g)
= FRP, FAST RESIDUE POOL (g-C/g)
```

Appendix 7 Vector Control Section continued...

```
= FHP, FAST HUMUS POOL (g-C/g)
= IHP, INTERMEDIATE HUMUS POOL(g-C/g)
= SHP, SLOW HUMUS POOL (g-C/g)
= NO3, NO3-N CONC. (ug/g)
= NH4, NH4-N CONC. (ug/g)
= P1S, PEST#1 CONC. ABSORB. (ug/g)
= P1W, PEST#1 CONC. IN SOL. (mg/ml)
= P1T, PEST#1 TOTAL MASS (ug/g of soil)
= P2S, PEST#2 CONC. ABSORB. (ug/g)
= P2W, PEST#2 CONC. IN SOL. (ug/ml)
= P2T, PEST#2 TOTAL MASS (ug/g of soil)
= P3S, PEST#3 CONC. ABSORB (ug/g)
= P3W, PEST#3 CONC. IN SOL. (mg/ml)
= P3T, PEST#3 TOTAL MASS (ug/g of soil)
= SOILT, SOIL TEMPERATURE (oC)
= SOC, TOTAL SOIL ORGANIC CARBON PER LAYER [Expressed as a
fraction] (g-C/g)             =
= SP,  SOIL PRESSURE HEAD (cm)
= N2O, N2O PRODUCTION (KG N/HA)
= NxO, NxO PRODUCTION (KG N/HA)
= CH4, CH4 PRODUCTION (KG N/HA)
= The following flags apply to the above variables.
…
=SWC SRP FRP FHP IHP SHP NO3 NH4 P1S P1W P1T P2S P2W P2T P3S P3W
P3T SoilT SOC SP N2O NxO CH4
===============================================================
  1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
```

The Vector Data Section…

```
======================================================================
=   MOISTURE CONTENT
=
=   line 1 no. of DATA POINTS (1 DATA POINT = 1 Date of Data),
=
=   AND SWC input type (0=volumetric and 1= gravimetric)
=
=   line 2...n HORIZON LAYER DATA
=
=
=
= DD  MM   YYYY  DEPTH(CM)  SWC (cm/cm or g/g)
=
======================================================================
11     0
 13   6   1985 30.000000 0.190900
 13   6   1985 60.000000 0.213300
 13   6   1985 90.000000 0.199000
 13   6   1985 120.000000 0.185000
 13   6   1985 150.000000 0.187700
 13   6   1985 180.000000 0.183700
 …
```

The Scalar Control Section…

```
=======================================================================
= MEASURED DAILY RESULTS
= No. of data points
= DAY  (dd mm yyyy)
= CROP YIELD (kg/ha)
= CROP BIOMASS (kg/ha)
= CROP N UPTAKE (kg/ha)
= GRAIN N UPTAKE (kg/ha)
= TILE FLOW SECTION
=    ACCUMULATED TILE FLOW (cm)
=    NITRATE-N IN TILE (mg/L)
=    PEST#1 CONC. IN TILE (mg/L)
=    PEST#2 CONC. IN TILE (mg/L)
=    PEST#3 CONC. IN TILE (mg/L)
= WATER TABLE (cm)
= SURFACE RUNOFF SECTION
=    DAILY RUNOFF  (cm)
=    NITRATE-N IN RUNOFF  (mg/L)
=    PEST#1 CONC. IN RUNOFF (mg/L)
=    PEST#2 CONC. IN RUNOFF (mg/L)
=    PEST#3 CONC. IN RUNOFF (mg/L)
= PLANT HEIGHT (cm)
= ET, from day1 to day2
= LAI
=YIELD,BIOMASS,PLANT N-UP,GRAIN-N-UP,TILE-FLOW,WATER
=TABLE,RUNOFF,HEIGHT,ET,LAI
=======================================================================
   1   1   0   0   0   0   0   1   1   1
=======================================================================
```

The Scalar Data Section…

```
=======================================================================
=   CROP YIELD     (LIMIT 100 data points)
=
=   HARVEST DATE (DD-MM-YYYY), YIELD, HARVEST EFFICIENCY, CROP#
=
=======================================================================
1
17       9        1985      9854.00   0.97      1
=======================================================================
=   CROP BIOMASS   (LIMIT 100 DATA POINTS)
=
=   DATE (DD-MM-YYYY), BIOMASS CROP#
=
=======================================================================
5
17       6        1985      435.00    1
2        7        1985      2877.00   1
15       7        1985      3396.00   1
23       7        1985      8241.00   1
21       8        1985      11258.00  1
```

... Continued for each scalar with flag =1 in the control section (height, LAI and ET not shown.

Note: Versions 2.1 and earlier Expdata.dat is accessed via a shell to WordPad from RZWQM2. WordPad File commands will save data or cancel in this file; it is not controlled by RZWQM2 File|Exit and Save or Do Not Save commands.

Tip: Double Check the record number entered matches the number of days of data. A mismatch will cause a statistical comparison not to run but not tell you why. If you have exceeded the records of a section limit a warning will display in the RZWQM2.LOG file.

Appendix 8:
RZWQM2 Science Scenario Output Files

File name	RZWQM Output Description
RZWQM2.LOG	Output daily report of each RZWQM2 simulation and errors on executing RZWQM2.
GLEAMS.LOG	Output record of the GLEAMS module execution if activated.
ACCWAT.OUT	Accumulated daily water storage, runoff, actual ET, seepage, drainage, macropore flow, and infiltration
AVG6IN.OUT	Average water content, nitrate and pesticide concentrations in six inch increments
CLEACH.OUT	Chemical leaching each day out of the bottom of the soil profile
HORIZON.OUT	Daily simulated soil layer output for variables that are present in experiment data.
MACRO.OUT	Detailed information on chemical mass balance after each infiltration
MANAGE.OUT	Recording of management practices, such as tillage, irrigation, fertilization, and planting and harvesting.
	Summary of plant harvesting information
MASSBAL.OUT	Mass balance problems for N, C, H2O, and Water.
MBLCARBON.OUT	Detailed carbon balance information
MBLNIT.OUT	Detailed nitrogen balance information
MBLP*.OUT	Mass balance information for each pesticide
MBLWAT.OUT	Detailed water balance information. The file can be extremely large based on user's option (debug level)
NEWINT.OUT	Recording of the system status at end of simulation runs
NUPTAK.OUT	Daily nitrogen uptake by plant, including passive and active uptake
NUTRI.OUT	Daily nitrogen status in organic and inorganic forms
PESTP.OUT	Pesticide distribution in the soil profile each day
PHENOLOGY.OUT	Simulated phenology dates for emergence, anthesis and maturity
PHOTOSYN.OUT	Daily Potential and Actual Photosynthesis and stresses
PLANT.OUT	Daily nitrogen balance of the plant

Appendix 8 continued...

PLPROD.OUT	Photosynthesis rate and corresponding plant population and LAI
PROFILE.OUT	Printing out important parameters associated with soil water movement
RZWQM.OUT	Geographic information of the site, soil properties by layer, summary of the simulation period, simulated daily nitrate breakthrough concentration, actual ET and transpiration
SEDIMENT.OUT	GLEAMS module output file for sediment loss
WEATHERD.OUT	Output of Daily Weather used in simulation
WEATHERH.OUT	Output of Hourly Weather used in simulation
DAILY.PLT	All the information to generate 2-D plots of user selected variables
LAYER.PLT	All the information to generate 3-D plots of user selected variables
NUPTAK.PLT	Contains same information as NUPTAK.OUT but for plotting purpose
NUTRI.PLT	Contains same information as NUTRI.OUT but for plotting purpose
PEST*.PLT	Total pesticide in the soil profile at the end of each day
PLANT0.PLT	Generic Plant Model daily plant biomass and nitrogen accumulation
PLANT1.PLT	Generic Plant Model Nitrogen concentration in plant tissues
PLANT2.PLT	Generic Plant Model recording of plant environmental fitness parameters
PLANT3.PLT	Generic Plant Model number of plants in each growth stage at end of each day
PLANT4.PLT	Generic Plant Model height and leaf area index
PLANT5.PLT	Generic Plant Model nitrogen demand and supply

Appendix 8 continued...

File name	DSSAT CSM Crop Growth Output Description
OVERVIEW.OUT	Overview file (default)
SOILC.OUT	Carbon File
PLANTGRO.OUT	Growth File (default)
PLANTN.OUT	Plant Nitrogen File (default)
SOILN.OUT	Soil Nitrogen File (default)
SOILWAT.OUT	Soil Water File (default)
SOILTEMP.OUT	Soil Temperature File
SUMMARY.OUT	Summary file

Appendix 9:
Scenario Comparison File – COMP2EXP.OUT

===

PLANT HEIGHT (CM)

DD MM YYYY JULIAN M-HEIGHT S-HEIGHT

17	6	1985	168	36.000	22.300
26	6	1985	177	56.000	47.000
2	7	1985	183	68.000	66.000
8	7	1985	189	97.000	74.300
15	7	1985	196	106.000	82.500
26	7	1985	207	179.000	135.000
2	8	1985	214	182.000	159.000
9	8	1985	221	189.000	178.000

RMSE	R2	SLOPE	INTERCEPT	T-SLOPE	T-5% CONFIDENCE
22.154	0.962	0.887	-5.752	-1.566	2.447

===

LAI

DD MM YYYY M-LAI S-LAI

17	6	1985	168	0.530	1.080
2	7	1985	183	1.590	2.370
15	7	1985	196	2.550	1.760
2	8	1985	214	3.250	3.050
21	8	1985	233	2.940	2.950

RMSE	R2	SLOPE	INTERCEPT	T-SLOPE	T-5% CONFIDENCE
0.561	0.693	0.622	0.891	-1.581	3.182

===

PLANT HARVEST YIELD (KG/HA)

DD MM YYYY JULIAN M-YIELD S-YIELD

17	9	1985	260	9854.000	9261.000	1

RMSE	R2	SLOPE	INTERCEPT	T-SLOPE	T-5% CONFIDENCE
593.000	-99.000	-99.000	-99.000	-99.000	-99.000

Appendix 9 continued...

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PLANT BIOMASS (KG/HA)

DD	MM	YYYY	JULIAN	M-BIOM	S-BIOM	Crop	M-Hindex	S-Hindex
17	6	1985	168	435.000	734.190	1	22.65	12.61
2	7	1985	183	2877.000	2417.820	1	3.43	3.83
15	7	1985	196	3396.000	3166.210	1	2.90	2.92
23	7	1985	204	8241.000	5299.220	1	1.20	1.75
21	8	1985	233	11258.000	13013.110	1	0.88	0.71

RMSE	R2	SLOPE	INTERCEPT	T-SLOPE	T-5% CONFIDENCE
1554.841	0.875	1.023	-437.848	0.105	3.182

COMPARED CONCENTRATION OF SWC OVER SAMPLING PERIOD(PPM FOR CHEMICALS)

DD	MM	YYYY	JULIAN	DEPTH	BD	EXP-SWC	PRED-SWC
13	6	1985	164	30.000	1.330	0.191	0.195
13	6	1985	164	60.000	1.320	0.213	0.224

Appendix 10:
RZWQM2 PEST INPUT AND OUTPUT FILES

Pest input and output files are located in a PEST directory under your scenario. This directory is created as soon as you access PEST from the main menu and enter input for weights and parameters. The rzzzzz.005 binary file is created in the scenario directory to store interface entered pest input data. This file is used to create the input files for PEST located in the PEST subdirectory in the scenario.

PEST Input Files:

- Template Files (*.TPL)
  - rzdat.tpl
  - cul1.tpl
  - Par2par.tpl - translation for soil parameters in RZWQM.DAT file
  - p2pcul1.tpl – translation for crop parameters in *.CUL file
- Instruction Files
  - Main.ins
- Input File
  - par2par.in
  - p2pcul1.in
- Main Pest Control File
  - Main.pst

PEST Output Files:

- PEST Pre-run Check file
  - Pstchk.txt
- Record of the Optimization Run
  - Main.rec
- Last best Optimized Parameters
  - Main.par
- Sensitivities
  - Parameters – Main.sen
  - Experiment Observations – Main.seo
- Residuals
  - Main.res
- Binary output not readable
  - Main.jco – jacobian matrix produced after optimization

PEST running Non-Linear Uncertainty generating parameter cases
Input:

- Cov.mat – Covariance matrix
- Case.jco – Jacobian matrix (binary)
- Case#.par – Parameters for the case generated (# is number of the case)

RZWQM Output Added

- Parameters.unc – lists by case # parameters for case
- Rmse.unc – lists by case # phi and rmse for each observation group (yield,lai…)
- Output.unc – lists by case # experimental observations

RZWQM Run with Uncertainty

Upon completion of the nonlinear uncertainty analysis steps you may return to the scenario view. Under Run you may choose Run with Uncertainty. This will create a directory under the scenario for output called ***UncFiles.*** Under UncFiles will be the statistics calculated for output variables in the layer.plt files and the *.ana files. A subdirectory called ***casedata*** will have the output files on which the statistics were calculated.

Input files:
PEST/Case#.par files - has the input parameters that produced the result

Intermediate Output Files:
Case#.ana
Case#.plt

Output files:
ANALYSIS_MEAN.ANA
ANALYSIS_RANGE.ANA
ANALYSIS_STDDEV.ANA
LAYER_MEAN.PLT
LAYER_RANGE.PLT
LAYER_STTDEV.PLT

Appendix 11:
RZWQMREADME.PDF

This document is located in the RZWQM2 <install>\ DOCUMENTATION directory and is updated with every web release. An excerpt of it is shown below.

RZWQM Readme

Root Zone Water Quality Model 2 Version 4.2
USDA ARS Rangeland Resources and Agricultural Systems
Research Unit

This readme contains the material from RZWQM2 What's New document along with more detailed information concerning bug fixes and new features. It covers all major issues and changes since the release of the original RZWQM98 program. While many of these features are explained in the RZWQM2 User's Guide in the Documentation directory, they are highlighted here for ease to new and current user alike.

Compatibility

RZWQM2 is Windows 7/8/10 compatible. Server Domain Security may require admin privileges to install and run from root directory. Private computers require only admin to install and one-time Beopest activation.

Installation Options

Register and Download RZWQM2 from the ARS Ag Software web site:
https://www.ars.usda.gov/research/software/download/?softwareid=412#downloadForm

RZWQM2 User Guide:

RZWQM2userguide.pdf located in …\RZWQM2\Documentation.

New Users:

Creating a New Project from default is recommended (RZWQM2 User Guide Chapter 4) or open example Projects and SAVE AS a new project. Store project outside the RZWQM2 install directory.

Previous Users:

CRITICAL: Before uninstalling RZWQM2, backup any projects or modified files you may have in the RZWQM2 install directories.

Uninstall Previous RZWQM2 application, If RZWQM2 directory still present RENAME it to "OLDRZ2", Install Current Version, Open Your Projects and Scenarios and "Import" the Scenario If Prompted

- Previous versions must be uninstalled before installing the current version of RZWQM2. The BIN, DATA, DATABASES, PROJECTS and STARTUP directories always install with RZWQM2. Do not put your projects in these directories.
- Users previous to version 2.6: DSSAT CSM Corn, Sorghum, and millet users must provide

2 additional height variables in the *.cul file for RZWQM2. Revisit screen to check default values.

- Users previous to version 2.0: perform an IMPORT of the RZWQM.DAT files when prompted: this updates all the scenario information to the current data formats. RUN and SAVE SCENARIO to finalize this update. After updating, the scenario is available for use within the current install
    - Previous Quick Plant/Turf users before version 1.8: you must re-visit Edit QCK* Parameters and check the new default entered input required and SAVE to rewrite the QCK*.DAT files. Then RUN and SAVE SCENARIO to upgrade.
    - Previous DSSAT CSM Millet (Pearl) Users before version 1.7: after IMPORT, you must re-edit DSSAT CSM parameters to remove previous millet tab with remove cultivar and add millet-prl on both DSSAT CSM dialogs O.K. Then reselect available crops for simulation and APPLY (for all management events to be updated; irrigation events must be updated by hand) and O.K. from management, before RUN and SAVE scenario.
    - RZWQM2 1.0 scenarios that used DSSAT3.5 CSM crops are not compatible with the current DSSAT4.0 CSM database. (RZWQM2 was updated to DSSAT4.0 CSM from DSSAT3.5 CSM). There are two options to fix this:

1. Manually add your crop to the DSSAT4.0 CSM Database (hand editing files)
(See CUSTOMZING A DSSAT DATABASE below)

Or

2. Delete the *.cul and *.rzx files, and then choose the same or a similar crop from the new DSSAT4.0 CSM database.
(See INPUT IMPORT DSSAT3.5 CSM item below)

Important Info:

- **RZWQM2 install always writes the most current version of its execution files (BIN), databases and data and projects examples to the install directory**
    - **Customized databases or projects you located in the install directory should be backed up BEFORE uninstalling RZWQM2.**
- **<u>Never use the example project in RZWQM2\Projects\ as your working copy or nest other projects in this directory</u>**.
    - See chapter 4 of the user guide for information related to setting up your own project and the initial setup of a scenario.
- If you would like to share your new cultivar (*.cul file) with the RZWQM2 application contact us at RZWQMsupport@ars.usda.gov.

What's New? : Version 4.2

- The sunflower, alfalfa, Bermuda grass modules from DSSAT are added.
- Brooks-Corey parameters may be estimated from 1/10 and 15 bar soil water contents.
- P module is added and tested by Dr. Zhiming Qi, McGill University, Canada.
- Chemical equilibrium module is alive and ready for testing.
- Auto-irrigation based on canopy temperature is added, but need testing.

What's New? : Version 4.0

- Hermes Sucros Crop Model and databases, N fixing nodules in Hermes and Generic Crops
- DSSAT CSM beet sugar (BS)
- Alfalfa with multiple harvest events (event 2 thru n has 0.0 plant population)

- Alternative to S-W PET Option to use ASCE Standardized reference crop (short or tall)
- SHAW AET as a plant water stress option
- GUI Bug fixes in Pesticides Ancestory Application Parent, Daughter, Granddaughter (WARNING No Copy Paste Allowed in Pesticides) and expdata below ground biomass entry
- Updates project summary, ana output, et deficit based on active rooting zone

What's New? : Version 3.41

- Project level parameter estimation/optimization (P-EST) across scenarios
- Update Science for plastic mulch

What's New? : Version 3.40

- Generic Alfalfa crop
- SHAW ET and plastic mulch

What's New? : Version 3.29

- Project summary updates in statistics computation and annual water/N mass balance output added to seasonal button.
- Harvest efficiency implementation for total aboveground biomass change
- Soil water redistribution occurs 24 hours a day

What's New? : Version 3.28

- GUI error fix project summary statistics
- Science update residue output

What's New? : Version 3.27

- GUI error fix on Meteorology|Edit *.met file and save
- Science update for over 100 years climate

What's New? : Version 3.25

- Science update for resetting soil water content in simulation and proso millet in rotation
- RZWQM2 P-EST update to optimize Potato and Millets
- GUI Site Description Horizon update for initializing P-EST horizons

What's New? : Version 3.20

- Science update for potato module
- GUI Project level Meteorology Create/Edit redesign

What's New? : Version 3.10

- Science update in comp2exp.out and add file horizon.out
- GUI vector variable output changed from Sat Hydr Conductivity to Hyd Conductivity, customize horizon.out depths in experiment data input.

What's New? : Version 3.00

- Public Version includes: Input Parameter Optimization, Non-Linear uncertainty analysis and running scenarios with uncertainty for confidence band statistics for output. Irrigation Rules with daily, seasonal, monthly or over year limitations. Sub irrigation available. Massbalance of Carbon, Nitrogen, Water and Pesticides. GHG emissions and CO2 effects. Options added SHAW ET model running with observed canopy or model, ability to reset to observed soil water during the run and new plant water stress algorithms. Sediment Erosion. Storm data smoothing tools. Multiple Crop Growth models. Observed vs. Predicted Performance statistics and graphics.

What's New? : Version 2.99

- Update Performance Graphs, GUI Text Changes, RZWQM2_PEST_run.bat

What's New? : Version 2.95/2.96

- Science Update Total Residue C replaces Total Humus N in the *.ANA file Variable #80

What's New? : Version 2.94

- Generic Crop Interface, Crop Selection screen upgrades

- Project summary average statistic excludes zero values
- Science update for the erosion model.

What's New? : Version 2.90

- Distributed Input of soil attributes
- Integration of the GLEAMS sediment erosion model

What's New? : Version 2.85

- Added planting options for 100cm soil profile water and planting window

What's New? : Version 2.84

- Science update for carbon mass balance
- New GUI tool to normalize breakpoint storm data in the meteorology section
- Nutrient Initialization Wizard input change to %OM by layer input (previously a fraction)
- New default of 0.0016 for the Nutrient Reaction Rates Nitrification N2O fraction.

What's New? : Version 2.82

- Science update for specific irrigation and sub-irrigation and water limitations
- Bug fixes for the multi-crop optimization and non-log transformed parameters uncertainty analysis

What's New? : Version 2.80

- Science updates
- Bug fixes for the uncertainty analysis
- New ANA values added to the science and GUI

What's New? : Version 2.70

- GUI and Science cosmetic bug fixes
- Additional variable of ambient CO2 added to the Site Description screen
- Simulated vs Observed and Time series performance graphs added
- Interface upgrades to Spread 8.0 and Graphic Essentials 7
- Hourly wind run default value changed, may impact soil water
- Soil surface resistance limit removed
- Updates in Science, Analysis.exe and Create breakpoint from hourly data
- In the Pesticides section: multiple formation pathways and percent values now allowed in Daughter Products
- Science update: Total CO2 added to output variable selection
- Performance improvement for "Run with Uncertainty" (Parallel processing)
  Experimental data input for plant LAI and Height expanded to 500 records
- Project level file export of scenarios to USDA CSIP platform
- Upgrades for compatibility with software restriction policies (SRP)
- "Run with Uncertainty" live
- Uncertainty sampling using either Latin hypercube or random sample methods
    - Experimental observations can be normal or log transformed.
- PEST parameter group name change for NH3 and CropGro crops.
- Precision expanded for porosity and Brooks-Corey soil properties
- Frozen soil and Greenhouse Gas Variable additions to project summary comparisons and *.ana file

What's New? : Version 2.60

- Experimental data soil water content (swc) can be reset for each horizon.
    - **This required Expdata.dat format change for the swc section**.
    - Please revisit Input Experiment Data Soil Water Content tab, re-select reset choice yes or no and O.K. to rewrite expdata.dat
- Irrigation:
    - Sub-irrigation type for all methods and depths

  - Minimum and maximum daily irrigation rate for rule based methods
  - Root Zone Depletion irrigation science bug fixed
- File Menu Save simplified to windows standard, EXIT will ask if you want to save before exit.
- Options added to specify Field Capacity and Wilting Point pressure heads.
- Phenology input and output
- Soil surface resistance updated
- More observed LAI and HT observed data allowed
- Dead-end pores allowed to be 0.0 with Macropore cracks
- PEST optimization and uncertainty analysis enabled

What's New? : Version 2.50

- PEST GUI, code and binary file I/O updated
- Expdata Soil Water Added Switch to reset soil water to measured values
- Irrigation option to limit irrigation to maximum monthly amounts, IO change
- YIELD IMPACT CHANGE:
  - Please EDIT DSSAT parameters if using Corn, Sorghum, or Millet
    - These crops now require 2 additional variables for height simulation.
    - Add values for these new variables in the DSSAT CSM screen and O.K. to rewrite the RZWQM2 *.cul file
- Science code now compiled using the Intel Visual FORTRAN 2011 compiler
- PET SHAW and PEST Options enabled
- New plant water stress options on PET Screen
- Bug fix: multiple iteration run was not reading the state.bin
- Bug fix: saves on Nutrient advanced parameters n20 and methane production
- N2O,NxO,and CH4 Added to Observed data and simulation output variables
- Variables added in QuickPlant and QuickTree
- Variables added for Greenhouse Gas simulation
- Value Range Change: Row space 1-1000cm, Stomatal Resistance 1-2000s/m

What's New? : Version 2.42

- Public Web update bug fix. PEST Disabled.

What's New? : Version 2.41

- Parameter Estimation Model (PEST) available in RZWQM2.

What's New? : Version 2.4

- Science change in Minimum Input method and Macropore code see below.
- Science change adds user access to soil surface resistance (PET screen) and minimum pressure head for water Movement (AdvRichardsEqu screen).
- Soil Physical Properties default microporosity changed from .1(or 0) to ratio using 15bar water content. New option to select the pressure head to calculate microporosity fraction
- New ET deficit rule based irrigation scheduling
- Interface includes EXPDATA.DAT data entry spreadsheets, click INPUT|
- EXPERIMENTAL DATA to load previous new scenario expdata.dat format and OK; a new standard format written (see "Input Experiment Data" below).
- Updated HTML format Help System, support for *.HLP files no longer needed
- RZWQM2 interface upgraded to MSDEV 2010 platform
- Project Summary has new scenario comparison features

….

Root Zone Water Quality Model Book Available at :
http://www.wrpllc.com/books/rzwqm.html

CUSTOMIZING A DSSAT40 CSM Cultivar Database

- Path to your RZWQM install directory\DATABASES\DSSAT and make a backup copy of the *040.cul file you wish to add to.
- Add a unique number for your cultivar and follow the format listed in the file for the crop. A maize example is below:
  @VAR# VRNAME.......... ECO# P1 P2 P5 G2 G3 PHINT
  PC0001 2500-2600 GDD IB0001 160.0 0.750 780.0 750.0 8.50 49.00
- If you are loading a previous RZ DSSAT Scenario the number and name added to the cultivar database must match exactly the tab name that appears in your scenario or a problem will be thrown. If this occurs exit the program. Use IMPORT and DELETE DSSAT35 FILES and EDIT DSSAT to select valid DSSAT40 crops from the crop selection, add cultivar, modify values, select simulation control by species and modify and O.K. on all dialogs. If the tab name matches, please return to edit DSSAT and check data and O.K. on DSSAT screens to write the *dssat.rzx file with the verified data.
- The RZWQM Interface will use the DSSAT40 cultivar files to load the combo box for the DSSAT Selection Screen.

RZWQM2 HELP System:

…

RZWQM2 PROJECT LEVEL:

…

RZWQM SCENARIO LEVEL:

…

Science Updates:

1. Maximum soil profile and Water Table is extended from 3 m to 30 m; there is a 300 numerical node limit.
2. Crop growth models are updated from DSSAT3.5 to DSSAT4.0.
3. A pseudo-lateral flow component is added for subsurface flow below drainage tiles. Pesticide may move directly to drained tiles via a small fraction of the macro pores.
4. Soil microbial growth may be disabled in Simulation Controls Shutdown Model
5. Brooks-Corey soil hydraulic property may be calculated from 1/3 and 15 bar soil water contents and from 1/3 bar only.
6. Plant root growth factor in DSSAT4.0 may be calculated from maximum rooting depth and an exponential constant.
7. Complete Soil Carbon balance and mass balance output
8. Update to reset run each year to initial soil condition of the simulation (Simulation Control Option)
9. Execute transplants with required data and harvest by date format
10. Science can compare predicted results to observed data in an experiment data file if present (example in STARTUP\Expdata.dat) or a DSSAT Crop has an observed data file (example Expdssat.*)
11. Equilibrium Chemistry Model Component currently deactivated.
12. Crop growth module/parameters added for Foxtail and proso-millet and Triticale.
13. User specified root zone depletion depth added to Irrigation Scheduling Default of 0.0 means the active rooting zone depth is the depletion zone.
14. Manure Fertilizer methods “incorporated surface broadcast” will default a tillage method with a field cultivator if the user did not specify one.

15. Updates for Quickplant, Quickturf, Quicktree and Pesticide modules.
16. Updates for Water, N and Pesticide mass balances.
17. Ver. 2.0 New routine for surface decomposition of residues replace worm-incorp, new routine to handle radiation coming in directly perpendicular to the surface handling incident radiation at high slope and aspect areas correctly. Determine Texture Code routine corrected to return a 27% clay as clay loam NOT loam.
18. Version 2.01: Error fixes in irrigation via root zone depletion with a user entered depletion depth, irrigation water required to meet recharge now includes credits for soil layers exceeding plant available water thus reducing the amount needed to recharge the depletion zone.
19. Version 2.1: Science simulation responds to Volatilization Constant input. Detailed Seepage and Drainage Output added in MBLP*.OUT (pesticide mass balance) files. Hourly ET calculation uses hourly soil temperature.
20. Version. 2.4: Code error corrected in minimum input method running with Minimum Input Radial checked, and in Macropore absorption algorithm affecting chemicals (pesticide or nitrogen).
21. Microporosity unchecked ran zero; and, .1 when checked initially. New default of -15bar applies for unchecked and newly checked microporosity box. Test projects show a <=1.2% affect on yield. Option added to select microporosity at a specified pressure head.
22. Soil Water Pressure variable added to EXPDATA enabling comparisons in COMP2EXP.OUT
23. Soil Surface resistance variable affecting S-W potential evaporation exposed to user control (default to 37) and option if -1 science adjusts resistance as a function of wind speed. User access to minimum soil pressure head added in Advanced Richards Equation dialog.
24. Version 2.5.0: Science added option to limit irrigation to a monthly maximum. A default of 0.0 for all 12 months of the year means no limits enforced.
25. Version 2.5.x:* Plant Height algorithm changed may affect previous yield predictions for corn, sorghum, and all millets. Max Plant height and plant weight at ½height required in the RZWQM2 cultivar files for these crops. Three options for plant water stress calculation added to PET section.
26. Version 2.6.x: Phenology added to comp2exp.out. Irrigation rule cumulative ET fix.
27. Version 2.60.5: Irrigation depletion first rule entry execution fixed.
28. Version 2.70.5: Science and GUI IO change Multiple Pathways and Percentages allowed in the formation of daughter products.
29. Version 2.70.6: Hourly wind run default change impacts soil water.
30. Version 2.82.x: Sub-irrigation mass balance error fix. Specific irrigation fix if starting simulation in the middle of the record of specific events, amounts were misaligned.
31. Version 2.83.x: C mass balance error fix.
32. Version 2.85.x: Planting date window and soil water content of 100cm.
33. Version 2.86.x: Complete GLEAMS sediment erosion hookup.
34. Version 2.93.x: Update GLEAMS code.
35. Version 2.95.x: Ana output variable #80 change to Total Residue C
36. Version 3.00.0: Public Release Compile
37. Version 3.10.0: Add simulated output by horizon for all soil layer variables measured.
38. Version 3.20.0: Potato module corrections, DSSAT CSM residue correction
39. Version 3.25.0: Corrections on Reset Soil Water Content During Simulation mass balance error and Proso Millet ana file yield in rotations
40. Version 3.27: Science update to handle over 100 years climate
41. Version 3.28: Residue Output added
42. Version 3.29: Soil Water redistribution not suspended during infiltration occurs 24 hours per day.
43. Version 3.40: Alfalfa and SHAW ET
44. Version 3.41: Bug fix on plastic mulch
45. Version 4.0 BETA: Add for Hermes Sucrose model testing, Add DSSAT CSM sugarbeet (BS), additional ana output, ET deficit rule change water applied base on soil water active rooting zone not whole soil profile and decline of LAI after maturity simulated.
46. Version 4.0 Beta Hermes update.
47. Version 4.0 Kappa Bug fix for Alfalfa
48. Version 4.0 Beta Hermes update.
49. Version 4.0 Kappa Bug fix for Alfalfa

50. Version 4.0 Lambda Update DSSAT-CSM Beet Sugar code and cultivar database
51. Version 4.0 Mu Update Generic Model plgen.dat for g/plant or kg/ha input units
52. Version 4.0 Nu Update Generic and Hermes Model Phenology output.

Important Issues/Limitations

- **You must be an administrator to install RZWQM2. Due to its root directory installation Server Domain Security may require you to RUN RZWQM2 as an Administrator.**
- For Maize, Sorghum, and Millets the user must enter valid Height at Max LAI and Biomass at ½ Max Height on cultivar (Variables in Addition to DSSAT-CSM). Defaults may not be applicable to your cultivar.
- You must have a an Administrator User Account to install/uninstall RZWQM2
- This is also required to take advantage of BEOPEST for optimization and uncertainty analysis for a one-time question to access a port. Beopest will use multiple cores to parallelize the pest process but within RZWQM2 it is hardcoded to use only the current machine IP address.
- RZWQM2 P-EST can optimize up to 5 different crops in Rotation. Only 1 cultivar per crop per optimization. Different cultivars of the same crop require their own optimization run.
- Extremely long paths (over 120 characters for the base project level) may cause component modules to fail.
- When deleting a DSSAT CSM crop from simulation, please remove from “Crops Selected” box BEFORE you “Edit DSSAT crops” and “Remove Cultivar” in the DSSAT edit dialog. REMOVE DSSAT cultivar will delete their files from your scenario and from the “Available Crops” selection screen; it does not remove it from “Crops Selected” for the run.
- AutoPlot files plant0.plt – plant5.plt, pltprod.plt and nuptak.plt were designed to run for the RZWQM generic plant model these will not display with plot data for DSSAT4.0 CSM crops.

Disclaimer

The use of trade, firm, or corporation names in any of the RZWQM2 model is for the information and convenience of the reader. Such use does not constitute an official endorsement or approval by the U.S.D.A. Agricultural Research Service, of any product or service to the exclusion of others that may be suitable.

***RZWQM2 is a research level simulation model for Agricultural Cropping Systems, proper training in its use is highly recommended. *****Last Update 10/5/2020**